

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Doris Johnson Hines
Administrative Law Judge**

In the Matter of

CERTAIN WEARABLE DEVICES

Investigation No. 337-TA-1478

COMPLAINANTS' IDENTIFICATION OF EXPERT WITNESSES

Pursuant to the Procedural Schedule set in Order No. 8, and Ground Rule No. 2, Complainants Samsung Electronics Co., Ltd. and Samsung Electronics America, Inc. (collectively, “Samsung” or “Complainants”), by and through their counsel, hereby provide identification of expert witnesses who may offer testimony at the hearing in this matter. As this Investigation is ongoing, Complainants reserve the right to identify additional expert witnesses and/or expected areas of testimony as further information becomes available, including in response to Respondents’ expert disclosures and expert reports. Complainants also reserve the right to call any expert witness that a party deposes, any expert identified or called to testify at the hearing by Respondent, and/or any expert for impeachment purposes, even if not identified below.

Complainants have retained the following experts to provide expert testimony on their behalf in this investigation. Complainants reserve the right to rely upon testimony from any of the experts identified herein.

1. Dr. Gregory D. Abowd

Dr. Gregory D. Abowd has a Ph.D. in electrical engineering from the University of Oxford, United Kingdom. Dr. Abowd is an expert in some of the technical matters at issue in this case, including but not limited to mobile and wearable devices, monitoring of health and biometric signals, sensor systems design (including for biosensors and light-based sensors), signal

processing, industrial and mechanical design, printed circuit board designs, wireless antenna and high frequency transmission circuits, and software for sensors and measurement and calculation of biometric signals. Dr. Abowd's curriculum vitae, attached hereto as Exhibit 1, sets forth his qualifications. Dr. Abowd is expected to testify regarding technical matters relating to, at least, one or more of the Asserted Patents, the subject matter of the Asserted Patents, the state of the art at the time of the invention of the Asserted Patents, the validity and/or enforceability of the Asserted Patents, the construction of claims and claim terms of the Asserted Patents, the functionality, design, and operation of products accused of infringement in this Investigation, Respondents' infringement of the Asserted Patents, and the proof of the required domestic industry for the Asserted Patents (technical prong). Additionally, Dr. Abowd may rebut expert witness testimony and reports of Respondents' experts.

2. Dr. Kimberly K. Cameron

Dr. Kimberly K. Cameron has a Ph.D. in mechanical engineering from Sanford University. Dr. Cameron is an expert in some of the technical matters at issue in this case, including but not limited to mobile and wearable devices, monitoring of health and biometric signals, sensor systems design (including for biosensors and light-based sensors), signal processing, industrial and mechanical design, printed circuit board designs, wireless antenna and high frequency transmission circuits, and software for sensors and measurement and calculation of biometric signals. Dr. Cameron's curriculum vitae, attached hereto as Exhibit 2, sets forth her qualifications. Dr. Cameron is expected to testify regarding technical matters relating to, at least, one or more of the Asserted Patents, the subject matter of the Asserted Patents, the state of the art at the time of the invention of the Asserted Patents, the validity and/or enforceability of the Asserted Patents, the construction of claims and claim terms of the Asserted Patents, the functionality, design, and operation of products accused of infringement in this Investigation, Respondents' infringement of

the Asserted Patents, and the proof of the required domestic industry for the Asserted Patents (technical prong). Additionally, Dr. Cameron may rebut expert witness testimony and reports of Respondents' experts.

3. Mr. Robert S. Plachno

Mr. Robert S. Plachno has a Master's degree in computer engineering from the University of Illinois. Mr. Plachno is an expert in some of the technical matters at issue in this case, including but not limited to mobile and wearable devices, monitoring of health and biometric signals, sensor systems design (including for biosensors and light-based sensors), signal processing, industrial and mechanical design, printed circuit board designs, wireless antenna and high frequency transmission circuits, and software for sensors and measurement and calculation of biometric signals. Mr. Plachno's curriculum vitae, attached hereto as Exhibit 3, sets forth his qualifications. Mr. Plachno is expected to testify regarding technical matters relating to, at least, one or more of the Asserted Patents, the subject matter of the Asserted Patents, the state of the art at the time of the invention of the Asserted Patents, the validity and/or enforceability of the Asserted Patents, the construction of claims and claim terms of the Asserted Patents, the functionality, design, and operation of products accused of infringement in this Investigation, Respondents' infringement of the Asserted Patents, and the proof of the required domestic industry for the Asserted Patents (technical prong). Additionally, Mr. Plachno may rebut expert witness testimony and reports of Respondents' experts.

4. Mr. Julia Rowe

Ms. Julia Rowe has a Master's degree in accountancy from DePaul University and is a Managing Director at Kroll. Ms. Rowe is an expert in the application of economics to intellectual property and international trade and has extensive professional experience in applying financial, accounting, and economic principles to complex business disputes, including significant

experience in disputes involving intellectual property and matters brought before the International Trade Commission. Ms. Rowe's curriculum vitae, attached hereto as Exhibit 4, sets forth her qualifications. Ms. Rowe is expected to testify regarding matters relating to, at least, the economic prong of domestic industry, remedy, bond, as well as issues of license and lack of objective evidence of non-obviousness. Additionally, Ms. Rowe may rebut expert witness testimony and reports of Respondents' experts.

5. Dr. Steve Warren

Dr. Steve Warren has a Ph.D. in electrical engineering from the University of Texas at Austin. Dr. Warren is an expert in some of the technical matters at issue in this case, including but not limited to mobile and wearable devices, monitoring of health and biometric signals, sensor systems design (including for biosensors and light-based sensors), signal processing, industrial and mechanical design, printed circuit board designs, wireless antenna and high frequency transmission circuits, and software for sensors and measurement and calculation of biometric signals. Dr. Warren's curriculum vitae, attached hereto as Exhibit 5, sets forth his qualifications. Dr. Warren is expected to testify regarding technical matters relating to, at least, one or more of the Asserted Patents, the subject matter of the Asserted Patents, the state of the art at the time of the invention of the Asserted Patents, the validity and/or enforceability of the Asserted Patents, the construction of claims and claim terms of the Asserted Patents, the functionality, design, and operation of products accused of infringement in this Investigation, Respondents' infringement of the Asserted Patents, and the proof of the required domestic industry for the Asserted Patents (technical prong). Additionally, Dr. Warren may rebut expert witness testimony and reports of Respondents' experts.

6. Dr. Gregory Francis Welch

Dr. Gregory Francis Welch has a Ph.D. in computer science from the University of North Carolina at Chapel Hill. Dr. Welch is an expert in some of the technical matters at issue in this case, including but not limited to mobile and wearable devices, monitoring of health and biometric signals, sensor systems design (including for biosensors and light-based sensors), signal processing, industrial and mechanical design, printed circuit board designs, wireless antenna and high frequency transmission circuits, and software for sensors and measurement and calculation of biometric signals. Dr. Welch's curriculum vitae, attached hereto as Exhibit 6, sets forth his qualifications. Dr. Welch is expected to testify regarding technical matters relating to, at least, one or more of the Asserted Patents, the subject matter of the Asserted Patents, the state of the art at the time of the invention of the Asserted Patents, the validity and/or enforceability of the Asserted Patents, the construction of claims and claim terms of the Asserted Patents, the functionality, design, and operation of products accused of infringement in this Investigation, Respondents' infringement of the Asserted Patents, and the proof of the required domestic industry for the Asserted Patents (technical prong). Additionally, Dr. Welch may rebut expert witness testimony and reports of Respondents' experts.

7. Dr. Darrin J. Young

Dr. Darrin J. Young has a Ph.D. in electrical engineering from the University of California at Berkeley. Dr. Young is an expert in some of the technical matters at issue in this case, including but not limited to mobile and wearable devices, monitoring of health and biometric signals, sensor systems design (including for biosensors and light-based sensors), signal processing, industrial and mechanical design, printed circuit board designs, wireless antenna and high frequency transmission circuits, and software for sensors and measurement and calculation of biometric signals. Dr. Young's curriculum vitae, attached hereto as Exhibit 7, sets forth his qualifications.

Dr. Young is expected to testify regarding technical matters relating to, at least, one or more of the Asserted Patents, the subject matter of the Asserted Patents, the state of the art at the time of the invention of the Asserted Patents, the validity and/or enforceability of the Asserted Patents, the construction of claims and claim terms of the Asserted Patents, the functionality, design, and operation of products accused of infringement in this Investigation, Respondents' infringement of the Asserted Patents, and the proof of the required domestic industry for the Asserted Patents (technical prong). Additionally, Dr. Young may rebut expert witness testimony and reports of Respondents' experts.

8. Dr. Matthew Schurman

Dr. Matthew Schurman has a Ph.D. in materials science and engineering from Rutgers University. Dr. Schurman is an expert in some of the technical matters at issue in this case, including but not limited to mobile and wearable devices, monitoring of health and biometric signals, sensor systems design (including for biosensors and light-based sensors), signal processing, industrial and mechanical design, printed circuit board designs, wireless antenna and high frequency transmission circuits, and software for sensors and measurement and calculation of biometric signals. Dr. Schurman's curriculum vitae, attached hereto as Exhibit 8, sets forth his qualifications. Dr. Schurman is expected to testify regarding technical matters relating to, at least, one or more of the Asserted Patents, the subject matter of the Asserted Patents, the state of the art at the time of the invention of the Asserted Patents, the validity and/or enforceability of the Asserted Patents, the construction of claims and claim terms of the Asserted Patents, the functionality, design, and operation of products accused of infringement in this Investigation, Respondents' infringement of the Asserted Patents, and the proof of the required domestic industry for the Asserted Patents (technical prong). Additionally, Dr. Schurman may rebut expert witness testimony and reports of Respondents' experts.

9. Mr. Gregory Kesden

Mr. Gregory Kesden has a master's degree in computer science from Clemson University. Mr. Kesden is an expert in some of the technical matters at issue in this case, including but not limited to mobile and wearable devices, monitoring of health and biometric signals, sensor systems design (including for biosensors and light-based sensors), signal processing, industrial and mechanical design, printed circuit board designs, wireless antenna and high frequency transmission circuits, and software for sensors and measurement and calculation of biometric signals. Mr. Kesden's curriculum vitae, attached hereto as Exhibit 9, sets forth his qualifications. Mr. Kesden is expected to testify regarding technical matters relating to, at least, one or more of the Asserted Patents, the subject matter of the Asserted Patents, the state of the art at the time of the invention of the Asserted Patents, the validity and/or enforceability of the Asserted Patents, the construction of claims and claim terms of the Asserted Patents, the functionality, design, and operation of products accused of infringement in this Investigation, Respondents' infringement of the Asserted Patents, and the proof of the required domestic industry for the Asserted Patents (technical prong). Additionally, Mr. Kesden may rebut expert witness testimony and reports of Respondents' experts.

10. Mr. Daniel Lanovaz

Mr. Daniel Lanovaz has a master's degree in computer science from the University of Alberta. Mr. Lanovaz is an expert in some of the technical matters at issue in this case, including but not limited to mobile and wearable devices, monitoring of health and biometric signals, sensor systems design (including for biosensors and light-based sensors), signal processing, industrial and mechanical design, printed circuit board designs, wireless antenna and high frequency transmission circuits, and software for sensors and measurement and calculation of biometric signals. Mr. Lanovaz's curriculum vitae, attached hereto as Exhibit 10, sets forth his qualifications.

Mr. Lanovaz is expected to testify regarding technical matters relating to, at least, one or more of the Asserted Patents, the subject matter of the Asserted Patents, the state of the art at the time of the invention of the Asserted Patents, the validity and/or enforceability of the Asserted Patents, the construction of claims and claim terms of the Asserted Patents, the functionality, design, and operation of products accused of infringement in this Investigation, Respondents' infringement of the Asserted Patents, and the proof of the required domestic industry for the Asserted Patents (technical prong). Additionally, Mr. Lanovaz may rebut expert witness testimony and reports of Respondents' experts.

11. Opinion Testimony from Other Witnesses

In addition to the expert witnesses identified above, Complainants reserve the right to offer opinion testimony from lay witnesses called at the hearing in this Investigation as to matters within the perception of the lay witness. Complainants further reserve the right to elicit testimony that may be deemed expert testimony from lay witnesses who have the requisite scientific, technical, or other specialized knowledge to provide that testimony and hereby designate those lay witnesses, including the witnesses identified on Complainants' upcoming Tentative List of Witnesses to be Called at the Evidentiary Hearing, as experts with regard to such testimony. These reservations include, but are not limited to, opinion or expert testimony regarding the design, operation, functionality, implementation, manufacture, marketing, sale, or importation of any relevant device, product, or technology.

Dated: April 8, 2026

Respectfully submitted,

/s/ D. Sean Trainor

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EXHIBIT 1

GREGORY D. ABOWD

DEAN

COLLEGE OF ENGINEERING

NORTHEASTERN UNIVERSITY

G.ABOWD@NORTHEASTERN.EDU

Fellow American Academy of Arts&Sciences

Fellow National Academy of Inventors

I. EARNED DEGREES

D. Phil., 1991, University of Oxford, United Kingdom, Computation. Thesis: *Formal Aspects of Human-Computer Interaction*, Mr. Bernard Sufrin (advisor).

M.Sc., 1987, University of Oxford, United Kingdom, Computation. Thesis: *Arabic Word Processing*, Dr. Michael Spivey (advisor).

B.S. (*summa cum laude*), 1986, University of Notre Dame, Honors Mathematics.

II. EMPLOYMENT HISTORY

Dean, College of Engineering, Northeastern University, 2021-present.

Professor, Department of Electrical and Computer Engineering, Northeastern University.

Affiliate appointments in Department of Health Sciences, Bouvé College of Health Sciences and Khoury College of Computer Sciences, 2021-present.

Associate Dean of Research and Space, College of Computing, Georgia Institute of Technology, 2019-2021.

Associate Dean of Off Campus and Special Initiatives, College of Computing, Georgia Institute of Technology, 2017 to 2020.

J.Z. Liang Endowed Chair, School of Interactive Computing, Georgia Institute of Technology, 2016-2021.

Regents' Professor, School of Interactive Computing, Georgia Institute of Technology, 2012-2022.

W. George Professor and Director, Health Systems Institute, Georgia Institute of Technology, 2008-2010.

Full Professor and Distinguished Professor, School of Interactive Computing, Georgia Institute of Technology, 2007-2022.

Associate Professor, College of Computing, Georgia Institute of Technology, 2000-2007.

Visiting Faculty, Intel Research Seattle, July 2004-June 2005.

Founding Director, Aware Home Research Initiative, Georgia Institute of Technology, 2000-2003. 2005-2008.

Associate Director for Broadband Institute in charge of Residential Laboratory, 1998-2003.

Assistant Professor, College of Computing, Georgia Institute of Technology, 1994-2000.

Visiting Scientist, Software Engineering Institute, Carnegie Mellon University, 1994-97.

Postdoctoral Research Associate, Computer Science Department and Software Engineering Institute, Carnegie Mellon University, 1992-1994.

Research Associate, Human-Computer Interaction Group, Computer Science Department, University of York, 1989-1992.

III. HONORS AND AWARDS

1993. SIGSOFT'93/FSE-1 conference paper commended for high technical merit and recommended for extension and submission to ACM TOSEM journal.
1994. One of two nominees by Georgia Tech for NSF Presidential Faculty Fellows Program.
1995. Received Georgia Tech nomination for Packard Foundation Fellowship.
1996. MobiCom'96 paper commended as one of top 9 papers in conference and recommended for submission to *ACM Wireless Networks* journal.
1997. Sigma Xi, Georgia Tech Chapter Young Faculty Research Award.
NSF CAREER award recipient.
College of Computing William A. "gus" Baird Outstanding Faculty Teaching Award.
1998. Intelligent User Interface'98 conference paper nominated as one of top papers at conference and recommended for extension and submission to *Knowledge Based Systems* journal. Georgia Tech, Outstanding Use of Innovative Educational Technology Award.
1999. Georgia Tech, College of Computing, Outstanding Junior Faculty Research Award.
2000. Schlumberger Foundation Technical Fellow, with \$35,000 award.
2001. IBM Research Faculty Fellowship, with \$20,000 award.
Schlumberger Foundation Technical Fellow, with \$25,000 award.
2002. Selected to attend National Academy of Engineering Symposium on Frontiers in Engineering, Irvine, CA.
2005. Best paper award at *Symposium On Usable Privacy and Security (SOUPS)*.
Best of CHI selection for paper presented to *ACM SIGCHI Conference on Human Factors and Computing (CHI 2006)*.
Best paper nominations at *The 4th International Conference on Pervasive Computing*.
Top Technology Idea from *New York Times Magazine*. Awarded to Capture Resistant Environment.
2006. College of Computing Senior Research Award.
Best paper nomination at the *ACM SIGCHI Conference on Computer Supported Cooperative Work (CSCW 2006)*.
2007. Best paper award and best presentation award at Ubicomp 2007.
ACM SIGCHI Social Impact Award.
Georgia Tech Don Bratcher Human Relations Award.
2008. Best paper award at Pervasive Computing 2008.
Elected to ACM SIGCHI Academy.
Georgia Tech Doctoral Dissertation Advisor Award.
Elected Fellow of the Association of Computing Machinery (ACM).
2009. ACM Eugene Lawler Humanitarian Award.
2017. Georgia Tech College of Computing Senior Faculty Research Award
State of Georgia Outstanding Achievement in Autism
2017. UbiComp conference 10-year Impact Award for 2007 publication [C115].
2018. College of Computing Faculty Mentor Award.
2018. College of Computing Dean's Award.
2018. UbiComp conference 10-year Impact Award for 2008 publication [C120].
2019. ISWC conference 20-year Impact Award for 1999 publication [C56].
2022. UbiComp conference 10-year Impact Award for 2012 publication [C150].
2023. ACM SIGCHI Lifetime Achievement Award for Research.
2024. ACM SIGCHI Best Paper Awardee for publication [C216].
2025. ACM SIGCHI Special Recognition "*For his extraordinary ability to inspire and mentor*

individuals from diverse backgrounds and his commitment to fostering collaboration, creativity, and impact.”

2025. Lebanese-American Club of Boston Engineer of the Year.

2025. Elected member of the American Academy of Arts & Sciences

2026. Elected member of National Academy of Inventors

IV. RESEARCH, SCHOLARSHIP, AND CREATIVE ACTIVITIES

*Student co-authors in **boldface**.*

A. PUBLISHED BOOKS, PARTS OF BOOKS, AND EDITED VOLUMES

A1. Books

- B1. Dix, Alan J., Janet E. Finlay, Gregory D. Abowd, and Russell Beale. (Principal contributions from Dix, Finlay and Abowd, with assistance from Beale) *Human-Computer Interaction*. Prentice Hall International, 1993. 2nd edition published in November 1997. 3rd edition published in 2004. **[Since early 2025, this book and its ~10,000 citations does not appear in Google Scholar.**

A2. Refereed Book Chapters

- BC1. Abowd, Gregory D. and Russell Beale. (Significant contributions from both authors) Contextualizing novel research in HCI. In Beale, R. and Finlay, J., editors, *Neural Networks and Pattern Recognition in Human-Computer Interaction*, Chapter 1. Ellis Horwood, 1992.
- BC2. Harrison, Michael, Gregory D. Abowd, and Alan J. Dix. (Significant contributions by all authors with Harrison as principal author) Analysing display oriented interaction by means of systems models. In Byerley, Paul F., Barnard, Philip J. and May, Jon, editors, *Computers, Communication and Usability: Design Issues, Research and Methods for Integrated Services*, pp. 147–163, Elsevier Science, 1993.
- BC3. Bass, Len. Paul Clements, Rick Kazman and Gregory D. Abowd. (significant contributions by all authors, principal authorship by Bass, Clements and Kazman). Analyzing Development Qualities at the Architectural Level: The Software Architecture Analysis Method (SAAM). *Software Architecture in Practice*. Chapter 8. by Addison-Wesley. March 1997.
- BC4. Bass, Len, Paul Clements, Rick Kazman, Gregory D. Abowd, Linda Northrop and Amy Zaremski. (significant contributions by all authors, principal authorship by Bass, Clements and Kazman). Architectural Reviews. *Software Architecture in Practice*. Chapter 9. Addison-Wesley. March 1997.
- BC5. Abowd, Gregory D. and Elizabeth Mynatt (equal contribution from both authors) Charting past, present and future research in ubiquitous computing. In *HCI in the New Millennium*, edited by Jack Carroll, Chapter 23. Addison-

- Wesley, 2001. Adapted from [J.12].
- BC6. **Dey, Anind K.**, Daniel Salber and Gregory D. Abowd. (principal research by Dey with joint authorship by all) A conceptual framework and a toolkit for supporting the rapid prototyping of context-aware applications. Book chapter adapted from [J.16]. Appears in book adapted from special three-issue Human-Computer Interaction (HCI) Journal. special issue on Context-Aware Computing, to appear 2002.
- BC7. Abowd, Gregory D. and **Jason Brotherton** (equal contribution by both authors) eClass.
Chapter 6 in Hazemi, Reza, Stephen Hailes and Steve Wilbur (eds.) *The Digital University: Building a Learning Community*, London: Springer Verlag, 2002, pp.252, ISBN 185233-478-9.
- BC8. **Anind K. Dey** and Gregory D. Abowd (principal authorship by Dey) Support for adapting applications and interfaces to context. In Seffah, Ahmed and Homa Javahery (eds.) *Multiple User Interfaces: Engineering and Application Frameworks*. John Wiley and Sons 2003.
- BC9. Ann L. Horgas and Gregory D. Abowd. (shared authorship). The Impact of Technology on Living Environments for Elderly Adults. In National Research Council report *Workshop on Adaptive Aging*. March 2003, pp. 230-252, ISBN 0-309-09116-0.
- BC10. Gregory D. Abowd and Elizabeth D. Mynatt (principal authorship by Abowd) Designing for the Human Experience in Smart Environments. In *Smart Environments: Technologies, Protocols and Applications*, Diane Cook and Sajal Das (editors), John Wiley. 2004.
- BC11. **Giovanni Iachello** and Gregory D. Abowd, Emerging Ubiquitous Computing Technologies and Security Management Strategy, in *Information Security: Policy, Processes, and Practices*, D.W. Straub, S. Goodman, R.L. Baskerville eds., M.E.Sharpe 2007.
- BC12. **Shwetak N. Patel, Khai N. Truong, Gillian R. Hayes, Giovanni A. Iachello, Julie A. Kientz**, and Gregory D. Abowd (principal research and authorship by all) The Personal Audio Loop: A Ubiquitous Audio-Based Memory Aid. *Handbook of Research on User Interface Design and Evaluation for Mobile Technology*, Joanna Lumsden, Editor.
- BC13. Mynatt, E.D., Abowd, G.D., **Mamykina, L., Keintz, J.A.**, (principal authorship by all) Understanding the Potential of Ubiquitous Computing for Chronic Disease Management, in *Health Informatics: A patient-centered approach to diabetes*, (Asprey, W., Hayes, B., editors), Chapter 2, pp. 85-106. MIT Press, Cambridge, MA, 2010.
- BC14. Abowd, Gregory D., Hayes, Gillian R., Kientz, Julie A., Arriaga, Rosa I., and **Nazneen** (principal research and authorship shared by all) Tools to Support Simplified Capture of Activities in Natural Environments. In Goodwin and Boser (eds.) *Technology and Autism*.
- BC15. N Nazneen, A Rozga, GD Abowd, CJ Smith, R Oberleitner, RI Arriaga, & JS Suri. Behavior Imaging. *Autism Imaging and Devices*, Chapter 16, pp. 345-353. CRC Press, Taylor & Francis Group, Boca Raton, FL. 2017.
- BC16. Thomaz, Edison & Essa, Irfan & Abowd, Gregory. (2017). Challenges and Opportunities in Automated Detection of Eating Activity. *Mobile Health:*

Sensors, Analytic Methods, and Applications. J. Rehg, S.A. Murphy, and S. Kumar (eds). Chapter 26, pp. 151-174. 10.1007/978-3-319-51394-2_9.

A3. Other Parts of Books

None

A4. Edited Volumes

IEEE Pervasive Computing magazine, Volume 1, Issue 3, July-September 2002. Guest Editor for special issue on Context-Aware with Maria Ebling, Guerney Hunt, Hui Lei and Hans Gellersen.

IEEE Pervasive Computing magazine, Volume 4, Issue 2, April-June 2005. Guest Editor for special issue on The Smart Phone: A First Platform for Pervasive Computing with Liviu Iftode and Helena Mitchell.

IEEE Pervasive Computing, Special Issue on Pervasive Computing for Successful Aging (with Sumi Helal and Andrew Sixsmith), vol. 3, no. 2, Apr-June 2004.

IEEE Computer, Special Issue 21st Century User Interface Design (with A. Oulasvirta), vol. 49. no. 7, July 2016.

B. REFEREED PUBLICATIONS AND SUBMITTED ARTICLES

B1. Published and Accepted Journal Articles

- J1. Abowd, Gregory D. and Kozak, John J. (Abowd is principal author, based on research work directed by Kozak) Pattern Development in Cellular Automata Triggered by Site- Specific Reactive Processes, *Physics Letters A*, **127**(3), February 15, 1988.
- J2. Abowd, Gregory D. and Dix, Alan J. (Equal contribution by both authors) Giving undo attention, *Interacting with Computers*, **4**(3):317:342, 1992.
- J3. Abowd, Gregory D., Robert Allen, and David Garlan. (Significant contribution by all authors with Abowd as principal author) Formalizing style to understand descriptions of software architecture. *ACM Transactions on Software Engineering and Methodology*. **4**(4):319-364, October 1995.
- J4. Abowd, Gregory D., Jonathan Engelsma, Luigi Guadagno and Okonon Okon. (Significant contribution from all authors, principal authorship by Engelsma and Abowd) Architectural Analysis of Object Request Brokers. *Object Magazine* special issue on distributed systems, March 1996, pp. 44-51.
- J5. Kazman, Rick., Abowd, Gregory D., Bass, Len. and Clements, Paul. (Significant contributions and authoring by all) Scenario-based analysis of software architecture, *IEEE Software*. **13**(6):47-56, November 1996.
- J6. Dix, Alan and Gregory D. Abowd. (Significant contribution from both authors, principal authorship by A. Dix) Moelling status and event behaviour of interactive systems. *Software Engineering Journal*. **11**(6)334-346, November 1996.

- J7. Abowd, Gregory D. Software Engineering and Programming Language Considerations for Ubiquitous Computing. *Computing Surveys*, 28A(4). in *Computing Surveys*, Vol. 28A, No. 4.
- J8. Abowd, Gregory D., Christopher G. Atkeson, **Anind Dey**, **Jason Hong**, **Sue Long**, **Rob Kooper** and **Mike Pinkerton**. (significant contributions from all authors, principal authorship by Abowd) Cyberguide: A mobile context-aware tour guide. *ACM Wireless Networks*. Volume 3, pages 421-433, November 1997.
- J9. **Dey, Anind**, Gregory D. Abowd, and **Andrew Wood**. Cyberdesk: A Framework for Dynamic Integration of Desktop and Network-based applications. *Knowledge Based Systems Journal*, Volume 11, pages 3-13. 1998.
- J10. Abowd, Gregory D., **Anind K. Dey**, **Robert Orr** and **Jason Brotherton** (Significant contributions by all; authorship by Abowd and Dey). *British VR Journal*, special issue on wearable computing, Volume 3, pages 200-11, 1998.
- J11. Gregory D. Abowd. Classroom 2000: An Experiment with the Instrumentation of a Living Educational Environment. *IBM Systems Journal*. Special issue on HCI / Pervasive computing, Volume 38, Number 4, pp. 508-530, October 1999. See <http://www.research.ibm.com/journal>.
- J12. Gregory D. Abowd and Elizabeth D. Mynatt. (equal contributions by both authors) Charting past, present and future research in ubiquitous computing. *ACM Transactions on Computer-Human Interaction*, special issue on HCI research in the new millenium. Volume 7, Number 1 pp. 29-58, March 2000.
- J13. Gregory D. Abowd and Sterbenz, J. (principal authorship by Abowd) Report on the Inter-Agency Workshop on Research Issues for Smart Environments. *IEEE Personal Communications*, Volume 7, Number 5, pp. 36-40, October 2000.
- J14. **Jennifer Mankoff**, Gregory D. Abowd and Scott E. Hudson (principal research and writing by Mankoff) OOPS: A toolkit supporting mediation techniques for resolving ambiguity in recognition-based interfaces. *Computers and Graphics*, Special issue on Calligraphic Interfaces: towards a new generation of interactive systems. Elsevier Science. Volume 24, Number 6, pp. 819-834, December 2000.
- J15. Maria da Graca Pimentel, Yoshihide Ishiguro, **Bolot Kerimbaev**, Gregory D. Abowd and Mark Guzdial (principal research and writing by Pimentel, Ishiguro and Abowd) Supporting long-term educational activities through dynamic web interfaces. *Interacting with Computers*, special issue on interfaces for the active Web. Volume 13, Number 3, pp. 353-374, February 2001.
- J16. **Anind K. Dey**, Daniel Salber and Gregory D. Abowd. (principal research by Dey with joint authorship by all) A conceptual framework and a toolkit for supporting the rapid prototyping of context-aware applications. *Human-Computer Interaction (HCI) Journal*. Anchor article of a special triple issue on Context-Aware Computing, Vol. 16, Numbers 2-4, 2001. Also published separately by Lawrence Erlbaum as edited book in January 2002, ISBN 0805896856.
- J17. Gregory D. Abowd, Elizabeth D. Mynatt and Tom Rodden. (principal writing by Abowd) The human aspect of ubiquitous computing. *IEEE Pervasive Computing*.

- Volume 1, Number 1, pp. 48-57, 2002.
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 - J19. **Anne R. Jacobs** and Gregory D. Abowd (equal contribution on research and writing) A framework for comparing perspectives on privacy and pervasive technologies. *IEEE Pervasive Computing Magazine*. Volume 2, Number 4, October-December, 2003, pp. 78-84.
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- C107. **Shwetak N. Patel, Khai N. Truong** and Gregory D. Abowd (principal research by Patel, writing by all) Powerline positioning: A practical sub-room-level indoor location system for domestic use. *International Symposium on Ubiquitous Computing (UbiComp 2006)*, Irvine, California, September 2006, Springer-Verlag LNCS Volume 4206, pp 123–140.
- C108. **Shwetak N. Patel, Julie A. Kientz, Gillian R. Hayes, Sooraj Bhat** and Gregory D. Abowd (principal research by Patel and Kientz, writing by all) Farther than you may think: An empirical investigation of the proximity of users to their mobile phones. *International Symposium on Ubiquitous Computing (UbiComp 2006)*, Irvine, California, September 2006, Springer-Verlag LNCS Volume 4206, pp 441–458.
- C109. **Julie A. Kientz, Shwetak N. Patel, Arwa Z. Tyebkhan, Brian Gane, Jennifer Wiley** and Gregory D. Abowd (principal work and writing by Kientz, Patel and Tyebkhan) Where's My Stuff? Design and Evaluation of a Mobile System for Locating Lost Items for the Visually Impaired . In *Proceedings of the Eighth International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2006)*, ACM Press, Portland, Oregon, October 2006, pp. 103–110.
- C110. **Julie A. Kientz, Gillian R. Hayes**, Gregory D. Abowd and Rebecca E. Grinter (principal research by Kientz, writing by all) From the war room to the living room: Decision support for home-based therapy teams. *Proceedings of Computer Supported Cooperative Work (CSCW '06)*, Banff, Canada, November, 2006, pp. 209–218. **One of only 7 overall submissions selected for consideration as Best Paper.**
- C111. **Moritz Köhler, Shwetak Patel, Jay Summet, Erich Stuntebeck** and Gregory D. Abowd (principal research and authorship by Köhler, Patel and Summet) TrackSense: Infrastructure-free precise indoor positioning using projected patterns. *Proceedings of The 5th International Conference on Pervasive Computing*. Toronto, Canada, May 13-16, 2007 Springer-Verlag LNCS Volume 4480, pp. 334-350.
- C112. **Julie A. Kientz, Rosa I. Arriaga, Marshini Chetty, Gillian R. Hayes, Jahmeilah Richardson, Shwetak N. Patel** and Gregory D. Abowd (principal research and authorship by Kientz) Grow and Know: Understanding record-keeping needs for the development of young children. *Proceedings of the ACM Human Factors in Computing Systems (CHI 2007)*, San Jose, California April 29-May 3, 2007, ACM

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- C113. Gillette, D. R., **Hayes, G. R.**, Abowd, G. D., Cassell, J., el Kaliouby, R., Strickland, D., and Weiss, P. Interactive technologies for autism. In *CHI '07 Extended Abstracts on Human Factors in Computing Systems* (San Jose, CA, USA, April 28 - May 03, 2007).
- C114. **Dounia Berrada, Mario Romero**, Gregory D. Abowd, Marion Blount and John Davis (principal research and authorship by Berrada and Romero) "Automatic administration of the get up and go test." International Conference On Mobile Systems, Applications And Services. Proceedings of the *1st ACM SIGMOBILE International Workshop on Systems and Networking Support for Healthcare and Assisted Living Environments*. ACM, July 2007.
- C115. **Shwetak N. Patel**, Tom Robertson, Julie A. Kientz, Matthew S. Reynolds, and Gregory D. Abowd (principal research by Patel and Robertson, authorship by all) At the flick of a switch: detecting and classifying unique electrical events on the residential power line. *International Symposium on Ubiquitous Computing (Ubicomp 2007)*, Innsbruck, Austria, September 16-19, 2007, Springer-Verlag LNCS Volume 4717, pp 271-288. **Best paper and Best presentation awards and 2017 Ten Year Impact Award Winner.**
- C116. **Shwetak N. Patel** and Gregory D. Abowd (principal research and authorship by Patel) BLUI: Low-cost Localized Blowable User Interfaces. In the *Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2007)*. Newport, RI. 2007.
- C117. **Erich Stuntebeck**, John Davis II, Gregory D. Abowd, Marion Blount. (principal research and authorship by Stuntebeck and Davis) HealthSense: Classification of Health-related Sensor Data through User-Assisted Machine Learning, *Workshop on Mobile Computing Systems and Applications (HotMobile '08)*, Napa, California, February 2008.
- C118. **Gillian R. Hayes, Lamar M. Gardere**, Gregory D. Abowd, and **Khai N. Truong** (principal research and authorship by Hayes) CareLog: a selective archiving tool for behavior management in schools. In *Proceeding of the Twenty-Sixth Annual SIGCHI Conference on Human Factors in Computing Systems (CHI 2008)*, Florence, Italy, April 05 - 10, 2008, ACM, New York, NY, pp. 685-694.
- C119. **Julie A. Kientz** and Gregory D. Abowd. (principal research and authorship by Kientz) When the Designer Becomes the User: Designing a System for Therapists by Becoming a Therapist. In *CHI '08 Extended Abstracts on Human Factors in Computing Systems* (Florence, Italy, April 05 - 10, 2008). ACM, New York, NY, 2071-2078.
- C120. **Shwetak N. Patel**, Mathew S. Reynolds, and Gregory D. Abowd (principal research by Patel, authorship by all) Detecting Human Movement by Differential Air Pressure Sensing in HVAC System Ductwork: An Exploration in Infrastructure Mediated Sensing. *Proceedings of The 6th International Conference on Pervasive Computing (Pervasive Computing 2008)*. Sydney, Australia May 19-22, 2008 Springer-Verlag LNCS Volume 5013, pp. 1-18. **Best paper award and 2018 Ten Year Impact Award Winner.**
- C121. **Gillian R. Hayes**, Gregory D. Abowd, Maria Ebling, John Davis, Marion Blount and Elizabeth Mynatt (principal research and authorship by Hayes) Opportunities for pervasive computing in chronic cancer care. *Proceedings of The 6th International Conference on Pervasive Computing (Pervasive Computing*

- 2008). Sydney, Australia May 19-22, 2008 Springer-Verlag LNCS Volume 5013, pp. 262-279.
- C122. Jaeseok Yun, **Shwetak Patel**, Matthew Reynolds and Gregory D. Abowd (principal research by Yun authorship by Yun, Reynolds and Abowd) A quantitative investigation of inertial power harvesting for human-powered devices. *International Symposium on Ubiquitous Computing (Ubicomp 2008)*, Seoul, Korea, September 21-24, 2008, ACM, New York, pp 74-83.
- C123. **Erich Stuntebeck**, Tom Robertson, **Shwetak N. Patel**, Matthew S. Reynolds and Gregory D. Abowd (principal research by Stuntebeck, authorship by all) Wideband powerline positioning for indoor localization. *International Symposium on Ubiquitous Computing (Ubicomp 2008)*, Seoul, Korea, September 21-24, 2008, ACM, New York, pp 94-103.
- C124. **Sunyoung Kim**, **Julie A. Kientz**, **Shwetak N. Patel** and Gregory D. Abowd (principal research and authorship by Kim and Kientz) Are you sleeping? Sharing portrayed sleeping status within a social network. *Proceedings of Computer Supported Cooperative Work (CSCW '08)*, San Diego, CA, November, 2008, ACM, New York, pp. 619-628 .
- C125. **Mario Romero**, **Jay Summet**, John Stasko and Gregory D. Abowd (principal research by Romero, authorship by all) Viz-a-Vis: Toward visualizing video through computer vision. *Proceedings of IEEE InfoVis 2008*, Columbus, OH, October 19-24, 2008, pp. 1261- 1268. Appears in *IEEE Transactions on Visualization and Computer Graphics*.
- C126. **Julie A. Kientz**, Rosa Arriaga and Gregory D. Abowd (principal research and authorship by Kientz) Baby Steps: Supporting Better Record-Keeping and Decision-Making for Parents of Young Children. In *Proceeding of the Twenty-Seventh Annual SIGCHI Conference on Human Factors in Computing Systems (CHI 2009)*, Boston, MA, April 04 - 9, 2009, ACM, New York, NY, pp. 1713-1722.
- C127. **Julie A. Kientz** and Gregory D. Abowd (principal research and authorship by Kientz) Experiences of participant observation when dealing with children with special needs. In *Proceedings of the Twenty-Seventh Annual SIGCHI Conference on Human Factors in Computing Systems (CHI 2009)*, Experience report. Boston, MA, April 04 - 9, 2009, ACM, New York, NY, pp. xxx-yyy.
- C128. **Julie A. Kientz**, Rosa Arriaga and Gregory D. Abowd (principal research and authorship by Kientz) KidCam. In *Proceedings of Pervasive Computing*, Nara, Japan, May 2009.
- C129. **Svetlana Yarosh**, S. Cuzzort, H. Mueller and G. D. Abowd. (principal research and authorship by Yarosh) Developing a Media Space for Remote Synchronous Parent-Child Interaction. In *Proceedings of the 8th International Conference on Interaction Design and Children (IDC 2009)*. ACM, June 3-5, 2009, Cuomo, Italy.
- C130. **Ping Wang**, Gregory D. Abowd and James M. Rehg. (principal research by Wang and Rehg, principal authorship by Wang and Rehg) Quasi-Periodic Event Analysis for Social Game Retrieval. In *Proceedings of the Twelfth IEEE International Conference on Computer Vision (ICCV 2009)*, Kyoto, Japan, September 29-October 4, 2009. One of 5% of submissions selected for oral presentation.
- C131. **Yi Han**, **Erich P. Stuntebeck**, John T. Stasko, Gregory D. Abowd (principal research by Yi Han, authorship by all) A Visual Analytics System for Radio

- Frequency Fingerprinting- based Localization. *IEEE Symposium on Visual Analytics Science and Technology (IEEE VAST 2009)*, pages 35-42, Atlantic City, NJ, October 2009.
- C132. **Fatima A. Boujarwah, Hwajung Hong**, Jackie Isbell, Rosa I. Arriaga, and Gregory D. Abowd (principal research and authorship by all) Training social problem solving skills in adolescents with high-functioning autism. In *Proceedings of the 4th International Conference on Pervasive Computing Technologies for Healthcare*, March 22-25, 2010, Munich, Germany.
- C133. **Tae-Jung Yun, Hee Young Jeong, Hee Rin Lee**, Rosa I. Arriaga, and Gregory D. Abowd (principal research and authorship by Yun, Jeong, Rin and Arriaga) Assessing asthma management practices through in-home technology probes. In *Proceedings of the 4th International Conference on Pervasive Computing Technologies for Healthcare*, March 22-25, 2010, Munich, Germany.
- C134. **Matthew Bonner, Jeremy Brudvik**, Gregory D. Abowd, and W. Keith Edwards (principal research and authorship by Bonner and Brudvik) No Look Notes: Accessible Eyes-Free Multi-Touch Text Entry. In *Proceedings of Pervasive Computing*, Helsinki, Finland, Springer-Verlag LNCS Volume 6030, pp. 409-426, May 2010.
- C135. **Karthir Prabhakar, Ping Wang, Sangmin Oh**, Gregory D. Abowd, and James M. Rehg. (principal research by Prabhakar and Rehg) Temporal Causality for the analysis of visual events. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR 2010)*. Oral presentation.
- C136. **Gabe Cohn, Erich Stuntebeck, Jagdish Pandey**, Brian Otis, Gregory D. Abowd, and Shwetak N. Patel. 2010. SNUPI: sensor nodes utilizing powerline infrastructure. In *Proceedings of the 12th ACM international conference on Ubiquitous computing (UbiComp '10)*. ACM, New York, NY, USA, 159-168. DOI=10.1145/1864349.1864377 <http://doi.acm.org/10.1145/1864349>.
- C137. **Nazneen, Fatima A. Boujarwah, Shone Sadler, Amha Mogus**, Gregory D. Abowd, and Rosa I. Arriaga (principal research by Nazneen, Boujarwah, Sadler and Mogus, principal authorship by Nazneen and Boujarwah) Understanding the challenges and opportunities for richer descriptions of stereotypical behaviors of children with ASD: A concept exploration and validation. In *Proceedings of the 13th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2010)*, October 25-27, 2010, Orlando, FL, pages 19-26.
- C138. **Svetlana Yarosh** and Gregory D. Abowd (principal research and authorship by Yarosh) Mediated parent-child contact in work-separated families. In *Proceedings of the 2011 annual conference on Human factors in computing systems (CHI '11)*. ACM, New York, NY, USA, pages 1185-1194.
- C139. **Mario Romero, Alice Vialard**, John Peponis, John Stasko, and Gregory D. Abowd. (principal research and authorship by Romero). Evaluating video visualizations of human behavior. In *Proceedings of the 2011 annual conference on Human factors in computing systems (CHI '11)*. ACM, New York, NY, USA, pages 1441-1450.
- C140. **Hee Young Jeong, Gillian R. Hayes, Tae-Jung Yun, Ja-Young Sung**, Gregory D. Abowd and Rosa Arriaga (principal research by all, primary authorship by Jeong, Hayes, Abowd and Arriaga). Act Collectively: Opportunities for technologies to support low-income children with asthma. In *Proceedings of the 25th BCS*

Conference on Human-Computer Interaction, Newcastle-upon-Tyne, UK, 4-8 July 2011.

- C141. **Fatima A. Boujarwah, Jennifer Kim**, Gregory D. Abowd and Rosa I. Arriaga (principal research and authorship by Boujarwah and Arriaga) Developing scripts to teach social skills: Can the crowd assist the author? *AAAI 2011 Workshop on Human Computation (WS-11-11)*, August 8-11, 2011, San Francisco, CA.
- C142. **Fatima A. Boujarwah, Nazneen, Hwajung Hong**, Gregory D. Abowd, and Rosa I. Arriaga (principal research by Boujarwah, Nazneen and Hong, authorship by all) Towards a framework to situate assistive technology design in the context of culture. In *Proceedings of the 13th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS 2011)*, October 24-26, 2011, Dundee, Scotland, pages 19-26.
- C143. **Tae-Jung Yun, Hee Young Jeong**, Tanisha Hill, Bert Lesnick, Randall Brown, Gregory D. Abowd, and Rosa I. Arriaga (principal research and authorship by all) Using SMS to provide continuous assessment and improve health outcomes for children with asthma. In *2012 ACM SIGHIT International Health Informatics Symposium (IHI 2012)*, Orlando, FL, January 28-30, 2012.
- C144. **Hwajung Hong, Jennifer Kim**, Gregory D. Abowd, and Rosa I. Arriaga (principal research and authorship by all) Designing a Social Network to Support the Independence of Young Adults with Autism. In *Proceedings of Computer Supported Cooperative Work (CSCW '12)*, pages 627–636 Seattle, WA, February 11-15, 2012.
- C145. **Fatima A. Boujarwah**, Gregory D. Abowd and Rosa I. Arriaga (principal research by Boujarwah, primary authorship by all) Socially computed scripts to support social problem solving skills. In *Proceedings of the Twenty-Seventh Annual SIGCHI Conference on Human Factors in Computing Systems (CHI 2012)*, Austin, TX, May April 5-10, 2012, ACM, New York, NY.
- C146. **Nazneen**, Gregory D. Abowd, Rosa I. Arriaga (principal research and authorship by Nazneen) Towards in-home collection of behavior specimens within the cultural context of autism in Pakistan. *Proceedings of the 6th International Conference on Pervasive Computing Technologies for Healthcare*, May 21-24, 2012, San Diego, CA.
- C147. **Caleb Southern, James Clawson, Brian Frey** Gregory D. Abowd, and Mario Romero, M. (principal research by Southern and Romero, principal writing by Southern and Romero) An evaluation of BrailleTouch: mobile touchscreen text entry for the visually impaired. *Proceedings of the 14th International Conference on Human-computer interaction with mobile devices and services (MobileHCI)*. September 24-27, 2012, pages 317-326, ACM, New York, NY.
- C148. **Edison Thomaz, Vinay Bettadapura, Gabriel Reyes, Megha Sandesh**, Grant Schindler, Thomas Ploetz, Gregory D. Abowd, and Irfan A. Essa. (principal research and authorship by Thomaz) Recognizing water-based activities in the home through infrastructure- mediated sensing. *International Symposium on Ubiquitous Computing (UbiComp 2012)*, Pittsburgh, PA, USA, September, 2012, ACM, New York, pp 85-94.
- C149. Thomas Ploetz, **Nils Hammerla**, Agata Rozga, Andrea Reavis, Nathan Call and

- Gregory D. Abowd. (principal research and authorship by all). Automatic Assessment of Problem Behavior in Individuals with Developmental Disabilities. *International Symposium on Ubiquitous Computing (Ubicomp 2012)*, Pittsburgh, PA, USA, September, 2012, ACM, New York, pp. 391-400. **(best paper honorable mention)**
- C150. Gregory D. Abowd. What next, UbiComp? Celebrating an intellectual disappearing act. *International Symposium on Ubiquitous Computing (Ubicomp 2012)*, Pittsburgh, PA, USA, September, 2012, ACM, New York, pp 31-40.
- C151. **Svetlana Yarosh, Anthony Tang, Sanika Mokashi** and Gregory D. Abowd (principal research and authorship by Yarosh) Almost Touching: Parent-Child Remote Communication Using theShareTable System. *Proceedings of Computer Supported Cooperative Work (CSCW '13)*, pages 181-192, San Antonio, TX, February 23-27, 2013.
- C152. **Hwajung Hong, Svetlana Yarosh, Jennifer G. Kim**, Gregory D. Abowd and Rosa I. Arriaga (principal research and authorship by Hong, Abowd and Arriaga) Investigating the Use of Circles in Social Networks to Support Independence of Individuals with Autism. In *Proceedings of the Twenty-Eighth Annual SIGCHI Conference on Human Factors in Computing Systems (CHI 2013)*, Paris, France, May April 27-May 2, 2013, ACM, New York, NY.
- C153. **Sanika Mokashi, Svetlana Yarosh** and Gregory D. Abowd (principal research and authorship by Mokashi and Yarosh) Exploration of Videochat for Children with Autism. In *Proceedings of the International Conference on Interaction Design and Children (IDC 2013)*. Short paper. June 24-27, 2013, New York, NY, pp. 320-323.
- C154. Rehg, James M., Gregory D. Abowd, Agata Rozga, Mario Romero, Mark A. Clements, Stan Sclaroff, Irfan Essa, Opal Ousley, **Yin Li, Chanh Kim, Hrishikesh Rao, Jonathan Kim, Liliana Lo Presti, Jianming Zhang, Denis Lantsman, Jonathan Bidwell, Zhefan Ye**, (principal research by all, principal authorship by Rehg, Abowd and Rozga) "Decoding Children's Social Behavior," *2013 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp.3414,3421, 23-28 June 2013. doi: 10.1109/CVPR.2013.438.
- C155. Yoshihiro Kawahara, Steve Hodges, Benjamin Cook, **Cheng Zhang**, Gregory D. Abowd (principal research and authorship by Kawahara) Instant Inkjet Circuits: *Lab-based Inkjet Printing to Support Rapid Prototyping of UbiComp Devices*. *2013 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '13)*, pp. 363-372, Sep 8- 12, 2013, Zurich, Switzerland. **Best Paper Award winner (one of 5 awarded)**.
- C156. **Edison Thomaz, Aman Parnami, Jonathan Bidwell**, Irfan Essa and Gregory D. Abowd (principal research by Thomaz and Parnami, authorship by all) Technological Approaches For Addressing Privacy Concerns When Recognizing Eating Behaviors With Wearable Cameras. *2013 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '13)*, pp. 739-748, Sep 8-12, 2013, Zurich, Switzerland.
- C157. **Edison Thomaz, Aman Parnami**, Irfan Essa, and Gregory D Abowd (principal research and authorship by Thomaz) Feasibility of identifying eating moments from first-person images leveraging human computation. *Proceedings of the 4th International SenseCam & Pervasive Imaging Conference*, pp. 26-33, San Diego,

CA, November 2013.

- C158. **Svetlana Yarosh**, Panos Markopoulos and Gregory D. Abowd (principal research and authorship by Yarosh) Towards a Questionnaire for Measuring Affective Benefits and Costs of Communication Technologies. *Proceedings of Computer Supported Cooperative Work (CSCW 2014)*, **Best Paper Award winner (one of 4 awarded at conference out of 134 accepted papers)**.
- C159. **Yangfen Ji, Hwajung Hong**, Rosa Arriaga, Agata Rozga, Gregory D. Abowd and Jacob Eisenstein (principal research and authorship by Ji, Hong, and Eisenstein) Mining themes and interests in the Asperger's and Autism community. *Proceedings of Association of Computational Linguistics (ACL 2014)*, June 2014.
- C160. Javier Hernandez, **Ivan Riobo**, Agata Rozga, Gregory D. Abowd, and Rosalind Picard (principal research by Hernandez and Riobo, principal authorship by all) Using electrodermal activity to recognize ease of engagement in children during social interactions. *2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '14)*, pp. 307-317, September 2014, Seattle, USA.
- C161. **Jonathan Bidwell**, Irfan A. Essa, Agata Rozga and Gregory D. Abowd (principal research by Bidwell, principal authorship by Bidwell, Rozga and Abowd) Measuring child visual attention using markerless head tracking from color and depth sensing cameras. *Proceedings of the 16th International Conference on Multimodal Interaction (ICMMI 2014)*, pp. 447-454, November 12, 2014, Istanbul, Turkey.
- C162. **Vinay Bettadapura, Edison Thomaz, Aman Parnami**, Irfan A. Essa, and Gregory D. Abowd (principal research and authorship by Bettadapura and Thomaz) Leveraging context to support automated food recognition in restaurants. *Proceedings of the Workshop on Applications of Computer Vision (WACV 2015)*.
- C163. **Hwajung Hong**, Gregory D. Abowd, and Rosa I. Arriaga (principal work by Hong, authorship by all) Towards designing social question-and-answer systems for behavioral support of individuals with autism. *Proceedings of the 2015 9th International Conference on Pervasive Computing Technologies for Healthcare (Pervasive Health 2015)*, pp. 17-24, IEEE.
- C164. **Edison Thomaz, Cheng Zhang**, Irfan Essa and Gregory D. Abowd. Inferring Meal Eating Activities in Real World Settings from Ambient Sounds: A Feasibility Study. *Proceedings of the 20th ACM International Conference on Intelligent User Interfaces (IUI 2015)*, pp. 427-431. Atlanta, GA. **Best paper awardee**
- C165. **Cheng Zhang, Anhong Guo, Caleb Southern** and Gregory D. Abowd BeyondTouch: Extending the Input Language with Built-in Sensors on Commodity Smartphones. *Proceedings of the 20th International Conference on Intelligent User Interfaces (IUI 2015)*, pp. 67-77. Atlanta, GA.
- C166. **Felicia Cordero, Daniel A. Epstein, Edison Thomaz**, Elizabeth Bales, Arvind Jagannathan, Gregory D. Abowd and James Fogarty (principal authorship by all, principal research by all) Barriers and Negative Nudges: Exploring Challenges in Food Journaling, *International ACM Conference on Human Factors in Computing Systems (CHI 2015)*, Seoul, Korea, May 2015. To appear. **Best paper honorable mention.**
- C167. **Hwajung Hong**, Eric Gilbert, Gregory D. Abowd, and Rosa Arriaga (principal

research and authorship by Hong) In-group Questions and Out-group Answers: Crowdsourcing Information and Advice for Individuals with Autism to Navigate Everyday Life.

International ACM Conference on Human Factors in Computing Systems (CHI 2015), pp. 777-786 Seoul, Korea, May 2015.

- C168. **Yi Han**, Agata Rozga, Nevena Dimitrova, Gregory D. Abowd and John Stasko, "Visual Analysis of Proximal Temporal Relationships of Social and Communicative Behaviors", *Computer Graphics Forum*, Volume 34, Issue 3, pages 51-60. *Proceedings of EuroVis 2015*.
- C169. **Arpita Bhattacharya, Mirko Gelsomini**, Patricia Perez-Fuster, Gregory D. Abowd and Agata Rozga (principal research and authorship by Bhattacharya, Gelsomini and Rozga) Designing motion-based activities to engage students with autism in classroom settings. In *Proceedings of the International Conference on Interaction Design and Children (IDC 2015)*. Pp. 69-78, ACM, June 21-24, 2015, Boston, MA, USA.
- C170. **Edison Thomaz**, Irfan Essa and Gregory D. Abowd (principal research and authorship by Thomaz) A practical approach for recognizing eating moments with wrist-mounted inertial sensing. In *2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '15)*, pp. 1029-1040, September 2015, Osaka, Japan.
- C171. **Daniel Castro, Steven Hickson, Vinay Bettadapura, Edison Thomaz**, Gregory D. Abowd, Henrik Christensen, and Irfan A. Essa (principal research by Castro, Hickson, Bettadapura and Thomaz) Predicting daily activities from egocentric images using deep learning. *Proceedings of the 2015 ACM International Symposium on Wearable Computing (ISWC 2015)*, pp. 75-82, September 2015, Osaka, Japan.
- C172. **Cheng Zhang**, Mitesh Patel, Senaka Buthpitiya, Kent Lyons, Beverly Harrison, and Gregory D. Abowd (principal research by Zhang, Patel, Buthpitiya, authorship by all) Driver classification based on driving behaviors. *Proceedings of the 21st International Conference on Intelligent User Interfaces (IUI 2016)*, Sonoma, CA, March 7-10, 2016. pp. 80-84. DOI: 10.1145/2856767.2856806.
- C173. **Cheng Zhang, AbdelKareem Bedri, Gabriel Reyes, Bailey Bercik**, Omer T. Inan, Thad E. Starner, and Gregory D. Abowd. 2016. TapSkin: Recognizing On-Skin Input for Smartwatches. *Proceedings of the 2016 ACM International Conference on Interactive Surfaces and Spaces (ISS '16)*, pp. 13-22. ACM, New York, NY, USA, 13-22. DOI: 10.1145/2992154.2992187.
- C174. **Aman Parnami, Apurva Gupta, Gabriel Reyes, Ramik Sadana**, Yang Li, and Gregory Abowd. 2016. Mogeste: mobile tool for in-situ motion gesture design. *Proceedings of the 2016 ACM International Joint Conference on Pervasive and Ubiquitous Computing: Adjunct (UbiComp '16)*. ACM, New York, NY, USA, pp. 345-348. DOI: 10.1145/2968219.2971395.
- C175. **Aman Parnami, Apurva Gupta, Gabriel Reyes, Ramik Sadana**, Yang Li, and Gregory D. Abowd. 2016. Mogeste: A Mobile Tool for In-Situ Motion Gesture Design. *Proceedings of the 8th Indian Conference on Human Computer Interaction (IHCI '16)*. ACM, New York, NY, USA, 35-43. DOI: 10.1145/3014362.3014365.
- C176. **Gabriel Reyes, Dingtian Zhang, Sarthak Ghosh, Pratik Shah, Jason Wu, Aman**

- Parnami, Bailey Bercik**, Thad Starner, Gregory D. Abowd and W. Keith Edwards (principal work and authorship by Reyes) Whoosh: Non-voice acoustics for low-cost, hands-free, and rapid input on smartwatches. In *Proceedings of the 2016 ACM International Symposium on Wearable Computers (ISWC '16)*. ACM, New York, NY, USA, 120-127. DOI: 10.1145/2971763.2971765.
- C177. **Weiren Wang, M Park, Y Fan**, T Starner, & GD Abowd. Face Card: An Information- sharing Framework on Google Glass. *CoRR*. 2016, abs/1606.08065.
- C178. **Cheng Zhang, Junrui Yang, Caleb Southern**, Thad E. Starner, and Gregory D. Abowd. (principal research by Zhang and Yang; principal authorship by Zhang). WatchOut: extending interactions on a smartwatch with inertial sensing. In *Proceedings of the 2016 ACM International Symposium on Wearable Computers (ISWC '16)*. ACM, New York, NY, USA, 136-143. DOI: <http://dx.doi.org/10.1145/2971763.2971775>.
- C179. Gregory D. Abowd. 2016. Hot or Not?: Moving Forward from Weiser's Vision of Ubiquitous Computing. *Proceedings of the 17th International Workshop on Mobile Computing Systems and Applications (HotMobile '16)*. ACM, New York, NY, USA, 1-1. DOI: 10.1145/2873587.2873606.
- C180. **Y. Han**, G.D. Abowd, & J. Stasko. Requirements for Visual Interaction Analysis Systems. *IEEE VIS 2016 Conference LIVVIL: Logging Interactive Visualizations & Visualizing Interaction Logs*. October 23, 2016.
- C181. **Yi Han**, Gregory D. Abowd, and John Stasko. Flexible Organization, Exploration, and Analysis of Visualization Application Interaction Events using Visual Analytics. *Proceedings of the IEEE VIS 2016 Workshop on Temporal & Sequential Event Analysis*. October 24, 2016. Baltimore, MD.
- C182. **J. Alcaidinho, G. Valentin**, G. Abowd, & M. Jackson. Training Collar-sensed Gestures for Canine Communication. *Proceedings of the Third International Conference on Animal Computer Interaction (ACI 2016)*. December 20, 2016.
- C183. **Caleb Southern, Yunnuo Cheng, Cheng Zhang**, and Gregory D. Abowd. (principal research all; principal authorship by Southern) Understanding the cost of driving trips. *International ACM Conference on Human Factors in Computing Systems (CHI 2017)*, Denver, Colorado, pp. 430-434. DOI: 10.1145/3025453.3025686.
- C184. Blake Wagner, III, **Elaine Liu**, Steven D. Shaw, Gleb Iakovlev, **Linlu Zhou, Christina Harrington**, Gregory Abowd, Carolyn Yoon, Santosh Kumar, Susan Murphy, Bonnie Spring, and Inbal Nahum-Shani. 2017. ewrapper: operationalizing engagement strategies in mHealth. *Proceedings of the 2017 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2017 ACM International Symposium on Wearable Computers (UbiComp '17)*. Workshop proceedings. ACM, New York, NY, USA. pp.790-798.
- C185. **Cheng Zhang, Xiaoxuan Wang, Anandghan Waghmare, Sumeet Jain**, Thomas Ploetz, Omer T. Inan, Thad E. Starner, and Gregory D. Abowd. 2017. FingerOrbits: interaction with wearables using synchronized thumb movements. *Proceedings of the 2017 ACM International Symposium on Wearable Computers (ISWC '17)*. ACM, New York, NY, USA, 62-65. DOI: 10.1145/3123021.3123041.
- C186. Edison Thomaz, Abdelkareem Bedri, Temiloluwa Prioleau, Irfan Essa, and

- Gregory D. Abowd. 2017. Exploring Symmetric and Asymmetric Bimanual Eating Detection with Inertial Sensors on the Wrist. *Proceedings of the 1st Workshop on Digital Biomarkers (DigitalBiomarkers '17)*. ACM, New York, NY, USA, 21-26. DOI: 10.1145/3089341.3089345.
- C187. **Cheng Zhang, Qiuyue Xue, Anandghan Waghmare, Ruichen Meng, Sumeet Jain, Yizeng Han, Xinyu Li**, Kenneth Cunefare, Thomas Plötz, Thad Starner, Omer Inan, Gregory D Abowd. FingerPing: Recognizing Fine-grained Hand Poses using Active Acoustic On-body Sensing. In the *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. Montreal QC, Canada, April 2018.
- C188. **Hyeokhyen Kwon**, Gregory D Abowd, Thomas Plötz. Adding structural characteristics to distribution-based accelerometer representations for activity recognition using wearables. In the *Proceedings of the 2018 ACM International Symposium on Wearable Computers (ISWC 2018)*. pp. 72-75. Singapore, October 2018. DOI: <https://doi.org/10.1145/3267242.3267258>.
- C189. **Hong Li**, Gregory D Abowd, Thomas Plötz. On specialized window lengths and detector based human activity recognition. In the *Proceedings of the 2018 ACM International Symposium on Wearable Computers Computers (ISWC 2018)*. pp. 68-71. Singapore, October 2018. DOI: <https://doi.org/10.1145/3267242.3267246>.
- C190. **Hong Li, Shishir Chawla, Richard Li, Sumeet Jain**, Gregory D Abowd, Thad Starner, **Cheng Zhang**, Thomas Plötz. Wristwash: towards automatic handwashing assessment using a wrist-worn device. In the *Proceedings of the 2018 ACM International Symposium on Wearable Computers (ISWC 2018)*. pp. 132, 139 Singapore, October 2018. DOI: <https://doi.org/10.1145/3267242.3267247>.
- C191. **Nivedita Arora** and Gregory D. Abowd) ZEUSSS: Zero Energy Ubiquitous Sound Sensing Surface Leveraging Triboelectric Nanogenerator and Analog Backscatter Communication. In the *Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology (UIST 2018)*. Poster. Berlin, Germany 2018. **Best poster award.**
- C192. **Fereshteh Shahmiri**, Chaoyu Chen, **Anandghan Waghmare, Dingtian Zhang, Shivan Mittal**, Steven Zhang, Yi-Cheng Wang, Zhong Lin Wang, Thad E. Starner, Gregory D. Abowd. Serpentine: A self-powered reversibly deformable cord sensor for human input. *International ACM Conference on Human Factors in Computing Systems (CHI 2019)*, Glasgow, Scotland. May 2019. To appear.
- C193. **Mehrab Bin Morshed, Koustuv Saha, Munmun De Choudhury**, Gregory D Abowd, Thomas Plötz. Measuring Self-Esteem with Passive Sensing. *PervasiveHealth '20: Proceedings of the 14th EAI International Conference on Pervasive Computing Technologies for Healthcare*, May 18-20 2020. Pp. 363-366. DOI: <https://doi.org/10.1145/3421937.3421952>. Atlanta, GA.
- C194. **Vedant Das Swain, Koustuv Saha**, Gregory D Abowd, **Munmun De Choudhury**. Social Media And Ubiquitous Technologies For Remote Worker Wellbeing And Productivity In A Post-pandemic World. *IEEE International Conference on Cognitive Machine Intelligence (CogMI)*, October 28-31 2020. DOI: 10.1109/CogMI50398.2020.00025. Atlanta, GA.
- C195. **Vanessa Oguamanam**, Taneisha Lee, Tom McKlin, **Zane Cochran**, Gregory D

- Abowd, Betsy DiSalvo. Cultural Clash: Exploring How Studio-based Pedagogy Impacts Learning For Students In Hci Classrooms. *DIS '20: Proceedings of the 2020 ACM Designing Interactive Systems Conference*, July 6-10 2020. Pp. 1131-1142. DOI: <https://doi.org/10.1145/3357236.3395544>. Eindhoven, Netherlands.
- C196. Julie A Kientz, Gillian R Hayes, Matthew S Goodwin, Mirko Gelsomini, Gregory D Abowd. Interactive Technologies And Autism, Second Edition. *Synthesis Lectures on Assistive, Rehabilitative, and Health-Preserving Technologies*, May 2020. 229 pages. DOI: <https://doi.org/10.2200/S00988ED2V01Y202002ARH013>.
- C197. Yogesh K Meena, Xing-Dong Yang, Marcus Löchtefeld, Matt Carnie, Niels Henze, Steve Hodges, Matt Jones, Nivedita Arora, Gregory D. Abowd. Self-sustainable: Self-powered Sustainable Interfaces And Interactions. *CHI'20 Extended Abstracts: CHI Conference on Human Factors in Computing Systems Extended Abstracts*, April 2020. DOI: <https://doi.org/10.1145/3334480.3375167>. Honolulu, HI.
- C198. **Vedant Das Swain**, Koustuv Saha, **Manikanta D Reddy**, **Hemang Rajvanshy**, Gregory D Abowd, Munmun De Chowdhury. Modeling Organizational Culture With Workplace Experiences Shared On Glassdoor. *ACM CHI Conference on Human Factors in Computing Systems*, April 2020. DOI: 10.1145/3313831.3376793. Honolulu, HI.
- C199. **Christopher Chen**, **David Howard**, **Stephen L Zhang**, **Youngwook Do**, **Sienna Sun**, **Tingyu Cheng**, **Zhong Lin Wang**, Gregory D. Abowd, Hyunjoo Oh. Spin (Self-powered Paper Interfaces): Bridging Triboelectric Nanogenerator With Folding Paper Creases. *TEI '20: Proceedings of the Fourteenth International Conference on Tangible, Embedded, and Embodied Interaction*. February 2020. Pp. 431-442. DOI: <https://doi.org/10.1145/3374920.3374946>. Sydney, NSW, Australia.
- C200. **Youngwook Do**, **Linh Thai Hoang**, **Jung Wook Park**, Gregory D Abowd, Sauvik Das. Spidey Sense: Designing Wrist-mounted Affective Haptics For Communicating Cybersecurity Warnings. *DIS '21: Designing Interactive Systems Conference 2021*, June 28-July 2 2021. Pp. 125-137. DOI: <https://doi.org/10.1145/3461778.3462027>. New York, NY.
- C201. **Youngwook Do**, **Siddhant Singh**, **Zhouyu Li**, **Stephen R Craig**, **Phoebe J Welch**, **Chengzhi Shi**, **Thad Starner**, Gregory D Abowd, Sauvik Das. Bit Whisperer: Enabling Ad-hoc, Short-range, Walk-up-and-share Data Transmissions Via Surface-restricted Acoustics. *UIST '21: The 34th Annual ACM Symposium on User Interface Software and Technology*, October 10-14, 2021. Pp. 1345-1357. DOI: <https://doi.org/10.1145/3472749.3477980>. New York, NY.
- C202. **Nivedita Arora**, **Ali Mirzazadeh**, **Injoo Moon**, **Charles Ramey**, **Yuhui Zhao**, **Daniela C Rodriguez**, Gregory D Abowd, **Thad Starner**. Mars: Nano-power Battery-free Wireless Interfaces For Touch, Swipe And Speech Input. *UIST '21: The 34th Annual ACM Symposium on User Interface Software and Technology*, October 10-14, 2021. Pp. 1305-1325. DOI: <https://doi.org/10.1145/3472749.3474823>. New York, NY.
- C203. **Youngwook Do**, **Linh Thai Hoang**, **Jung Wook Park**, Gregory D Abowd, Sauvik Das. Spidey Sense: Designing Wrist-mounted Affective Haptics For Communicating Cybersecurity Warnings. *DIS '21: Designing Interactive Systems Conference 2021*, June 28-July 2 2021. Pp. 125-137. DOI: <https://doi.org/10.1145/3461778.3462027>. New York, NY.
- C204. **Jung Wook Park**, Alishan Hassan, **Tingyu Cheng**, Rosa Arriaga, Gregory Abowd. A

- Simulation and Prototyping Toolkit for Airflow Energy Harvesting in Vehicles. *SenSys '21: Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems*, November 15-17, 2021. Pp 588-589. DOI: <https://doi.org/10.1145/3485730.3493359>. Coimbra, Portugal.
- C205. Kaely Hall, **Dong Whi Yoo**, Wenrui Zhang, **Mehrab Bin Morshed**, **Vedant Das Swain**, Gregory D Abowd, Munmun De Choudhury, Alex Endert, John Stasko, Jennifer G Kim. Supporting the Contact Tracing Process with WiFi Location Data: Opportunities and Challenges. *CHI '22: Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, April 29-May 5, 2022. Pp 1-14. DOI: <https://doi.org/10.1145/3491102.3517703>. New Orleans, LA.
- C206. **Tingyu Cheng**, **Jung Wook Park**, **Jiachen Li**, **Charles Ramey**, Hongnan Lin, Gregory D Abowd, Carolina Brum Medeiros, HyunJoo Oh, Marcello Giordano. PITAS: Sensing and Actuating Embedded Robotic Sheet for Physical Information Communication. *CHI '22: Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, April 29-May 5, 2022. Pp 1-16. DOI: <https://doi.org/10.1145/3491102.3517532>. New Orleans, LA.
- C207. **Vedant Das Swain**, Victor Chen, Shrija Mishra, Stephen M Mattingly, Gregory D Abowd, Munmun De Choudhury. Semantic Gap in Predicting Mental Wellbeing Through Passive Sensing. *CHI '22: Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, April 29-May 5, 2022. Pp 1-16. DOI: <https://doi.org/10.1145/3491102.3502037>. New Orleans, LA.
- C208. **Mehrab Bin Morshed**, Harish Kashyap Haresamudram, Dheeraj Bandaru, Gregory D Abowd, Thomas Plötz. A Personalized Approach for Developing a Snacking Detection System using Earbuds in a Semi-Naturalistic Setting. *ISWC '22: Proceedings of the 2022 ACM International Symposium on Wearable Computers*, September 11-15, 2022. Pp 11-16. DOI: <https://doi.org/10.1145/3544794.3558469>. Cambridge, United Kingdom.
- C209. **Nivedita Arora**, Vikram Iyer, HyunJoo Oh, Gregory D Abowd, Josiah D Hester. Circularity in Energy Harvesting Computational "Things". *SenSys '22: Proceedings of the 20th ACM Conference on Embedded Networked Sensor Systems*, November 6-9, 2022. Pp 931-933. DOI: <https://doi.org/10.1145/3560905.3568106>. Boston, MA.
- C210. **Tingyu Cheng**, **Zhihan Zhang**, **Bingrui Zong**, **Yuhui Zhao**, Zekun Chang, Yejun Kim, Clement Zheng, Gregory D Abowd, HyunJoo Oh. SwellSense: Creating 2.5 D Interactions with Micro-Capsule Paper. *CHI '23: Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, April 23-28, 2023. Pp. 1-13. DOI: <https://doi.org/10.1145/3544548.3581125>. Hamburg, Germany.
- C211. **Tingyu Cheng**, **Taylor Tabb**, **Jung Wook Park**, Eric M Gallo, Aditi Maheshwari, Gregory D Abowd, HyunJoo Oh, Andreea Danielescu. Functional Destruction: Utilizing Sustainable Materials' Physical Transiency for Electronics Applications. *CHI '23: Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, April 23-28, 2023. Pp. 1-16. DOI: <https://doi.org/10.1145/3544548.3580811>. Hamburg, Germany.
- C212. **Vedant Das Swain**, Lan Gao, William A Wood, Srikruthi C Matli, Gregory D Abowd, Munmun De Choudhury. Algorithmic Power or Punishment: Information Worker Perspectives on Passive Sensing Enabled AI Phenotyping of Performance and Wellbeing. *CHI '23: Proceedings of the 2023 CHI Conference on Human Factors in*

- Computing Systems*, April 23-28, 2023. Pp. 1-17. DOI: <https://doi.org/10.1145/3544548.3581376>. Hamburg, Germany.
- C213. **Tingyu Cheng**, Gregory D Abowd, HyunJoo Oh, Josiah Hester. Transient Internet of Things: Redesigning the Lifetime of Electronics for a More Sustainable Networked Environment. *HotCarbon '23: Proceedings of the 2nd Workshop on Sustainable Computer Systems Conference 2023*, July 9, 2023. Pp. 1-8. DOI: <https://doi.org/10.1145/3604930.3605723>. Boston, MA.
- C214. **Youngwook Do, Nivedita Arora, Ali Mirzazadeh, Injoo Moon**, Eryue Xu, **Zhihan Zhang**, Gregory D Abowd, Sauvik Das. Powering for Privacy: Improving User Trust in Smart Speaker Microphones with Intentional Powering and Perceptible Assurance. *USENIX Security '23: Proceedings of the 32nd USENIX Security Symposium*, August 9-11, 2023. Pp 2473-2490. Anaheim, CA.
- C215. **Dong Whi Yoo**, Hayoung Woo, Viet Cuong, Nguyen, Michael Birnbaum, Kaylee Payne Kruzan, Jennifer G. Kim, Gregory D. Abowd, Munmun De Choudhury. Patient Perspectives on AI-Driven Predictions of Schizophrenia Relapses: Understanding Concerns and Opportunities for Self-Care and Treatment. *CHI '24: Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, May 11, 2024. DOI: <https://doi.org/10.1145/3613904.3642369>.
- C216. **Vedant Das Swain**, Lan Gao, Abhirup Mondal, Gregory D. Abowd, Munmun De Choudhury. Sensible and Sensitive AI for Worker Well-Being: Factors that Inform Adoptions and Resistance for Information Workers. *CHI '24: Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, May 11, 2024. DOI: <https://doi.org/10.1145/3613904.3642716>. **Best Paper Awardee (top 1% of submitted papers).**

B3. Other refereed material

B4. Submitted Journal Articles (with date of submission)

C. OTHER PUBLICATIONS

List all other publications that are not otherwise included in Sections IV. A. and B.

- Harrison, M. D., Gregory D. Abowd and Alan J. Dix. (Significant contribution by all three authors, presented by Harrison) State of the art: Formal aspects of user interfaces. Invited presentation at Eurographics'90 conference, May 1990.
- Abowd, Gregory D. (extended abstract reviewed by program committee) Using formal methods for the specification of user interfaces. In Selby, R., editor, *Proceedings of the 2nd Annual Irvine Software Symposium — ISS'92*, pp. 109– 130. Irvine Research Unit in Software, University of California, Irvine, 1992.
- Abowd, Gregory D. Using formal methods of software engineering in HCI. Position paper for Basic Research Symposium at INTERCHI'93 conference, Amsterdam, April 1993.
- Abowd, Gregory D. and Bonnie E. John. (Significant contribution by Abowd through research directed by John) Using formal methods to improve the usability of Soar. Poster presented at HCI Consortium Winter Workshop, Fraser, Colorado, February 1994.

- Abowd, Gregory D. and Bonnie E. John. (Significant contribution by Abowd through research directed by John) Using formal methods to improve the usability of Soar. Paper presented at Soar-13 Workshop, Columbus, Ohio, March 1994.
- Abowd, Gregory D. Interface refinement. Position paper for Basic Research Symposium at CHI'94 conference, Boston, MA, April 1994.
- Abowd, Gregory D. Defining reference models and software architectural styles for cooperative systems. Position paper for CSCW'94 workshop on software architectures for cooperative systems, October 1994.
- Abowd, Gregory D. Automated dialogue verification. Position paper for Basic Research Symposium at CHI'95 conference, Denver, CO, May 1995.
- Dey, Anind and Gregory Abowd. (principal contributions and authorship by Dey) Position paper on wearable computing. Submitted to CHI'97 Workshop on Wearable Computing, January 1997.
- Dey, Anind and Gregory D. Abowd (Significant contributions from Dey, principal authorship by Dey). Cyberdesk demonstration. Presented at the User Interface Software Technology Symposium — UIST'97. October 1997.
- Appelbe, B. and Abowd, G.D. (Equal contributions by both authors) Beyond objects: A response. *Software Engineering Notes*, **20**(3):45–48.
- Abowd, Gregory D. What's in a name? Article in Beyond the Desktop: considering the future of interaction. In *SIGCHI Bulletin*, 2000.
- Dey, Anind K. and Gregory D. Abowd (principal research and authorship by Dey and Abowd) Towards a Better Understanding of Context and Context-Awareness. Presented at the CHI 2000 Workshop on The What, Who, Where, When, Why and How of Context-Awareness, April 1-6, 2000.
- Dey, Anind K. and Gregory D. Abowd (principal research and authorship by Dey) The Context Toolkit: Aiding the Development of Context-Aware Applications. In Workshop on Software Engineering for Wearable and Pervasive Computing, workshop affiliated with ICSE 2000 conference, Limerick, Ireland, June 6, 2000.
- O'Connell, Thomas, Peter Jensen, Anind Dey and Gregory D. Abowd. Location in the Aware Home. Position paper for Workshop on Location Modeling for Ubiquitous Computing, at Ubicomp 2001 conference, September 30, 2001. See <http://www.teco.edu/locationws>.
- Richter, Heather and Gregory D. Abowd. Evaluating capture and access through authentic use. Position paper for Workshop on Evaluation Methodologies for Ubiquitous Computing, at Ubicomp 2001 conference, September 30, 2001. See <http://zing.ncsl.nist.gov/ubicomp01>.
- Abowd, Gregory D. Challenges to construction beyond the desktop. Article in Beyond the Desktop: considering the future of interaction. In *SIGCHI Bulletin*, 2001.
- Scholtz, Jean, Heather Richter and Gregory D. Abowd.. Report from Ubicomp 2001 Workshop: Evaluation methodologies for ubiquitous computing. Article in Beyond the Desktop: considering the future of interaction. In *SIGCHI Bulletin*, 2001.
- Abowd, Gregory D. A search is a search... The impact of awareness technologies on privacy litigation. Article in Beyond the Desktop: considering the future of interaction. In *SIGCHI Bulletin*, 2001.
- Abowd, Gregory D. Is that a PDA in your pocket, or are you just happy to see me? Article in Beyond the Desktop: considering the future of interaction. In *SIGCHI*

Bulletin, 2002.

- Abowd, Gregory D. Sometimes we aim too high. Article in *Beyond the Desktop: considering the future of interaction*. In *SIGCHI Bulletin*, 2002.
- Edwards, Keith, Beki Grinter and Gregory D. Abowd. Smart Homes or homes that smart? Article in *Beyond the Desktop: considering the future of interaction*. In *SIGCHI Bulletin*, 2002
- Nagel, Kristine S. and Gregory D. Abowd (principal research and authorship by Nagel). Developing a Context-Aware Audio System Supporting Interpersonal Communication Presented at the HCI Consortium Annual Workshop, January, 2002, Fraser, Colorado.
- Iachello, Giovanni and Gregory D. Abowd (principal research and writing by Iachello) Security requirements for environmental sensing technology, 2nd Workshop on Ubicomp Security, in conjunction with Ubicomp 2003. Oct. 2003, Seattle, WA, USA.
- White, David R., Jose Antonio Camacho-Guerrero, Khai N. Truong, Gregory D. Abowd, Michael J. Morrier, Pooja C. Vekaria and Diane Gromala (principal research by White and Camacho-Guerrero, principal authorship by White and Abowd) Mobile capture and access for assessing language and social development in children with autism. Ubicomp 2003 companion proceedings. Demo paper. Oct. 2003, Seattle, WA, USA.
- Abowd, Gregory D., Gaetano Borriello and Gerd Kortuem (equal contributions by all authors). Report from the Ubicomp education workshop. In *IEEE Pervasive Computing Magazine*, Volume 3, Number 1, January-March 2004, pp. 94-98.
- Gillian R. Hayes, and Gregory D. Abowd. Assessment of Children with Autism Spectrum Disorders through Data Collection in the Natural Environment Workshop on Health Issues in HCI at *CHI'05* (April 2-7, Portland, OR, USA), 2005.
- Sebastian Boring, Julie A. Kientz, Gregory D. Abowd, and Gillian R. Hayes. Abaris: Capture and Access for Structured One-on-One Educational Settings. In the Extended Abstracts of *UBICOMP 2005: The 7th International Conference on Ubiquitous Computing*, September 11-14, Tokyo, Japan, 2005.
- Julie A. Kientz, and Gregory D. Abowd. Collective Remembering in Evidence-Based Care. Workshop on Collective Remembering at *CHI 2006* (April 22-27, Montreal, Canada), 2006.
- Gillian R. Hayes, Juane Heflin, Gregory D. Abowd, Lamar M. Gardere, Ellen Matthews, Julie A. Kientz, Ron Oberleitner (principal work by Hayes and Gardere) Evaluating a Selectively Archived Video Recording System for Functional Behavior Assessment in Schools. Poster abstract presented at *International Meeting for Autism Research (IMFAR 2006)*, Montreal, Canada, June 2006.
- Ron Oberleitner, Uwe Reischl, James Ball, John Harrington, Suhas Pharkute, and Gregory Abowd (principal writing by Ron Oberleitner) Remote autism evaluation through telehealth technology. Poster abstract presented at *International Meeting for Autism Research (IMFAR 2006)*, Montreal, Canada, June 2006.
- Hayes, G.R., Gardere, L.M., de Fazio, C., Abowd, G.D., Oberleitner, R. ABA Autism 2007 CareLog. In 2007 Progress and Challenges in the Behavioral Treatment of Autism Conference. Boston, Massachusetts. 2007.
- Kientz, J.A. and G.D. Abowd. Abaris: Technology to Support Collaboration and Decision Making for Discrete Trial Training. In 2007 Progress and Challenges in the

- Behavioral Treatment of Autism Conference. Boston, Massachusetts. 2007.
- Kientz, J.A., Hayes, G.R., Arriaga, R., Abowd, G.D. Designing and Developing Technology for Caregivers of Individuals with Autism CHI 2008 Workshop on Technology in Mental Health. Florence, Italy. April 2008.
- Svetlana Yarosh and G. D. Abowd. Segmented Institutionalism as a Perspective for Understanding Family Technologies. *Designing for Families Workshop. Ext. Abst. Of CSCW*. ACM (2008).
- Yarosh, S., & Abowd, G. D. Embodied Interaction for Mediated Communication between Children and Parents. *Children and Embodied Interaction Workshop. Ext. Abst. of Interaction Design and Children*. ACM (2009).
- Ping Wang, Gregory D. Abowd and James M. Rehg (principal research and authorship by Wang) Automatic retrieval of mother-infant social games from unstructured videos. *International Meeting for Autism Research (IMFAR)*, poster presentation, Chicago, IL, 2009.
- Matthew N. Bonner, Jeremy T. Brudvik, Hee Young Jeong, H.A. Mohamed, and Gregory D. Abowd (principal research and authorship by Bonner, Brudvik, Jeong and Mohamed) MEW: Coping with pediatric transplants. *Supplemental Proceedings of the 11th International Conference on Ubiquitous Computing (Ubicomp 2009)*, Poster presentation, Orlando, FL.
- Ping Wang, Tracy Westeyn, Gregory D. Abowd and James M. Rehg (principal research and authorship by Wang and Westeyn) Automatic classification of parent-infant social games from video. *International Meeting for Autism Research (IMFAR)*, Philadelphia, PA, poster presentation, 2010.
- Fatima A. Boujarwah, Hwajung Hong, Jackie Isbell, Rosa I. Arriaga, and Gregory D. Abowd (principal research and authorship by Boujarwah, Hong, and Isbell) Training social problem solving skills in adolescents with high-functioning autism. *International Meeting for Autism Research (IMFAR)*, May 20-22, 2010, Philadelphia, PA, poster presentation.
- Nazneen, Fatima A. Boujarwah, Amha Mogus, Shone Sadler, Mohammed Habibula, Rosa I. Arriaga, and Gregory D. Abowd (principal research and authorship by Nazneen, Boujarwah, Mogus, Sadler and Habibula) Understanding the context of stereotypical behaviors. *International Meeting for Autism Research (IMFAR)*, May 20-22, 2010, Philadelphia, PA, poster presentation.
- Fatima A. Boujarwah, Mark O. Riedl, Gregory D. Abowd, Rosa I. Arriaga (principal research and authorship by Boujarwah) REACT: Intelligent authoring of social skills instructional modules for adolescents with high-functioning autism. *SIGACCESS Newsletter*, Spring 2011.
- Fatima A. Boujarwah, Nazneen, Hwajung Hong, Gregory D. Abowd, and Rosa I. Arriaga (principal research and authorship by Boujarwah, Nazneen, and Hong) Cross-cultural comparisons of social expectations of individuals with autism. *International Meeting for Autism Research (IMFAR)*, May 12-14, 2011, San Diego, CA, poster presentation.
- Fatima A. Boujarwah, Jennifer G. Kim, Mark O. Riedl, Rosa I. Arriaga, and Gregory D. Abowd (principal research and authorship by Boujarwah, and Kim) Building a knowledge base to support the authoring of social skills instructional modules. *International Meeting for Autism Research (IMFAR)*, May 12-14, 2011, San Diego, CA, poster presentation.

Fatima A. Boujarwah, Nazneen, Hwajung Hong, Rosa I. Arriaga, and Gregory D. Abowd (principal research and authorship by Boujarwah, Nazneen and Hong) Cross-cultural comparisons of social expectations of individuals with autism and other intellectual disabilities. *40th Annual Meeting of the Society for Cross- Cultural Research*, February 16-19, 2011, Charleston, SC.

Aisling Ann O'Kane, Amy Hurst, Gerrit Niezen, Nicolai Marquardt, Jon Bird, and Gregory Abowd. 2016. Advances in DIY Health and Wellbeing. *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '16)*. ACM, New York, NY, USA, pp.3453-3460. DOI: 10.1145/2851581.2856467.

D. PRESENTATIONS

List all conference presentations (separate keynote and invited from submitted), testimony before legislative committees or other public bodies, invited seminars, performances of creative works, etc. (Do not list a presentation here if it is listed elsewhere.)

D1. Keynotes and Invited Distinguished Lectures

- K1. The Impact of New Technology in Education: Classroom 2000 and Beyond. Keynote presentation at Mobility Foundation Annual Conference, Los Angeles, CA. March 1, 1998.
- K2. Building Software for Mobile and Ubiquitous Computing Applications. Invited presentation at IEEE CS Annual Workshop on VLSI: System Level Design. Orlando, FL, April 16, 1998.
- K3. Building a digital library of captured experiences. International Conference on Digital Libraries. Invited keynote address, Kyoto, Japan, November, 2000.
- K4. Designing and Building Living Laboratories for Ubiquitous Computing Research: Opportunities for Software Engineering Research. Keynote address for Promodis (Programming of Modular and Distributed Systems) Swedish National conference, December 11, 2000.
- K5. Computing in living laboratories: Beyond dreaming beyond the desktop. State of Louisiana Center for Advanced Computing Studies Distinguished Lecture, Lafayette, March 8, 2002.
- K6. Computing in living laboratories: A grand challenge. Invited keynote address at the High Performance and Distributed Computing Conference (HPDC 2002), Edinburgh, Scotland, July 22-24, 2002.
- K7. Research challenges in ubiquitous computing. Invited keynote speaker at National Academy of Engineering Joint North America/Japan Frontiers in Engineering Symposium, Japan October 2002.
- K8. Abowd, Gregory D. Programming environments...literally: Ubicomp's grand challenge for software engineering. Opening keynote at ACM SIGSOFT Foundations of Software Engineering conference, Charleston, S.C., Nov. 20-22, 2002.
- K9. Challenges for applications research in ubiquitous computing. University of California, Irvine Institute for Research on Software (IROS) Distinguished

- Lecture, Irvine, CA, Jan. 17, 2003.
- K10. Ubiquitous computing: Information technology challenges for the next 40 years. Invited keynote at 40th anniversary of the Chicago chapter of ACM, Chicago, IL, April 23, 2003.
 - K11. Programming Environments: A grand challenge for ubiquitous computing. Invited keynote address at ENC 2003, the Fourth Mexican International Conference on Computer Science, September 10, 2003.
 - K12. Programming Environments: A grand challenge for ubiquitous computing. Invited keynote address to UCS 2003, the first national Japanese conference on ubiquitous computing, Kyoto, Japan, Nov. 17, 2003.
 - K13. Realizing the dreams of ubiquitous computing: It's all in the family. Invited keynote address at *Graphics Interface* 2004, London, Ontario, Canada, May 19, 2004.
 - K14. The Aware Home Research Initiative: Balancing technology and applications research. Invited keynote address to ICOST 2004, Singapore, September 17, 2004.
 - K15. Technologies to Support the Care of Children With Autism. Invited keynote at Idaho Conference on Telehealth and Autism, March 10, 2006.
 - K16. From Whimsical to Medical: Why I care about automating the capture of everyday experiences. Distinguished Lecture for Department of Computer Science, University of Illinois at Urbana-Champaign. April 3, 2006.
 - K17. Hold That Thought: Why I care about automating the capture of everyday experiences. Distinguished Lecture, Faculty of Information Science, Cornell University, September 27, 2006.
 - K18. Hold That Thought: Why I care about automating the capture of everyday experiences. Technology and Social Behavior Series, Northwestern University, October 12, 2006.
 - K19. Make IT Matter. Keynote talk at Ambient Intelligence 2007, Darmstadt, Germany, November 7-10, 2007.
 - K20. Measuring behavior in the home. Keynote talk at Measuring Behavior 2008, Maastricht, The Netherlands, August 26-29, 2008.
 - K21. Make IT Matter: How Computing Can Make a Difference. Closing lunch plenary at SIGCSE 2009, Chattanooga, TN, March 7, 2009.
 - K22. Make IT Matter: The opportunities and responsibilities of pervasive computing research. Keynote talk at IEEE PerCom 2009, Galveston, TX, March 10, 2009.
 - K23. Make IT Matter: We Can Be Clever and Make a Difference. Keynote at ETC'09, Emerging Technologies Conference, Iowa State University, April 2, 2009.
 - K24. Ubiquitous Care: Case studies in human-centered technology for health and education. Virginia Tech Department of Computer Science Distinguished Lecture. February 19, 2010.
 - K25. Computing and Autism: An odyssey borne of passion. The 2010 International Symposium on Collaborative Technology and Systems (CTS 2010). Keynote, Chicago, May 17, 2010.
 - K26. Computing and Autism. Invited Distinguished Lecture, University of Newcastle, September 2010.
 - K27. Computing and Autism. Stanford University, HCI Distinguished Lecture Series, October 15, 2010.

- K28. Computing and Autism, Emory University, Department of Computer Science Distinguished Lecture, October 25, 2010.
- K29. Computing and Autism. Distinguished lecture, Arizona State University, February 2011.
- K30. Computing Where IT Matters: Reflections on the Relevance of HCI Research. Invited closing keynote address for British HCI Conference, July 2011, Newcastle, United Kingdom.
- K31. Pervasive Healthcare: The role of ubiquitous computing in chronic conditions Xi'an Northwestern Polytechnical University Distinguished Lecture, Xi'an, China, September 2011.
- K32. Oh, the Ubiquity! Don't look back, but there is a LOT of data headed your way! Keynote address, 2011 American Medical Informatics Association, October 2011, Washington, D.C.
- K33. Using Computing Technologies to Support the Challenges of Autism: How can a computational approach impact DIR? International Council on Developmental and Learning Disorders invited keynote address and panelist, November 2011, Washington, D.C.
- K34. Computing and Autism: How Computer Science and Personal Passion Influence Each Other Keynote, SwSTE 2012 (IEEE CS International Conference on Software Science, Technology, and Engineering) 12 June 2012 in Herzliya Israel. Lecture delivered remotely.
- K35. Computing and Autism: How a real-world challenge drives a computing research agenda. Invited distinguished lecture, Microsoft Research Cambridge, November 27, 2012.
- K36. Computing and Autism: How a real problem can drive research in a different field, Swansea (Wales) University Distinguished Lectures in Computer Science, November 29, 2012.
- K37. Computing and Autism: How real-world challenges drive a research agenda. Invited keynote lecture at Kuwaiti e-Systems International Conference, December 18, 2012, Kuwait City, Kuwait.
- K38. Computing and Autism: How a real problem drives multimodal activity recognition research. Invited keynote address for 2013 IEEE Face and Gesture Conference, April 23, 2013, Shanghai, China.
- K39. Ubicomp and the Home. Invited keynote for HomeSys 2013 workshop, collocated with Ubicomp 2013 conference. Zurich, Switzerland, Sep. 9, 2013.
- K40. Ubicomp and Health: Merging Research Agendas. Cornell University School of Information Sciences Distinguished Lecture, October 28, 2013, Ithaca, NY.
- K41. Computing and Autism: How a real-world challenge drives a computing research agenda. ACM Distinguished Speakers talk for American University of Athens, Athens, Greece, December 13, 2013.
- K42. Ubicomp and Health: Merging Research Agendas. ACM Distinguished Speakers talk for ACM SIGCHI Greek chapter, University of Athens, Athens, Greece, December 13, 2013.
- K43. Approaches to Ubiquitous Computing Research. Keynote talk at Broadening Participation in Computing workshop at Ubicomp 2014 conference. September 14, 2014, Seattle, WA.

- K44. On being an applied computer scientist. University of Wisconsin-Madison Computer Science Department Distinguished Lecture, October 16, 2014.
- K45. Beyond ubiquitous computing: The fourth generation of computing is already here! University of Wisconsin WHCI+D Distinguished Lecture, October 17, 2014.
- K46. HCI: What does the future hold for the human experience? Invited keynote address for IEEE Rebooting Computing meeting in Santa Clara, CA, October 23, 2014.
- K47. On being an applied computer scientist. College of William and Mary Distinguished Lecture, Nov. 18, 2014.
- K48. Health and technology: How to balance the research agendas effectively. Invited keynote address to Workshop on Interactive Systems for Health 2014, held in conjunction with the AMIA Annual Meeting, Washington, DC, Nov. 15, 2014.
- K49. Beyond Ubicomp: Recognizing the next generation of computing technologies and applications. Distinguished Lecture at SMU Department of Computer Science, Dallas, TX, March 18, 2015.
- K50. Autism and Technology: An autobiographical account. Invited keynote and tutorial at the Irvine Thompson Center for Autism, March 23, 2015.
- K51. On being an applied computer scientist. Distinguished Lecture, CalTech Department of Computer Science. March 24, 2015.
- K52. A new generation of computing and its potential for PATCH. Invited keynote address to PATCH workshop at the Intelligent User Interface conference, March 29, 2015.
- K53. Applied computing research: Challenges and opportunities in a world of ubiquitous computing and beyond. Distinguished lecture at University of Texas Department of ECE, April 27, 2015.
- K54. Computing and health: An interdisciplinary approach in an era of mobile and ubiquitous computing. Invited keynote address at Symposium on Patient Safety with Interactive Medical Devices, Reading, England, June 24, 2105.
- K55. A new generation of computing and the promise for new social interactions. Keynote address for Mobile Systems for Computational Social Science workshop, in conjunction with the Ubicomp conference, Osaka, Japan, September 7, 2015.
- K56. Beyond ubiquitous computing: Technology advances and applications. ACM Distinguished Speaker address at the College of New Jersey, October 20, 2015.
- K57. Technology and autism: An autobiographical account. Geisinger-Bucknell Autism Center Distinguished Lecture, December 8, 2015.
- K58. Beyond Weiser's ubiquitous computing: 2 hopeful ideas. Toronto User Experience (TUX) Sanders Series Distinguished Lecture, January 12, 2016.
- K59. Hot or Not? Moving forward from Weiser's vision of ubiquitous computing. Invited keynote address for HotMobile, St. Augustine, FL, February 23, 2016.
- K60. On being an applied computer scientist. University of Michigan CSE Distinguished Lecture, April 19, 2016.
- K61. Interactive Computing Beyond Weiser's Ubicomp. University of Michigan School of Information Distinguished Lecture, April 20, 2016.

- K62. On Being an Applied Computer Scientist. University of Michigan School of Information Distinguished Lecture, April 20, 2016.
- K63. On being an applied computer scientist. University of Newcastle Doctoral Summer School keynote lecture, July 20, 2016.
- K64. Today's and tomorrow's interactive technologies: Collective computing and computational skins. University College London Symposium on HCI Grand Challenges, September 12, 2016.
- K65. Mental health: A driving application for 4th generation computing. Closing keynote for Ubicomp 2016 Mental Health workshop, Sep. 13, 2016.
- K66. Beyond Weiser: From ubiquitous computing to collective computing. University of Notre Dame Dept. of Computer Science Distinguished Lecture, Nov. 18, 2016.
- K67. Moving Beyond Weiser's Ubiquitous Computing. Keynote address at the 11th International Conference on Ubiquitous Computing and Ambient Intelligence, Nov. 8, 2017. Villanova University, Philadelphia.
- K68. The COSMOS Conjecture: Dreaming and doing in a post-Moore's Law era, January 16, 2018. Distinguished Lecture at Milan Politecnico.
- K69. The Material of the 21st Century: Moving beyond Moore and Weiser. March 21, 2018. MIT Media Lab.
- K70. The Material of the 21st Century: Moving beyond Moore and Weiser. Dartmouth University, Department of Computer Science Distinguished Lecture. Hanover, NH. September 28, 2018.
- K71. The Material of the 21st Century. Distinguished lecture at Nanyang Technological University, Singapore. October 8, 2018.
- K72. The Material of the 21st Century. Refereed and invited vision talk. ACM Symposium on User Interface Software Technology (UIST 2018). October 16, 2018. <https://www.youtube.com/watch?v=cNIZBtgceS4>.
- K73. The Material of the 21st Century. Distinguished Lecture, Tianjin University, Tianjin, China. December 13, 2018.
- K74. The Material of the 21st Century: Towards self-sustainable computational materials. Google X Tech Talk. January 30, 2019.
- K75. The Material of the 21st Century: Moving beyond Moore and Weiser. Stanford University Seminar on People, Computers, and Design. March 8, 2019.
- K76. The Material of the 21st Century: Moving beyond Moore and Weiser. ESE Invited Lecture at University of Virginia. March 22, 2019.
- K77. The Material of the 21st Century: Moving beyond Moore and Weiser. University of Lancaster Distinguished Lecture, May 10, 2019.
- K78. The Material of the 21st Century: Moving beyond Moore and Weiser. ACM. MobiSys conference keynote. June 19, 2019.
- K79. The Internet of Materials: The next logical step or a paradigm shift? University College London UCLIC Distinguished Lecture September 6, 2019.
- K80. The Internet of Materials: The next logical step or a paradigm shift? University of Swansea Distinguished Lecture, September 7, 2019.
- K81. Past, present, and future of computing and health: Perspectives from a ubicomp researcher. IEEE/ACM CHASE conference keynote. September 26, 2019.

- K82. The Internet of Materials: The next logical step or a paradigm shift? UC Berkeley EECS Colloquium Series. December 9, 2019.
- K83. The Internet of Materials: The next logical step or a paradigm shift? University of Minnesota Cray Distinguished Lecture. December 19, 2019.
- K84. Using digital technologies to support Covid-19 response on campus: A case study in the use of WiFi data. ACM ACM Bioinformatics, Computational Biology, and Health Informatics (ACMBCB2020) Conference Keynote, September 23, 2020.
- K85. The Internet of Materials: The next logical step or a paradigm shift? Indiana University Distinguished Lecture, October 23, 2020.
- K86. The Internet of Materials: The next logical step or a paradigm shift? UCSD Computer Science Distinguished Lecture, October 28, 2020.
- K87. Ambient Computing: A 21st Century Take on Disappearing. Google Ambient Computing Keynote. August 6, 2021.
- K88. The Internet of Materials: The next logical step or a paradigm shift? MIT CSAIL HCI Seminar Series. November 2020.
- K89. How engineering influences the future of computational devices. Distinguished lecture at Bose. September 20, 2022.
- K90. The Internet of Materials: Rethinking the future of computing. Distinguished lecture, Boston University Metropolitan Campus. Oct. 3, 2022.
- K91. The Internet of Materials: Rethinking the future of computing. Distinguished lecture, UCLA CS/ECE/HCI Distinguished Lecture. November 9, 2022.
- K92. The Internet of Materials: Rethinking the future of computing. Distinguished lecture, USC CS/HCI series. November 10, 2022.
- K93. Ignorance is Bliss: ACM SIGCHI Lifetime Achievement Award for Research talk. April 26, 2023.
- K94. Ubiquitous computing considered harmful: A two-act story of getting things right for the wrong reason. CMU HCII Seminar. September 9, 2023.
- K95. Rethinking the manufacturing of computational devices to address sensing and systems for digital health. IEEE BSN 2023 keynote. October 9, 2023.
- K96. Ubiquitous Computing Considered Harmful: Why the maker movement is key to changing the entire computing industry. Keynote address, UIST Workshop on Sustainability. October 29, 2023.
- K97. Doing the right thing for the wrong reason: How a vision for ubiquitous computing can be reconciled to have better impact. Columbia University/Cornell Tech distinguished lecture. November 3, 2023.
- K98. Doing the right thing for the wrong reason: How a vision for ubiquitous computing can be reconciled to have better impact. Hong Kong University, Department of Computer Science Distinguished Lecture. Nov. 27, 2023.

D2. Other presentations

- P1. Harrison, Michael D. and Gregory D. Abowd. (Significant contribution by both authors and equal presentation) Formal methods in human-computer interaction: a tutorial. Tutorial presentation for CHI'91 conference, New Orleans, LA, April 1991.
- P2. Dix, Alan J. and Gregory D. Abowd. (Significant contributions by both authors and equal presentation) Applying formal methods to HCI research. Tutorial presentation for HCI'92 conference, York, England, September 15, 1992.
- P3. Abowd, Gregory D. Evaluating user interface tools. Tutorial presented at HCI'94 conference, Glasgow, Scotland, August 23, 1994.
- P4. Dix, Alan J. and Gregory D. Abowd. (Significant contributions by both authors and equal presentation) Formal methods and HCI. Tutorial presented at HCI'94 conference, Glasgow, Scotland, August 23, 1994.
- P5. Abowd, Gregory D. Models and notations for interactive systems design and analysis.
Tutorial presented at SIGSOFT'94 conference, New Orleans, LA, December, 1994.
- P6. Abowd, Gregory D., Colin Potts and Spencer Rugaber. MORALE: Integrating techniques to support the evolution of software systems. Tutorial presented at the *International Conference on Software Maintenance — ICSM'99*. Oxford, England, August 30, 1999.
- P7. (with Jason A. Brotherton) Classroom 2000: An experiment in automated capture and access in an educational environment. Invited lecture at Microsoft Research, January 15, 1999.
- P8. Classroom 2000: An experiment in automated capture and access in an educational environment. Distinguished Lecture, Computer Science Department, Brown University. March 12, 1999.
- P9. Research experiences in automated capture applied to an educational environment.
Distinguished Lecture Vassar University, April 15, 1999.
- P10. (with Barry Boehm, John Knight and David Parnas) Panel: The relevance to industry of software engineering research in academia. Motorola Software Engineering Symposium— SES'99, Ft. Lauderdale, FL, June 23, 1999.
- P11. (with Richard N. Taylor) Panel: Software Architecture trends. Motorola Software Engineering Symposium — SES'99, Ft. Lauderdale, FL, June 23, 1999.
- P12. Gregory D. Abowd. Panel: Technology Enhanced Learning. Internet2 Sociotechnical Summit. Ann Arbor, MI, Sept. 13-15, 1999.
- P13. Gregory D. Abowd and Anind Dey (moderators); Peter J. Brown, Nigel Davies, Mark Smith and Pete Steggles (panelists). Panel: Towards a better understanding of context and context-aware computing. International Symposium on Handheld and Ubiquitous Computing — HUC'99. Karlsruhe, Germany, September 27-29, 1999.
- P14. Classroom 2000: Investigating automated capture in an educational setting. Carnegie Mellon HCI Institute Seminar, Pittsburgh, PA, October 13, 1999.
- P15. Computing Beyond Your Dreams: Opportunities and Challenges for Cyberenvironments. Georgia Tech College of Computing John P. Imlay Dean's Lecture, November 4, 1999.
- P16. Automated capture in a living laboratory: Classroom 2000. University of

Washington

Computer Science Seminar, Seattle, WA, November 22, 1999.

- P17. Ubiquitous computing environments: Research accomplishments and future visions. Oregon Graduate Institute Center for Human-Computer Communication Distinguished Lecture Series on the Future of Human-Computer Interaction, February 4, 2000.
- P18. Examining ubiquitous computing and automated capture and access in a living laboratory. Michigan-Ohio SIGCHI Chapter, invited lecture, February 11, 2000.
- P19. Future Computing Environments Research. Invited talk to Andersen Consulting (now Accenture) Center for Strategic and Advanced Research (CSTaR), June 5, 2000. Joe McCarthy host.
- P20. Research Challenges in Ubiquitous Computing. Invited lecture as Schlumberger Technical Fellow, September 15, 2000. Meyer Bengio host.
- P21. Research in Future Computing Environments. Invited presentation at Sony Computer Science Labs, Tokyo, Japan, November 13, 2000. Jun Rekimoto host.
- P22. Research in Future Computing Environments. Invited presentation at ATR MIC Labs, Kyoto, Japan, November 17, 2000. Kenji Mase host.
- P23. Research in Future Computing Environments. Invited presentation at NEC Research Labs, Kyoto, Japan, November 17, 2000. Mikiya Tani and Yoshihide Ishiguro hosts.
- P24. Research in Future Computing Environments. Invited presentation at the Interactive Institute, Gothenberg, Sweden, December 12, 2000. Lars-Erik Holmquist host.
- P25. Abowd, Gregory D. Beyond Dreaming Beyond the Desktop: Challenges in Building and Evaluating Ubiquitous Computing Applications. Invited talk at IBM Research, Watson talk, May 2001.
- P26. Abowd, Gregory D. Beyond Dreaming Beyond the Desktop: Challenges in Building and Evaluating Ubiquitous Computing Applications. Invited talk at Lucent Labs, Illinois, June 11 2001.
- P27. (with James Rowan). Intel Smart Home Forum invited talk: As We May Live: Building an Aware Home. August 2, 2001.
- P28. As We May Live: The Georgia Tech Broadband Institute Residential Laboratory and the Aware Home Research Initiative. Invited 45-minute presentation as part of panel for American Society of Interior Designers meeting, New Orleans, LA, Feb. 16, 2002.
- P29. If a classroom could listen, would anyone care? Experiences with eClass. Invited lecture at University of Toronto Knowledge Media Design Institute Spring 2002 Seminar Series, March 6, 2002.
- P30. Building a research program on ubiquitous computing in the home. Invited lecture at University of Texas, Arlington, March 25, 2002.
- P31. Applications research in ubiquitous computing. Invited talk for National Academy of Engineering Joint Japan America Frontiers of Engineering Symposium, Nov. 2002.
- P32. (joint with T. Rodden, W. Gaver, J. Wejchert) Panel on ubiquitous

- computing in the home. At Ubicomp 2002 conference, Gothenberg, Sweden, October 2002.
- P33. As we may live. Invited talk to Michigan-Ohio ACM SIGCHI Special Interest Group (MOCHI), Ann Arbor, MI, Feb. 12, 2003.
 - P34. As we may live. Invited talk to Microsoft Research, Seattle, WA, March 3, 2003.
 - P35. As we may live. Invited talk to Intel Research, Seattle, WA, March 4, 2003.
 - P36. The Role of Technology for Healthy Aging. Statement prepared and presented by Gregory D. Abowd to U.S. Senate Special Committee on Aging, hearing on Baby Boomers at the Gate: Enhancing Independence through Innovation and Technology. May 20, 2003. Full statement available at <http://aging.senate.gov/hearings/hr101ga.pdf>
 - P37. The Role of Technology for Healthy Aging. Presentation to the National Governor's Association annual meeting, New Orleans, October 11, 2003.
 - P38. Ubiquitous Computing: An overview of research accomplishments and challenges. Delivered to Samsung Electronics Research, Seoul, Korea, Nov. 18, 2003.
 - P39. eClass: A retrospective. Invited presentation to Web Group at University of Notre Dame, Indiana, March 2, 2004.
 - P40. Realizing the dream of ubiquitous computing: It's all in the family. Invited seminar to CSE Department, University of Notre Dame, Indiana, March 2, 2004.
 - P41. Progress update on the Aware Home Research Initiative. Invited presentation to Intel, Hillsborough, OR, March 11, 2004.
 - P42. Pervasive Technologies and applications for a Smart Home. Invited presentation and panel participation at George Washington University, May 20, 2004.
 - P43. Technologies to Track Interventions for Autism. Invited presentation to Cure Autism Now Innovative Technologies for Autism Think Tank, June 11-13, 2004, San Francisco, CA.
 - P44. Application-led research in ubiquitous computing. Panel presentation in UbiApp workshop, held in conjunction with Pervasive Computing Conference, May 11, 2005. Munich, Germany. Other panelists included Gaetano Borriello, William Newman, and Abigail Sellen.
 - P45. Better living through technology for persons with disabilities: The Aware Home. Lunch speaker at 11th Annual Georgia Epilepsy Symposium, November 11, 2006.
 - P46. Abowd, Gregory D. Evaluation of real deployments in ubicomp. Tutorial presented at the *International Conference on Pervasive Computing (Pervasive 2007)*, Toronto, Canada, May 16, 2007.
 - P47. Abowd, Gregory D. Pervasive computing: Introduction to research in the area. Tutorial presented at the *International Conference on Pervasive Computing (Pervasive 2008)*, Sydney, Australia, May 22, 2008.
 - P48. Panel on the history and future of Ubicomp, *International Symposium on Ubiquitous Computing (UbiComp 2008)*, Seoul, Korea, September 21-24, 2008.
 - P49. HCI Research: Does it really matter? Carnegie Mellon University HCI Institute Seminar Series, Wednesday, October 1, 2008.

- P50. Computing and Autism. Newcastle University Creative Play Group Lecture series, October 1, 2010.
- P51. Computing and Autism. Stanford University Distinguished HCI Lecture series, October 15, 2010.
- P52. Career Assistance for Autism Through High Technology and Science. American Academy of Pediatrics Georgia Chapter 2010 Annual Fall meeting. October 21, 2010.
- P53. Computing and Autism. Emory University Computer Science Lecture series, October 22, 2010.
- P54. Computational Approaches to Measuring Behavior, Autism Speaks Workshop on Outcome Measures for Clinical Trials with Individuals with ASD: Challenges and Opportunities, Washington, DC, January, 2011.
- P55. Reflection on 20 years after Mark Weiser's Scientific American article. At Ubicomp 2011, Beijing, China.
- P56. TEDxGeorgiaTech talk, April 2011.
- P57. Center for Discovery talk, April 2011
- P58. The COSMOS Conjecture: Dreaming and doing in a post-Moore's Law era. GVU Center Brown Bag talk, November 10, 2016.
- P59. The Future of Computing: Interesting challenges facing mobile computing. RADIO Atlanta: VMWare/AirWatch internal research and development Conference, September 21, 2017, Atlanta, Georgia.
- P60. Autism and Technology: An autobiographical account. March 21, 2018. MIT Media Lab. Lecture in course taught by Rosalind Picard on autism and technology.
- P61. There is more than one way to expand on computing. Georgia Tech IEN Technical Exchange Conference keynote for The Promises and Challenges of High Performance Electronics Everywhere. May 22, 2018.
- P62. Short course on Ubiquitous Computing and HCI. July 13 – July 21, 2018. Peking University, Beijing, China.
- P63. Confessions of an Applied Computer Scientist. GVU Center Brown Bag talk, January 10, 2019.

E. GRANTS AND CONTRACTS

E1. AS PRINCIPAL INVESTIGATOR

List all funded grants and contracts as principal and co-principal investigator. List PI and Co-PI for each grant, with total grant funding followed by sub amount allocated to candidate. Proposals pending may be included, but must be listed separately. Do not include grants and contracts not funded.

Title of Project: An architectural case study in telecommunications
 Agency/Company: Bell Northern Research
 Total Dollar Amount: \$20,000
 Role: PI
 Period of Contract: 6/1995-5/1996

Title of Project: Software architectures for global information systems
Agency/Company: Motorola University Partnerships in Research Program
Total Dollar Amount: \$27,389
Role: PI
Period of Contract: 9/1996-8/1997

Title of Project: Classroom 2000
Agency/Company: NEC
Gregory Abowd and Christopher Atkeson
GVU Industrial Affiliates Program funding from NEC
Amount awarded: \$30,000 for one year, beginning January 1996.

Title of Project: Ubiquitous Computing
Agency/Company: FX-PAL
Total Dollar Amount: \$25,000
Role: PI
Collaborators: Christopher Atkeson (co-PI)
Period of Contract: 9/1996-8/1997

Title of Project: Applications of Java to the monitoring and control of physical systems
Agency/Company: Siemens
Total Dollar Amount: \$30,000
Role: PI
Collaborators: Christopher Atkeson (co-PI)
Period of Contract: 4/1997-3/1998

Title of Project: Future Computing Environments
Agency/Company: Mitsubishi Electronic Research Laboratory
Total Dollar Amount: \$30,000
Role: PI
Period of Contract: 1/1997-12/1997

Title of Project: Service creation and development environments for audio interfaces.
Agency/Company: Motorola Cellular Infrastructure Group
Total Dollar Amount: \$100,000
Role: PI
Collaborators: Colin Potts (co-PI)
Period of Contract: 1/1997-12/1997

Title of Project: Research in Mobile and Ubiquitous Computing
Agency/Company: Mobility Foundation
Total Dollar Amount: \$38,000
Role: PI
Collaborators: C. Atkeson (co-PI)
Period of Contract: 3/1997-2/1998

Title of Project: NSF IRI-9703384 CAREER: Investigating research issues in ubiquitous computing: The capture, integration, and access problem
Agency/Company: National Science Foundation
Total Dollar Amount: \$438,376
Role: PI
Period of Contract: 9/1997-8/2002

Title of Project: Classroom 2000 infrastructure
Agency/Company: Sun Microsystems
Total Dollar Amount: \$270,000 in kind equipment
Role: PI
Period of Contract: 1997

Title of Project: Research in capture, integration and access
Agency/Company: Corporation for National Research Initiatives
Total Dollar Amount: \$25,000 cash and \$8000 equipment
Role: PI
Period of contract: 1998

Title of Project: ESS: Automated Understanding of Captured Experience
Agency/Company: National Science Foundation
Total Dollar Amount: \$850,000
Role: PI
Collaborators: C. Atkeson and I. Essa
Period of Contract: 7/1998-6/2001

Title of Project: Future Computing Environments research
Agency/Company: Hitachi Corporation
Total Dollar Amount:
Role: PI
Period of Contract: 8/1998-7/1999

Title of Project: Future Computing Environments research
Agency/Corporation: NEC Corporation
Total Dollar Amount: \$50,000
Role: PI
Period of Contract: 10/1998-9/1999

Title of Project: Augmenting the Capture and Understanding of Everyday Experiences
Agency/Corporation: National Science Foundation CISE Infrastructure Grant
Total Dollar Amount: \$120,000
Role: PI
Collaborators: Atkeson, Essa, MacIntyre, Mynatt, Potts, Ramachandran, Ribarsky, Rugaber and Starner (co-PIs)
Period of Contract: 1/1999-12/2001

Title of Project: Inter-Agency Workshop on Smart Environments
Agency/Corporation: National Science Foundation
Total Dollar Amount: \$27,496
Role: PI
Period of Contract: August-December 1999

Title of Project: Development and Understanding of Automated Capture
Environments to Support Long-Term Use
Agency/Company: National Science Foundation and CNPq Brazil
Total Dollar Amount: \$201,670
Role: PI
Collaborators: Maria da Graca Pimentel (co-PI through CNPq in Brazil)
Period of Contract: 7/2000-6/

Title of Project: Handling sensed context in a ubiquitous computing setting
Agency/Company: DARPA ITO Ubiquitous Computing/Expeditions
Total Dollar Amount: \$200,000
Role: PI
Period of Contract: 1/2000-12/2000

Title of Project: Battlefield Visualization System
Agency/Company: Army Research Labs
Total Dollar Amount: \$100,000 subcontract to GTRI
Role: PI
Collaborators: Kirk Pennywitt co-PI
Period of Contract: 1/2000-12/2000

Title of Project: The Aware Home Research Initiative
Agency/Company: Intel, HP, MERL, Motorola, Accenture CSTAR, Visteon Total
Dollar Amount: \$705,000
Role: PI
Collaborators: Aaron Bobick, Irfan Essa, Blair MacIntyre, Elizabeth Mynatt, Thad
Starner (co-PIs)
Period of Contract: 5/2000-12/2003

Title of Project: ITRI: The Aware Home: Sustaining the Quality of Life for an Aging
Population
Agency/Company: National Science Foundation Information Technology Research
Initiative
Total Dollar Amount: \$1,600,000
Role: PI
Collaborators: Aaron Bobick, Irfan Essa, Elizabeth Mynatt and Wendy Rogers
(Psychology) (co-PIs)
Period of Contract: 9/2001-8/2006

Title of Project: Developing automated capture applications for the Personal Server.
Agency/Company: Intel Research

Total Dollar Amount: \$50,000 PI:
Gregory Abowd
Period of Contract: 12/2003-11/2004

Title of Project: Using Visual Analytics to Support Data Analysis at the Center for
Discovery
Agency/Company: Center for Discovery
Total Dollar Amount: \$130,000
Role: PI:
Collaborators: John Stasko (co-PI), Johanna Lantz (co-PI)
Period of Contract: 3/1/2014-12/31/2015

Title of Project: Autism Center of Excellence award from State of New York
Agency/Company: State of New York through Center for Discovery subcontract
Total Dollar Amount: \$30,000
Role: PI for subcontract
Collaborators: John Stasko (co-PI), Terry Hamlin (PI for total award at CfD)
Period of Contract: 1/2016-9/2017

Title of Project: Atlanta Autism Consortium
Agency/Company: Federal earmark subcontract from Children's Healthcare of
Atlanta
Total Dollar Amount: \$75,000
Role: PI
Period of Contract: 3/1/2011-12/2011

Title of Project: Intel Science and Technology Center on Pervasive Computing
Agency/Company: Intel
Total Dollar Amount: \$800,000
Role: PI
Collaborators: James M. Rehg (PI)
Period of Contract: 2011-2016
Candidates' Share: \$400,000

Title of Project: Using visual analytics to support data analysis at CfD
Agency/Company: Center for Discovery
Total Dollar Amount: \$800,000
Role: PI
Collaborators: John Stasko (GT co-PI), Johanna Lantz (CfD), Matthew Northrop
(CfD), Terry Hamlin (CfD)
Period of Contract: 2015-2016
Candidates' Share: \$70,000

Title of Project: Applying a maker philosophy to teaching computer science
Agency/Company: VMWare Foundation
Total Dollar Amount: \$25,000
Role: PI

Period of Contract: January 2017-
Candidates' Share: \$25,000

Title of Project: Exploring notification feedback in VR systems
Agency/Company: HP Labs
Total Dollar Amount: \$30,000
Role: PI
Period of Contract: 2016-2017
Candidates' Share: \$30,000

Title of Project: Wearable approaches to eating detection
Agency/Company: Semiconductor Research Corporation
Total Dollar Amount: \$300,000
Role: PI
Period of Contract: Jan 1, 2018-Dec. 2020
Candidate's Share: \$300,000

Title of Project: NSF INCLUDES Alliance: Engineering PLUS
Agency/Company: National Science Foundation
Total Dollar Amount: \$10,000,000
Role: PI
Collaborators: Michael Silevitch, Claire Duggan, Richard Harris (co-PIs)
Period of Contract: 8/1/2024-7/31/2026

E2. AS CO-PRINCIPAL INVESTIGATOR

Title of Project: MORALE: Mission Oriented Architectural Legacy Evolution
Agency/Company: DAPRA BAA #95-05 Evolutionary Design of Complex Software Total
Dollar Amount: \$1,226,108
Role: co-PI
Collaborators: Ashok Goel, W. Michael McCracken, Melody Moore, Colin Potts,
Spencer Period of Contract: 7/1996-4/2000

Title of Project: The Home Information Infrastructure Lab
Agency/Company: Intel
Total Dollar Amount: \$60,000
Role: co-PI
Collaborators: K. Calvert (PI), D. Howard (co-PI)
Period of Contract: 1/1997-12/1997

Title of Project: Educational innovation
Agency/Company: Hewlett-Packard
Total Dollar Amount: \$1.3 million deep-discounted equipment
Role: co-PI Collaborators: Tom Barnwell (PI) Period of
Contract: 1997
Candidate's Share: \$289,360

Title of Project: Collaborative environments for capturing military tactics
Agency/Company: Army Research Laboratory
Total Dollar Amount: \$135,000 subcontract with GTRI
Role: co-PI
Collaborators: Kirk Pennywitt
Period of Contract: 5/1998-9/1999

Title of Project: NMI: Exploration of Middleware Technologies for Ubiquitous Computing with Applications to Grid Computing
Agency/Company: National Science Foundation Middleware Initiative
Role: co-PI
Collaborators: Umakishore Ramachandran (PI), Raj Kumar (HP Labs, co-PI), Sujoy Basu (HP Labs, co-PI)
Period of Contract: 9/2003-8/2006

Title of Project: Development of a home-based system for biobehavioral recording or individuals with autism
Agency/Company: Simons Foundation Autism Research Initiative
Total Dollar Amount: \$1,065,607
Role: co-PI
Collaborators: PI: Matthew Goodwin (Northeastern); co-PI: James Rehg (Georgia Tech)
Period of Contract: 4/2013-3/2016
Candidate's Share: \$204,029

Title of Project: Designing and studying maker oriented learning to transform advanced computer science education
Agency/Company: National Science Foundation
Total Dollar Amount: \$718,753
Role: co-PI
Collaborators: PI: Betsy DiSalvo
Period of Contract: 9/1/2014-8/31/2016

Title of Project: Expeditions: Computational Behavioral Science: Modeling, Analysis and Visualization of Social and Communicative Behavior
Agency/Company: National Science Foundation
Total Dollar Amount: \$10,000,000
Role: co-PI
Collaborators: PI: James Rehg, co-PIs: Gregory D. Abowd, Mark Clements, Agata Rozga with collaborators from Northeastern (Goodwin), Boston U. (Sclaroff), MIT (Picard), Illinois (Forsyth, Karahalios) CMU (Dey, Kanade), USC (Narayanan)
Period of Contract: 9/2010-8/2015
Candidate's Share: \$3.6M to Georgia Tech

E3. AS SENIOR PERSONNEL OR CONTRIBUTOR

List all funded grants and contracts on which you were not Principal or Co-Principal investigator but were listed as senior personnel. Proposals pending may be included, but do not include grants and contracts not funded.

Title of Project: Research Infrastructure Grant
Agency/Company: National Science Foundation
Total Dollar Amount: \$2,000,000
Role: Key Personnel
Collaborators: Ramachandran (PI)
Period of Contract: 7/1999-
8/2002

Title of Project: DARPA ASSIST
Agency/Company: DARPA ITO Total
Dollar Amount: \$2,000,000 Role: Key
Personnel
Collaborators: PI: Starner (GT) Pentland (MIT); Co-Pis: Essa, Isbell, Picard (MIT)
Period of Contract: 6/2005-11/2006

Title of Project: Using Selective Archiving to support Functional Behavior
Assessment in Schools
Agency/Company: NICHD SBIR Phase 1
Total Dollar Amount: \$100,000
Role: Consultant
Collaborators: Ron Oberleitner, eMerge Medical Systems (PI)
Period of Contract: 1/2006-6/2006

Title of Project: Using Selective Archiving in telehealth settings
Agency/Company: NICHD SBIR Phase II
Total Dollar Amount: \$2,400,000
Role: Consultant
Collaborators: Ron Oberleitner (Behavior Imaging Solutions, Inc.), co-PI: Agata Rozga
(GT),
Period of Contract: 6/2007-5/2009

Title of Project: Using smartphones to support naturalistic diagnostic and
monitoring services for autism
Agency/Company: NIMH SBIR Phase II renewal Total
Dollar Amount: \$3,000,000
Role: Consultant
Collaborators: Ron Oberleitner (Behavior Imaging Solutions, Inc.), co-PI: Agata Rozga
(GT),
Period of Contract: 6/2013-5/2016

Title of Project: Center for Excellence for Mobile Sensor Data-to-Knowledge (MD2K)
Agency/Company: NIH

Total Dollar Amount: \$10,800,000

Role: Budgeted collaborator

Collaborators: PI: S. Kumar (Memphis), co-PI: J. Rehg (Georgia Tech) Period of Contract: 9/29/2014-5/31/2018

E3. PROPOSALS UNDER REVIEW

Title of Project: NSF INCLUDES Alliance: Engineering PLUS (supplement)

Agency/Company: National Science Foundation

Total Dollar Amount: \$400,000 for 1 year (informally approved in Nov 2024 and on hold)

Role: PI

Collaborators: NU: Michael Silevitch, Richard Harris, Claire Duggan.

Title of Project: FMSG: CYBER: RADMat: Rapid Assembly for Digital Materials.

Agency/Company: National Science Foundation

Total Dollar Amount: \$500,000/year for 3 years

Role: co-PI

Collaborators: NU: Ruobing Rao (PI), Jeffrey Lipton (co-PI), Ben Davaji (co-PI)

Title of Project: Track 4: Phase II: The Autism Self-Advocacy Center for Equity and Neurodiversity in Engineering (A-SCENE)

Agency/Company: National Science Foundation

Total Dollar Amount: \$900,000/year for 3 years (informally approved in Nov 2024 and on hold)

Role: co-PI

Collaborators: Vanderbilt: Krish Roy (PI); UConn: Kazem Kaserounian (co-PI); Fisk University: Kent Wallace (co-PI)

F. OTHER SCHOLARLY ACCOMPLISHMENTS

List all other scholarly accomplishments such as software, patents, invention disclosures, start-up companies, etc.

Intellectual Property

- IP1. Shwetak N. Patel, Gregory D. Abowd, Khai N. Truong, Jay Summet. Systems and Methods for Disabling Recording Features of Cameras. US Patent Application US2007013552 A1. Publication date: May 10, 2007. Priority date: Nov. 10, 2005. Pending.
- IP2. Gillian R. Hayes, Khai N. Truong, Lamar M. Garder and Gregory D. Abowd. Systems and Methods for Archiving of Continuous Capture Buffers. US Patent application US20070214292 A1. Publication date: Sep. 13, 2007, Priority date: Feb. 24, 2006. Pending.
- IP3. Julie Kientz, Sebastian Boring, Gregory D. Abowd. Method and computer program product for synchronizing, displaying and providing access to

- data collected from various media. US Patent 8,275,243 B2. Publication date: September 25, 2012. Priority date: August 31, 2006.
- IP4. Shwetak N. Patel Thomas Robertson, Matthew Reynolds, Gregory D. Abowd. Detecting actuation of electrical devices using electrical noise over a power line. US Patent 8,094,034 B2. Publication date: January 10, 2012. Priority date: September 18, 2007.
Also published as US 8334784 B2.
- IP5. James M. Rehg, Karthir Prabhakar, Sangmin Oh, Ping Wang, Gregory D. Abowd. Systems and methods for retrieving causal set of events from unstructured signals. US Patent application US201220301105 A1. Publication date: Nov. 29, 2012. Priority date: Mar 22, 2011. Pending.
- IP6. Shwetak Patel, Gregory D. Abowd, Matthew S. Reynolds, Thomas Robertson, Erich Stuntebeck. Sub room level indoor location system using wideband power line positioning. US Patent 8,494,762 B2. Publication date: July 23, 2013. Priority date: June 28, 2006.
- IP7. Shwetak N. Patel, Matthew S. Reynolds, Gregory D. Abowd. Motion Detecting Device, Method of Providing the Same, and Method of Detecting Movement. US Patent application US20130289930 A1. Publication date: Oct 31, 2013. Priority date: May 12, 2009. Pending.
- IP8. Gregory D. Abowd, Peter S. Abowd, Wesley Horner, Steven Tengler, Yi Han, and Chen Chen. Systems and methods for utilizing micro-interaction events on computing devices to administer questions. Pending US patent application US20140298260 A1. Patent not granted.
- IP9. EP Stuntebeck, TM Robertson, GD Abowd, SN Patel. Method and apparatus for using in- home power lines to support lower power wireless sensors and to extend the range of low-power wireless devices. US Patent 9,385,783.
- IP10. **Cheng Zheng**, Gregory D Abowd, Junrui Yang. Extending Interactions Of A Portable Electronic Device. US patent application US10684694B2. Publication date: August 3, 2018. Priority date: August 29, 2016.
- IP11. **Nivedita Arora**, Gregory D Abowd, **Mohit Gupta**, **Diego Osorio**, **Seyedeh F Shahmiri**, Thad Starner, Yi-Cheng Wang, Zhengjun Wang, Zhong Lin Wang, Steven L Zhang, Peter McAughan, Qiuyue Xue, Dhruva Bansal, Ryan Bahr, Emmanouil Tentzeris. Thin And Flexible Self-powered Vibration Transducer Employing Triboelectric Nanogeneration. US application US20190373375 A1. Publication date: December 5, 2019. Priority date: May 29, 2019.
- IP12. Cheng Zhang, Gregory D Abowd, Omer Inan, Pranav Kundra, Thomas Ploetz, Yiming Pu, Thad Eugene Starner, Anandghan Waghmare, Xiaoxuan Wang, Kenneth A Cunnefare, Qiuyue Xue. Systems, Methods and Devices for Gesture Recognition. US application US20210109598 A1. Publication date: April 15, 2021. Priority date: September 6, 2018.
- IP13. Nivedita Arora, Gregory D Abowd, Mohit Gupta, Diego Osorio, Seyedeh Fereshteh Shahmiri, Thad Eugene Starner, Yi-Cheng Wang, Zhengjun Wang, Zhong Lin Wang, Steven L Zhang, Peter McAughan, Qiuyue Xue, Dhruva Bansal, Ryan Bahr, Emmanouil Tentzeris. Thin and Flexible Self-powered Vibration Transducer Employing Triboelectric Nanogeneration. US application US20210281955 A1. Publication date: September 9, 2021. Priority date: January 18, 2021.

- IP14. Hyeokhyen Kwon, Gregory D Abowd, Harish Kashyap Haresamudram, Thomas Ploetz, Eu Gen Catherine Tong, Yan Gao, Nicholas Lane. Method And System For Automatic Extraction Of Virtual On-Body Inertial Measurement Units. US Patent 12,236,616 Publication date: February 2, 2025. Priority date: September 1, 2021.
- IP15. Cheng Zhang, Gregory D Abowd, Omer Inan, Thad Eugene Starner. Electronic Device and Method of Controlling Same. US application US11389084 B2. Publication date: July 19, 2022. Priority date: August 15, 2017.
- IP16. Seyedeh Fereshteh Shahmiri, Chaoyu Chen, Gregory D Abowd, Shivan Mittal, Thad Eugene Starner, Yi-Cheng Wang, Zhong Lin Wang, Dingtian Zhang, Steven L Zhang, Anandghan Waghmare. Flexible sensing interface systems and methods. U.S. Patent 11962296. Publication date: April 16, 2024.

Software Systems

- SW1. Zen-Star 1.0. Software system for automated capture, integration and access of university lectures. Produced as part of the Classroom 2000 project. Designers: Gregory Abowd, Jason Brotherton, Christopher Atkeson and Janak Bhalodia.
- SW2. The Context Toolkit 1.0. Software framework for the development of context-aware computing applications. Designers: Gregory Abowd, Anind Dey and Daniel Salber.
- SW3. StuPad 1.0. Software system that works with Zen-Star system [S/W.1], to provide personalized note-taking by students in Classroom 2000.
- SW4. OOPS toolkit. Software system that supports reusable mediation strategies to handle ambiguity in recognition-based interfaces. Designers: Jennifer Mankoff, Scott E. Hudson (CMU) and Gregory D. Abowd.
- SW5. INCA toolkit. Software framework for development of automated capture and access applications. Designers: Khai Truong and Gregory D. Abowd.
- SW6. TeamSpace. Software for distributed meeting capture. Designers: Heather Richter, Khai Truong, Ludwin Fuchs (Boeing), Werner Geyer (IBM/Lotus) and Gregory Abowd
- SW7. The Location Service. System to support fusion of separate location-sensing technologies. Designers: Thomas O'Connell and Agathe Battestini and Gregory Abowd.
- SW8. The Walden automated tracking system prototype. Prototype capture application for mobile analysis of behaviors of children with autism. Designers: David White,
- SW9. The Family Video Archive. System to support annotation and browsing of digital home movies. Designers: Gregory Abowd, Matthias Gauger, Andreas Lachenmann and Shwetak Patel.
- SW10. The Personal Audio Loop. Application developed for Motorola i-series handheld phones to support near-term audio reminders. Designers: Khai Truong, Shwetak Patel, Gillian Hayes, Julie Kientz, Giovanni Iachello, Rob Farmer and Gregory Abowd.
- SW11. Abaris. An automated capture application to support the discrete trial intervention therapy of ABA. Designers: Gregory Abowd. David White, Julie Kientz and 3911 Senior Design team (Aldrian Harjati, Mark Bernardi, Pavel Kremer, Chris Widjaja). Version 2.0 developed in Fall 2004 by Gregory D. Abowd, Julie Kientz and Sebastian Boring.

- SW12. CareLog. A system using the Intel Personal Server to support monitoring, diagnosis and intervention treatments of behavioral and learning disabilities in children. Designers: Gillian Hayes, Gregory Abowd and Trevor Pering (Intel).
- SW13. GT Sound Source Location system An indoor 16-microphone array system to deliver location of sound sources. Designers: Xuehai Bian, Jim Rehg and Gregory Abowd.
- SW14. The ContextCam. Automated video annotation system. Designers: Shwetak Patel and Gregory Abowd.
- SW15. Multi-user Card Game Engine for the DiamondTouch. Designers: Shwetak Patel. Gregory D. Abowd and 3911 Senior Design team (John Bunch, Kyle Forkner, Logan Johnson, Tiffany Johnson, Michael Rosack).
- SW16. LoCoL: a multiplayer location-aware game for mobile phones. Designers: Julie Kientz, Shwetak Patel and Jose Zagal.
- SW17. Gvu Procams Toolkit. Designers: James Rehg, Jay Summet and Matt Flagg. Offers a programming interface designed to allow programmers quickly build projector/camera applications on a Win32 platform. Virtual Rear Projection system variants have been produced with this toolkit, under the advisement of Rehg and Abowd.
- SW18. The Capture Resistant Environment: Khai Truong, Jay Summet, Shwetak Patel and Gregory D. Abowd. Camera-projector system that prevents CCD-CMOS based phones and camcorders from being able to capture sensitive information in the environment. Selected as New York Times Magazine 2005 Top Idea of the Year and patent pending.
- SW19. Experience Buffers, Bufferware, CareLog FBA: Gillian Hayes, Khai Truong, Lamar Gardere, Ellen Matthews. Selective archiving infrastructure used to support informal sensor data capture with annotation. Subject of pending utility patent.
- SW20. Powerline positioning system: Shwetak N. Patel, Khai N. Truong and Gregory D. Abowd. Indoor positioning system leveraging domestic powerline infrastructure and fingerprinting. Invention disclosure April 2006 and provisional patent filed 2006.
- SW21. TrackSense: Moritz Köhler, Shwetak Patel, Jay Summet, Erich Stuntebeck, Gregory D. Abowd. Infrastructure-free indoor localization system. Invention disclosure November 2006.
- SW22. Continuous Flagging and Recording Technology (CRAFT): Nazneen, Yi Han. 2008
- SW23. Refl-ex and REACT system. Fatima A. Boujarwah, 2011
- SW24. Quedget: Lockscreen interaction to obtain quick answers to simple questions. Part of commercialization effort.

Startup Companies

DominINC LLC, technical co-founder 2006, working on camera-blocking technologies
uSenso/Zensi LLC, technical co-founder 2009, infrastructure-mediated sensing technologies. Acquired by Belkin in 2010.
BrailleTouch LLC, technical co-founder 2012, Braille input on smartphones.
Behavior Imaging Technologies, Inc. (originally Caring Technologies), Chief Research Office 2006-2015, telehealth solutions for the autism market

LSQ (dba Quedget), technical co-founder 2013, input techniques on smartphones

Consulting

Behavior Imaging Solutions, Inc. (formerly Caring Technologies, Inc.) 2005-2015. I serve as Chief Research Officer for this start-up company and direct activities for grant funding of research involving information technologies and autism. Several successful SBIR grants (NICHD, Phase 1 and Phase 2, NIMH, Phase 2 renewal) have been received and executed upon.

Ultra Large Scale Software Systems panel, Carnegie Mellon University Software Engineering Institute, August 2005-April 2006.

NICTA, Sydney Australia. Visiting researcher teaching a course on HCI and Ubiquitous Computing, June 2005.

External member of Board of Directors, HomeCom Communications, Inc. 1997-1999.

Motorola Satellite Communications Division, Software architecture evaluation, April 10–11, June 28–29 1995.

Ford Motor Company, Electronics Division, Software architecture evaluation, April 28, 1995.

Motorola Cellular Infrastructure Group, Software architecture evaluation, August 22, 1995.

Software Engineering Institute, Software architecture evaluation report for Sandia Labs, two days consulting over summer 1995.

HomeCom Communications, Inc., Future interests in the Internet, May 6, 1995.

HomeCom Communications, Inc. Board of directors, September 1996–present.

Schlumberger, architecture consulting, January 1999.

Intel Microprocessor Research Labs (MRL) Board of Advisors, September 2000–present.

National Science Foundation Science and Technology Center at UCLA (Deborah Estrin, PI), Architectural Advisory Board, 2001–present.

University of Florida, consulting on development of laboratory for home technologies, September 2003.

Samsung Electronics, consulting on commercial potential of automated capture applications, Sep-Dec. 2003.

University of Florida CREATE Center academic advisory board, 2004-2008.

Advisory Board for Autism Speaks (formerly Cure Autism Now) Innovative Technologies for Autism (ITAB), 2004-2010.

Intel Research Seattle Visiting Faculty, July 2004-June 2005.

Center for Discovery, IT consulting for supporting research and educational practices, January 2012-2015. 1-6 hours per week.

LSQ Mobile LLC, Technical co-founder advising on mobile market research activities, April 2013-present.

Disney Research Pittsburgh, consulting

Expert Witness experience

Since 2008, I have assisted numerous clients in intellectual property litigation. I have

prepared declarations and expert reports in support of various trials. I have been deposed (3 times) and testified in court (1 time) in the US.

2008 Consulting expert witness for Sidley Austin LLP, Microsoft v. Tom-Tom. Summer 2008. Did claims analysis for a set of patents relating to navigation interfaces and services. Attorneys: Douglas Lews and John McBride.

2008-2010 Georgia Bureau of Investigation, request to provide an expert witness account on a case against former Georgia Tech faculty members accused of inappropriate use of sponsored research funds. 2008-2010.

2010 Subpoenaed by the International Trade Commission as an expert witness in a case between Nokia and Apple and I was deposed by lawyers representing both sides for about 90 minutes. I was brought in to discuss details of some of my work from the 1990's that was being claimed as prior art. July 2010. Uncompensated.

2011 For Nokia with Alston & Bird LLP to assist in evaluating certain claims and issues Nokia had against Apple Inc. relating to Apple's iPhones, iPads, Macs, and other electronic devices This case was settled. Attorney contact: Adam Swain.

2011 For Openwave with Alston & Bird LLP in case against RIM/Google/Android/LG. Attorney contact: David Ben-Meir. Letter of engagement signed on July 15, 2011. Openwave decided in November 2011 to change representation, and I completed no work.

2011-2012 For Samsung with Quinn Emmanuel in International Trade Commission Investigation No. 337-TA-794. Attorneys: Alan Whitehurst, Anthony Zappin, Marissa Ducca. Wrote expert reports, was deposed and testified in court.

2012 For Samsung with Quinn Emmanuel in United States District Court, Northern District of California, San Jose Division, case number 11-CV-01846-LHK . Wrote expert reports. Attorney: Kenneth Suh. Patent dropped from case before deposition.

2013 For CooTek with Finnegan, Henderson, Farabow, Garrett and Dunner in International Trade Commission Investigation No. 337-TA-864 . Nuance Communications, Inc., Swype, Inc., Tegic Communications, Inc., and ZI Corporation of Canada, Inc., (collectively, "Nuance") v. Shanghai HanXiang (CooTek). Prepared expert reports and rebuttals and was deposed and prepared written expert testimony for trial. Case settled prior to court date. Attorneys: Aidan Skoyles and Qingyu Yin.

2014 For Learner's Digest International with Perkins Cole LLP in Astute Technology vs. Learner's Digest International, Eastern District of Texas case. Prepared evidence for invalidity arguments. Case settled. Attorney: David R. Pakarek Krohn.

2016 For Immersion Corp. with Irell & Manelli LLP vs. Apple Corp and AT&T Mobility in International Trade Commission Investigation Nos. 337-TA-1004 and 337-TA-990 (Consolidated) "Certain mobile and portable electronic devices incorporating haptics (including smartphones and laptops) and components thereof". Preparing expert reports, deposition and witness statement. Attorneys: Gavin Snyder and Rich Birnholz. Went to trial with cross examination.

2017 For ARRIS Enterprises LLC with Fish & Richardson P.C. in Sony Corp. v. ARRIS Enterprises LLC, IPR2017-01695 and IPR2017-01803. Preparing expert reports. Attorneys: Erin Alper and Noah Graubart. Client dropped law firm prior to submission of IPR preliminary reports.

2018 For Samsung Electronics with Arnold & Porter Kaye Scholer LLP in KEYnetik, Inc. v.

- Samsung Electronics Co., Ltd. et al., Case No. 17-cv-2794-JLL (District of New Jersey). Retained as expert witness for invalidity of 2 patents. Also involved in related IPR case with Paul Hastings as counsel.
- 2019** For Google with Wolf and Greenfield Google v. Ziplt. Details of case withheld by request of counsel. Retained as expert witness for IPR case.
- 2021** For Samsung Electronics America with Paul Hastings LLP in Gesture Technology Partners, LLC v. Samsung Electronics Co., Ltd. et al., 2-21-cv-00041 (Eastern District of Texas. Attorney Anderson To.
- 2021** For T-Mobile with Perkins Coie LLP in AGIS Software Development LLC v. T-Mobile, Case No. 21-CV-00072 (E.D. Tex.). Attorney Ryan McBrayer.
- 2021** Apple v. Maxell. Retained as fact witness by DLA Piper.
- 2022** For Snap Inc. With Munger, Tolles & Olson LLP in You Map, Inc. v. Snap Inc., No. 20-162-CFC (D. Delaware). IPR review. Attorney David Brightman.
- 2022** For Immersion with Folio Law Group PLLC in Immersion Corporation v. Meta Platforms, Inc., f/k/a Facebook, Inc., Case No 6:22-cxv-541, United States District Court for the Western District of Texas.
- 2022-23** For Cisco and HP with Winston & Strawn in BillJCo v. Cisco and HP Case No. 2:21-cv-0081-JRG (Eastern District of Texas). Attorneys David Lin and Krishnan Padmanabhan.
- 2024** For Samsung Electronic Co., Ltd. and Samsung Electronics America, Inc. (collectively, "Samsung") with O'Melveny & Myers LLP in litigation against Slyde Analytics LLC ("Slyde") in the cases captioned as Slyde Analytics LLC v. Samsung Elecs. Co., Ltd., Case No. 2:23-CV-83 (E.D. Tex.) and in certain matters expected to be filed at the United States Patent and Trademark Office. Attorney Bradley Berg.
- 2024** For Lenovo (United States), Inc., Lenovo Group Limited, and Lenovo Information Products (Shenzhen) Co., Ltd. (collectively, "Lenovo") with O'Melveny & Myers LLP in IngenioSpec, LLC (IngenioSpec) v. Lenovo Certain Electronic Eyewear Products and Components Thereof, Investigation No. 337-TA-1383 (ITC). Attorneys Grace McFee and Nicholas Whilt.

G. SOCIETAL AND POLICY IMPACTS

Present a brief list of the broader impacts of your scholarship, and elaborate on them in your personal statement.

I started the Aware Home project in 1998 with funding from the Georgia Research Alliance. This effort has greatly influenced national and international efforts on aging in place, i.e., technologies for an aging population, as well as technologies for health in general. Many other universities followed Georgia Tech's lead by creating their own living home laboratories. I was invited to testify before the U.S. Senate Special Committee on Aging, hearing on baby Boomers at the Gate on May 20, 2003 (<http://aging.senate.gov/hearings/hr101ga.pdf>). I was also invited to write a chapter for the National Research Council on Technologies for Aging (see book chapter BC9).

In 2008, I founded the Atlanta Autism Consortium, a non-profit organization that facilitates

broader understanding and communication between the research community and all stakeholder communities related to autism in the Greater Atlanta area. This effort started as an off-shoot of my own research in technology and autism. I received the 2017 State of Georgia Outstanding Achievement in Autism Award for my efforts in leading the Consortium and the overall impact it has had on the community.

At Northeastern, I am lead PI for the NSF INCLUDES Engineering PLUS Alliance, looking to increase URM/women participation in engineering undergraduate and graduate programs. This work connects national organizations, such as ASEE and the 50K Alliance.

At Northeastern, I have initiated a Neurodiversity Initiative to support workforce development for university students who identify as neurodivergent. This effort has resulted in a collaboration with Vanderbilt University, the University of Connecticut, and Fisk University in a NSF proposal for Broadening Participation in Engineering. I look to use Northeastern's co-op program as a model for ensuring a resilient post-graduate career for this population, and in turn create a more inclusive environment at Northeastern and other universities that also results in more inclusive work environments.

V. TEACHING

A. COURSES TAUGHT

List course taught at Georgia Tech. (Most recent first and include the last six years)

Fall, 2018	CS/PSYC 3750	User Interface Design	70
Spring, 2018	CS/PSYC 3750	User Interface Design	70
Spring, 2017	CS 4605/7470	Mobile and Ubiquitous Computing	73
Fall, 2016	CS/PSYC 3750	User Interface Design	68
Fall, 2015	CS/PSYC 3750	User Interface Design	67
Fall, 2015	CS 6452	Prototyping Interactive Systems	22
Spring, 2015	CS 4605	Mobile and Ubiquitous Computing	40
Fall, 2014	CS 6452	Prototyping Interactive Systems	28
Fall, 2013	CS 4911 C4G	Computing for Good	60 (w/Zegura)
Spring, 2013	CS 4605/7470	Mobile and Ubiquitous Computing	85 (w/Starner)
Fall, 2012	CS 4911 C4G	Computing for Good	45 (w/Vempala)
Fall, 2011	CS/PSYC 6750	Human-Computer Interaction	40
Fall, 2011	CS 4605/7470	Mobile and Ubiquitous Computing	65
Fall, 2010	CS 7001	Introduction to Graduate Studies	40 (w/Feamster)
Fall, 2010	CS 4605/7470	Mobile and Ubiquitous Computing	65
Spring, 2010	CS 8803 TAU	Technology and Autism	15 (w/Arriaga)
Fall, 2009	CS 4605/7470	Mobile and Ubiquitous Computing	60
Spring, 2009	CS/PSYC 6750	Human-Computer Interaction	35

B. INDIVIDUAL STUDENT GUIDANCE

List all Postdoctoral Fellows, Ph.D. students, M.S. Thesis students, and undergraduate students supervised/advised. Explicitly indicate any co-advisement relationships. For Ph.D. and M.S. Thesis students, include date of graduation and title of thesis and, if known, the current position of the graduate students. For graduate students currently supervised, indicate the semester advisement began, their progression through appropriate exams, title of their project/dissertation, and current position if available. Provide any indicators you have of the quality of your mentorship.

B1. Postdoctoral Fellows

1. Daniel Salber, October 1997-September 1999. Worked on ubiquitous computing and context-aware computing. Work has resulted in conference publications. Work resulted in Context Toolkit software system. Formerly a Research Scientist at IBM T.J. Watson Labs, currently back in France.
2. Jaeseok Yun, 2006-2009. Postdoctoral research scientist. Working on sensing in a home environment.
3. Mario Romero, 2009-2012. Postdoctoral research scientist. Working on visualization of human behavior. Currently Associate Professor at KTH in Sweden.
4. Thomas Ploetz, 2011-2012, Visiting Research Scientist from Germany. Work on behavior imaging and pattern recognition of on-body sensing. Currently Senior Lecturer at University of Newcastle.

B2. Ph.D. Students

Graduated

1. Richard Kurt Stirewalt, (co-advised with S. Rugaber) 1994-1997. Thesis "Automatic Generation of Interactive Systems from Declarative Models." Computer Science Ph.D. program, College of Computing, Georgia Tech. NSF CAREER Award winner 2000. Formerly Associate Professor (with tenure) at Michigan State University. Currently VP Application Architecture at LogicBlox.
2. Anind Dey, 1995-2000. Thesis: "Providing Architectural Support for Building Context-Aware Applications." Named 2000 College of Computing Outstanding Graduate Research Assistant. Was Professor and Charles M. Geschke Director of the Human-Computer Interaction Institute at Carnegie Mellon University. Currently Dean and Professor University of Washington iSchool.
3. Jennifer Mankoff, 1996-2001. Thesis: "An architecture and interaction techniques for handling ambiguity in recognition-based input." Named 2001 College of Computing Outstanding Graduate Research Assistant. Sloan Fellow. IBM Faculty Fellow. Was Assistant Professor in Computer Science at UC Berkeley. Was Professor in the Human-Computer Interaction Institute at Carnegie Mellon University. Currently Richard E. Ladner Endowed Professor in Computer Science and Engineering at the Allen School of

- Computer Science and Engineering at the University of Washington.
4. Jason Brotherton, 1996-2001. Thesis: "Enriching Everyday Activities through the Automated Capture and Access of Live Experiences: eClass: Building, Observing and Understanding the Impact of Capture and Access in an Educational Domain." Working on automated capture and access for live experiences. Was Assistant Professor at Ball State University, 2001-2002. Awarded British Society Postdoctoral Fellowship 2002-2006. Now living as Clio Soleil.
 5. Robert Waters, 1997-2004. Thesis: "Obtaining Architectural Descriptions from Legacy Systems: The Architectural Synthesis Process (ASP)." Currently a full-time instructor in the College of Computing at Georgia Tech.
 6. Heather (Richter) Lipford, 1997-2005. Thesis: "Designing and Evaluating Meeting Capture and Access Services." Currently a Full Professor in Dept. of Information Science at University of North Carolina Charlotte.
 7. Khai Truong, 1998-2005. Thesis: "INCA: An Infrastructure to Support the Generation, Preservation & Use of Memories from Everyday Life." Currently a tenured Associate Professor and Associate Chair of Computer Science at University of Toronto.
 8. Lonnie Harvel, 1999-2005. Thesis: "Using Student Generated Notes as an Interface to a Digital Repository." Formerly Vice President of Educational Technology, Georgia Gwinnett College. Deceased: November 4, 2010.
 9. Giovanni Iachello, 2003-2006. Thesis: "Privacy and Proportionality." NSF Graduate Fellow, 2003-2006. Currently Head of International & New Markets at LinkedIn.
 10. Kris Nagel, 2000-2006. Thesis: "Using Availability Indicators to Enhance Context-Aware Family Communication Applications." 2005 Google Anita Borg Fellowship. Formerly a tenured Associate Professor at Georgia Gwinnett College. Currently Academic Professional in the School of Electrical and Computer Engineering at Georgia Tech.
 11. Jay Summet, 2002-2007. (co-advisor Jim Rehg) Thesis: "Virtual Rear Projection: Improving the User Experience with Multiple Redundant Projectors." Currently a full-time instructor with the College of Computing, Georgia Institute of Technology.
 12. Gillian Hayes, 2003-2007. Thesis: "Documenting and Understanding Everyday Activities through the Selective Archiving of Live Experiences." Awarded IBM Research Fellowship and Anita Borg Fellowship in 2006. Currently a Professor and Robert A. and Barbara L. Kleist Chair in Informatics at University California, Irvine.
 13. Julie Kientz, 2004-2008. Thesis: "Supporting Data-based Decision-Making for Caregivers through Embedded Capture and Access." NSF Graduate Fellow, 2004-2007. Google Anita Borg Fellowship in 2007. PEO Fellow 2007. Currently a tenured Associate Professor in the Department of Human-Centered Design and Engineering, University of Washington.
 14. Shwetak Patel, 2003-2008. Thesis: "Infrastructure Mediated Sensing." 2007 College of Computing Outstanding GRA. NSF Graduate Fellow, 2005-2008. MacArthur Genius Grant awardee, 2011. Currently Washington Research Foundation Entrepreneurship Endowed Professor in the Computer Science and

- Electrical Engineering Departments, University of Washington.
15. Mario Romero, 2006-2009. Thesis: "Supporting Human Interpretation and Analysis of Activity Capture Through Overhead Video." School of Interactive Computing CS Ph.D. program. Currently Associate Professor in Computing at KTH in Sweden.
 16. Ping Wang, co-advisor with Jim Rehg, 2006–2010, Thesis: "Social Game Retrieval from Unstructured Videos." School of Interactive Computing CS Ph.D. program. Currently working as Research Scientist at ObjectVideo.
 17. Tracy Westeyn, co-advisor (with Thad Starner), 2006–2010. Thesis: "Child's Play: Activity Recognition for Monitoring Children's Developmental Progress with Augmented Toys." School of Interactive Computing CS Ph.D. program. Currently working for NSA.
 18. Erich Stuntebeck, 2006–2010. Thesis: "An Analysis of the Domestic Power Line Infrastructure to Support Indoor Real-Time Localization." School of Electrical and Computer Engineering PhD student. Currently a VP of Research for AirWatch in Atlanta, GA a division of VMWare.
 19. Fatima Boujarwah, co-advisor (with R. Arriaga), 2009–2012. Thesis: "Facilitating the Authoring of Multimedia Social Problem Solving Skills Instructional Modules." School of Interactive Computing CS Ph.D. program. Currently Assistant Professor at Kuwait University.
 20. Lana Yarosh, 2007–2012. Thesis: "Supporting Remote Synchronous Communication Between Parents and Young Children." School of Interactive Computing Human-Centered Computing Ph.D. program. 2010 recipient of IBM Graduate Research Fellowship. Currently Assistant Professor at University of Minnesota.
 21. Tae-Jung Yun, co-advisor (with R. Arriaga) 2007– 2012. Thesis: "Using Ubiquitous Communication Technology to Improve Pediatric Asthma Management." School of Interactive Computing Human-Centered Computing Ph.D. program. Currently Research Scientist with Samsung Electronics.
 22. Nazneen, co-advisor (with R. Arriaga) 2009–2014, Thesis: "Supporting in-home collection and sharing of behavior specimens for diagnostic assessment of children with Autism." School of Interactive Computing Computer Science Ph.D. program. Currently User Experience Researcher at Google.
 23. Hwajung Hong, co-advisory (with R. Arriaga) 2010-2015, Thesis: "Specializing Social Networking Services to Support the Independence of Adolescents and Adults with Autism." School of Interactive Computing Human-Centered Computing Ph.D. program. Currently Assistant Professor at Seoul National University, Korea.
 24. Edison Thomaz, (with I. Essa), 2011-2015, Thesis: "Automatic Eating Detection in Real-World Settings with Commodity Sensing." School of Interactive Computing, Human Centered Computing Ph.D. program. Work on health informatics and the home. Currently Assistant Professor at University of Texas, Austin.
 25. Arri Ciptadi, (with J. Rehg), 2010-2016, Thesis: "Interactive Tracking and Action Retrieval to Support Human Behavior Analysis." School of Interactive Computing

- Computer Science Ph.D. program. Machine Learning Research Scientist at Amazon Lab126.
26. Yi Han, (with J. Stasko), 2011-2016, Thesis: "Understanding Visual Analysis Processes from User Interactions using Visual Analytics." School of Interactive Computing, Computer Science Ph.D. program. Currently working for Delta Electronics.
 27. Aman Parnami, (with B. DiSalvo) 2013-2017, Thesis: "Enabling In Situ & Context-Based Motion Gesture Design." School of Interactive Computing, Computer Science Ph.D. program. Currently Assistant Professor IIT Delhi.
 28. Joelle Alcaidinho, (with M. Jackson) 2016-2017, Thesis: "The Internet of Living Things: Enabling Increased Information Flow In Dog-Human Interactions." School of Interactive Computing, Human Centered Computing Ph.D. program. Currently a Research Scientist at Intel Labs.
 29. Gabriel Reyes, (with K. Edwards), 2010-2017, Thesis: "Enabling One-Handed Input for Wearable Computing." School of Interactive Computing, Computer Science Ph.D. program. Currently Research Scientist at Thalmic Labs.
 30. Cheng "Allen" Zhang, (with O. Inan in ECE) since August 2012-2018, Thesis: "Novel gestures for wearables." School of Interactive Computing, Computer Science Ph.D. program. Currently Assistant Professor at Cornell University.
 31. Dingtian "Alan" Zhang, 2016-2021, Thesis: "Ubiquitous Self-Powered Ambient Light Sensing Surfaces", School of Interactive Computing, Computer Science Ph.D. program. Currently working at startup Hyperfine.
 32. Jung Wook Park, (with R. Arriaga) 2016-2021, Thesis: "Compute-proximal Energy Harvesting for Mobile Environments: Fundamentals, Applications, and Tools" School of Interactive Computing, Computer Science Ph.D. program. Currently Principal and Manager of the Future Technologies R&D group at Accenture Labs.
 33. Hyeokhyen "Allen" Kwon, (with T. Plötz) 2016-2021, Thesis: "Deriving Sensor-based Complex Human Activity Recognition Models Using Videos" School of Interactive Computing, Computer Science Ph.D. program. Currently Assistant Professor, Emory University Department of Biomedical Informatics.
 34. Mehrab Bin Morshed, (with T. Plötz) 2018-2021, Thesis: "Eating Behavior In-The-Wild and Its Relationship to Mental Well-Being" School of Interactive Computing, Computer Science Ph.D. program. Currently Sr. Research Engineer at Samsung Research America.
 35. Nivedita Arora, (with T. Starner) August 2016-2023, Thesis: "Sustainable Interactive Wireless Stickers - From Materials to Devices to Applications" School of Interactive Computing, Computer Science Ph.D. program. Currently Assistant Professor and & Allen K. and Johnnie Cordell Breed Jr. Professor of Design, Dept. of Electrical and Computer Engineering, Northwestern University.
 36. Vedant Das Swain (with M. De Choudhury) 2018-2023, Thesis: "Passive Sensing Frameworks for the Future of Information Worker" School of Interactive Computing, Computer Science Ph.D. program. Distinguished Postdoctoral Fellow in the Khoury College of Computer Sciences at Northeastern University.

37. Dong Whi Yoo (with M. De Choudhury) 2018-2023, Thesis: "Understanding Social Requirements for Social Media Powered Artificial Intelligence (SoMPAI) for Mental Health Care" School of Interactive Computing, Human-Centered Computing Ph.D. program. Currently Assistant Professor in the School of Information at Kent State University.
38. Youngwook Do (with S. Das) 2018-2023. Thesis: "Physicalizing Security and Privacy: Exploring Physically Intuitive Design to Build Users Trust in Using Ubiquitous Sensors" School of Interactive Computing, Computer Science Ph.D. program. Currently Research Scientist with Global Technology Applied Research, JPMorgan Chase.

Current

1. Tingyu Cheng (with H. Oh) August 2019-present. School of Interactive Computing, Human-Centered PhD program. Working on sustainable computational materials.

B3. M.S. Students (Indicate thesis option for each student)

2018-2019 Madhuri Bhavana (MS HCI project), Xiaomeng (Michelle) Ma (MS HCI project), Peter McAughan (MS CS independent research), Yuhui Zhao (MS CS independent research)

2017-2018 Caleb Southern (MS CS with HCC specialization), Yilin Liu (MS HCI project), Ruichen Meng (MS HCI project), Linlu Zhou (MS HCI project), Kshitish Milind (MS CS project), Qiuyue Xue (MS CS project), Vedant das Swain (MS HCI project), Anandghan Waghmare (MS CS project), Yuhui Zhao (MS CS), Richard Li (MS CS), Diego Osorio (MS HCI)

2016-2017 Shao-Yu Chen (MS HCI project), Yunnuo Cheng (MS HCI project), Sarthak Ghosh (MS HCI project), Aakanksha Mirdha (MS HCI project), Yiming Pu (MS HCI project), Tara Ramanan (MS HCI project), Felix Tener (MS HCI project), Sumeet Jain (MS independent study), Nikita Juneja (MS independent study), Pranav Kundra (MS independent study), Wei Yang Quek (MS independent study), Pratik Pravinkumar Shah (MS independent study), Vedant Das Swain (MS independent study), Anandghan Waghmare (MS CS project)

2015-2016 Manasvi Lalwani (MS HCI project), Abdelkareem Bedri (MS CS project), Nivedita Arora (MS HCI project), Jonathan Shaw (MS HCI project), John Hinkel (MS HCI project), Sarthak Ghosh (MS CS project), Lawrence Chan (MS HCI project), Apoorva Verleker (MS HCI project)

2013-2015 Arpita Bhattacharya (MS CS project), Rushil Khurana (MS HCI project)

2012-2013 Anhong Guo (MS HCI project)

2012 Aman Parnami (MS HCI project)

2012 Ravi Karkar (MS CS project)

2011- Alex Lynch (MS CS project)

2010-2012 Jennifer Kim (MS HCI project)

2010-2011 Gregory Garrett (MS CS project)

2010 Dani Rayan (MS CS project)

2007-2009 Sidhant Gupta (MS CS project), Mayank Garg (MS CS project), Mayank Goel

(MS CS project), Hyorim Park (MS HCI project), Yi Han (MS CS project), Sumit Savla (MS InfoSec project), Sidarsun Kannan (MS CS project), Hwajung Hong (MS HCI project)

2008 Jan Gillesen, Roel Hendriks and Bart van den Boogard. Visiting masters students from Technical University Eindhoven.

2007-2008 Alicia Nachmann (MS HCI project).

2006-2008 Sunyoung Kim (MS HCI project), Dounia Berrada

2006 Moritz Köhler, Visiting student from Technical University of Munich

2005 Jennifer Wiley (MS HCI project) Ja-Young Sun (MS HCI project), Janna Kimmel (MS HCI project), Yelena Nakhimovsky (MS HCI project)

2004-2005 Sebastian Boring, Visiting student from LMU, Munich Germany

2004-2005 Lamar Gardere (MS CS project)

2004 Michael Biebl. Visiting student from Karlsruhe University, Germany.

2004-2005 Jeremy Johnson (MS CS project), Peter Jensen (MS CS)

2003 Matthias Gauger and Andreas Lachenmann, Visiting students from Germany

2002-2003 David White (MS HCI project), Molly Stevens (MS IDT thesis), Palavi Garg (MS CS project)

2001-2003 Ramswaroop Somani (MS CS project), Gabe Hoffman (MS CS project), Venkataswaraman Ramachandran (MS CS project), Arpit Agarwal (MS CS project), Agathe Battestini (CS MS project)

2000-2002 Kuleen Mehta (MS CS project), Thomas O'Connell (MS CS project)

1999-2000 Gregory Krohne (MS CS project)

1998-1999 Vishal Dalal (MS CS project)

1997-1998 Roy Rodenstein (MS CS project)

1996-1997 Wasim Khan (MS CS project), James Seymour (MS CS project)

1996-1997 Mike Pinkerton, (MS CS Thesis: "Ubiquitous Computing: Extending Access To Mobile Data". GVV Technical Report GIT-GVV-97-09.)

1996-1997 Lein Ton, Visiting master's student from Delft University.

1995-1996 Sue Long (MS CS project), Rob Kooper (MS CS project), Savita Chandran (MS CS project), Yoosuf Goolamabbas (MS CS project), Ami Feinstein (MS CS project)

1995-1996 Nitin Sawhney (MS IDT project), Kipp Jones (MS CS project), Harinarayanan Balakrishnan (MS CS project)

1995-1996 Dietmar Aust, Visiting master's student from Germany

B4. Undergraduate Students

2018-2019 Ruixan Sun (undergraduate thesis), Anmol Lal (research capstone), Ruihan Xu (undergraduate thesis), Dhruva Bansal (undergraduate research), Shivan Mittal (undergraduate research)

2017-2018

2016-2017 Victor Chen (independent study), Anmol Lai (independent study), Saqib Ali (undergrad research), Casey Barnette (undergrad research), Bailey Bercik (undergrad research), Beiwen Liu (undergrad research), Joshua Morton (undergrad research), Jatin Nanda (undergrad research), James Park (undergrad research), Jason Wu (undergrad research), Jinming Yu (undergrad research).

2010-2011. Alex Lynch

2009. Pamela Overman, Jason Bennett
2008. Caitlin East.
2006. Roman Savaryn, Kate Rosier, Arwa Tyebkhan.
2005. Tejesh Patel, Ellen Matthews, Roman Savaryn, Ping Taing, Tina Chou, Asif Ladak, Jonathan McAbee Reher, Sunil Venkataram, Lauren Griffin, Forest Andrew Skaggs, Derek DeRaps, Tim Dorr, Kevin Kane , Nate Padgett, Eric Derenne, Axel Molina, Ping Taing, In Jae yi, Robert, Soe Htet, Will Clarke, Akash Shah, and Jarvis Greene, Shamsuddim Lakha
2004. Brian Adle, Patrick Jarrett, John Ndukuba, Forest Andrew Skaggs, Carlo Tambuatco, Jaime Yap, Irfan Kassam. Matt Balaun, Ray Cole, James Gaythwaite, Cory Jacobsen, William Allen, James Robert Farmer, Trevor Mann, Abiodun Otolorin, Ian Port, T. Scott Saponas, John Bunch, Kyle Forkner, Logan Johnson, Tiffany Johnson and Michael Rosack.
2003. Mohammad Ashraf, Meghan Byrne, Adam Gent, Christopher Gooley, Pavel Kremer, T. Scott Saponas, Wen Tian, Timothy Morton, Jonathan Razza, David Sharpe, Kevin Wong, William Allen, Kanishk Kapur, Ian Port, John Woehler, Mark Bernardi, Aldrian Harjati, Pavel Kremer and Christianto Widjaja, Shwetak Patel, Jesse Shieh, Nasir Barday, Tim Hardcastle, Scott Saponas, Trayton Otto, Adrian Abraham, Branden Hughes, Robert Martoncik, Dino Tufekcic.
2002. Shwetak Patel, Jesse Shieh, Nasir Barday, Scott Saponas, Trayton Otto, Samir Kaushal, Harold Owen Noll, Borna Safabaksh, Shuo Wang, Johnny S. Yen, Mohammed Ashraf, Lloyd Engebretsen, Lev Parnas, Randall Hutchison, Kamal Patel, Shwetak Patel, Jesse Shieh, Shuo Wang, Liang Wen, Johnny Yen, Gooley, Seth Lilavivat, Belinda Nambooz, T. Scott Saponas, Timothy Bethea, Logan Hauenstein, Branden Hughes, Robert Martoncik, James McDuffie, Tali Padan, Hossein Sharifi, Maxwell Speyer, Andrew Wightwick, Christopher Geisler, Peter Jensen.
1999. Cheryl Kay Holifield and Jamonica DeRamus, Cory Kidd, Mark Chapman.
1998. Gregor Altvater, John Desetto, Colin Waters, Holloway, Thinh To, Lynn Bacher and Jimmy Nesmith.
1997. Janak Bhalodia, Ahmad Aslami, Khai Truong, Dale Everett, Vergilia Chin and Paul Gulley, Darin Heuerman, Jonathan Somers, Jason Anderson, Jason Pierce, Kristin, Chris Reynolds, Caleb Billingsley, Tristan Jackson, Stephen Rushton, Thierry Ways, Kyle Phillips, Yonatan Feldman, Paul Gulley, Jonathan Somers, Jason Hong.
1996. David Chow, Darin Heurmann, Jason Hong, Gulley, Dale Everett, Scott Campbell.
1995. Greg Hankins, Casey Powell, Dilorio and John Lindsey.

B5. Service on thesis or dissertation committees

Georgia Tech

Annie Anton, Krishna Bharat, Erika Poole, Elaine Huang, Michael Terry, Kent Lyons, James Clawson, Andrea Forte, Temiloluwa Olubanjo (ECE), Maia Jacobs

External

Micah Rye, MIT Media Lab. Advisors: Mitchell Resnick and Rosalind Picard. Thesis topic: collaborative story creation and the impact on children with autism. External committee member, Defended December 2014.

João Sousa, Carnegie Mellon University School of Computer Science. Advisor: David Garlan. Thesis topic: Saving users from the distractions of ubiquity: an architectural framework. External committee member. Fall 2002-2005.

Antti Aaltonen, University of Groningen, Finland. 2007.

Remi Emoney, University of Grenoble, France. 2009. Advisor: James Crowley.

Andrea Tartaro, Northwestern University, 2010. Advisor: Justine Cassell.

Saunya Williams, Georgia Tech, School of Electrical and Computer Engineering. 2011. Advisor: Nikil Jayant.

Joshua Hailpern, University of Illinois, Department of Computer Science, 2012. Advisor: Karrie Karahalios.

B6. Mentorship of research scientists and visiting scholars

Mikiya Tani, September 1996-August 1997. Visiting Scientist from NEC in Japan. Work on retrieval interfaces for Classroom 2000.

Masayasu Futakawa, August 1998-August 1999. Visiting Scientist from Hitachi, Ltd. in Japan. Working on context-aware computing.

Yoshihide Ishiguro, since October 1998-September 1999. Visiting Scientist from NEC in Japan. Working on content-based retrieval for Classroom 2000.

Maria da Graca Pimentel, October 1998-January 2000. Visiting Scientist from University of San Paolo at Sao Carlos, Brazil. Working in automated support for capture and access to live experiences in the Classroom 2000 project.

Renata Fortes, January 2000-January 2001. Visiting Scientist from University of San Paolo, Brazil. Working in automated support for capture and access to meetings. Collaboration has also led to a joint NSF/CNPq grant from July 2000 – June 2003.

Cory D. Kidd, May 2000-May 2001. Research Scientist in charge of managing Residential Laboratory.

Thomas O'Connell, May 2001-August 2003. Research Scientist in charge of managing Residential Laboratory.

Annie Jacobs, January 2001-September 2002. Research Scientist examining socio-legal implications of research in the Aware Home.

Lamar Gardere, June 2005-January 2006. Research programmer employed to develop CareLog FBA system for use in schools.

Hideyuki Nakanishi, 2005. Visiting faculty from Keio University. Work on wearable computing and language support.

Rosa Arriaga, 2006-2007. Research Scientist in developmental psychology. Work on technologies for developmentally delayed and caregivers. Currently Senior Research Scientist in School of Interactive Computing at Georgia Institute of Technology.

Tom Robertson, 2006-2009. Research Scientist working on sensing solutions on the powerline.

Rongheng Lin, 2008-2009. Visiting Ph.D. student from Chinese Academy. Work on

data collection techniques for center-based special education.

Jong Choi, 2010-2011, Visiting Research Scientist from Korea. Worked jointly with Rosa Arriaga on collaboration technologies for chronic health and location-aware social networking.

Agata Rozga, 2010-2015, Research Scientist II. Work on technology to support analysis of children with autism. Currently Senior Research Scientist in School of Interactive Computing at Georgia Institute of Technology.

Kaya de Barbaro, 2015-2017, Research Scientist II. Working on Computational Behavioral Science associated with the CampusLife project. Currently Assistant Professor at University of Texas Austin.

C. OTHER TEACHING ACTIVITIES

I co-authored the textbook, *Human-Computer Interaction*, published by Prentice-Hall in 1993. Second edition published in November 1997. Third edition published in 2004. This text is one of the most popular HCI textbooks in the world, and has currently been translated into German, Chinese, Italian and Greek. A complete companion Web site has been developed for the book at <http://www.hcibook.com>, including answers to all exercises, overhead slides for every chapter, additional case studies, and project suggestions.

I have designed a course on Mobile and Ubiquitous Computing jointly with Thad Starner. The first instance of this course was given in the Spring Quarter of 1999 and it is now part of the regular undergraduate/graduate curriculum as CS 4605/7470. In Spring 2003, I taught this course jointly with Kishore Ramachandran, bringing in much more influence from the systems perspective. I have also been involved over the past year with efforts within the Ubiquitous and Pervasive Computing research communities to assemble and understand best practices for teaching these topics to undergraduates and graduates. This effort has resulted in the book *Foundations of Ubiquitous Computing*.

In Spring 2007, I designed a new course on the topic of technology and autism. This course was sponsored by the Cure Autism Now Foundation. The course has been offered twice, and is now merging into a more Atlanta-wide teaching and practicum experience.

In collaboration with Prof. Betsy DiSalvo in the School of Interactive Computing at Georgia Tech as part of a NSF-funded project, I have begun to redesign senior undergraduate computing classes to include more active learning materials through hands-on studio-style lectures. I have applied this model of learning to three different courses in the School of Interactive Computing CS 6452 (*Prototyping Interactive Systems*), CS 4605 (*Mobile and Ubiquitous Computing*) and CS 3750 (*User Interface Design*).

I have taught a 3-4 day summer course on Ubiquitous Computing and HCI at the University of Peking in Beijing, China in July 2015 and July 2018.

VI. SERVICE

A. PROFESSIONAL CONTRIBUTIONS

List all national and international contributions of service and positions of leadership in the profession.

Professional society membership

Member, IEEE Computer Society, 1992-present.

Member, ACM SIGSOFT, 1993-present. Member, ACM SIGCHI, 1994-present.

Member, IFIP Working Group 2.7 (User Interface Engineering), 1990-1998, elected position.

Member, Phi Beta Kappa, 1985-present, elected position.

Conference committee activities

Chair, CHI'94 Basic Research Symposium, 1994

Panels review committee, CHI'94, 1994.

Program committee, IFIP 2.7 Working Conference on Engineering for Human-Computer Interaction, 1992, 1995.

External reviewer, Design, Specification and Verification of Interactive Systems—DSV-IS'95.

External reviewer CSEE'96.

Workshop organizer, CHI'96 workshop on formal methods for interactive systems.

Program committee, BCS Formal Aspects of Human–Computer Interaction, BCS-FAHCI, 1996.

Program committee, Design, Specification and Verification of Interactive Systems—DSV-IS'96.

SIGSOFT'96 Symposium on Foundations of Software Engineering (FSE-4), Tutorials and Panels Chair.

ICSE'97 International Conference on Software Engineering, Student Volunteers co-chair.

1st International Symposium for Wearable Computing, program committee, 1997.

CHI'97 Workshop on Ubiquitous Computing, co-organizer, 1997.

NSF Program on Experimental Systems, grant reviewer, 1997.

SIGGRAPH'98 paper reviewer.

UIST'98 and UIST'99 panels committee.

AAAI 1998 Spring Symposium on Intelligent Environments, program committee. British HCI'98 conference, paper reviewer.

Foundations of Software Engineering, FSE'98, program committee.

ICSE'99, program committee and doctoral consortium committee.

DARPA/NSF/NIST Inter-Agency Workshop on Smart Environments. Atlanta, GA July 25-26, 1999. Local organizer.

CHI'2000, papers and panels reviewer.

Program chair for First International Workshop on Software Engineering Issues for Wearable and Pervasive Computing, held in conjunction with ICSE'2000 in Limerick, Ireland, June 2000.

Handheld and Ubiquitous Computing Symposium (HUC'2000) program committee, September 2000.

IEEE Workshop on Mobile Computing Systems and Applications (WMCSA 2000), program committee, December 2000.

Ubicomp 2001 (formerly HUC conference series), General Chair, September 2001.

CHI 2002 Associate Papers chair, Doctoral Consortium faculty panel, 2002.

IEEE Workshop on Perceptive User Interfaces (PUI 2001), program committee, November 2001

IEEE Workshop on Mobile Computing Systems and Applications (WMCSA 2002), program committee, 2002.

AAAI 2002 conference, program committee, 2002. UIST

2003, Program Chair (joint with Blair MacIntyre)

Workshop on Mobile Computing Systems and Applications (WMCSA '03) program committee, 2003.

Pervasive Computing 2004 conference program committee.

MobiSys 2004 program committee

Ubicomp conference series, Chair and Founding member of Executive Steering Committee, October 2003-present.

Ubicomp 2005 conference program committee.

Pervasive Computing 2006 conference program committee.

Ubicomp 2006 conference program committee, Doctoral Consortium Chair, Open Session Chair.

CHI 2007 Associate Papers chair.

Ubicomp 2007 co-Program Chair.

Pervasive Computing 2007 program committee

Ubicomp 2008 program committee

Pervasive Computing 2008 Doctoral Colloquium co-Chair.

CHI 2009 Subcommittee program chair

UbiComp/ISWC 2020 Technical Program Chair

Editorial positions

ACM SIGCHI Bulletin editor for Beyond the Desktop column on challenges of future computing technologies, 2001-present. Write a column for bi-monthly publication.

HCI Journal, Associate Editor 2001-present

IEEE Pervasive Computing Magazine, founding editorial board member and Associate Editor, 2001-2005.

Foundations and Trends in HCI, founding editor in chief, 2005-2009.

Proceedings of the ACM on Interactive, Mobile, Wearable, and Ubiquitous Technologies (IMWUT), Founding Editor-in-Chief, 2016-present.

B. PUBLIC SERVICE

List all public service activities that are professionally related.

C. COMMUNITY SERVICE

List all community contributions that are professionally related.

Presentation on GVV Center for Parents Weekend, Fall 1996.

FOCUS program speaker, January 16, 1997.

Co-organizer of Georgia Tech College of Computing weekly Hack Fest, Fall 1996 to present.

Georgia Tech Battle of the Bands judge, October 1997.

Rhodes Foundation, Georgia State Selection Committee, 1998-2000

Autism Society of America, Greater Georgia Chapter. Elected board of directors May 2006.

Founding President, Atlanta Autism Consortium, 2008-present

Marcus Autism Center, Board of Trustees, 2013-2017

Emory Autism Center, Board of Advisors, 2014-present

State of Georgia Autism Advisory Council, 2014-present

D. INSTITUTE CONTRIBUTIONS

As Dean of the College of Engineering at Northeastern, I have served on numerous extended and ad hoc committees to address long- and medium-term needs of the university.

Since, 2022, I have served on the Governing Board of the Northeastern University London campus.

EXHIBIT 2



KIMBERLY K. CAMERON, Ph.D., P.E.

PRINCIPAL

kkcameron@engsys.com

Dr. Kimberly Cameron is a Principal for ESi in the Mechanics and Materials practice. She specializes in design, failure analysis, and risk assessments of engineering structures and components. She has conducted hundreds of investigations on a wide variety of engineering structures, from miniature biomedical devices to large scale process equipment. She has also taught classes for engineers preparing to take the fundamentals of engineering exam and the professional engineering licensing examination in both Civil Engineering and Mechanical Engineering.

Dr. Cameron has performed multidisciplinary investigations on biomedical devices, pipelines, consumer electronics, consumer products, aircraft structures, industrial machinery, heat exchangers and cooling fins, refrigeration products, power-plant components, solar panels, pressure vessels, and vehicle components. The common thread in each of Dr. Cameron's investigations is the application of the fundamentals of metallurgy, materials science, engineering physics, computational mathematics, and engineering mechanics to help understand and solve complex problems.

Dr. Cameron is a registered patent agent, with experience in both the patent application process, as well as intellectual property disputes. In particular, she has experience in biomedical, automotive, electrical, mechanical, and materials science fields and has provided numerous invalidity and infringement reports. She has testified in federal and state court as well as in front of the ITC and in arbitration.

Areas of Specialization

Metallurgy	Product Design	Biomedical Devices
Plastics & Composites	Structural Engineering	Oil & Gas
Corrosion	Product Recalls	Energy & Power
Materials Selection	Root Cause Investigations	Consumer Products & Electronics
Mechanical Engineering	Regulatory Compliance	Automotive & Aviation
Intellectual Property Matters	Failure Analysis	Electrical Equipment

Education

Ph.D. Mechanical Engineering, minor Materials Science & Engineering, Stanford University, CA, 2004

M.S. Mechanical Engineering, Stanford University, CA, 2000

B.S.E. Mechanical and Aerospace Engineering. Certificates: Engineering Physics, Applied & Computational Mathematics, Materials Science & Engineering, Woodrow Wilson School of Public Policy & International Affairs, Princeton University, NJ, 1999

Licensed Professional Engineer (P.E.)

Mechanical Engineer, State of CaliforniaLicense No. 33732

Metallurgical Engineer, State of California.....License No. 1969

Professional Affiliations/Honors

American Society of Mechanical Engineers

Member

TMS

Member

Department of Defense

Fellowship

National Science Foundation

Fellowship

Lucent Technologies

Graduate Fellowship

Positions Held

Engineering Systems Inc., Sunnyvale, California

Principal, Mechanics, 2011 – Present

Exponent, Menlo Park, California

Senior Engineer, Mechanics & Metallurgy, 2004 – 2011

Publications/Presentations

“Crack Growth and Arrest in Steam Turbine Casings,” EPRI 3002003504, EPRI, Palo Alto, CA, December 2014.

M. H. Anderson, D. Cislo, J. Saavedra, and **K. Cameron**, Why International Inventors Might Want to Consider Filing Their First Patent Application at the United States Patent Office & the Convergence of Patent Harmonization and E-Commerce, 30 Santa Clara High Tech. L.J. 555 (2014).

“Increasing Reliability of Small Punch Fracture Toughness Testing with Acoustic Emission Monitoring,” Proceedings, 3rd International Conference on Small Sample Test Techniques, September 2014. (with D. Purdy and J. Foulds).

“Small Punch Fracture Toughness Evaluation of Combustion Turbine Materials,” EPRI 3002001468, EPRI, Palo Alto, CA, December 2013. (with D. Purdy, J. Foulds and J. Rodgers).

“Small Punch Testing for Fracture Toughness” by J. Foulds, J. Rodgers, **K. Cameron** and P. Sullivan, EPRI, Palo Alto, CA: 2013. 3002000250.

“The Effect of Low Sulfur Content on the Weldability of Linepipe Steel” by **K. Cameron** and A.M. Pettinger, Proceedings of the 9th International Pipeline Conference, September 2012.

- “Effectiveness of Hydrostatic Testing for High Strength Pipe Material” by **K.K. Cameron** & A.M. Pettinger, Proceedings, 8th International Pipeline Conference, October 2010.
- “Axial Loads from Soil Movement Challenge Pipeline Integrity” by **K.K. Cameron** & A.M. Pettinger, PipeLine Gas Technology, November 2009.
- “Assessing Pipeline Integrity Using Fracture Mechanics and Currently Available Inspections Tools” by **K.K. Cameron** & A.M. Pettinger, Journal of Pipeline Engineering, October 2009.
- “Assessing Pipeline Integrity Using Fracture Mechanics and Currently Available Inspections Tools” by **K.K. Cameron** & A.M. Pettinger, Proceedings, 2008 Evaluation and Rehabilitation of Pipelines Conference, Clarion Technical Conferences, Scientific Surveys Ltd., Prague, Czech Republic, October 2008.
- “Fatigue Damage in Bulk Metallic Glass I: Simulation” by **K.K. Cameron** & R. Dauskardt, Scripta Materialia 2006; 54(3):349–353.
- “Fundamentals of Engineering in a Flash” by **K.K. Cameron**, PPI, August 2006.
- “Transmission Electron Microscopy Structure and Platinum-like Temperature Coefficient of Resistance in a Ruthenate-Based Thick Film Resistor with Copper Oxide” by **K.K. Cameron**, G. Crosbie, J. Jiang, & X. Pan, J. Appl. Phys. 2000; 88:1124–1128.

Selected Project Experience

Failure analysis, design analysis, material selection, mechanical testing, finite element analysis, fatigue and fracture evaluation, and development of various implantable medical devices including cardiovascular implants (heart valves, stents, catheters, pacemakers), orthopedic implants (screws, plates, pins, hip implants, spinal implants), cranial implants, etc.

Design and failure analysis of various fluid and drug delivery systems including implantable pumps, drug eluting stents, syringes, patches, microneedles, CPAP machines, anesthesia machines, insulin and other drug delivery pumps, balloon respirators, etc.

Design and failure analysis of other types of medical devices including surgical tools, scooters, hospital beds, physical therapy equipment, refrigerated portable coolers, etc.

Performed analyses to support FDA submissions.

Evaluated the design, pipeline integrity management program, in-line inspection tools, leak detection systems, control room management and operation and maintenance practices, of both liquid and gas pipeline systems under both 49 CFR 195 and 192.

Evaluated pipeline construction practices of various pipeline systems and performed pipeline failure investigations after various landslide events.

Assisted the Inter-American Development Bank with a pipeline integrity study and supervision of implementing a pipeline integrity management program. This included geotechnical site investigations, fracture mechanics, metallurgical investigations, in-line inspection data, and external inspection data.

Evaluated welding procedures and determined the effect of welding defects and other construction factors on pipeline integrity.

Evaluated pipe-soil interactions, mechanical and welded joints, welding procedures, in-line inspection tools, regulatory compliance, and the overall integrity of pipeline systems.

Failure analysis of various insulated steel oil and water pipeline systems, including behavior of the polyurethane foam and jacket materials.

Design evaluation and failure analyses of pipe couplings and fittings, including finite element analyses, as well as analysis of various plumbing fixtures and components.

Performed design and failure analyses of various gas distribution systems made of polyethylene and steel pipe, including the evaluation of compression couplings.

Retained as an expert for the United States Patent and Trademark Office in the evaluation of a patent claim dispute relating to large diameter pipes.

Evaluated the design of a hydrogenerator system. Performed a finite element analysis and fatigue analysis of the rotors, thrust bridge, and head cover to assess the design.

Evaluated the design of various pumps, rotors, bearings, seals and insulation systems.

Thermal analyses of industrial heat exchangers and cooling fins, nuclear reactor heads, and welds.

Failure analysis of various refrigeration systems and components and HVAC system components.

Performed design analysis of various solar panel system components and assisted in design changes to meet standardized safety testing.

Developed testing protocols and conducted failure analyses and design reviews for various consumer electronic devices and accessories, electrical connectors, power cords, and jacket sleeves.

Design of electrical connectors and various components of several consumer electronic devices including phones, tablets, and computers.

Design evaluation of various consumer products, including washing machines, refrigerators, various child containment products, hand and power tools, security and display systems, coffee machines, exercise machines, cooling sleeves and pads, consumer electronic accessories, breast pumps, baby bottles, etc.

KIMBERLY K. CAMERON, Ph.D., P.E.

PROJECT

STATE

Fluidmaster, Inc. v. Danco, Inc. IPR2024-00634 USPTO Patent Trial and Appeal Board	D.C.	(deposition)
Stephen Barick v. Puli Pravin Reddy and Paizly, LLC. Civil Action No.: 2023-CV-382050 Superior Court of Fulton County, State of California	CA	(deposition)
DBR Finance, Inc. v. Rugged Cross Hunting Blinds, LLC IPR 2024-00291 Before the USPTO Patent Trial and Appeal Board	D.C.	(deposition)
Insulet Corporation v. EOFlow Co., et al. Civil Action No. 1:23-cv-11780-FDS United States District Court for the District of Massachusetts	MA	(deposition, trial)
Julie And Mark Nygren, et al. v. DCP Operating Company LP, et al. Case No.: 2020CV30400 Weld County District Court, Colorado	CO	(deposition)
In the Matter of Certain Liquid Transfer Devices Investigation No. 337-TA-1362 United States International Trade Commission	D.C.	(deposition, trial)
Eugene Davis, in his capacity as Liquidating Trustee of the Venoco Liquidating Trust v. Plains Pipeline, L.P., et al. Case No.: 16CV01319 Superior Court of the State of California, County of Santa Barbara	CA	(deposition, trial)
In the Matter of Certain Photovoltaic Connectors Investigation No. 337-TA-1365 United States International Trade Commission	D.C.	(deposition, trial)
Novo Nordisk Inc., and Novo Nordisk A/S v. Orbicular Pharmaceuticals C.A. No. 22-856 (CFC) United States District Court for the District of Delaware	DE	(deposition)
Smarte Carte Inc. v. Innovative Vending Solutions, LLC C.A. No. 1:19-cv-08681-ESK-AMD United States District Court for the District of New Jersey	DE	(deposition)
Baby Bjorn, AP, v. The Ergo Baby Carrier, Inc. Case No. 2:23-CV-00446AB U.S. District Court, Central District of California	CA	(deposition)
Grey Fox, LLC. v. Plains All American Pipeline, L.P. Civil Action No. 2:16-cv-03157 U.S. District Court, Central District of California	CA	(deposition)

California State Lands Commission et al v. Plains Pipeline, L.P. Case No.: 18CV02504 Superior Court of the State of California, County of Santa Barbara	CA	(deposition)
Hoya Corporation et. al. v. Alcon Inc. et. al. Civil Action No. 3:20-cv-03629-M US District Court for the Northern District of Texas	TX	(deposition)
National Products Inc. v. Innovative Intelligent Products, LLC D/B/A GPS Lockbox Case No. 2:20-cv-00428-DGE United States District Court for the Western District of Washington	WA	(deposition)
EIS Inc. v. IntiHealth Ger GmbH, WOW Tech USA, Ltd. et. al. Civil Action No. 19-CV-12279 (GBW)(MPT) United States District Court for the District of Delaware	DE	(deposition, trial)
Novo Nordisk Inc. and Novo Nordisk AS v. Teva Pharmaceuticals Civil Action. No. 21-1782 (CFC) United States District Court for the District of Delaware	DE	(deposition)
In the Matter of Compact Wallets and Components Thereof Investigation No. 337-TA-1288 United States International Trade Commission	D.C.	(deposition)
Hoya Corporation and Hoya Surgical Optics, Inc. v. Alcon Inc. Case No. 3:20-cv-03629 5 th Circuit, US District Court for the Northern District of Texas	TX	(deposition)
Envirotainer AB v. Doubleday Acquisitions, LLC IPR2022-00864, 00851, 00292, 00239 USPTO Patent Trial and Appeal Board	D.C.	(deposition)
Victualic v. ASC Engineered Solutions, LLP Case No. 1-20-CV-00887-CFC United States District Court for the District of Delaware	DE	(deposition, trial)
Echologics, LLC, Mueller International, LLC, and Mueller Canada, LTD. d/b/a Echologics v. Orbis Intelligent Systems, Inc. Case No. 3:21-cv-01147-RBM-AHG United States District Court Southern District of California	CA	(deposition)
Wonderland Switzerland AG v. Evenflo Company, Inc. Civil Action No. 1:20-cv-00727-JPM United States District Court for the District of Delaware	DE	(deposition, trial)
In the Matter of Certain Playards and Strollers Investigation No. 337-TA-1288 United States International Trade Commission	D.C.	(deposition, trial)
Timothy Johnson v. LA Fitness; Fitness International LLC Case No. 30-2019-01059520 Superior Court for the State of California for the County of Orange	CA	(deposition, trial)

Ultra-Mek, Inc. v. United Furniture Industries, Inc. et. al. Case No. 1:18-cv-281-WO-JEP United States District Court for the Middle District of North Carolina	NC	(deposition)
Unirac, Inc. v. Ecofasten Solar, LLC and ESDEC, Inc. Case No. 1:21-CV-000058, 1:21-CV-000059 United States District Court for the District of Delaware	DE	(deposition)
Transcontinental Gas Pipe Line Company LLC v Latex Construction Case No. CJ-2019-3611 In the District Court in and for Tulsa County	OK	(deposition)
Antonio Barrera and Martha Barrera v. Southern California Edison Case No. CIVDS1700188 Superior Court of the State of California	CA	(deposition)
Regeneron Pharmaceuticals, Inc. v. Novartis Pharma AG Case No. IPR2021-00816 Patent Trial and Appeal Board	D.C.	(deposition)
Novo Nordisk Inc. and Novo Nordisk AS v. Sandoz Inc. Case No. 20-747-CFC United States District Court for the District of Delaware	DE	(deposition)
Marathon Pipe Line LLC. v. Energy Transfer L.P, et. al. Case No. 2021CH14 Circuit Court Fourth Judicial Circuit in Marion County	IL	(deposition)

EXHIBIT 2

Robert S. Plachno
Patent Litigation Technical Analyst
24550 Miller Hill Rd., Los Gatos, CA 95033
bob@elqc.com

Career Skills

- Over 80 litigation code reviews and expert witness matters {C++, C, Java, RTL/VHDL, etc.}.
- Software code written {dynamic timing logic simulator, design-rule-independent layout, timing characterization, library development, project management tools, etc., C++, C, Python}
- Software and Hardware manager {embedded software, application software, driver software, mixed-signal design, logic design, analog design, layout, circuit design, physical design, customer support.
- Technical expertise {software and hardware, embedded processors, transistor level design - both digital and analog, logic design, memory design, CAD/CAE software, etc.}.

Work Experience (2010-2026)

For Litigation Consulting Work (2010-2026):

I am an independent code reviewer paid by multiple clients. My direct consultant work for Patent Litigation matters have been paid (2010 to present) by:
Precedia Associates LCC. in Palo Alto, CA, Finnegan Henderson Farabow Garrett & Dunner LLP in Palo Alto, CA, Google in Mountain View, CA, UnitedLex in Overland Park KS, O'Melveny & Myers LLP in San Francisco Ca, Glaser Weil in Los Angeles CA, Teklicon Inc. in San Jose CA, HID Global in Austin Tx, Morgan Lewis in Washington DC, Keker Van Nest in San Francisco CA, Baker Botts LLP in New York, NY, Intuity Consultants, Inc. in Morrisville, NC, Fabricant LLP in Short Hills, New Jersey, AT&T in Dallas, Tx.

The patent, trade secret, copyright, etc. litigation matters are:

- *UCT v. HP*, Case 4:24-cv-04097-YGR, United States District Court for the Northern District of California, filed 11/22/24, *Shook, Hardy & Bacon L.L.P.*, representing HP
 - Hardware and Software review representing HP, 2026
- *Groq, Inc. v. Positron AI*, JAMs Arbitration, *O'Melveny & Myers LLP*, filed 2/13/2025, representing Groq
 - Hardware and Software review representing Groq, 2025
- *Entropic v. Comcast*, Case 2:23-cv-01049-JWH-KES, United States District Court Central District of California, *Winston & Strawn LLP*, filed 10/31/23, representing Comcast
 - Software review representing Comcast, 2025
- *Gamba v. Samsung*, Case 2-25-cv-00438, United States District Court Eastern District of Texas, filed 4/25/2025, *Baker Botts*, representing Samsung
 - Software review representing Samsung, 2025
- *Oracle v. Procore*, Case 4:24-cv-7457-JST, United States District Court Northern District of California Oakland Division *Quinn Emanuel Urquhart & Sullivan*, filed 04/10/25, representing Procore
 - Software review representing Procore, 2025
- *Granite Vehicle Ventures v. Tesla*, Case 2:24-cv-01007-JRG, United States District Court Eastern District of Texas Marshall Division, filed 12/06/2024, representing Tesla
 - Software review representing Tesla, 2025
- *TPV v. DivX*, Case 37-2022-00024579-CU-CO-CTL, Superior Court of the State of California County of San Diego, filed 06/23/2022, *O'Melveny & Myers LLP*, representing TPV
 - Hardware and Software review representing TPV, 2025

- *UCT v. Lenovo*, Case 2:23-cv-00449-JRG, United States District Court Eastern District of Texas Marshall Division, filed 06/24/24, *Kilpatrick Townsend & Stockton LLP*, representing Lenovo
 - Hardware and Software review representing Lenovo, 2025
- *Cerence Operating Company v. Samsung*, Case 2:23-CV-0482-JRG-RSP, United States District Court Eastern District of Texas Marshall Division, filed 10/13/2023, *O'Melveny & Myers LLP*, representing Samsung
 - Software review representing Samsung, 2025
- *Sandpiper CDN v Comcast*, Case 2:24-cv-00886-JRG-RSP, United States District Court Eastern District of Texas Marshall Division, filed 11/1/2024, *Davis Polk*, representing Comcast
 - Software review representing Comcast, 2025
- *Wildseed Mobile v Google*, Case 6:21-cv-01245-ADA, United States District Court Western District of Texas Waco Division, filed 2/28/2022, *Keker Van Nest & Peters LLP*, opposing Wildseed
 - Software review opposing Wildseed, 2025
- *Mullen v. Samsung*, Case 2:24-cv-00049-JRG, (E.D. Tex.), filed 1/26/2024, *O'Melveny & Myers LLP*, representing Samsung
 - Software review representing Samsung, 2024-2025
- *Averon v. AT&T*, Case 1:22-cv-01341-TMH, Delaware District Court filed 4/08/2024, *O'Melveny & Myers LLP*, representing AT&T
 - Software review representing AT&T, 2024-2025
- *Attila Csupo, Andrew Burke, Kerry Hecht v. Google*, Case 19-cv-352557 Superior Court of the State of California County of Santa Clara filed 8/9/2019, *Cooley LLP*, opposing Attila Csupo, Andrew Burke, Kerry Hecht.
 - Software review opposing Attila Csupo, Andrew Burke, Kerry Hecht, 2024
- *Medtronic v. Axonics*, 337-TA-1396 filed 2/28/2024, United States International Trade Commission, *Winston & Strawn LLP*, representing Medtronic.
 - Software reviews representing Medtronic, 2024
- *Nokia v. Amazon*, Matters: 337-cv-1379 and 337-cv-1380 filed 10/31/2023, United States International Trade Commission, *Perkins Coie LLP*, representing Amazon.
 - Hardware and Software reviews representing Amazon, 2024
- *Eyesmatch v. Samsung*, Case 2:23-cv-00363 (E.D. Tex.) filed 8/8/2023, *Baker Botts LLP*, representing Samsung.
 - Software review representing Samsung, 2024
- *Active Wireless v. T-Mobile, Nokia, Ericsson*, Case 2:23-cv-00261-JRG (E.D. Tex.) filed 11/19/2023, *Fabricant LLP*, representing Active Wireless
 - Software review representing Active Wireless, 2024
- *Agis v. Samsung*, Case 2:22-cv-00263-JRG (E.D. Tex.) filed 7/14/2022, *Agis v. Lenovo*, Case 2:22-cv-00445-JRG (E.D. Tex.) filed 11/18/2022, *Google v. AGIS*, Case 3:23-cv-03624-PHK (N.D. Cal.) filed 7/21/2023, *O'Melveny & Myers LLP, Baker Botts LLP*, opposing AGIS.
 - Software review opposing AGIS, 2023
- *Nantworks v. Bank of America*, Case 2:20-cv-07872-GW-PVCx (C.D. Cal.) filed 8/27/2020, *Quinn Emanuel Urquhart & Sullivan*, representing Nantworks.
 - Software review representing Nantworks, 2023
- *Nariste Networks v. Samsung*, Case 2:23-cv-00031, filed 1/27/2023, United States District Court Eastern District of Texas Marshall Division, *O'Melveny & Myers LLP*, representing Samsung.
 - Software review, representing Samsung, 2023

- *Confidential arbitration in the semiconductor industry*, American Arbitration Association San Francisco Regional Office, *Keker Van Nest & Peters LLP*.
 - Software and Hardware review, 2023
- *Multimodal Media v. Samsung.*, Case 2:22-cv-00462-JRG-RSP, filed 11/29/2022, United States District Court Eastern District of Texas Marshal Division, *O'Melveny & Myers LLP*, representing Samsung.
 - Software review, representing Samsung, 2023
- *Webroot v. CrowdStrike*, Case 6:22-cv-00243-ADA-DTG, filed 3/15/2023, United States District Court Western District of Texas Waco Division, *Quinn Emanuel Urquhart & Sullivan*, representing CrowdStrike.
 - Software review representing CrowdStrike, 2023
- *AGIS v Google*, Case 337-TA-1347, filed 11/16/2022, United States International Trade Commission, *Wolf Greenfield & Sacks*, opposing AGIS.
 - Software review opposing AGIS, 2023
- *Inquisient v ServiceNow*, Case 1:22-cv-00900-UNA, filed 07/05/22, United States District Court for the District of Delaware, *Quinn Emanuel Urquhart & Sullivan*, representing ServiceNow.
 - Software review representing ServiceNow, 2023
- *HID Global Corporation v. Vector Flow, Inc.*, Case 1:21-cv-01769-VAC, filed 12/17/2021, Delaware District Court, *Morgan Lewis, O'Melveny & Myers LLP*, representing HID Global Corporation
 - Software review , representing HID Global Corporation, 2022-2023
- *Google Inc. v. Sonos Inc.*, Case T-952-20, filed 8/21/2020, Federal Court Toronto Ontario Canada, *Goodmans LLP*, opposing Sonos.
 - Expert Witness & Software review opposing Sonos, 2022
- *NXP v. Mediatek*, Case 37-TA-1287, filed 11/01/2021, United States International Trade Commission, *Quinn Emanuel Urquhart & Sullivan*, representing Mediatek.
 - Hardware review representing Mediatek, 2022
- *InfoGation v Google*, Case 3:21-cv-00843-H-JLB, filed 9/23/2021, United States District Court Southern District of California San Diego Division, *O'Melveny & Myers LLP*, opposing InfoGation.
 - Software review opposing InfoGation, 2021, 2022
- *Wisk v Archer*, Case 3:21-cv-02450-WHO, filed 10/06/2021, United States District Court Northern District of California San Francisco Division, *Quinn Emanuel Urquhart & Sullivan*, representing Wisk.
 - Software and Hardware review representing Wisk, 2021, 2022
- *Mediatek v. NXP*, Case 37-TA-1272, filed 06/21/2021, United States International Trade Commission, *Quinn Emanuel Urquhart & Sullivan*, representing Mediatek.
 - Hardware and software review representing Mediatek, 2021
- *Rubrik Inc v Commvault Systems Inc*, Case 6:21-cv-00135-ADA, filed 02/09/2021, Texas Western District Court, *Quinn Emanuel Urquhart & Sullivan*, representing Commvault Systems.
 - Software review representing Commvault Systems, 2021
- *Neural Magic, Inc. v. Facebook, Inc. and Aleksandar Zlateski*, Case No. 1:20-cv-10444-DJC, Filed 3/4/2020, United States District Court for the District of Massachusetts, *Quinn Emanuel Urquhart & Sullivan LLP*, representing Neural Magic, Inc., 2021.
 - Software review representing Neural Magic, 2021
- *Sonos Inc v Google LLC*, Case 6:20-cv-00881-ADA, filed 9/29/2020, United States District Court for the Western District of Texas Waco Division, *Quinn Emanuel Urquhart & Sullivan*, opposing Sonos.
 - Software review opposing Sonos, 2021

- *Sonrai Memory v Google*, Case 6:21-cv-00167-ADA, filed 2/23/2021, United States District Court, Western District of Texas Waco Division, *O'Melveny & Myers LLP*, opposing Sonrai.
 - Hardware review opposing Sonrai, 2021
- *Commvault Systems Inc. v. Cohesity Inc.*, Case 1:20-cv-00525-UNA, filed 04/21/20, United States District Court for the District of Delaware, *Quinn Emanuel Urquhart & Sullivan*, representing Commvault Systems.
 - Software review representing Commvault Systems, 2021
- *CoreOptical Technologies, LCC*, Case 8:19-cv-02190-JAK-RAO, filed 02/02/2021, United States District Court for the Central District of California, *Glaser Weil*, representing Core Optical.
 - Hardware review representing Core Optical, 2021
- *Nevro Corp v. Boston Scientific*, Case 18-644-CFC-CJB CONSOLIDATED, filed 6/22/2020, United States District Court for the District of Delaware, *Akin Gump*, representing Boston Scientific.
 - Hardware review representing Boston Scientific, 2020
- *Google v Sonos*, Case 3:20-cv-03845-EMC, filed 6/11/2020, California Northern District, *Quinn Emanuel Urquhart & Sullivan*, opposing Sonos.
 - Software review opposing Sonos, 2020-2021
- *AGIS v Samsung*, Case 2:19-cv-361-JRG, filed 11/04/2019, United States District Court, Eastern District of Texas Marshal Division, *Baker Botts LLP*, representing Samsung.
 - Software review representing Samsung, 2020
- *AGIS v Google*, Case 2:19-cv-361-JRG, filed 11/04/2019, United States District Court, Eastern District of Texas Marshal Division, *O'Melveny & Myers LLP*, opposing AGIS.
 - Software review opposing AGIS, 2020
- *ProofPoint v Vade*, Case 3:19-cv-04238, filed 07/24/2019, United States District Court, Northern District California San Francisco Division, *Quinn Emanuel Urquhart & Sullivan* representing ProofPoint,.
 - Software review representing ProofPoint, 2019-2020
- *SuperInterconnect Technologies LLC v. Google LLC.*, Case TXED-2-18-cv-00463, filed: 11/10/2018, Texas Eastern District Court, *O'Melveny & Myers LLP*. opposing SupperInterconnect.
 - Hardware and software source code review opposing SupperInterconnect, 2019-2020
- *Motorola Solutions Inc. v. Hytera Communications Corp. Ltd.*, Case NSD1283/2017, filed: 7/31/2017, Federal Court of Australia, New South Wales District Registry, *Shelston IP Lawyers Pty Ltd.* representing Hytera, 2019
 - Software source code review representing Hytera, 2019
- *Magento. v. Express Mobile*, Case 3.17-cv-02605-RS, filed 07/13/2018, United States District Court, Northern District California San Francisco Division, *Devlin Law Firm LCC* representing Express Mobile, 2019.
 - Software review representing Express Mobile, 2019
- *X2Y Attenuators, LLC. v. Intel Corp*, Case 3:18-cv-1394-HZ, filed: 1/4/2019, United States District Court, District of Oregon, Portland Division, *Dovel & Luner* representing X2Y, 2019
 - Hardware review representing X2Y, 2019
- *Hytera Communications Corp. Ltd., Inc. v. Motorola Solutions Inc.*, Case 3:17-CV-01973, filed: 9/20/2018, United States District Court for the Northern District of Illinois Eastern Division, *Steptoe & Johnson, LLP* representing Hytera, 2018
 - Software source code review representing Hytera, 2018

- *Qualcomm Inc. v. Apple Inc. et al*, Case No. 3:17-CV-02398-DMS-MDD, United States District Court Southern District of California, *Quinn Emanuel Urquhart & Sullivan*, 2018
 - Hardware source code review representing Qualcomm, 2018
- *Hytera Communications Corp. Ltd., Inc. v. Motorola Solutions Inc.*, PTAB Case IPR2018-00128, filed: 10/26/2017, United States District Court for the Eastern District of Virginia, *Calfee, Halter & Griswold LLP* representing Hytera, 2018
 - Software source code review representing Hytera, 2018
- *Qualcomm Inc. v. Apple Inc. et al*, Case No. 3:17-CV-1375-DMS-MDD, United States District Court Southern District of California, *Quinn Emanuel Urquhart & Sullivan*, 2018
 - Hardware and Software source code review representing Qualcomm, 2018
- *Qualcomm Inc. v. Apple Inc. et al*, Investigation No. 337-TA-1093, United States International Trade Commission, *Quinn Emanuel Urquhart & Sullivan*, 2018
 - Hardware and Software source code review representing Qualcomm, 2018
- *Qualcomm Inc. v. Apple Inc. et al*, Investigation No. 337-TA-1065, United States International Trade Commission, *Quinn Emanuel Urquhart & Sullivan*, 2018
 - Hardware and Software source code review representing Qualcomm, 2018
- *Qualcomm Inc. v. Apple Inc. et al*, 37-2017-00041389-CU-BC-NC, Superior Court of the State of California, San Diego County, *Quinn Emanuel Urquhart & Sullivan*, 2017-2018
 - Software source code review representing Qualcomm, 2017-2018
- *Waymo LLC v. Uber Technologies, Inc. et al.*, No. C 17-00939 WHA, United States District Court for the Northern District of California, *Quinn Emanuel Urquhart & Sullivan*, 2017
 - Software source code review representing Waymo, 2017
- *X One, Inc. v. Uber Technologies, Inc.*, Case No. 5:16-CV-6050-LHK (SVK), Filed: 08/19/2016, United States District Court for the Northern District of California San Jose Division, *Finnegan Henderson Farabow Garrett & Dunner LLP* representing X One, Inc., 2017
- *Hytera Communications Corp. Ltd., Inc. v. Motorola Solutions Inc.*, Case No. 1:17-CV-01794-DCN, filed: 11/03/2017, United States District Court for the Northern District of Ohio Eastern Division, *Calfee, Halter & Griswold LLP* representing Hytera, 2017-2018
 - Software source code review representing Hytera, 2017-2018
- Certain Two-Way Radio Equipment Systems, Related Software and Components Thereof, Investigation No. 337-TA-1053, United States International Trade Commission and Case No. 1:17-cv-1972, Filed: 03/14/2017 United States District Court for the Northern District of Illinois, *Finnegan Henderson Farabow Garrett & Dunner LLP*
 - Software source code review representing Hytera, 2017
- Certain Digital Video Receivers and Hardware and Software Components Thereof, Investigation No. 337-TA-1001, United States International Trade Commission, *Winston & Strawn, LLP*
 - Software source code review representing Comcast, Inc., 2016
- In the Matter of Certain Electronic Devices, Including Wireless Communication Devices, Computers, Tablet Computers, Digital Media Players, and Camera, Investigation No. 337-TA-952, United States International Trade Commission, *Winston & Strawn, LLP*
 - Software source code review representing Ericsson, Inc., 2015
- Certain Wireless Standard Compliant Electronic Devices, Including Communication Devices and Tablet Computers, Investigation No. 337-TA-953, United States International Trade Commission, *Winston & Strawn, LLP*
 - Software source code review representing Ericsson, Inc., 2015
- Ericsson Inc. and Telefonaktiebolaget LM Ericsson v. Apple Inc., Case No. 2:15-CV-293, United States District Court Eastern District of Texas, *Winston & Strawn, LLP*
 - Software source code review representing Ericsson, Inc., 2015

- *OpenTV, Inc. v. Apple Inc.*, District of Northern California, San Francisco Division, Civil Action No. 3:14-cv-01622-JST, 2015.
 - Software source code review for OpenTV
- *Enterprise v. Apple, Samsung, LG, HTC, et al*, Investigation No. 337-TA-925, United States International Trade Commission, 2015.
 - Software source code review against Enterprise
- *Oblon Spivak McClelland Maier and Neustadt LLP*
 - Various patent prior art searches 2013-2014.
- *Optical Devices, LLC v. Samsung, LG. et al*, Investigation No. 337-TA-897, United States International Trade Commission; Through ESS Technologies using K&L Gates LLP, 2014
 - Expert witness for Samsung
- *S3 Graphics Co., LTD. v. ATI Technologies, ATI International SRL, and Advanced Micro Devices, Inc.*, Civil Action No. 11-1298-LPS, United States District Court for the District of Delaware, Winston & Strawn LLP representing S3 Graphics, 2013-2015
 - Expert witness and HW/SW code reviews for S3 (provided deposition 1/28/2015)
- *Intertrust Technologies Corporation v. Apple, Inc*, Civil Action No. 5:13-cv-01235-HRL, United States District Court Northern District of California, *Quinn Emanuel Urquhart & Sullivan LLP representing Intertrust*, 2013
 - Software code reviews and analysis for Intertrust
- *Black Hills Media, LLC v. LG Electronics, Inc. et al*, Investigation No. 337-TA-882, United States International Trade Commission; *Finnegan Henderson Farabow Garrett & Dunner LLP representing LG Electronics*, 2013
 - Software code reviews for LG
- *Apple v Samsung*, US Case No. 12-cv-00630-LHK, (Regarding USPN 7,761,414 & 6,847,959)
 - Software code reviews and analysis for Samsung
- *VIASAT v Loral*, US Case No. 3:12-cv-00260-H-WVG (Regarding USPN 6,400,696 & 6,879,808)
 - Prior art, software code reviews and analysis for VIASAT
- *HTC v Apple*, US Case No. 11-785 (GMS) (D. Del) (Regarding USPN 5,630,152)
 - Hardware and Software code reviews and analysis for HTC
- *S3G v NVIDIA & Apple*, US ITC, 2011-337-TA-813
 - Hardware Code reviews (graphics/video) and analysis for S3G
- *HTC v Apple*, US ITC, 2011-337-TA-808 (Regarding USPN 7,765,414)
 - Software code reviews and analysis for HTC
- *Linex v Apple*, US ITC, 2011-337-TA-775 (Regarding USPN 6,757,322)
 - Performed code reviews (Verilog RTL & VHDL) on behalf of Linex
- *HTC v Apple*, US ITC, 2010-337-TA-721 (Regarding USPN 6,999,800)
 - Power management study for HTC relating to prior art documents
- *Fenner v Microsoft*, E.D.Tx, 2007-6:07-CV-00008 (Regarding USPN 6,297,751)
 - Consulted on behalf of Microsoft (through ESS) and I was deposed on issues relating to prior art documents.

Intellectual Property Analysis

- Provided technical product review. This was done through Intuity Consultants, Inc. and Winston & Strawn LLP on behalf of Snap in 2024.
- Provided technical product review. This was done through Precedia and Quinn Emanuel on behalf of Qualcomm in 2019.
- Provided technical review of products. This was done through Precedia on behalf of Spotify in 2019.
- Reviewed and compared two Class-D amplifier designs regarding copyright/trade-secret issues. This was done through Teklicon on behalf of Maxim in 2012.

For Software Code Development (2010-2023):

I have written a collection of C++/C software CAD/CAE programs for designing integrated circuits with dramatically improved cost/speed/power-dissipation using a novel proprietary design flow.

- http://www.elqc.com/RobertPlachno/RP_Min_Pdf.pdf (sample PDF file writer C++ code - scroll down and click on any highlighted name in **BOLD**)
- <http://www.elqc.com/RobertPlachno/engsw.htm> (CAD software overview)
- I presently use Microsoft Visual Studio C++ for CAD/CAE software development.
- Interactive programs such as the GDSII layout viewer, GSL editor and the logic waveform viewer use the Fat-Finger touch screen interface using OpenGL.

For Design Consulting Work (2010-2014):

My direct design consultant work for was paid for by:

Silicon Light Machines, in Sunnyvale, Ca, 2013

Transistor level design for a high-speed massively parallel LVDS DDR interface. My personal work encompassed schematics, simulations, layout, verification, and documentation.

For ESS Technology (2010 to 8/2/2019):

Milpitas & Fremont, Ca

Distinguished Technologist and HW/SW Program Manager

I worked on various integrated circuit design concerns including block compilers and cost/power reductions of high-volume products. Design compilers include analog & digital filters and a low power modulator. I was also a program manager (both hardware and software) for the engineering groups in Beijing, Canada, and California.

Work Experience (prior to 2010)

ESS Technology Dec 2000 to Jan 2010 Fremont, Ca

VP of IC Design

Mixed Signal, Audio, DVD, VoIP

- Vice President in charge of IC design & software groups located in Beijing, Canada, and California.
- The Canadian group designed the industry's highest performing audio DAC (the Sabre DAC) with a DNR of 135dB and a THD+N of 120dB. These audio DACs are in production and are sold into many high-end audiophile consumer products. The audio product line consists of a family of parts ranging from the high-performance high-cost parts to fully integrated low-cost and low-BOM products.
- The California group is responsible for the audio/video System-On-Chip integrated circuits. The SOC products are mixed signal designs that include a 64-bit SIMD engine, a 32-bit RISC and multiple 32-bit DSPs. This group designs chips for speaker bars, streaming audio, consumer audio, DVDs, photo-frame and HDMI support chips.
- The Canadian group is also designing a low-cost RF silicon tuner for TV and radio markets, a high-end audio ADC and supports the company with analog IP. These products use highly innovative and disruptive technologies.
- The Beijing group designed a VoIP chip with the industry's lowest BOM due to integrated Ethernet and (internally designed) SLICs. The VoIP software for the embedded RISC and DSP was internally developed and is supported by the software group.
- As VP of Engineering, I manage software groups writing embedded software for various consumer products. For the DVD products this includes the video and audio software required for the internal SIMD, RISC, and DSPs. Also, there is embedded controller software for communication including the Ethernet, HDMI and USB interfaces. The audio software includes various DSP algorithms for sound effects as required by the customers.

ESS Technology Dec 2000 to Dec 2006

VP of Communication IC Engineering
Communication, DVD, Analog, DSP

- I managed 4 engineering groups that designed communication and/or video integrated circuits. This included a circuit group in Fremont, a logic group in Fremont, an analog group in Fremont and another analog group in Canada.
- The Fremont logic group designed video chips for DVD, other video consumer products, home networking, LAN, modem, DSPs, and RISCs.
- The Fremont analog group designs transceivers for communication products and for recordable DVD products.
- The Fremont circuit group supports the other groups for place and route, layout, pads, PLLs, etc.
- The Canadian analog group designs analog circuits for video consumer products. This includes specialized ADCs, DACs, RF, servo, and Class-D amplifiers.
- My personal technical work was on library characterization, a DSP for the communication products and product debug.

Summit Semiconductor Dec 1999 to Dec 2000 Los Gatos, Ca

IC Design, CAD software

- I incorporated myself into this company for contract work during this period specializing in custom circuit design, design intellectual property, and engineering services.

MKNet Corp Sept 1999 to Dec 1999 San Jose, Ca

VP of Engineering
VFIR PHY and MAC

- I managed the engineering groups for a new infra-red start-up which required the development of a Very Fast Infra-Red (VFIR) 16Mb/s solution that includes a digital controller chip, an analog transceiver chip and software.

ESS Technology 1993 to 1999 Fremont, Ca

VP of Audio, VP of IC, and VP of R&D

Audio Codec, FM & Wavetable Music synthesizers, Embedded RISC

- I managed the IC group for the development of the PC audio product line (mixed signal). This product line was very successful, and ESS went public. The company grew from 40 to over 400 employees during my VP management time-period and the production volume grew to over 2.5 million mixed signal audio chips shipped per month.
- The audio design hardware group I managed was less than 20 engineers that produced two high volume IC products each year and several minor designs. This group contained architects, logic designers, circuit designers, firmware engineers, patch developers, layout, IC routing, PCB design, and technicians.
- I also managed the software group (about 15 software engineers) that were responsible for the Windows drivers and application software.
- I personally designed the RISC processor as well as the standard cells, pads, some memories, and some other logic sections used within the designs. The RISC processor was a true single cycle design that improved the software performance by about 6x, improved the operating power supply range, and eliminated royalty payments.
- This group also designed general MIDI wavetable chips and 3D sound effect chips.
- I gave a deposition, and we successfully defended the company in a major patent lawsuit with a much larger competitor.

NARA 1986 to 1993 San Jose, Ca

VP of R&D

Microprocessor Peripheral Logic

- This was a small engineering service company that made money by reverse engineering ICs and designing PC chip sets. I worked on the 82xx peripherals including the DMA

controller, counter/timer, interrupt controller, real time clock, VGA, XGA, UARTs, floppy disk controller, embedded controllers, and x86 processors. Using a semi-custom design approach, I was able to reduce the transistor counts (gate count) of these peripherals by a minimum of 1/2 to a maximum of 1/7 of the previous designs.

- I wrote many software programs including a tool set for the automation of the reverse engineering of chips from scanned images to high level behavioral. I retain partial and full ownership of some software for dynamic timing logic simulation, design rule independent layout and other CAD/CAE software.
- I was used for several legal expert witness contracts including the AMD-INTEL co-processor lawsuits and for multiple processor patent matters.

American Information Technology 1984 to 1986 San Jose, Ca
Processor Design

- I managed the circuit design group (as a consultant) for American Information Technology.
- At AIT I personally designed the execution unit datapath (transistor level) of a 64-bit integrated processor and managed the engineers and layout designers for the rest of the processor and the required support chips (including graphics). I personally did much of the memory design, standard cell design and the place & route at AIT.

Zytrex 1981 to 1984 Sunnyvale, Ca
VP of Engineering, Director of Engineering
SSI, MSI logic

- After Texas I moved back to California to help start up a minor semiconductor company. The company grew from a 7-engineer living-room operation to a 200-man company including our own fab and assembly. Zytrex went public while I was VP and was listed on NASDAQ.
- After the first year, the company switched to using a unique high speed CMOS process (2μ metal gate SATO devices) created by Frank Wanlass. I hired and managed a small group of engineers and layout designers that designed a total of more than 200 74HCTxx SSI and MSI part numbers using about 50 different mask sets. We also worked on high-speed multipliers and microprocessor 82xx peripheral products.
- For the first year of operation the company tried to supply BI-CMOS ROMs. I designed other chips using “watch-chip” technology.
- I was responsible for IC design and many other activities in the start-up including the purchasing, initial operation and programming of the IC test equipment, proto-type assembly equipment, and CAD equipment.

Mostek Carrollton, TX
Design Engineer
DRAM, SRAM

- I was one of two design engineers working on a 32K pseudo static RAM. This design used a unique three transistor memory cell that was sustained using a charge pump.
- I received three US circuit design patents and presented a paper for the pseudo static at the 1981 ISSCC.
- I was the project manager of a 64K static RAM and worked on the 128K dynamic RAM project.

American Microsystems, Inc. Santa Clara, Ca
Design Engineer
SRAM, DRAM, ROM

- My first product design out of school was a 150ns, 1K by 4 VMOS static RAM (AMI 1560). I was then given a project manager lead for a high speed, 55ns, 1K by 4 VMOS static RAM (AMI 1599). The analog sense amp I designed for the 1599 was used for years at various companies.

- Then I was the project manager for the first 16K static RAM in the industry (AMI 1600).
- Also, while I was at AMI, I worked on ROMs, dynamic RAMs and did some minor process development. The technology used for these designs were 4.5 micron VMOS transistors supplemented by 2.5 micron SATO (self-aligned by thick oxide) NMOS transistors with resistor loads. This product group was managed by T. J. Rodgers who later founded Cypress.

Education

BS in Computer Engineering
University of Illinois, Urbana, Illinois 1973-1976
MS graduate study in Electrical Engineering
University of Illinois, Urbana, Illinois 1976-1977

Patents

[USP 4,379,974](#)
[USP 4,360,903](#)
[USP 4,347,448](#)

Publications

"A 32K Static RAM utilizing a 3-Transistor Cell", Paper was presented at ISSCC in New York.
[ISSCC 81](#)
"A True Single Cycle RISC Processor without Pipelining",
[RP RISC Paper](#)

Web Pages

[Robert S Plachno Engineering Software Web Site](#)
[Robert S Plachno on Google Scholar](#)
[Robert S Plachno on LinkedIn](#)

EXHIBIT 4

Julia Rowe

Vice President

MA, Accountancy
DePaul University

BA, Business Administration
University of Michigan

Julia Rowe brings a unique blend of public policy, valuation, and litigation expertise to the valuation of intellectual property. She has provided deposition and trial testimony and advises clients in a wide range of settings—including licensing negotiations, business strategy, financial reporting, the International Trade Commission, Federal Court, State Court and Arbitration. Julia has performed business and asset valuations, economic and financial analyses, and licensing-related consulting services. Her industry experience includes computer hardware and software, e-commerce, medical devices, telecommunications, and much more.

In addition to her deposition and testimony experience, Julia has also served as a consultant through trial and jury verdict in several high-profile cases involving patent infringement, trademark infringement, copyright infringement, and theft of trade secrets. She has extensive experience analyzing international trade issues (including public interest, domestic industry and bonding), lost profits, reasonable royalties, and unjust enrichment damages. Julia has also valued privately-held interests and has served in a variety of commercial litigation matters involving breach of contract, unfair competition, and corporate fraud.

Prior to joining CRA, Julia was a consultant at Kroll. She has a bachelor's degree in business administration from the University of Michigan and a master of accountancy degree from DePaul University. Julia is a registered CPA in Illinois.

Speeches, Presentations & Publications

"What Every IP Lawyer Needs to Know International Licensing," ACI's 5th Annual Summit for Women Leaders in IP Law, June 2025

"Advanced Damages Techniques," 2023 ABA-IPL Section Annual Meeting, April 2023

"Strategies for Realizing Value from Patents Without Litigation," Continuing Legal Education Program for the Georgia Bar Patent Committee, January 2021

"ITC Investigations: Economic Issues" Continuing Legal Education Program for Faegre Drinker Biddle & Reath LLP, November 2020

"IP Value And the Benefits Of Cross-Department Collaboration," Duff & Phelps Webinar, November 2019

"IP Of the Future: Issues For Disruptive Technologies," Duff & Phelps 2018 IP Value Summit, November 2018

"The Growing Role Of Economics In IP-based Investigations At the ITC," Annual Technology & Law Seminar, Kirkland Ellis LLP-Chicago, May 2018

“Hot Topics In Patent Licensing And Litigation,” Licensing Executives Society, Silicon Valley Chapter Meeting, March 2018

“Competitors And the Right To Exclude,” Duff & Phelps 2017 IP Value Summit, December 2017

“Connecting DC With Wall Street—How IP Investors View the Legal And Policy Landscape,” Intellectual Asset Management Patent Law and Policy Conference, November 2016

“Remedies In the Defend Trade Secrets Act,” Association of Corporate Counsel (Nationwide teleconference), July 2016

“Valuing Intellectual Property & Trade Secrets For Financial Reporting Purposes,” Michigan Association of Certified Public Accountants, May 2015

“Eastern District Of Texas (EDTX) General Order Regarding Alternative Procedure for Case Management,” Duff & Phelps IP Client Update, February 2014

“The Use Of Surveys For U.S. Patent Litigation,” Continuing Legal Education Program for Covington & Burling LLP (Nationwide teleconference), October 2013

“Recent Trends In IP Damages Law,” Licensing Executives Society, Inc. Chicago Chapter Meeting, June 2013

“Recent Trends In IP Damages Law and International Trade Commission Issues,” Continuing Legal Education Program For Duane Morris (Boston), October 2012

“Valuing Limited Partnership Interests,” Trusts & Estates, October 2010

Professional & Business History

2024–Present	<i>Vice President</i> , Charles River Associates, Chicago, IL
2011–May 2024	<i>Managing Director</i> , Kroll (formerly Duff & Phelps) <i>Managing Director</i> , 2021–2024 <i>Director</i> , 2013–2021 <i>Vice President</i> , 2011–2013 <i>Senior Associate</i> , 2011
2009–2011	<i>Associate</i> , CRA International, Chicago, IL <i>Analyst</i> , 2008–2009

Professional Associations and Certifications

Certified Public Accountant (CPA), Registered in Illinois

Certified Licensing Professional (CLP)

Member, Illinois CPA Society, Licensing Executives Society, Chicago Women in IP (“ChiWIP”)

Julia Rowe **Expert Testimony – Prior Four Years**

Client/party denoted with *

International Trade Commission (Washington, DC)
Certain Wi-Fi Routers, Wi-Fi Devices, Mesh Wi-Fi Network Devices and Components Thereof (Inv. No. 337-TA-1454)

Estelgia LLC v. ASUSTeK Computer Inc.,* ASUS Computer International,* D-Link Systems, Inc.,* D-Link Corporation,* Plume Design, Inc.*
Report, Deposition

International Trade Commission (Washington, DC)
Certain Balloon Dilation Devices, Systems, and Components Thereof (Inv. No. 337-TA-1449)
Stryker Corporation,* Stryker Sales, LLC,* Entellus Medical, Inc.* v. Fiagon AG Medical Technologies, Fiagon GmbH, Fiagon NA Corporation, Hemostasis, LLC.
Report, Deposition, Hearing

AAA Arbitration
Beam Therapeutics Inc. v Prime Medicine, Inc. * (Case Nos. 01-25-0001-8995 and 01-25/0001-9252)
Reports, Deposition, Hearing

United States District Court, Eastern District of Texas, Marshall Division
Aspen Networks, Inc. v. T-Mobile USA, Inc.* (Case No. 2:23-cv-00558)
Report, Deposition

International Trade Commission (Washington, DC)
Certain Foreign-Fabricated Semiconductor Devices, Products Containing the Same, and Components Thereof (Inv No. 337-TA-1443)
Longitude Licensing, Ltd. v. Taiwan Semiconductor Manufacturing Company Limited,* Apple Inc.,* Broadcom Inc.,* Lenovo (Shanghai) Electronics Technology Co., Ltd.,* Lenovo PC International Limited,* Lenovo PC HK Limited,* Lenovo Information Products (Shenzhen) Co., Ltd.,* Lenovo Beijing Co., Limited,* Lenovo (United States Inc.,* Motorola (Wuhan) Mobility Technologies Communication Company Limited,* Motorola Mobile Communication Technology Ltd.,* OnePlus Technology (Shenzhen) Co., Ltd.* and Qualcomm Incorporated*
Reports, Deposition, Trial

United States District Court, Southern District of California, San Diego
iGolf, Inc. v. Bushnell Holdings, Inc. * (Case No. 3:23-cv-0595)
Report

United States District Court, Southern District of Texas, Houston
Braden Group B.V.,* Braden Americas, Inc.* v. EnergyLink Corporation (Case No. 4:23-cv-00626)
Reports, Deposition

United States District Court, Eastern District of Texas, Marshall Division
Unilin BV* v. Vilox Sweden AB, et al. (Case No. 2:24-cv-00040)
Reports

International Trade Commission (Washington, DC)
Certain Flash-Spun Nonwoven Materials and Products Containing Same (Inv. No. 337-TA-1424)
DuPont de Nemours, Inc., DuPont Safety & Construction, Inc., and DuPont Specialty Products USA, LLC v. Xiamen Dangs New-Materials Co., Ltd. AKA Dawnsens New Materials Co., Ltd., Beijing Dangsheng Technology Co., Ltd., Xiamen Dangsheng Technology Co., Ltd., Kingwills New Material Technology Co., Ltd, Zhejiang Qingyun New Material Co., Ltd.,* Jiangsu Qingyun New Materials Co., Ltd. AKA Jiangsu Kingwills New Materials Co. Ltd.,* Shanghai Qingyun New Material Technology Co., Ltd.,* Kingwills International Ltd.*
Report, Deposition, Witness Statement, Hearing

In the Court of Common Pleas State of South Carolina, County of Spartanburg
American Credit Acceptance, LLC v. Nick Weingartner, Prog Leasing, LLC d/b/a Progressive Leasing, Gradient Financial LLC, Mark Scott and Grey Register* (Civil Action No. 2021-CP-23-00325)
Report, Deposition

United States District Court, Eastern District of Texas, Marshall Division
Vasu Holdings, LLC v. Samsung Electronics Co., Ltd.,* Samsung Electronics America, Inc.* (Case No. 2:24-cv-00034)
Report, Deposition

United States District Court, Eastern District of Texas, Marshall Division
Cerence Operating Company v. Samsung Electronics Co., Ltd.,* Samsung Electronics America, Inc.* (Case No. 2:23-cv-00482)
Reports, Deposition

United States District Court, Eastern District of Texas, Marshall Division
Collision Communications, Inc. v. Samsung Electronics Co., Ltd.,* Samsung Electronics America, Inc.* (Case No. 2:23-cv-00587)
Report, Deposition, Trial

United States District Court, District of Delaware, Wilmington
LiTL LLC v. HP Inc.* (Case No. 1:23-cv-00120)
Reports, Deposition

United States District Court, District of Delaware, Wilmington
LiTL LLC v. ASUSTek Computer Inc.,* Asus Global Pte. Ltd.* (Case No. 1:23-cv-00122)
Reports, Deposition

International Trade Commission (Washington, DC)
Certain Rechargeable Batteries And Components Thereof (Inv. No. 337-TA-1421)
LithiumHub, LLC, LithiumHub Technologies, LLC, and Mr. Martin Koebler v. Bass Pro Outdoor World LLC,* Cabela's LLC,* Relion Battery (Shenzhen) Technology Co.,* Renogy New Energy Co., Ltd.,* RNG International Inc.,* Clean Republic SODO LLC,* Shenzhen Fbtech Electronics Ltd.,* Shenzhen LiTime Technology Co., Ltd.,* and MillerTech Energy Solutions LLC*
Report

International Trade Commission (Washington, DC)
Certain Photodynamic Therapy Systems, Components Thereof, and Pharmaceutical Products Used in Combination with the Same (Inv. No. 337-TA-1411)
Sun Pharmaceutical Industries, Inc. v. Biofrontera Inc.,* Biofrontera Pharma GmbH,* Biofrontera Bioscience GmbH,* Biofrontera AG*
Reports, Deposition, Witness Statements, Hearing

International Trade Commission (Washington, DC)
Certain Sensors with Pixels and Products Containing the Same (Inv. No. 337-TA-1403)
SiOnyx, LLC v. Samsung Electronics America, Inc.,* Samsung Electronics Co., Ltd.,* Samsung Semiconductor, Inc.*
Reports, Deposition, Witness Statement

International Trade Commission (Washington, DC)
Certain High-strength Aluminum or Aluminum Alloy-coated Steel, and Automotive Products and Automobiles Containing Same (Inv. No. 337-TA-1402)
ArcelorMittal v. VinFast Auto, LLC,* VinFast Auto Ltd.,* VinFast Trading and Production JSC,* VinFast USA Distribution, LLC,* Vingroup USA, LLC,*
Report

International Trade Commission (Washington, DC)
Certain Surface Cleaning Devices and Components Thereof (Inv. No. 337-TA-1404)
SharkNinja Operating LLC, Omachron Alpha Inc., Omachron Intellectual Property Inc. v. Dyson, Inc.,* Dyson Technology Limited,* Dyson Canada Limited*
Report, Deposition

International Trade Commission (Washington, DC)
Certain Medical Programmers with Printed Circuit Boards, Components Thereof, And Products and Systems for Use With The Same (Inv. No. 337-TA-1396)
Medtronic, Inc.,* Medtronic Logistics, LLC,* Medtronic Puerto Rico Operations Co.,* and Medtronic USA, Inc.* v. Axonics, Inc. f/k/a Axonics Modulation Technologies, Inc.
Reports, Depositions, Trial

International Trade Commission (Washington, DC)
Certain Capacitive Discharge Ignition Systems, Components Thereof, And Products Containing Same (Inv. No. 337-TA-1390)
Altronic, LLC v. MOTORTECH GmbH* and MOTORTECH Americas, LLC*
Report, Deposition

International Trade Commission (Washington, DC)
Certain Smart Ceiling Fans, Components Thereof, And Associated Systems And Software (Inv. No. 337-TA-1374)
Wangs Alliance Corporation d/b/a WAC Lighting v. Minka Lighting, LLC,* Tech Lighting LLC,* and VC Brands, LLC*
Report, Deposition

United States District Court, Western District of Texas, Waco Division
*paSafeShare LLC v. Microsoft Corporation** (Civil Action No. 6:20-cv-0397)
Reports, Deposition

United States District Court, Eastern District of Texas, Marshall Division
*Mojo Mobility Inc. v. Samsung Electronics Co., Ltd., and Samsung Electronics America, Inc.** (Civil Action No. 2:22-cv-00398)
Report, Deposition, Trial

United States District Court, District of Delaware
*Graco Inc. and Graco Minnesota Inc. v. Carlisle Construction Materials, LLC** (Civil Action No. 21:245)
Report, Deposition

United States District Court, Northern District of Illinois Eastern Division
*Gesture Technology Partners, LLC v. Motorola Mobility LLC** (Civil Action No. 1-22-cv-03535)
Report

United States District Court, Northern District of California
*NantWorks, LLC and Nant Holdings IP, LLC v. Niantic, Inc.** (Civil Action No. 3:20-cv-06262)
Report, Deposition

United States District Court, District of Delaware
*Cleveland Medical Devices, Inc. v. ResMed Inc.** (Civil Action No. 1:22-cv-00794)
Report, Deposition

International Court of Arbitration of the International Chamber of Commerce
SoftSmile, Inc. v. Institut Straumann AG* (ICC Case No.: 27375/PDP)
Reports

International Trade Commission (Washington, DC)
Certain LED Lighting Devices, LED Power Supplies, Components Thereof, And Products Containing Same (Inv. No. 337-TA-1371)
Signify North America Corporation and Signify Holding B.V. v. Current Lighting Solutions, LLC*
Report, Deposition, Trial

International Trade Commission (Washington, DC)
Wi-Fi Routers, Wi-Fi Devices, Mesh Wi-Fi Network Devices, and Hardware and Software Components Thereof (Inv. No. 337-TA-1361)
Netgear, Inc. v. TP-Link Research Institute USA Corp.,* TP-Link Research Institute USA Corp. d/b/a TP-Link Research America Corp.,* TP-Link Technologies Co., Ltd.,* TP-Link USA Corporation*
Report, Deposition, Trial

United States District Court, Central District of California
Microsoft Corporation v. MediaPointe, Inc. and AMHC, Inc.* (Civil Action No. 2:22-cv-1009)
Report, Deposition

International Trade Commission (Washington, DC)
Components for Certain Environmentally-Protected LCD Digital Displays and Products Containing Same (Inv. No. 337-TA-1349)
Samsung Electronics Co., Ltd.,* Samsung Electronics America, Inc.,* Samsung Research America, Inc.,* and Samsung International, Inc.* v. Manufacturing Resources International, Inc.
Report, Deposition, Trial

United States District Court, District of Delaware
*Groove Digital, Inc. v. Jam City, Inc.** (Civil Action No. 1:18-cv-01331)
Report, Deposition

International Trade Commission (Washington, DC)
Certain Soft Projectile Launching Devices, Components Thereof, Ammunition, And Products Containing Same (Inv. No. 337-TA-1325)
Hasbro, Inc. and Spin Master, Inc. v. Gel Blaster, Inc.,* Shenzhen Yi Jin Electronics Science (a/k/a “Shenzhen Yi Jin Electronics”),* Guangdong Yu Lee Technology Corporation,* Yu Lee Company Ltd.,* Splat-R-Ball, LLC,* Daisy Manufacturing Company (d/b/a “Daisy Outdoor Products”),* S-Beam Precision Products Ltd.,* Prime Time Toys LLC,* Prime Time Toys Ltd.,* and Easebon Services Ltd.*
Reports, Deposition, Trial

International Trade Commission (Washington, DC)
Certain Power Semiconductors, and Mobile Devices and Computers Containing Same (Inv. No. 337-TA-1308)
Arigna Technology Ltd. v. Apple Inc.,* TTE Technology, Inc.,* TCT Mobile (US) Inc.,* TCL Communication Limited,* *et. al.*
Report, Deposition, Trial

International Trade Commission (Washington, DC)
Certain Computer Network Security Equipment And Systems, Related Software, Components Thereof, And Products Containing The Same (Inv. No. 337-TA-1314)
Centripetal Networks, Inc. v. Keysight Technologies, Inc.*
Reports, Deposition, Trial

United States District Court, Eastern District of Louisiana
Wilco Marsh Buggies & Draglines, Inc. v. Weeks Marine, Inc.* (Civil Action No. 2:20-cv-3135)
Reports, Deposition

International Trade Commission (Washington, DC)
Certain Knitted Footwear (Inv. No. 337-TA-1289)
Nike, Inc.* v. adidas AG, adidas North America, Inc., and adidas America, Inc.
Report

AAA Arbitration

Doerfer Corp. v. Intelliport Corp. & Mel D. Terry**
Report

International Trade Commission (Washington, DC)

Certain Casual Footwear and Packaging Thereof (Inv. No. 337-TA-1270)

Crocs, Inc. v. Orly Shoe Corp., Hobby Lobby Stores, Inc., Walmart Inc., and Quanzhou ZhengDe Network Corp., d/b/a Amoji*

Reports, Deposition, Witness Statement, Trial

International Trade Commission (Washington, DC)

Certain Barcode Scanners, Mobile Computers With Barcode Scanning Capabilities, Scan Engines, And Components Thereof (Inv. No. 337-TA-1285)

Honeywell International Inc., Hand Held Products, Inc.,* and Metrologic Instruments, Inc.* v. Zebra Technologies Corporation and Symbol Technologies, Inc.*

Report, Deposition

United States District Court, Western District of Texas, Waco Division

*WSOU Investments, LLC d/b/a Brazos Licensing and Development v. Microsoft Corporation** (Civil Action No. 6:20-cv-00457)

Report

United States District Court, Central District of California, Southern Division

Glam and Glits Nail Design, Inc. v. iGel Beauty, LLC, Daniel Bui, Tu Bui, Alvin Bui, Andrew Bui* (Case No. CV-20-00088-JVS)

Report, Deposition, Declaration

United States District Court, Eastern District of Texas, Marshall Division

Solas OLED Ltd. v. Samsung Electronics Co., Ltd., Samsung Electronics America, Inc.** (Civil Action No. 2:21-cv-00105)

Report, Deposition

United States District Court, Western District of Texas, Waco Division

*WSOU Investments, LLC d/b/a Brazos Licensing and Development v. Microsoft Corporation** (Civil Action No. 6:20-cv-00463)

Report

International Trade Commission (Washington, DC)

Certain Active Matrix OLED Display Devices and Components Thereof (Inv. No. 337-TA-1243)

Solas OLED Ltd. v. Samsung Electronics Co., Ltd., Samsung Electronics America, Inc.,* and Samsung Display Co., Ltd.**

Report, Deposition

International Trade Commission (Washington, DC)

Certain Robotic Floor Cleaning Devices and Components Thereof (Inv. No. 337-TA-1252)

iRobot Corporation v. SharkNinja Operating LLC, SharkNinja Management LLC, SharkNinja Management Co., SharkNinja Sales Co., EP Midco LLC, SharkNinja Hong Kong Co. Ltd.*

Reports, Trial

International Trade Commission (Washington, DC)

Certain IP Camera Systems Including Video Doorbells and Components Thereof (Inv. No. 337-TA-1242)

SkyBell Technologies, Inc., SB IP Holdings, LLC, and Eyetalk365, LLC v. Vivint Smart Home, Inc., SimpliSafe, Inc.,* and Arlo Technologies, Inc.**

Report, Deposition

United States District Court of Harris County, Texas, 215th Judicial District
SP Silica Sales, LLC f/k/a Preferred Sands of Texas, LLC v. NexTier Completion Solutions, Inc., successor to Keane Frac, LP* (Cause No. 2018-79957)
Disclosures, Deposition, Trial

Arbitration
*SNI Solutions, Inc. and Natural Alternatives, LLC v. Univar USA, Inc.**
Report, Deposition, Trial

United States District Court, Northern District of California, Oakland Division
*Lambda Labs v. Lambda, Inc.** (Case No. 4:19-cv-04060)
Report

United States District Court, Northern District of Illinois, Chicago
*Shure Incorporated v. ClearOne Inc.** (Case No. 1:17-cv-03078)
Declaration

United States District Court, District of Delaware, Wilmington
*Shure Incorporated and Shure Acquisition Holdings, Inc. v. ClearOne Inc.** (Case No. 1:19-cv-001343)
Reports, Deposition

United States District Court, Eastern District of Texas, Marshall Division
Joe Andrew Salazar v. HTC Corporation, HTC America, Inc., AT&T Mobility LLC, Sprint/United Management Company, T-Mobile USA, Inc.* and Cellco Partnership d/b/a Verizon Wireless* (Case No. 2:20-cv-00004)
Report

EXHIBIT 5

EDUCATION

The University of Texas at Austin, Austin, TX 78712

Ph.D. in Electrical Engineering, May 1994, G.P.A. 3.833/4.0

Kansas State University, Manhattan, KS 66506

M.S. in Electrical Engineering, May 1991, G.P.A. 4.0/4.0

B.S. in Electrical Engineering, May 1989, G.P.A. 3.934/4.0 (Summa Cum Laude)

DOCTORAL DISSERTATION

Combined Intravascular Ultrasound and Fluorescence Spectroscopy for the Diagnosis of Atherosclerosis

- Developed a catheter-based diagnostic system for atherosclerosis by interfacing a laser-induced fluorescence system to commercial intravascular ultrasound imaging equipment
- Wrote a Monte Carlo radiation transport code to simulate photon transport and fluorescence emission in tissue
- Initiated a clinical study to correlate restenosis rates of patients undergoing arterial atherectomy with the chemical composition of the atherosclerotic tissue removed

MASTER'S THESIS

An Empirical Study of the Dynamic Hysteresis Characteristics of Ferromagnets

- Developed a frequency-dependent, wide-bandwidth ferromagnetic hysteresis model suitable for circuit simulation programs
- Accurately simulated low and high-frequency hysteresis in ferrite toroidal cores
- Created an exhaustive annotated bibliography on mathematical hysteresis modeling

EXPERIENCE

Professor

February 2017 – Present

Associate Professor

February 2002 – January 2017

Assistant Professor

August 1999 – January 2002

Kansas State University, Department of Electrical and Computer Engineering

- Received ABET accreditation for the new KSU Biomedical Engineering Program
- Served as the Program Director for the new KSU Biomedical Engineering Program
- Designed and resourced a new KSU Biomedical Education and Innovation Laboratory
- Established a new KSU undergraduate program in Biomedical Engineering
- Established and directed the KSU Medical Component Design Laboratory
- Received grant support for biomedical component research/education
- Initiated multi-disciplinary teaching and research relationships with industry and academia
- Represented Kansas State through invited presentations, conferences, and short courses
- Supported and advised graduate students
- Integrated undergraduate honors students into a medical component research program
- Remained active in professional societies
- Served on KSU Faculty Senate and University-level academic/administrative committees
- Founded a new Midwest Professional Chapter of the IEEE EMBS
- Founded and advised a new KSU Student Chapter of the IEEE EMBS
- Instructed courses in the Biomedical, Electrical, and Computer Engineering curricula
- Helped to redesign the EE Bioengineering Option
- Designed real-world course projects that often required industry input

- Obtained internships for capable undergraduate students
- Advised students regarding academic and career goals

Professor**Associate Professor**

Kansas State University, Department of Gerontology

February 2017 – Present

July 2006 – January 2017

Principal Member of the Technical Staff

1998 – July 1999

Sandia National Laboratories, Albuquerque, NM

TELEMEDICINE PROJECT (Principal Investigator - \$3.2M effort) – “*The Role of Technology in Reducing Health Care Costs*”

- Designed a secure, plug-and-play information architecture for telemedicine
- Assembled a high-performance, multi-disciplinary team to modify a commercial telemedicine system and demonstrate its effectiveness within a secure framework
- Empowered team members to pursue innovative ideas
- Participated in invitation-only workshops dedicated to advanced home care technology

MONTE CARLO RADIATION TRANSPORT PROJECT – “*Monte Carlo Radiation Transport in Computer-Aided-Design (CAD) Geometries*”

- Presented ITS-CAD, a CAD-based Monte Carlo radiation transport code, to the high-performance computing community for the first time
- Prepared material to market this capability to the scientific community
- Completed initial algorithms for electron trapping within solid bodies

Senior Member of the Technical Staff

March 1996 – October 1998

Sandia National Laboratories, Albuquerque, NM

TELEMEDICINE PROJECT (Principal Investigator)

- Performed an industry-wide technology survey on modular telemedicine systems
- Received an individual performance award for the phase I final report
- Assisted with the definition of a clinical study for determining the cost-effectiveness and diagnostic feasibility of telemedicine applied to hypertension monitoring
- Established relationships with leaders in the telemedicine community

MONTE CARLO RADIATION TRANSPORT PROJECT

- Designed a method to incorporate the ACIS boundary representation modeler into the Integrated Tiger Series of legacy Monte Carlo radiation transport codes
- Incorporated C++ classes for analyzing complex hierarchical assemblies
- Developed efficient algorithms to track particles in these geometries
- Implemented a heterogeneous grid structure to increase code efficiency and make simulation times less dependent on geometrical complexity
- Applied the ACIS-based transport code to a neutron generator dose deposition problem

OTHER

- Studied the feasibility of using infrared imaging technology to remotely monitor vital signs and physiological indicators of malevolent intent
- Presented proposals to various customers regarding light-based, wearable sensors to determine human physiological status and to ascertain a wearer's identity
- Surveyed burn depth measurement technologies for a military customer
- Promoted high-school science in New Mexico through regional science competitions

Post-Doctoral Member of the Technical Staff
Sandia National Laboratories, Albuquerque, NM

July 1994 – February 1996

PERSONAL STATUS MONITOR PROJECT

- Completed an extensive literature search on pulse oximetry, motion artifact, and non-invasive vital sign measurement via light
- Performed Monte Carlo simulations to determine ideal light-based sensor configurations for minimizing motion artifact
- Produced a prototype of an ideal reflectance sensor incorporating a vertical cavity laser array and a large area photodiode
- Developed a MATLAB-based graphical user interface for time-frequency analysis
- Received an individual performance award for the final report
- Designed instrumentation, defined measurement parameters, and proposed operational criteria for the Emergency/Distress Monitor

MODEL BASED DESIGN PROJECT

- Electromagnetic Subproject Leader
 - Planned the budget and technical approach for magnetic field analyses on the W88 magnetic stronglink
 - Conducted an extensive review of commercial, state-of-the-art signal integrity and 3D electromagnetic modeling tools
- Demonstrated model-based design principles to multiple outside organizations, including the Technologies Enabling Agile Manufacturing thrust leaders, Boeing Aerospace, Proctor & Gamble, Motorola, Allied Signal, Schlumberger, and Ansoft
- Co-designed Sandia's Engineering Integration Facility
- Wrote the statement of work and budget proposal for the Sandia/Boeing virtual prototyping cooperative research and development agreement

Graduate Research Assistant

June 1992 – June 1994

The University of Texas at Austin, Department of Electrical and Computer Engineering

- Designed a system to integrate laser-induced fluorescence data from coronary artery cross-sections with two-dimensional intravascular ultrasound images
- Conducted a clinical study with the University of Texas Medical Branch in Galveston to determine the efficacy of this approach
- Performed experiments to characterize the laser-induced fluorescence of human coronary artery and aorta at 476 nm excitation
- Developed a Monte Carlo simulation code to predict excitation and emission profiles (absorption, reflection, and transmission) given an incident laser beam
- Created an algorithm to correct fluorescence spectra obtained with non-contact excitation/emission fiber bundles
- Prepared and published my own histology specimens
- Optimized an arterial tissue fluorescence model that determines arterial chemical composition based on laser-induced fluorescence spectra at 476 nm excitation
- Initiated a clinical effort with Ochsner Clinic in New Orleans to assess likelihood of restenosis based on chemical composition of atherectomy specimens
- Mentored undergraduate students performing honors projects

Graduate Teaching Assistant**August 1991 – May 1992***The University of Texas at Austin, Department of Electrical and Computer Engineering*

- Taught *Pascal Programming* (Spring 1992)
- Monitored the computer laboratory for *Pascal Programming* and *Introduction to Computation*
- Taught one *Electrical Engineering Lab I* section and monitored a second (Fall 1991)
- Graded midterm exams, final exams, and oral reports for *Electrical Engineering Lab I*

Research Assistant**June 1991 – August 1991***Lawrence Livermore National Laboratory, Human Genome Flow Cytometry Unit, Livermore, CA*

- Developed a Pascal interface between a Hewlett-Packard workstation and a Summagraphics digitizer board to determine dyed chromosome separation in a DNA mapping procedure
- Developed post-processing software for the high-speed cell sorter

Graduate Teaching Assistant**August 1989 – May 1991***Kansas State University, Department of Electrical Engineering, Manhattan, KS*

- Taught *Electrical Engineering Lab I* for two semesters
- Graded assignments for *Electromagnetic Theory I* and *Electronics I*
- Lectured during the instructor's absence

Research Assistant**June 1989 – August 1989***Lawrence Livermore National Laboratory, Nuclear Safety Systems Division, Livermore, CA*

- Experimentally determined the high-frequency hysteresis behavior of ferrite cores used in pulse transformer technology
- Obtained a government security clearance

Teaching Assistant**August 1988 – May 1989***Kansas State University, Department of Electrical Engineering, Manhattan, KS*

- Graded assignments for *Electromagnetic Theory I* and *Intro. to Computer Engineering*

JOURNAL PUBLICATIONS

- Suliman, Ahmad, Md Rakibul Mowla, Alaleh Alivar, Charles Carlson, Punit Prakash, Balasubramaniam Natarajan, Steve Warren, and David E. Thompson. "Effects of Ballistocardiogram Peak Detection Jitters on the Quality of Heart Rate Variability Features: A Simulation-Based Case Study in the Context of Sleep Staging," *Sensors*, Vol. 23, March 2023, pp. 2693–2707.
- Lee, Taehee, Bala Natarajan, and Steve Warren. "Pen-Type Electrodermal Activity Sensing System for Stress Detection Based on Likelihood Ratios," *IEEE Transactions on Biomedical Circuits and Systems*, Vol. 15, No. 6, December 2021, pp. 1467-1476.
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- Alivar, Alaleh, Charles Carlson, Ahmad Suliman, Steve Warren, Punit Prakash, David E. Thompson, and Balasubramaniam Natarajan. "Smart Bed Based Daytime Behavior Prediction in Children with Autism Spectrum Disorder - A Pilot Study," *Medical Engineering & Physics*,

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- Ade, Carl J., Ryan M. Broxterman, Jesse C. Craig, Susanna J. Schlup, Samuel L. Wilcox, Steve Warren, Phillip Kuehl, Dana Gude, Chen Jia, and Thomas J. Barstow. "Prediction of Lunar- and Martian-Based Intra- and Site-to-Site Task Performance," *Aerospace Medicine and Human Performance*, Vol. 87, No. 4, April 2016, pp. 367–374. Available at <https://stephenson.pure.elsevier.com>.
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- Kuhn, William, Thomas Barstow, Dwight Day, Don Gruenbacher, Balasubramaniam Natarajan, Tim Sobering, and Steven Warren. “Biosensor Networks and Telecommunication Subsystems for Long-Duration Missions, EVA Suits, and Robotic Precursor Scout Missions,” NNX11AM05A, Final Performance Report, 10/1/2011 – 6/30/2015, July 16, 2015, 39 pages.
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- Carpenter, Kenneth H. and Steve Warren. **Modeling of Ferrite Core Magnetic Couplers and Design of Transformers for Use with Voltage Multipliers**, Final Report Under Subcontract 1507303: Regents of the University of California with Kansas State University and Lawrence Livermore National Laboratory.

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- Warren, Steve and Charles Carlson. “Work in Progress: A Formal Medical Device Teardown as a Biomedical Engineering Learning Experience,” *2025 Annual Conference and Exposition, American Society for Engineering Education*, Montreal Convention Center, Montreal, Quebec, Canada, June 22–25, 2025.
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Engineering, University of Missouri at Kansas City, 546 Robert H. Flarsheim Hall, Kansas City, MO, 64110, March 6, 2007. (Invited)

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INVITED PRESENTATIONS

- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering: Wellness Tools for Children with Autism Spectrum Disorder,” *AAI 860 – Systems and Data in Animal Health*, Kansas State University, Olathe Campus, Via Zoom from Kansas State University, December 11, 2025. (Invited)
- Warren, Steve, Charles Carlson, Don Gruenbacher, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “New KSU Undergraduate Biomedical Engineering Degree Program,” Guest Lecture, Kansas State University, Engineering Hall, Room DUE 0093, *DEN 160 – Introduction to Engineering*, April 22, 2025. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering: Wellness Tools for Children with Autism Spectrum Disorder,” *AAI 860 – Systems and Data in Animal Health*, Kansas State University, Olathe Campus, Via Zoom from Kansas State University, November 21, 2024. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering: Wellness Tools for Children with Autism Spectrum Disorder,” *AAI 860 – Systems and Data in Animal Health*, Kansas State University, Olathe Campus, Via Zoom from Kansas State University, November 15, 2023. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Applied Telehealth Tools for Neurotypical and Special Needs Populations,” MGH/Harvard Medical School Biomedical Engineering Seminar Series, Via Zoom from Kansas State University, January 13, 2023. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering,” Guest Lecture, Kansas State University, *DEN 160 – Introduction to Engineering*, October 25, 2022. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering and Public Health,” Guest Lecture, Kansas State University, *KIN 110 – Introduction to Public Health*, October 20, 2022. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering,” Guest Lecture, Kansas State University, *DEN 160 – Introduction to Engineering*, October 22, 2021. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering and Public Health,” Guest Lecture, Kansas State University, *KIN 110 – Introduction to Public Health*, October 14, 2021. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering,” Kansas State University, Pre-Health Ambassadors, Zoom, April 27, 2021. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Biomedical Engineering and Public Health,” Guest Lecture, Kansas State University, *KIN 110 – Introduction to Public Health*, April 22, 2021. (Invited)

- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Bed-Based Systems to Track the Nighttime Wellness of Children with Autism Spectrum Disorder,” *2020 One Health Research Symposium: Biosensors*, Virtual Event Sponsored by BioNexus KC, Kansas City, MO, November 16-18, 2020, <https://bionexuskc.org/event/2020-one-health-research-symposium-biosensors/>. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “Applied Telehealth Tools for Neurotypical and Special Needs Populations,” University of North Florida, College of Computing, Engineering, and Construction, Jacksonville, FL, October 6, 2020. (Invited)
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, Bala Natarajan, Taehee Lee, Kelsey Warren, Sophie Braynock, and Dallas Rice. “Biomedical Engineering Endeavors at Kansas State University,” University of North Florida, College of Computing, Engineering, and Construction, Jacksonville, FL, November 12, 2019. (Invited)
- Warren, Steve and Charles Carlson. “Biomedical Engineering (BME) Program Overview,” Kansas State University, Electrical and Computer Engineering, College of Engineering Advisory Council, October 25, 2019. (Invited)
- Warren, Steve. “Plans for the Reichenberger Cornerstone Teaching Scholar Award,” Kansas State University, College of Engineering Named-Award Celebration with the Carl R. Ice Family, October 18, 2019. (Invited)
- Warren, Steve. “Biomedical Engineering (BME) Program Overview,” Kansas State University, College of Engineering, Engineering Ambassadors, November 5, 2018. (Invited)
- Warren, Steve. “Biomedical Engineering (BME) Program Overview,” Kansas State University, Electrical and Computer Engineering, ECE Advisory Council Exchange, October 12, 2018. (Invited)
- Warren, Steve. “Tools to Quantify Well-Being in Children with Severe Disabilities,” University of North Florida, College of Computing, Engineering, and Construction, Jacksonville, FL, October 9, 2018. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected Wireless World,” Biomedical Science Seminar, Derby High School, Derby, KS, November 1, 2017. (Invited)
- Warren, Steve. “Research and Technology to Quantify Human and Animal Health,” University of North Florida, College of Computing, Engineering, and Construction, Jacksonville, FL, October 10, 2017. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected Wireless World,” *Future City Leadership Retreat*, Lawrence College and Career Center, Lawrence, KS, September 22, 2017. Middle school student presentations. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected Wireless World,” *ID 545 – Interior Design Studio 7, ID 651 – Design for Supportive Environments*, Kansas State University, College of Human Ecology, Manhattan, KS, September 15, 2017. (Invited)
- Warren, Steve. “Medical Device Design in the KSU Medical Component Design Laboratory,” KSU College of Engineering, Netsmart Executives, November 14, 2016. (Invited.)
- Warren, Steve, Punit Prakash, David Thompson, Bala Natarajan, Charles Carlson, Ahmad Suliman, Shangxian Wang, Taishan Li, Kim Fowler, Ed Brokesh, Jack Xin, Wayne Piersel, Janine Kesterson, and Steve Stoffregen. “Experiences and Lessons Learned: Design Projects Intended to Aid and Quantify Children with Severe Disabilities,” University of North Florida, Department of Electrical Engineering, October 25, 2016. Interactions with Dr. Juan Aceros, Dr. Mary Lundy, and other UNF faculty. Supported by the National Institutes of Health. (Invited)
- Steve Warren, Punit Prakash, David Thompson, Bala Natarajan, Charles Carlson, Ahmad Suliman, Tianyu Lin, and Shangxian Wang. “NSF GARDE Project: Overview and Next Steps,” KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society, KSU College of Engineering, March 7, 2016. (Invited)

- Warren, Steve, Punit Prakash, and David Thompson. "Technology for Successful Aging," Meadowlark Hills Advisory Board, Meadowlark Hills Retirement Community, Manhattan, KS, November 11, 2014. (Invited)
- Warren, Steve. "Biomedical Research in the K-State Department of Electrical & Computer Engineering: Wearable Wireless Medical Devices and Applications," Combined Grand Rounds, Kansas Medical Education Foundation, Topeka, KS, October 18, 2014. (Invited)
- Warren, Steve. "Bioengineering Option in Electrical Engineering," ECE 015 – ECE Orientation, Kansas State University, Manhattan, KS, October 14, 2014. (Invited)
- Larsen, Jace, Job Shiach, and Steve Warren. "KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society – Top 3 Reasons to Join EMBS," ECE 015 – ECE Orientation, Kansas State University, Manhattan, KS, September 16, 2014. (Invited)
- Warren, Steve. "Med/Bio Projects Supported by the KSU Medical Component Design Laboratory," Manhattan High School, Medical Explorers Club, Mercy Hospital Conference Room, Manhattan, KS, February 6, 2014. (Invited)
- Warren, Steve. "Med/Bio Projects Supported by the KSU Medical Component Design Laboratory," CIS 450 – *Computer Architecture and Organization*, Department of Computing & Information Sciences, Kansas State University, November 4, 2013. (Invited)
- Warren, Steve. "Med/Bio Projects Supported by the KSU Medical Component Design Laboratory," KSU Student Chapter of the IEEE EMBS, Kansas State University, September 19, 2013. (Invited)
- Warren, Steve, Morgan Hoffman, and Kaleb Anderson. "KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society – Top 3 Reasons to Join EMBS," ECE 015 – ECE Orientation, Kansas State University, Manhattan, KS, September 17, 2013. (Invited)
- Warren, Steve. "Biomedical Research and Technology for a Connected World," KSU Presidential Lecture Series, Delta Kappa Gamma, Meadowlark Hills, Manhattan, KS, April 17, 2013. (Invited)
- Warren, Steve. "Biomedical Research and Technology for a Connected World," KSU Presidential Lecture Series, Haskell Indian Nations University, Lawrence, KS, April 3, 2013. (Invited)
- Warren, Steve. "NASA-Funded KSU Efforts to Assess/Predict Astronaut Performance," ECE 015 – New Student Assembly, Kansas State University, Manhattan, KS, November 14, 2012. (Invited)
- Warren, Steve. "Biomedical Research and Technology for a Connected World," Meadowlark Hills Lecture Series, Sponsored by the KSU Center on Aging, Manhattan, KS, November 8, 2012. (Invited)
- Warren, Steve. "Biomedical Research and Technology for a Connected World," KSU Presidential Lecture Series, Sports Medicine, Athletic Training, and Health Careers Class, Olathe North High School, Olathe, KS, September 27, 2012. (Invited)
- Warren, Steve. "Biomedical Research and Technology for a Connected World," KSU Presidential Lecture Series, Life Enrichment Group, Butler Community College, El Dorado, KS, September 18, 2012. (Invited)
- Warren, Steve. "Biomedical Device Design for a Connected Wireless World," 2012 IEEE International Conference on Computer Science and Automation Engineering (CSAE2012), Zhangjiajie, China, May 25-27, 2012, <http://www.ieee-csae.org/keynote.asp>. Invited keynote presentation.
- Warren, Steve. "Biomedical Device Design for a Connected Wireless World," Kansas State University Medical Component Design Laboratory, Research Overview and Recruiting Discussion, Beihang University, Beijing, China, May 23, 2012. (Invited)
- Warren, Steve. "Biomedical Device Design for a Connected Wireless World," Kansas State University Medical Component Design Laboratory, Research Overview and Recruiting Discussion, Jiaotong University, Dalian, China, May 22, 2012. (Invited)

- Warren, Steve. “Biomedical Research and Technology for a Connected World,” KSU Presidential Lecture Series, 7th–12th Grade Gifted Student Seminar, Community Church of God, Phillipsburg, KS, March 28, 2012. (Invited)
- Warren, Steve. “Bioengineering Option in Electrical Engineering,” ECE 015 – ECE Orientation, Kansas State University, Manhattan, KS, November 8, 2011. (Invited)
- Warren, Steve. “Wearable/Ingestible Sensors to Acquire Animal-Level State of Health in Cattle Herds,” Frontiers in Multiscale Computational Modeling for Zoonotic Epidemics, Embassy Suites Kansas City Plaza, Kansas City, MO, October 11, 2011. (Invited)
- Warren, Steve. “Biomedical Research and Technology for a Connected World,” KSU Presidential Lecture Series, Gifted and AP Student Seminar, Topeka West High School, Topeka, KS, September 30, 2011. (Invited)
- Warren, Steve. “Update on the Newly Funded NSF RAPD (GARDE) Project: KSU Senior Design Projects to Aid Heartspring Children,” Heartspring, Wichita, KS, Heartspring Board Meeting, September 28, 2011. (Invited)
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Kansas State University, Electrical and Computer Engineering, ECE Advisory Council Exchange, April 8, 2011. (Invited)
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” UMKC Student Chapter of the IEEE Engineering in Medicine and Biology Society, Department of Computer Science and Electrical Engineering, School of Computing and Engineering, University of Missouri at Kansas City, 546 Robert H. Flarsheim Hall, Kansas City, MO, 64110, March 11, 2011. (Invited)
- Warren, Steve. “Biomedical Research and Technology for the Future,” 2011 Future Cities Competition, K-State Union Forum Hall, Kansas State University, Manhattan, KS, January 15, 2011. (Invited Keynote Speaker)
- Warren, Steve. “KSU Medical Component Design Laboratory: Biomedical Research and Veterinary Opportunities,” Information exchange with Hill’s Pet Nutrition, ECE Garmin Conference Room, Kansas State University, Manhattan, KS, November 29, 2010. (Invited)
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Information exchange with Robert Bestgen, ECE Garmin Conference Room, Kansas State University, Manhattan, KS, November 19, 2010. (Invited)
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Industrial and Manufacturing Systems Engineering Graduate Seminar, Kansas State University, Manhattan, KS, November 3, 2010. (Invited)
- Warren, Steve. “Bioengineering Option in Electrical Engineering,” ECE 015 – ECE Orientation, Kansas State University, Manhattan, KS, October 12, 2010. (Invited)
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Eta Kappa Nu Student Body Meeting, Kansas State University, Manhattan, KS, September 15, 2010. (Invited)
- Warren, Steve and Jason Orr. “KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society – Top 3 Reasons to Join EMBS,” ECE 015 – ECE Orientation, Kansas State University, Manhattan, KS, September 14, 2010. (Invited)
- Warren, Steve. “Focus Areas in the KSU Medical Component Design Laboratory,” Information exchange with Robert Bestgen, Information Exchange with Honeywell Federal Manufacturing and Technologies, Kansas City Plant, 2000 East 95th St., Kansas City, MO 64131-6159, August 19, 2010. (Invited)
- Warren, Steve. “Thoughts from a Former Student, Mentee, and Colleague,” Richard G. Gallagher Retirement Dinner, KSU Alumni Center Ballroom, Kansas State University, Manhattan, KS, May 7, 2010. (Invited)
- Warren, Steve. “Biomedical Research and Education Focus Areas at Kansas State University,” NSF CCLI Information Exchange, East Carolina University, Greenville, NC, December 17, 2009. (Invited)

- Warren, Steve. “Physical Device Opportunities for the KSU/Cerner Partnership,” Discussion with Adam Cross, Cerner Corporation, Kansas State University, Manhattan, KS, December 15, 2009. (Invited)
- Warren, Steve. “Physical Device Opportunities for the KSU/Cerner Partnership,” Cerner Innovation Campus, Kansas City, MO, November 20, 2009. (Invited)
- Warren, Steve. “Intelligent Pervasive Technology Applied to Home Health Care,” Discussion with Cerner Executives, Cerner Health Conference 2009, Kansas City, Missouri, October 5, 2009. (Invited)
- Warren, Steve. “Plug-and-Play Monitoring Systems: Thought Progression and Prototype Implementation,” NSF CPS Team Information Exchange, Cerner Health Conference 2009, Kansas City, Missouri, October 5, 2009. (Invited)
- Warren, Steve. “KSU Research in Wearable Telemonitoring Technology,” Aging related research presentation for the Center on Aging, Justin Hall Room 254, Kansas State University, Manhattan, KS, March 27, 2009. (Invited)
- Warren, Steve, Daniel Andresen, Donald Gruenbacher, and Bala Natarajan. “KSU Research in Wearable Telemonitoring Technology,” Information Exchange with Abhi Ray, Digital Rays, Kansas State University, Manhattan, KS, August 5, 2008. (Invited)
- Warren, Steve, Daniel Andresen, Donald Gruenbacher, and Bala Natarajan. “KSU Research in Wearable Telemonitoring Technology,” Information Exchange with Colonel Wayne Parks, Great Plains Cyber Electronic University Consortium, Kansas State University, Manhattan, KS, July 9, 2008. (Invited)
- Warren, Steve. “A Partnership Toward Pervasive Health Care Delivery Environments,” Program Development Exchange with Cerner Corporation, Kansas State University, May 5, 2008. Presentation and white paper. (Invited)
- Warren, Steve. “Pervasive Home Health Care Technology,” KSU Epicenter Information Exchange with Dr. Tim Study, IBM Consulting Client Executive, America's Healthcare and Life Sciences, Kansas State University, January 31, 2008. (Invited)
- Warren, Steve. “Plug-and-Play Monitoring Systems: Thought Progression and Prototype Implementation,” Program Development Exchange with Cerner Corporation, Kansas State University, January 23, 2008. (Invited)
- Warren, Steve. “Intelligent Pervasive Technology Applied to Home Health Care,” Information Exchange with Cerner Corporation, Kansas City, MO, December 17, 2007. (Invited)
- Warren, Steve. “Trends in Personalized Health Care Technology,” Graduate Seminar, Department of Industrial and Manufacturing Systems Engineering, Kansas State University, November 16, 2007. (Invited)
- Warren, Steve. “Pervasive Home Health Care Technology,” Health Care Advisory Committee, Department of Industrial and Manufacturing Systems Engineering, Kansas State University, October 25, 2007. (Invited)
- Warren, Steve. “Intelligent Care Systems for the Home,” KSU Student Chapter of the IEEE EMBS, Kansas State University, March 8, 2007. (Invited)
- Warren, Steve. “Home Health Monitoring Devices,” UMKC Student Chapter of the IEEE, Department of Computer Science and Electrical Engineering, School of Computing and Engineering, University of Missouri at Kansas City, 546 Robert H. Flarsheim Hall, Kansas City, MO, 64110, March 6, 2007. (Invited)
- Warren, Steve. “In Pursuit of Intelligent Wearable Health Monitoring Systems,” Kansas City Section of the IEEE, Wyndham Garden Hotel, Overland Park, KS, November 16, 2006. (Invited)
- Warren, Steve, Daniel Andresen, Kevin Smith, Angel Martinez, Roland Craddolph, Patrick York, Howard Erickson, Tammi Hildreth, and David Poole. “Wearable and Wireless: Distributed, Sensor-Based Telemonitoring Systems for State of Health Determination in Cattle,” Information Exchange with Greatbatch, Inc., Tadtman Board Room, Alumni Center, Kansas State University, Feb. 7, 2006. (Invited)

- Warren, Steve and Emil Jovanov. "The Need for Rules of Engagement Applied to Wireless Body Area Networks," *IEEE Consumer Communications and Networking Conference*, Harrah's Las Vegas Hotel & Casino, Las Vegas, NV, January 8–10, 2006. (Invited paper/presentation.) (16% acceptance)
- Warren, Steve, Daniel Andresen, Luke Nagl, Scott Schoenig, Balakumar Krishnamurthi, Howard Erickson, Tammi Hildreth, David Poole, and Mark Spire. "Wearable and Wireless: Distributed, Sensor-Based Telemonitoring Systems for State of Health Determination in Cattle," Anatomy & Physiology Department Seminar, Kansas State University, Practice Management Center, Trotter Hall, September 27, 2004.
- Warren, Steve, Jianchu Yao, Ryan Schmitz, and Jeff Lebak. "Reconfigurable Point-of-Care Systems Designed with Interoperability Standards," *26th Annual Conference of the IEEE EMBS*, Westin St. Francis Hotel, San Francisco, CA, September 1–5, 2004, pp. 3270–3273. (Invited)
- Warren, Steve. "Wearable Medical Device Networks Designed with Interoperability in Mind," U.S. Food and Drug Administration Staff College, 9200 Corporate Blvd., Rockville, MD, November 3, 2003.
- Warren, Steve. "Beyond Telemedicine: Infrastructures for Intelligent Home Care Technology," *Pre-ICADI Workshop on Technology for Aging, Disability, and Independence*, The Royal Academy of Engineering, 29 Great Peter Street, Westminster, London SW1P 3LW (England), June 26–27, 2003, <http://www.cise.ufl.edu/~helal/preICADI>.
- Warren, Steve. "Commodity Building Blocks for Home Health Care," Community Health Council, Mercy Hospital, Manhattan, KS, September 11, 2002. Summarized in the September 2002 Community Health Council bulletin, <http://www.chcnet.info>.
- Warren, Steve and Ronald P. Kensek. "Monte Carlo Transport in Computer-Aided Design (CAD) Geometries," *Spatial Insider's Summit 2002*, Westminster, CO, May 13–15, 2002.
- Warren, Steve, Ruth Miller, Daniel Marcus, Thomas Barstow, Howard Erickson, David Poole, and Daniel Andresen. "Biomedical Sensor Webs for Human and Animal Monitoring," Kansas State University, February 13, 2002. Meeting with Daniel L. Schmoldt, National Program Leader for the U.S. Department of Agriculture Instrumentation and Sensors Program.
- Warren, Steve. "Bioengineering Option in Electrical Engineering," IEEE Student Chapter Meeting, Kansas State University, Manhattan, KS, October 9, 2001.
- Warren, Steve. "Beyond Telemedicine: Home Care Technologies," *Informational Meeting: Research to Aid Persons with Disabilities (RAPD)*, National Science Foundation, Arlington, VA, June 28, 2001.
- Warren, Steve. "Medical Component Confederacies and Home-Based Sensor Webs," *FDA Workshop on Home Care Technology Trends*, U.S. Food and Drug Administration, Rockville, MD, January 25, 2001.
- Warren, Steve. "Smart Health Care Technology and the Home of the Future," *Biomedical Engineering Seminar*, Department of Biomedical Engineering, University of Minnesota, Twin Cities Campus, November 30, 1999. Broadcast via UNITE instructional television.
- Warren, Steve. "The Health Care Facelift: How Telecommunications Technology, Distributed Computing, and Intelligent Medical Devices will Redefine the Patient-Physician Relationship," *Fall 1999 Seminar Series*, Department of Electrical and Computer Engineering, Wichita State University, October 22, 1999.
- Warren, Steve. "The Vital Signs of a Dynamic Engineering Curriculum," Kansas State University, Department of Electrical and Computer Engineering, April 2, 1999.
- Warren, Steve. "Radiation Transport in Complex CAD/CAM/CAE Geometries," Sandia National Laboratories, Center 6500 Management Team Briefing, March 2, 1998.
- Love, Jerry T. and Steve Warren. "The Personal Status Monitor Project," for the Warfighter Personal Status Monitor project team, Combat Casualty Care Research Program, Sandia National Laboratories, August 16, 1996.
- Warren, Steve. "Photo-Plethysmographic Signal Acquisition and Motion Artifact," Sandia National Laboratories, Center 9400 Technology Exchange Meeting, August 22, 1996.

- Warren, Steve. “Pulse Oximetry Measurement and Motion Artifact,” for Colonel William P. Wiesmann, Director, Combat Casualty Care Research Program (RAD II), U.S. Army Medical and Research Material Command, June 17, 1996.
- Warren, Steve. “Burn Depth Measurement,” for Colonel William P. Wiesmann, Director, Combat Casualty Care Research Program (RAD II), U.S. Army Medical and Research Material Command, June 17, 1996.
- Warren, Steve. “Model Based Design at Sandia National Laboratories,” The University of Texas at El Paso, *MFG 390 – Special Topics in Manufacturing*. El Paso, TX, October 25, 1994.
- Warren, Steve. “Agile Product Realization at Sandia National Laboratories,” *Chesapeake Electronics Show*, Holiday Inn, Greenbelt, MD. October 13, 1994.
- Warren, Steve. “Combined Ultrasound and Fluorescence Spectroscopy for Physico-Chemical Imaging of Atherosclerosis,” Los Alamos National Laboratory, Chemical and Laser Sciences Division. Los Alamos, New Mexico, February 24, 1994.
- Warren, Steve. “Combined Ultrasound and Fluorescence Spectroscopy for Physico-Chemical Imaging of Atherosclerosis,” Sandia National Laboratories, Program Development Department. Albuquerque, New Mexico, January 23, 1994.

OTHER PRESENTATIONS

- Warren, Steve, Don Gruenbacher, Crystal Strauss, Elizabeth Goetz, John Baybutt, and Charles Carlson. “KSU Undergraduate Biomedical Engineering Degree Program,” Women in Engineering Spring Event, DUE 3099, Carl R. Ice College of Engineering, Kansas State University, February 28, 2025.
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “The New Kansas State University Biomedical Engineering Degree Program,” 2024 Kansas HOSA Fall Leadership Conference, Future Health Professionals, Kansas State University, Manhattan, KS, October 17, 2024.
- Warren, Steve, Charles Carlson, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, and Bala Natarajan. “The New Kansas State University Biomedical Engineering Degree Program,” 2024 Kansas HOSA Spring Leadership Conference, Future Health Professionals, Kansas State University, Manhattan, KS, March 27, 2024.
- Warren, Steve, and Charles Carlson. “New KSU Undergraduate Biomedical Engineering Degree Program,” 2023 Health Professions Summer Program, DUE 3099, Carl R. Ice College of Engineering, Kansas State University, July 25, 2023.
- Garcia, Damian and Steve Warren. “The Potential for EEG Waves to Examine Attentiveness in Children With Special Needs,” Undergraduate Research Showcase, Office of Undergraduate Research and Creative Inquiry, Kansas State University, Student Union Ballroom, April 13, 2023. Poster session.
- Garcia, Damian, Charles Carlson, and Steve Warren. “Needs Exploration at the No Stone Unturned Therapeutic Learning Center,” Undergrad Research Showcase, Office of Undergraduate Research and Creative Inquiry, Kansas State University, Student Union Ballroom, July 28, 2022. Poster session.
- Warren, Steve, and Charles Carlson. “New KSU Undergraduate Biomedical Engineering Degree Program,” 2022 Health Professions Summer Program, DUE 3099, Carl R. Ice College of Engineering, Kansas State University, July 11 and July 28, 2022.
- Warren, Steve, Charles Carlson, Don Gruenbacher, Dong Ren, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, Bala Natarajan, Thomas Barstow, and Carl Ade. “KSU Undergraduate Biomedical Engineering Degree Program, Plus Potential Joint KSU/Garmin Endeavors,” Discussion with Garmin Representatives, Kansas State University, Carlo R. Ice College of Engineering, April 20, 2022.
- Warren, Kelsey, Charles Carlson, and Steve Warren. “Hands-On Undergraduate BME Design Projects,” *Engineering Undergraduate Research and Creative Inquiry Showcase*, Kansas State University, Carl R. Ice College of Engineering, April 17-27, 2021.

- Rice, Dallas and Steve Warren. “Autonomous Sanitation/Disinfection Drone,” *Engineering Undergraduate Research and Creative Inquiry Showcase*, Kansas State University, Carl R. Ice College of Engineering, April 17-27, 2021.
- Warren, Steve, Punit Prakash, David Thompson, Bala Natarajan, Charles Carlson, Daniel Andresen, John Hatcliff, Thomas Barstow, Carl Ade, and Howard Erickson. “Human and Animal Biomedical Research in the KSU Medical Component Design Laboratory,” KSU/NBAF Biomedical Collaboration Meeting, Kansas State University, College of Engineering, Manhattan, KS, March 29, 2019.
- Steve Warren, Punit Prakash, David Thompson, Bala Natarajan, Charles Carlson, and Don Gruenbacher. “A Snapshot of KSU ECE Endeavors with National Instruments Toolsets,” Igor Alvarado, National Instruments, Kansas State University, College of Engineering, Manhattan, KS, February 21, 2019.
- Carlson, Charles, Ahmad Suliman, and Dong Ren. “Bed-Based Unobtrusive Sleep Quality Monitoring,” General Meeting, Kansas State University Student Chapter of the IEEE Engineering in Medicine and Biology Society, KSU College of Engineering, February 12, 2019.
- Ren, Dong, Charles Carlson, and Steve Warren. “Algorithm to Differentiate the Pulse Rates of Two individuals that Share a Sensor-Laden Bed,” *Research and the State*, Kansas State university, November 8, 2018.
- Steve Warren, David Thompson, Charles Carlson, and Dong Ren. “KSU Biomedical Engineering Program Overview,” Shawnee Mission School District, Biotechnology Signature Program STEM Visit, Kansas State University, Department of Electrical & Computer Engineering, Manhattan, KS, November 1, 2018.
- Carlson, Charles and Ahmad Suliman. “Overview of the GARDE Project,” General Meeting, Kansas State University Student Chapter of the IEEE Engineering in Medicine and Biology Society, KSU College of Engineering, October 8, 2018
- Carlson, Charles, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, Bala Natarajan, and Steve Warren. “A Pilot Study of an Unobtrusive Bed-Based Sleep Quality Monitor for Severely Disabled Autistic Children,” K-State GRAD Forum, Kansas State University, Regnier Hall, College of Architecture, March 29, 2018.
- McKittrick, Andrew, Shangxian Wang, Charles Carlson, Punit Prakash, David Thompson, Bala Natarajan, Wayne Piersel, Janine Kesterson, Steve Stoffregen, and Steve Warren. “Enuresis Detection Using Thermal Sensing Methods,” Kansas State University, College of Engineering, Undergraduate Research Poster Forum, April 26, 2018. Poster presentation.
- Washington, Derrius, Charles Carlson, and Steve Warren. “Improving Sleep Quality for Children with Autism Spectrum Disorder,” Developing Scholars Poster Forum, Kansas State University, K-State Union Ballroom, April 15, 2018.
- Carlson, Charles, Alexandra Lyle, Jake Chappell, Mariah Brown, Brie Phillips, and Steve Warren. “The KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society,” Engineering Scholars Day, KSU College of Engineering, Manhattan, KS, October 24, 2017.
- Wang, Shangxian, Charles Carlson, Punit Prakash, David Thompson, Bala Natarajan, Wayne Piersel, Rachel Boulivard, Steve Stoffregen, and Steve Warren. “Enuresis Detection Via Thermocouple Data Analysis,” Kansas State University, College of Engineering, Undergraduate Research Poster Forum, April 27, 2017. Poster presentation.
- Carlson, Charles and Ahmad Suliman. “Unobtrusive Bed Based Ballistocardiogram Monitor for Severely Disabled Autistic Children,” Kansas State University, College of Engineering, Open House, March 31 – April 1, 2017.
- Warren, Steve. “Biomedical Research in the K-State Department of Electrical and Computer Engineering: Wearable Wireless Medical Devices and Applications,” Kansas State University, Electrical & Computer Engineering, August 29, 2016. Information exchange with KSU Animal Sciences.
- Ortega, Martin, Steve Warren, and Charles Carlson. “Ambient Sensor Research for Sleep Quality Environmental Assessment,” Research Immersion: Pathways to STEM (RiPS) Poster Session, KS-LSAMP Program, -Kansas State University, Union Ballroom, July 27, 2016.

- Wang, Shangxian, Taishan Li, Charles Carlson, Punit Prakash, David Thompson, Bala Natarajan, Wayne Piersel, Janine Kesterson, Steve Stoffregen, and Steve Warren. "Enuresis Detection Using Thermal Sensing Methods," Kansas State University, College of Engineering, Undergraduate Research Poster Forum, April 15, 2016. Poster presentation.
- White, Austin, Charles Carlson, and Steve Warren. "Electrodermal Activity," Kansas State University, College of Engineering, Undergraduate Research Poster Forum, April 15, 2016. Poster presentation.
- Carlson, Charles, Ahmad Suliman, Punit Prakash, David Thompson, and Steve Warren. "Sensor Beds for Tracking Sleep in Severely Disabled Children," Kansas State University, Electrical & Computer Engineering, General Engineering Course, March 8, 2016.
- Wang, Shangxian, Charles Carlson, Punit Prakash, David Thompson, Bala Natarajan, Wayne Piersel, Janine Kesterson, and Steve Warren. "Enuresis Detection by Way of Thermocouple Data Analysis," Kansas State University, College of Engineering, Undergraduate Research Poster Forum, December 3, 2015. Poster presentation.
- Warren, Steve, Punit Prakash, David Thompson, Bala Natarajan, Charles Carlson, Ahmad Suliman, Tianyu Lin, and Shangxian Wang. "NSF GARDE Project: Overview and Next Steps," Heartspring, Wichita, KS, November 13, 2015.
- Warren, Steve. "American Society for Engineering Education: Overview and KSU Involvement," Kansas State University, College of Engineering, ECE Garmin Conference Room, May 7, 2015.
- Shiach, Job, Julia Nyiro, Megan Richards, and Steve Warren. "KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society," ECE Advisory Council, Kansas State University, March 27, 2015.
- Warren, Steve. "Trip Debrief: IEEE EMBC 2014, Chicago, IL," Bio/Edu Seminar, Kansas State University, Electrical & Computer Engineering, Manhattan, KS, October 15, 2014.
- Warren, Steve. "Med/Bio Projects Supported by the KSU Medical Component Design Laboratory," Meadowlark Hills Administration, KSU ECE, Manhattan, KS, August 18, 2014.
- Warren, Steve. "Med/Bio Projects Supported by the KSU Medical Component Design Laboratory," Dimensional Innovations, KSU ECE, Manhattan, KS, July 14, 2014.
- White, Austin, Sergio Ortiz, Punit Prakash, and Steve Warren. "Monitoring Nighttime Well Being of Children with Disabilities," *14th Annual Research Poster Symposium*, Kansas State University Developing Scholars Program, April 13, 2014, K-State Union Ballroom, Manhattan, KS.
- Ortiz, Sergio, Austin White, Punit Prakash, and Steve Warren. "Using Commercial Tools to Validate a Bed Sensor Suite for Nighttime Monitoring of Severely Disabled Children," *14th Annual Research Poster Symposium*, Kansas State University Developing Scholars Program, April 13, 2014, K-State Union Ballroom, Manhattan, KS.
- Luan, Shiwei and Steve Warren. "Modularized PCA Pump Design," KSU Engineering Scholars Day, KSU College of Engineering, Manhattan, KS, October 30, 2013.
- Anderson, Kaleb and Steve Warren. "KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society," ECE Advisory Council, Kansas State University, April 5, 2013.
- Warren, Steve, Xiongjie Dong, Chen Jia, and Tianyu Lin. "Biomedical Devices for Space Applications," Bioengineering Session, E3 Workshop for Science Teachers Hosted by the KSU College of Engineering, Manhattan, KS, Bette Grauer, Organizer, May 30, 2013.
- College of Engineering Open House, Kansas State University, Manhattan, KS, April 19–20, 2013. NASA EPSCoR and NASA Crew Performance Projects.
- White, Austin, Hannah Gray, Steve Warren, Wen Song, Kaleb Anderson, Gregory Nitcher, and Neelou Hadavandifard. "Monitoring Night-Time Well Being of Disabled Children at Heartspring," *13th Annual Research Poster Symposium*, Kansas State University Developing Scholars Program, April 21, 2013, K-State Union Ballroom, Manhattan, KS.
- Gray, Hannah, Steve Warren, Punit Prakash, Wen Song, Kaleb Anderson, Gregory Nitcher, Austin White, and Neelou Hadavandifard. "Bed Sensor Suite Project: Nighttime Wellness

Monitoring for Children at Heartspring,” *13th Annual Research Poster Symposium*, Kansas State University Developing Scholars Program, April 21, 2013, K-State Union Ballroom, Manhattan, KS.

- Garman, Kevin, Timothy Gustafson, James Thoennes, Punit Prakash, and Steve Warren. “Weight Distribution Analysis for Patients Targeted with Unilateral Lower-Limb Orthotics or Prosthetics,” *13th Annual Research Poster Symposium*, Kansas State University Developing Scholars Program, April 21, 2013, K-State Union Ballroom, Manhattan, KS.
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Suman Saripalli, KalScott Engineering, Intellispeak, Kansas State University, Electrical & Computer Engineering, April 9, 2013.
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Jim Baxendale, Whiteboard to Boardroom Regional Commercialization Program, UMKC Innovation Center, Kansas State University, Electrical & Computer Engineering, March 7, 2013.
- Kuhn, Bill, Amelia Hodges, Xiongjie Dong, and Steve Warren. NASA Project Overview, ECE Department, Engineering Scholarship Day, February 15, 2013.
- Li, Kejia Li, Xiongjie Dong, Dana Gude, Chenyu Zheng, Devon Krenzel, Steve Warren, and Don Gruenbacher. Bioengineering Session, E3 Workshop for Science Teachers Hosted by the KSU College of Engineering, Bette Grauer, Organizer, May 31, 2012.
- Barstow, Tom, Carl Ade, Ryan Broxterman, Steve Warren, Dana Gude, Wen Song, Tommy Nelson, Dale Schinstock, and Tim Moulam. “KSU Research Update: Standardized “Pre-Flight” Exercise Tests to Predict Performance During Extravehicular Activities in a Lunar Environment,” KSU Student Chapter of the IEEE EMBS, KSU College of Engineering, April 5, 2012.
- Warren, Steve and Timothy Gustafson. “KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society,” EECE Advisory Council, Kansas State University, March 30, 2012.
- Warren, Steve. “Kansas State University Department of Electrical & Computer Engineering,” KSU Electrical & Computer Engineering, March 27, 2012. Information exchange with National Instruments (Robert Canik, Brian Platt, Mark Troutfetter, and Jim Schwartz).
- Krenzel, Devon, Steve Warren, Bala Natarajan, and Gurdip Singh. “Cerner Slips and Falls Project,” Manhattan, KS, March 1, 2012.
- Warren, Steve, Dana Gude, Tommy Nelson, Wen Song, Devon Krenzel, and Kejia Li. “Standardized “Pre-Flight” Exercise Tests to Predict Performance During Extravehicular Activities in a Lunar Environment - KSU Electrical & Computer Engineering Activity Summary,” NASA HRP Mid-Contract Review, Kansas State University, Electrical & Computer Engineering, January 26, 2012.
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Kim Fowler, Kansas State University, Electrical & Computer Engineering, November 14, 2011.
- Warren, Steve. “Service Design Projects Sponsored by the IEEE EMBS Student Chapter at Kansas State University”, Gerontology Alumni Open House, Meadowlark Hills, Manhattan, KS, October 28, 2011.
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” Carl Mayer and Chris Justice, RBC Medical Innovations, Kansas State University, Electrical & Computer Engineering, October 27, 2011.
- Warren, Steve, Caterina Scoglio, Bala Natarajan, Noel Schulz, and Sanjoy Das. “Kansas State University Department of Electrical & Computer Engineering,” The University of Texas at Austin, Department of Electrical Engineering, Engineering and Science Building, October 3, 2011. Recruiting presentation for the ECE open position.
- “Biosensor Networks and Telecommunication Subsystems for Long-Duration Missions, EVA Suits, and Robotic Precursor Scout Missions - Coordinated Research Spanning Early-Stage Innovations through Game Changing Technologies,” Kansas State University, Electrical &

Computer Engineering, September 16, 2011. NASA EPSCoR project presentation/discussion with Steve Bowen, NASA astronaut.

- Li, Kejia, Steve Warren, and John Hatcliff. CIMIT Open House Demonstrations: “MDCF Demo 1: Platform-Oriented Device” and “MDCF Demo 2: Online Reconfigurable Device,” Boston, MA, Sept. 6-8, 2011. “Medical Platform Oriented Device Demo” (video) at <http://www.youtube.com/watch?v=fj-rOFNy6kk> created in association with the 3rd Annual Medical Device Connectivity Conference and Exhibition, Joseph B. Martin Conference Center at Harvard Medical School, Boston, MA, Sept. 8–9, 2011.
- Warren, Steve. “NSF RAPD Proposal Summary & Early Thoughts Regarding KSU Senior Design Projects to Aid Heartspring Children,” Heartspring, Wichita, KS, August 26, 2011.
- Mendenhall, Kekoa, Xiongjie Dong, and Steve Warren. “Controlling Future Devices ... With Your Eyes?,” Developing Scholars Program, College of Engineering, Kansas State University, April 17, 2011.
- Warren, Steve. “Biomedical and Educational Research in the KSU Medical Component Design Laboratory,” William Graveman, Biomedical Devices of Kansas, March 29, 2011.
- Warren, Steve. “KSU Medical Component Design Laboratory Focus Areas,” Heartspring Visit, Kansas State University, Electrical & Computer Engineering, September 24, 2009.
- Warren, Steve. “KSU Medical Component Design Laboratory Focus Areas,” Shulz Family Tour of Engineering Laboratories, Kansas State University, Electrical & Computer Engineering, September 23, 2009.
- *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 17–18, 2009.
 - Warren, Steve. “The Bioengineering Option in Electrical Engineering”
 - Moratto, Zachary, Tyler Kreybiel, Paul Edgington, Steve Taylor, Zane Pauls, and Zachary Maier, and Steve Warren. “Introduction to Computer Graphics”
 - Hoskins, Seth and Steve Warren. “Wireless Magnetic Link for a Cattle Physiology Monitoring Pill”
 - Li, Kejia and Steve Warren. “Wireless Pulse Oximeter”
 - Nagaraja, Ashvin, Kejia Li, Travis Upah, Kentaro Takamatsu, Chen Jia, Grant Damas, and Steve Warren. “Vital Sign Sensing Pad”
 - Gruber, Lucinda, Adriann Sullivan, Connor Griffith, Ethan Young, and Steve Warren. “The KSU Student Chapter of the IEEE EMBS”
 - Takamatsu, Kentaro and Steve Warren. “Greenhouse Monitoring Project”
- Warren, Steve, Connor Griffith, Lucinda Gruber, and Ethan Young. “KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society,” EECE Advisory Council, Kansas State University, April 3, 2009.
- Hatcliff, John, Daniel Andresen, Andrew King, Sam Proctor, and Steve Warren. “A Publish-Subscribe-Based Device Integration Framework,” Food & Drug Administration, Rockville, MD, July 29, 2008.
- Bala Natarajan and Steve Warren. “Applied Biosecurity Research in the KSU Electrical & Computer Engineering Department,” Kansas State University, Biosecurity Research Institute, Sceptor Industries, April 30, 2008.
- Rome, Nicholas. “Reflective Pulse Oximetry: Creation of a Recording VI,” Poster Session, Campus Student Internship Program, Kansas State University, Fiedler Auditorium, April 23, 2008.
- Warren, Steve. “KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society,” EECE Advisory Council, Kansas State University, April 4, 2008.
- Gruenbacher, Don, Bala Natarajan, and Steve Warren. “Applied Biosecurity Research in the KSU Electrical & Computer Engineering Department,” Kansas State University, Biosecurity Research Institute, Raytheon Corporation, March 26, 2008.
- Warren, Steve. “Electrical & Computer Engineering,” *DEN 160 – Engineering Concepts*, Paslay Lecture Hall, Kansas State University, March 25, 2008.

- Warren, Steve. "The Biomedical Option in Electrical Engineering," *EECE 015 – New Student Assembly*, Paslay Lecture Hall, Kansas State University, November 7, 2007.
- Warren, Steve. "Pulse Oximeter Designs in the Medical Component Design Laboratory," Medical Component Design Laboratory, Kansas State University, October 9, 2007.
- Kevin Smith, Angel Martinez, Patrick York, Roland Craddolph, Steve Warren, Daniel Andresen, Tammi Hildreth, and Howard Erickson. "Veterinary Telemonitoring for Disease Prevention in Cattle Herds," *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 7–8, 2006.
- Steve Warren. "The Bioengineering Option in Electrical Engineering," *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 7–8, 2006.
- Steve Warren. "The Multimedia Option in Computer Engineering," *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 7–8, 2006.
- Warren, Steve and Daniel Andresen. "Human and Animal Health Monitoring," High School Programming Contest, Computing & Information Sciences, KSU Union Ballroom, Kansas State University, November 4, 2005.
- Warren, Steve. "Health Monitoring Efforts in the Medical Component Design Laboratory," Graduate Student Recruitment Fair, Kansas State University Graduate School, Oct. 14-15, 2005.
- Warren, Steve, Jianchu Yao, Kevin Smith, Jeffrey Lebak, Balakumar Krishnamurthi, and Daniel Andresen. "Embedded Monitoring Systems for Human & Animal Health Assessment," Research Experiences for Teachers Workshop, Medical Component Design Laboratory, Kansas State University, June 30, 2005.
- Steve Warren. "The Bioengineering Option in Electrical Engineering," *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 8–9, 2005.
- Kevin Smith, Angel Martinez, Scott Schoenig, Patrick York, Roland Craddolph, Dominic Gelinas, Steve Warren, Daniel Andresen, Tammi Hildreth, David Poole, and Howard Erickson. "Veterinary Telemonitoring for Disease Prevention in Cattle Herds," *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 8–9, 2005.
- Wareing, Austin and Steve Warren. "Optimization of Reflectance-Mode Pulse Oximeter Sensors," *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 8–9, 2005.
- Schoenig, Scott, Kevin Smith, Dominic Gelinas, Balakumar Krishnamurthi, Tammi Hildreth, Howard Erickson, Mark Spire, David Poole, Daniel Andresen, and Steve Warren. "Veterinary Telemonitoring for Disease Prevention in Cattle," *Capitol Research Forum*, Capitol Building, Topeka, KS, March 1, 2005. Poster presentation.
- Yang, Yong, and Steve Warren. "Automated Homework Generation and Assessment for Engineering Courses," *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 2–3, 2004. Poster presentation.
- Luke Nagl, Scott Schoenig, Balakumar Krishnamurthi, Roland Craddolph, Steve Warren, Daniel Andresen, Tammi Hildreth, David Poole, Mark Spire, and Howard Erickson. "Veterinary Telemonitoring for Disease Prevention in Cattle Herds," *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 2–3, 2004. Poster presentation and equipment demonstration.
- Yao, Jianchu and Steve Warren. "Wearable and Wireless Medical Devices Designed with Interoperability Standards," *College of Engineering Open House*, Kansas State University, Manhattan, KS, April 2–3, 2004. Poster presentation and equipment demonstration.
- Luke Nagl and Tammi Hildreth. "BMOO: Bovine Mobile Observation Operation," Research Experience for Teachers seminar, Mitch Nielsen, Organizer, June 25, 2003.
- Luke Nagl, Ryan Schmitz, Jianchu Yao, Steve Warren, Ravi Kollarapu, Daniel Andresen, Tammi Hildreth, David Poole, and Howard Erickson. "Veterinary Telemonitoring for Disease Prevention in Cattle Herds," *College of Engineering Open House*, Kansas State University, Manhattan, KS, March 28–29, 2003. Poster presentation.

- Warren, Steve. “Telemedicine Project Overview,” Telemedicine and Advanced Technology Research Center, U.S. Army Medical Research and Materiel Command, Fort Detrick, Frederick, MD, March 19, 1999.
- Warren, Steve. “Telemedicine Vision and Project Summary,” Sandia National Laboratories, For Gerry Yonas, Vice President - Systems, Science, and Technology Division, February 15, 1999.
- Warren, Steve. “Portable Owner-Aware Biometric Access Control Device,” Sandia National Laboratories LDRD Surety Review, July 22, 1998.
- Warren, Steve, Ronald P. Kensek, and Leonard J. Lorence. “ITS-ACIS: Radiation Transport in CAD/CAM/CAE Geometries,” Sandia National Laboratories, Quarterly Meeting of the Neutron Generator Hostile Environment Project, April 22, 1998.
- Sill, Anthony E. and Steve Warren. “The Role of Technology in Reducing Health Care Costs,” Sandia National Laboratories, February 19, 1998.
- Warren, Steve. “Model Based Design in Defense Systems: High-Fidelity Electromagnetic Analysis,” Boeing Aerospace. Satellite link video conference, March 28, 1995.
- Warren, Steve. “Model-Based Design at Sandia National Laboratories,” Ansoft Corporation, Pittsburgh, PA, February 21, 1995.
- Warren, Steve. “Distress Monitor Conceptual Review: Requirements and Theory,” Sandia National Laboratories, September 26, 1994.
- Warren, Steve, and Randall B. Smith. “Virtual Prototyping at Sandia National Laboratories,” Workshop on U.S. Electronics Manufacturing, National Electronics Manufacturing Initiative, Georgetown University, Washington, D.C., July 25, 1994.

INVITED WORKSHOPS/MEETINGS

- **Information Exchange with Matt Dodge**, Biosense Webster, Kansas City, MO, October 25, 2022, January 20, 2023
- **Information Exchange with Rachelle White**, TRIMEDX, Salina, KS January 20, 2023
- **Information Exchange with IEEE EMBS Student Chapter**, University of North Florida, Jacksonville, FL, November 7, 2022
- **Information Exchange with Ceres Tag Representatives**, Kansas State University, Research Park Circle, Manhattan, KS, August 9, 2022.
- **Information Exchange with Karl McCombs**, Children’s Mercy Hospital, Kansas City, MO, July 7, 2022
- **Information Exchange with Garmin Representatives**, Kansas State University, Carl R. Ice College of Engineering, Manhattan, KS, April 20, 2022.
- **Information Exchange with Ron Pellegrini**, Vice President, Head of Field Service LD/POC, Siemens Healthineers, Kansas State University, College of Engineering, Manhattan, KS, October 14, 2021.
- **Information Exchange with Nick Fowler**, Mechanical Engineering, Engenious Design, Zoom Meeting, KS, October 13, 2021.
- **Information Exchange with Lee G. Willis**, Senior Project Manager and Safety System Engineer, Johnson Space Center, NASA, Kansas State University, College of Engineering, Manhattan, KS, October 7, 2021.
- **Information Exchange with Mike Guilliame**, Site Manager, TRIMEDX, Ascension Via Christi, Manhattan, KS, September 23, 2021.
- **Information Exchange with John Hrynuk**, U.S. Army Research Laboratory, Kansas State University, College of Engineering, Manhattan, KS, April 23, 2019.
- **Information Exchange with Dr. Steve Kappes, Dr. Beth Lautner, Diana Whipple, and Dr. Pat Glas**, U.S. Department of Agriculture, KSU/NBAF Biomedical Collaboration Opportunities, Kansas State University, College of Engineering, Manhattan, KS, March 29, 2019.
- **Information Exchange with Igor Alvarado**, National Instruments, KSU College of Engineering, Manhattan, KS, February 21, 2019.

- **Information Exchange with Dr. Steve Eckels**, KSU Mechanical & Nuclear Engineering, Manhattan, KS, February 6, 2017.
- **Information Exchange with Dr. Tim Rozell and Dr. Ashley Rhodes**, KSU Animal Sciences, Manhattan, KS, January 30, 2017.
- **Information Exchange with Netsmart**, KSU College of Engineering, November 14, 2016.
- **Information Exchange with Dr. Lindsey Hulbert**, KSU Animal Sciences, Manhattan, KS, August 29, 2016.
- **Information Exchange with SORA Medical**, Kansas State University, Olathe Campus, Olathe, KS, April 27, 2016. Opportunities related to the SORA Medical ApolloMED clinician app.
- **Information Exchange with i3D Plus**, Salina, KS, June 7, 2015. True 3D imaging technology and biomedical applications.
- **Information Exchange with Dennis and Malinda Dillman**, KSU ECE Department, September 30, 2014. KSU NASA projects.
- **Healthcare Robotics Expert Panel**, Kansas City Chapter of the IEEE EMBS, Embassy Suites, Overland Park, KS, September 11, 2014. K-State Salina, K-State Manhattan, Pittsburg State University.
- **Information Exchange with Meadowlark Hills**, Meadowlark Hills Retirement Community, July 16, 2014; August 4, 2014; August 18, 2014.
- **Information Exchange with Kevin Rooney, Dimensional Innovations**, KSU ECE Department, July 14, 2014.
- **Information Exchange with National Instruments**, KSU ECE Department, September 24, 2013.
- **Information Exchange with KalScott Engineering**, KSU ECE Department, April 9, 2013.
- **Information Exchange with National Instruments**, KSU ECE Department, April 2, 2013.
- **Information Exchange with Jim Jaax**, KSU ECE Department, March 29, 2013.
- **Information Exchange with UMKC Innovation Center**, KSU ECE Department, March 7, 2013.
- **NSF RAPD Design Project Reviews**, KSU ECE Department, October 9, 2012.
- **Information Exchange with National Instruments**, KSU ECE Department, March 27, 2012.
- **Information Exchange with SANY Corporation**, Changsha, China, May 29, 2012.
- **Information Exchange with Beihang University**, Beijing, China, May 23, 2012.
- **Information Exchange with Jiaotong University**, Dalian, China, May 22, 2012.
- **Mid-Contract Review**, NASA Johnson Space Center (Linda Loerch, Lori Ploutz-Snyder, and Jeffrey Ryder), KSU ECE Department, January 24, 2012.
- **Information Exchange with Kim Fowler**, KSU ECE Department, November 14, 2011.
- **Information Exchange with Carl Mayer and Chris Justice, RBC Medical Innovations**, KSU ECE Department, October 27, 2011.
- **Information Exchange with Michael Duncan, NASA Flight Surgeon**, KSU ECE Department, September 30, 2011.
- **Information Exchange with Steve Bowen, NASA Astronaut**, KSU ECE Department, September 16, 2011.
- **Information Exchange with Colorado Engineering**, KSU ECE Department, May 23, 2011.
- **Information Exchange with William Graveman, Biomedical Devices of Kansas**, KSU ECE Department, March 29, 2011.
- **Information Exchange with Hill's Pet Nutrition**, KSU ECE Department, November 29, 2010.
- **Information Exchange with Robert Bestgen**, KSU ECE Department, November 19, 2010.
- **Information Exchange with Black & Veatch**, KSU ECE Department, September 29, 2010.
- **Information Exchange with Honeywell**, Honeywell Federal Manufacturing and Technologies, Kansas City Plant, 2000 East 95th St., Kansas City, MO 64131-6159, August 19, 2010.
- **Funding Opportunities with Cerner Corporation**, Cerner Innovation Campus, Kansas City, MO, November 20, 2009.

- **Information Exchange with Heartspring Corporation**, KSU ECE Department, September 24, 2009.
- **Medical Component Design Laboratory Overview**, Heartspring, KSU ECE Department, September 24, 2009.
- **Medical Component Design Laboratory Overview**, Kirk Shulz Family Tour, KSU Engineering Laboratories, September 23, 2009.
- **Advisory Council Exchange with Heartspring**, Heartspring Campus, Wichita, KS, April 6, 2009.
- **Information Exchange with Rick McMullen**, Kansas University Director of Research Computing, Medical Component Design Laboratory, January 21, 2009.
- **MDBus Dialogue with Cerner Corporation**, 236 Nichols Hall, Kansas State University, December 17-18, 2008.
- **KSU Engineering Research Focus Meeting**, Edwards Conference Room, Fiedler Hall, Kansas State University, Manhattan, KS, September 2, 2008. Research investment areas for the KSU College of Engineering.
- **Opportunity Discussion with Digital Rays** (Abhi Ray, CEO), KSU Health IT Working Group, KSU Lafene Health Center, and Meadowlark Hills, Manhattan, KS, August 5, 2008. Opportunities related to Manhattan as a home base for the Digital Rays company.
- **Project Discussion with Meadowlark Hills Retirement Community** (Steve Shields, CEO) **and Elite Care**, Manhattan, KS, July 29, 2008. Discussion regarding the requirements for the implementation of a home care and medical information system at Meadowlark Hills.
- **Information Exchange with Colonel Wayne Parks**, Great Plains Cyber Electronic University Consortium, Kansas State University, Manhattan, KS, July 9, 2008.
- **Information Exchange with Meadowlark Hills Retirement Community** (Steve Shields, CEO), Manhattan, KS, May 19, 2008.
- **Information Exchange with the Kansas Foundation for Medical Care** (KFMC, Larry Pitmann, President and CEO) **and Lead Horse Technologies**, (Junction City, KS), Kansas State University, May 8, 2008.
- **Information Exchange with Cerner Corporation and Meadowlark Hills**, Kansas State University, Health IT Working Group, May 5, 2008.
- **Information Exchange with Sceptor Industries**, Kansas State University, Biosecurity Research Institute, April 30, 2008.
- **Information Exchange with Raytheon Corporation**, Kansas State University, Biosecurity Research Institute, March 26, 2008.
- Meeting with **HeartSpring** in Wichita, KS, March 14, 2008. Discussions regarding sensor fusion and the assessment of health and behavior in severely disabled children.
- **Architecture Dialogue with Cerner Corporation**, 236 Nichols Hall, Kansas State University, January 22-24, 2008.
- **Information Exchange for Possible Health Care Joint Ventures**, Rebecca S. MacKinnon and Michie P. Slaughter, 102 Fairchild Hall, Kansas State University, January 22, 2008.
- **Information Exchange with Cerner Corporation**, Kansas City, MO, December 17, 2007. Potential KSU-Cerner collaborations.
- **Health Care Advisory Committee Meeting**, Department of Industrial and Manufacturing Systems Engineering, Edwards Conference Room, Kansas State University, October 25, 2007.
- Host for Robert Wald, Professional Performance Award, **Seaton Society Banquet**, October 21–22, 2006.
- Meeting with **HeartSpring** in Wichita, KS, June 13, 2006. Discussions regarding how technology can help children that have severe mental and physical disabilities.
- **Meeting with the Provost and Associate Provost**, February 28, 2006. Lunch meeting to discuss research opportunities and barriers.
- **Leadership Appreciation Reception**, Blue Key Honor Society, Beach Museum of Art, January 31, 2006.

- **NSF IGERT Workshop**, Wichita State University, November 14, 2005.
- Trip to **Midwest Research Institute**, November 5, 2005. Day-long meeting to discuss biosecurity work between MRI and KSU.
- **Pre-ICADI Workshop on Technology for Aging, Disability, and Independence**, The Royal Academy of Engineering, 29 Great Peter Street, Westminster, London SW1P 3LW (England), June 26-27, 2003, <http://www.cise.ufl.edu/~helal/preICADI>.
- **Information Meeting. Research to Aid Persons with Disabilities**, National Science Foundation, Arlington, VA, June 28-29, 2001.
- **FDA Workshop on Home Care Technology Trends**, U.S. Food and Drug Administration, Rockville, MD, January 25, 2001.
- Warren, Steve. "The Health Care Facelift: How Telecommunications Technology, Distributed Computing, and Intelligent Medical Devices will Redefine the Patient-Physician Relationship," **Four-hour short course on smart health care technology** prepared for the Department of Biomedical Engineering, University of Minnesota, Twin Cities Campus, December 1, 1999.
- **Workshops on Future Medical Devices: Home Care Technologies for the 21st Century**. Sponsored by the National Science Foundation, the Catholic University of America, and the Food and Drug Administration. Mini-Workshop: Center for Devices and Radiological Health (CDRH), Rockville, MD, December 6-7, 1998. Full Workshop: Department of Health and Human Services (DHHS), Rockville, MD, April 7-9, 1999. **Committee Chair: Smart Devices and the Home of the Future**. Smart devices and advanced technologies (civilian and government) that the National Science Foundation should fund through 2010.
- **Texas A&M Program Development Meetings** with Upper Management at Sandia National Laboratories, December 11, 1998. Discussions with Jim Wall, Ph.D., Director of Computing and Information Technology, TEES Texas Center for Applied Technology. Information architecture and surety needs for telemedicine systems used for home health and emergency response.
- **High-Performance Healthcare**. HOLON ATP Program Review sponsored by the Koop Foundation, University of Maryland Shady Grove Conference Center, Rockville, MD, June 17-18, 1998. Internet and security technology dedicated to remote health care delivery.
- **Computational Science Initiative Workshop**, Department of Energy, Office of Energy Research, Lawrence Berkeley National Laboratory Office in Washington, D.C., Jan. 28-29, 1998. Fiscal year 2000 initiative for advanced computation in the Department of Energy.
- **Army Natick Research, Development & Engineering Center Program Summary**, Sandia National Laboratories, Albuquerque, NM, February 26, 1998. Battlefield monitoring of soldier state-of-health and chemical-biological preparedness.
- Dr. John Rather, Director of Innovative Development, **Karmanos Cancer Institute**, Meeting with Upper Management of Sandia National Laboratories, August 28-29, 1997. The need for the national laboratories to apply substantial computational resources to high-resolution body imaging and subsequent cancer prevention.
- **Workshop on Lower Limb Prosthetics**, Co-sponsored by the National Center for Medical Rehabilitation Research (NCMRR), and National Institute of Child Health & Human Development, the National Institute of Health (NIH), and Sandia National Laboratories, Sandia National Laboratories, Albuquerque, NM, November 8-10, 1995. Technologies for assisted living.

INVITED SPEAKERS

- Leet, Corey. "NAMSA," NAMSA, Minneapolis, MN, Kansas State University, KSU IEEE EMBS Student Chapter, April 10, 2025.
- Eastling, Marissa. "Integrating Language Acquisition Motor Planning (LAMP) into Literacy Education," Special Education Teacher, Farmington School District, Farmington, MN, *BME 200 – Introduction to Biomedical Engineering*, Kansas State University, March 31, 2025.

- Armstrong, Lauren and Mason Diebold. “Vantage MedTech,” Vantage MedTech, Lenexa, KS, Kansas State University, KSU IEEE EMBS Student Chapter, March 27, 2025.
- Cerutti McGinnis, Allison Kathleen. “Orthotics and Prosthetics Careers,” Northwestern University Prosthetics-Orthotics Center, Chicago, IL, Kansas State University, KSU IEEE EMBS Student Chapter, March 13, 2025.
- Griffin, Tanner. “Engenious,” Engenious Design, Prairie Village, Kansas City, KS, Kansas State University, KSU IEEE EMBS Student Chapter, February 27, 2025.
- Bledsoe, Heather, Melissa Frock, and Adam Frock. “Spinal Simplicity,” Spinal Simplicity, Overland Park, KS, Kansas State University, KSU IEEE EMBS Student Chapter, January 30, 2025.
- Dear, Adam. “Biosense Webster Opportunities,” Biosense Webster, Kansas City, MO, Kansas State University, BME 575 – Clinical Systems Engineering, December 5, 2024.
- Colling, Tom. “DEKA Projects and Life After College,” DEKA Research and Development Corporation, Kansas State University, BME 575 – Clinical Systems Engineering, November 22, 2024.
- Fischer, Ken. “Learn About Graduate School and KU Bioengineering,” The University of Kansas, Kansas State University, BME 575 – Clinical Systems Engineering, October 14, 2024.
- Prusener, Derek, Erin Piacetelli, and Camden Johnson, “Meet Medtronic,” Medtronic, Inc., Kansas State University, KSU IEEE EMBS Student Chapter, September 12, 2024.
- Kodanaz, Tyler and Megan Kinnane, “Opportunities with Engenious Design,” Engenious Design, Kansas State University, KSU IEEE EMBS Student Chapter, April 4, 2024.
- Bradley, Janae, “Engineering Efforts at Hill’s Pet Nutrition,” Hill’s Pet Nutrition, Kansas State University, KSU IEEE EMBS Student Chapter, March 21, 2024.
- Leet, Cory, “Medical Device Testing at NAMSA,” NAMSA, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Presentation, February 22, 2024.
- Frizzell, Bradie, “Xenon MRI,” Kansas University Medical Center, Kansas State University, KSU IEEE EMBS Student Chapter, February 8, 2024.
- Griffith, Craig, “IEEE: Who We Are and How We Can Help,” IEEE, Tucson, AZ, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, November 30, 2023.
- Warren, Kelsey, “Clinical Engineering at Children’s Mercy Hospital,” Children’s Mercy Hospital, Kansas City, MO, Kansas State University, KSU IEEE EMBS Student Chapter, November 9, 2023.
- Shane, Douglas, “Technology for Animal Health,” Kansas State University, Olathe, KS, Kansas State University, KSU IEEE EMBS Student Chapter, October 19, 2023.
- Dodge, Matt, “Electrophysiology” Biosense Webster, Kansas State University, KSU IEEE EMBS Student Chapter, October 5, 2023
- Jansen, Michael, “Remote Patient Monitoring,” Biointellisense, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, September 21, 2023
- Medtronic, “The Role of an Engineering in a Medical Device Company,” Kansas City, KS, Kansas State University, KSU IEEE EMBS Student Chapter, May 4, 2023.
- Reagan, Corbin and Jeff Chininis, “Complex Medical Devices,” RBC Medical Innovations, Olathe, KS, Kansas State University, KSU IEEE EMBS Student Chapter, April 6, 2023.
- Friedrichs, Dan, “Medical Device Industry Operations, Minnetronix, St. Paul, Minnesota, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, February 23, 2023.
- Synexsis, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, January 26, 2023.
- Gould, Grant and Taylor Rogalia, “Biosense Webster,” Biosense Webster, Kansas City, KS, Topeka, KS, Kansas State University, KSU IEEE EMBS Student Chapter, November 10, 2022.
- Griffith, Craig, “IEEE: Who We Are and How We Can Help,” IEEE, Tucson, AZ, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, October 13, 2022.

- Golway, Michael, “3D Tissue Printing Systems,” President and CEO, Advanced Solutions, Louisville, KY, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, December 2, 2021.
- Baird, Keith, “You’ve Got a Product Idea. So What ...,” President, Biomedical Devices of Kansas, Tonganoxie, KS, Kansas State University, KSU IEEE EMBS Student Chapter, November 18, 2021.
- Dear, Adam, “Products and Software for Cardiac Ablation Surgery,” Regional Technology Specialist, Biosense Webster, Kansas City, KS, Kansas State University, KSU IEEE EMBS Student Chapter, October 7, 2021.
- Ybarra, Anthony, “KSU Career Center,” Kansas State University, BME 001 – New Student Assembly, Online, October 5, 2021.
- Chappell, Jake, “Infection Control Technology,” Mechanical Engineer, HEPACART, Lenexa, KS, Kansas State University, KSU IEEE EMBS Student Chapter, September 23, 2021.
- Weiss, Mark, “Machine Learning for Biologists,” Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, April 22, 2021.
- Glick, Stephen, “Emerging Technologies in X-Ray Imaging of Breast Cancer,” Office of Science and Engineering Laboratories, Center for Devices and Radiological Health, U.S. Food and Drug Administration, Silver Spring, MD, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, April 8, 2021.
- Lee, Taehee, “Early Detection of Psychiatric Disorders,” iMediSync, Seoul, South Korea, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, March 11, 2021.
- Crosby, Petra, Katie Ostmeyer, and Gilbert Magallanes, “NSUTLC Services and Technology Needs,” No Stone Unturned Therapeutic Learning Center, Manhattan, KS, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, March 11, 2021.
- Griffith, Craig, “IEEE Resources, Activities, and Student Membership,” IEEE, San Francisco, CA, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, February 11, 2021.
- Ybarra, Anthony, Kansas State University, “KSU Career Center,” BME 001 – New Student Assembly, Online, December 6, 2020.
- Hatesohl, Brian, “Biotechnology and the Drive to Create,” Engenious Design, Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, November 19, 2020.
- Steele, Brian and Mark Meyer, University of Kansas School of Medicine, “Engineering in Medicine and Biology Society,” Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, November 6, 2020.
- Keogh, Kelly, No Stone Unturned Foundation, “Welcome KSU EMBS Students,” Kansas State University, KSU IEEE EMBS Student Chapter, Zoom Seminar/Discussion, October 29, 2020.
- Lee, Taehee. “Pen-Type Electrodermal Activity Sensor,” Experimental Session, ECE 571 – Introduction to Biomedical Engineering, January 30, 2020.
- Hopper, Ben. “Building a Generation of Biomedical Engineers,” *K-Stater*, Winter 2019, pp. 26-29.
- Thompson, David, Kansas State University, “Neural Interfacing,” Kansas State University, BME 200 – Introduction to Biomedical Engineering, Fiedler Auditorium Room 1107, November 1, 2019.
- Carlson, Charles, Kansas State University, “Sensor Beds for Children with Severe Disabilities,” Kansas State University, BME 200 – Introduction to Biomedical Engineering, Fiedler Auditorium Room 1107, November 1, 2019.
- DeLuca, Anne, Kansas State University, “KSU Career Center,” BME 001 – New Student Assembly, Fiedler Auditorium Room 1107, September 3, 2019.
- Fattahi, Niloufar, Kansas State University, “Screening Diffusive, Antagonistic Bacterial Interactions Using Photoreleasable Hydrogels,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 0093, April 25, 2019.

- Aceros, Juan, University of North Florida, “Adaptive Toys, Neurodevelopment, and Play,” Kansas State University, Electrical & Computer Engineering Distinguished Lecture Series, KSU IEEE EMBS Student Chapter, Fiedler Hall Room 1109, April 17, 2019.
- Aceros, Juan, University of North Florida, “Neural System Development and Play,” Kansas State University, *ECE 571 – Introduction to Biomedical Engineering*, Rathbone Hall Room 1052, April 17, 2019.
- Martinez, Andres, Vanderbilt University, “Control of Powered Lower-Limb Prostheses and Exoskeletons,” Kansas State University, *ECE 571 – Introduction to Biomedical Engineering*, Rathbone Hall Room 1052, March 29, 2019.
- Martinez, Andres, Vanderbilt University, “Center for Rehabilitation Engineering and Assistive Technology,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 0093, March 28, 2019.
- Dear, Adam, Biosense Webster, Kansas State University, *BME 200 – Introduction to Biomedical Engineering*, Engineering Hall Room 3099, October 26, 2018.
- Lyle, John. “Experiences with an Implanted Defibrillator,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 0093, October 25, 2018.
- Citron, Paul, National Academy of Engineering, “Medical Devices: Saving and Improving Lives,” Kansas State University, *BME 200 – Introduction to Biomedical Engineering*, Engineering Hall Room 0093, September 28, 2018.
- Griffith, Craig, IEEE Client Services Manager. “Introduction to IEEE,” Kansas State University, *BME 200 – Introduction to Biomedical Engineering*, KSU IEEE EMBS Student Chapter, Engineering Hall Room 0093, September 26-27, 2018.
- Radke, Donald, Heartland O&P, Inc., “Life with a Prosthetic,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 0093, September 12, 2018.
- Snyder, Larry, KSU Veterinary Medicine, “Stem Cell Therapy in Veterinary Medicine,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 3099, March 28, 2018.
- Monroe, Tyrone, Hangar Prosthetics and Orthotics, “Prosthetic Design and Manufacturing,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 3099, March 15, 2018.
- Dear, Adam, Biosense Webster, Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 3099, April 24, 2018.
- Ade, Carl, KSU Kinesiology, “How Human Spaceflight Has, and Will, Advance Medicine,” Kansas State University, KSU IEEE EMBS Student Chapter, Engineering Hall Room 3099, February 8, 2018.
- Zollman, Dean. “KSU Center for Research and Innovation in STEM Education (CeRISE),” Seminar sponsored by the KSU College of Engineering and the American Society for Engineering Education, Engineering Hall Room 0093 (Phase IV), April 26, 2016.
- Yang, Lydia. “Fundamentals of Education-Related Research,” Seminar sponsored by the KSU College of Engineering and the American Society for Engineering Education, Engineering Hall Room 0093 (Phase IV), March 29, 2016.
- Schwartz, Carole. “Ageism and Society,” Kansas State University, Introduction to Biomedical Engineering Class, Rathbone Hall, April 17, 2015. 2015 K-State Global Campus Alumni Fellow, Department of Gerontology.
- Grande-Allen, Jane. “Experimental Framework for Analysis of Heart Valve Mechanobiology,” Kansas State University, College of Engineering, February 11, 2015. Sponsored by the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society, the KSU Student Chapter of the IEEE, and the KSU Department of Electrical & Computer Engineering.
- Grande-Allen, Jane. “Heart Valve Mechanics,” Kansas State University, *ECE 571 – Introduction to Biomedical Engineering*, February 11, 2015.
- Cox, Jessica. “Think Outside the Shoe,” Kansas State University, College of Engineering, November 6, 2013. Sponsored by the KSU Student Chapter of the IEEE Engineering in

Medicine & Biology Society, the KSU Student Chapter of the IEEE, and the KSU Department of Electrical & Computer Engineering.

- Gallagher, Richard. “The History of Biomedical Engineering,” Kansas State University, College of Engineering, October 9, 2013. Sponsored by the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Gay, Nicholas and Sam Bruner. Garmin Day speakers for *ECE 512 – Linear Systems*, Sept. 3, 2013.
- Prakash, Punit. “Technologies for Non- and Minimally-Invasive Image-Guided Surgery,” Kansas State University, College of Engineering, April 26, 2013. Sponsored by the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Singleton, Gary and Riley Dutton. “You Can Make a Difference: The Future of Engineering and Children with Special Needs,” Heartspring, Wichita, Kansas, Kansas State University, Paslay Lecture Hall, October 9, 2012. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Fowler, Kim. “Bringing a Medical Device to Market,” Kansas State University, November 15, 2011. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Mayer, Carl and Chris Justice. “Medical Device Development: Engineering for Life,” RBC Medical Innovations, Kansas State University, October 27, 2011. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Thompson, David. “Neural Interfacing,” Kansas State University, October 14, 2011. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Nelson, Tommy and Phillip Schumm. “EMBC 2011 and Epidemic Modeling,” Kansas State University, September 26, 2011. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Gustafson, Tim. “Prosthetics Technology,” Kansas State University, September 9, 2011. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Larson, Brian. “Software Verification and Medical Device Regulation,” Kansas State University, February 11, 2011, *ECE 571 – Introduction to Biomedical Engineering*.
- Chikan, Viktor. “Magnetic Hyperthermia: Integrating Nanotechnology with Biology for Cancer Therapy,” Kansas State University, November 19, 2010. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Coetzee, Hans. “Use of Electronic Tools to Measure Pain in Cattle,” Kansas State University, September 16, 2010. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Weiss, Mark L. “Engineering Cells, Tissues, and Critters,” Kansas State University, February 15, 2010. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Singleton, Gary and Chris Vacek. “Technology to Improve the Quality of Life for Disabled Children,” Heartspring, Wichita, Kansas, ECE Graduate Seminar, Kansas State University, September 24, 2009. Sponsored by the KSU Department of Electrical & Computer Engineering and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.
- Derakhshani, Reza. “Biometrics: An Introduction to Current Technologies and Future Trends,” EECE Graduate Seminar, Kansas State University, October 25, 2007. Sponsored by the KSU Department of Electrical & Computer Engineering, the Midwest Section of the IEEE Engineering in Medicine & Biology Society, and the KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society.

- Perkins, Leigh and David Lingerfelt. “Opportunities at Cerner Corporation,” KSU Student Chapter of the IEEE Engineering in Medicine & Biology Society, Kansas State University, October 8, 2007.

GRANTS

Total award amount since 1999: **\$13,995,910**; ECE allocation: **\$6,322,523**

- (Submitted) Kump, Paul (Marty), Charles Carlson, Steve Warren, and Amy Betz. “Research Initiation: Integrating Ethical Design Challenges Across Electrical, Computer, and Biomedical Engineering to Foster Professional Identity,” National Science Foundation, NSF 20-558: PFE: Research Initiation in Engineering Formation (PFE: RIEF), 06/01/2026–05/31/2028, \$199,919.
- (Declined) Zhu, Ninghao, Charles Carlson, and Steve Warren. “Biomedical Engineering Research Hub (BME Hub),” K-State Applied Learning Experiences (ALE) Incentive Grants, K-State Program Curriculum Exploration, Kansas State University, 08/01/2025–07/31/2027, \$20,000.
- (Declined) Carlson, Charles, Steve Warren, and Briana Nelson Goff. “Meeting the Needs of Children with Disabilities Through Community Engaged Applied Human Sciences and Biomedical Student Design Projects,” K-State Applied Learning Experiences (ALE) Incentive Grants, K-State Transdisciplinary Applied Learning, Kansas State University, 06/01/2026–05/31/2027, \$39,548.
- (Declined) Goff, Briana Nelson, Charles Carlson, and Steve Warren. “Meeting the Needs of Children with Disabilities Through Community Engaged Applied Human Sciences and Biomedical Student Design Projects,” Harold and Letha Reser Family Community Innovation Award, Kansas State University, College of Health and Human Sciences, 06/01/2025–05/31/2026, \$29,843.
- (Submitted) Carlson, Charles, Steve Warren, and Briana Nelson Goff. “Biomedical Design Projects and Community Engagement Experiences Informed by the Needs of Children with Disabilities,” National Science Foundation, NSF 23-510: Improving Undergraduate STEM Education: Directorate for STEM Education (IUSE: EDU), Engaged Student Learning (Level I), 08/01/2025–07/31/2027, \$399,738.
- (Submitted) Thompson, David, Carl Ade, Heather Bailey, Charles Carlson, and Steve Warren. “Bed-Based Sensors for Predictive Care Management of Daytime Behavior in Alzheimer's Disease Populations,” National Institutes of Health, R01 Research Project Grant, PAR-22-093: Research on Current Topics in Alzheimer's Disease and Its Related Dementias (R01 Clinical Trial Optional), 07/01/2025–06/30/2030, \$3,619,621.
- (Awarded) Robert and Becca Reichenberger Cornerstone Teaching Scholar, Kansas State University College of Engineering, May 1, 2024 – April 30, 2027, \$28,500.
- (Declined) Carlson, Charles, Steve Warren, Briana Goff, Punit Prakash, David Thompson, and Don Gruenbacher. “Biomedical Design Projects and Community Engagement Experiences Informed by the Needs of Children with Disabilities,” National Institutes of Health, R25 Education Project, PAR-19-215: Team-Based Design in Biomedical Engineering Education, 5/15/2022–5/14/2027, \$204,000.
- (Awarded) Rice, Dallas and Steve Warren. “Sanitation Drone for Large Indoor Venues,” OURCI Research Award, Kansas State University, Office of Undergraduate Research & Creative Inquiry, 5/3/2021–12/03/2021, \$500.
- (Declined) Carlson, Charles, Steve Warren, Punit Prakash, David Thompson, and Don Gruenbacher. “Biomedical Design Projects and Community Engagement Experiences Informed by the Needs of Children with Disabilities,” National Institutes of Health, R25 Education Project, PAR-19-215: Team-Based Design in Biomedical Engineering Education, 5/15/2021–5/14/2026, \$204,000.
- (Awarded) “Biomedical Engineering Teaching Laboratory,” Tong Foundation, 5/01/2020 – 4/30/2023, \$750,000, Kansas State University, Carl R. Ice College of Engineering, \$750,000 match. Total award: \$1,500,000.

- (Awarded) Doll, Gayle, Migette Kaup, and Laci Cornelison. “PEAK 2.0 Evaluation,” Kansas Department of Health and Environment, 9/01/2019 - 8/31/2021, \$560,292. Role: key participant and advisory council member.
- (Awarded) Kim, Jungkwun, Don Gruenbacher, Steve Warren, Charles Carlson, Dwight D. Day, and Yingying Wu. “Advanced Expert Training Program for Industrial Convergence of Wearable Smart Devices,” Korea Institute for Advanced Technology (KIAT), Korea Display Industry Association (KDIA), Project no. P0002397, 1/4/2020–1/11/2020, \$13,163.
- (Awarded) Robert and Becca Reichenberger Cornerstone Teaching Scholar, Kansas State University College of Engineering, September 1, 2019 – June 30, 2024, \$47,500.
- (Declined) Zollman, Dean, Lydia Yang, Lisa Wilken, and Amy Betz. (Warren role: senior personnel) “IGE: Providing Graduate Students with Interdisciplinary Skills to Succeed in Discipline-Based Education Research,” National Science Foundation, Innovations in Graduate Education (IGE) Program, 6/1/2019–5/31/2022, \$499,423.
- (Submitted) Zacharakis, Jeffrey, Charles Rice, and Steve Warren. “IGE Leadership Development for STEM Graduate Students,” National Science Foundation, Innovations in Graduate Education (IGE) Program, 7/1/2019–6/30/2022, \$493,446.
- (Declined) Diaz, Eduardo, Grant Anderson, Steve Warren, and Michael Sawyer. “Incorporating ‘Individual’ Functional Characteristics with Bio-Sensors to Support Optimal Human Health and Performance,” NASA TRISH BRASH1801 Program, 1/1/2019–12/31/2020, \$390,232, (KSU: \$199,274).
- (Awarded) Carlson, Charles, Ahmad Suliman, Alaleh Alivar, Punit Prakash, David Thompson, Balasubramaniam Natarajan, and Steve Warren. “A Pilot Study of an Unobtrusive Bed-Based Sleep Quality Monitor for Severely Disabled Autistic Children,” *40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, Hawaii Convention Center, Honolulu, HI, USA, July 17-21, 2018, KSU Graduate Student Council, July 2018, \$450.
- (Awarded) Suliman, Ahmad, Charles Carlson, Punit Prakash, Steve Warren, and David Thompson. “Performance Evaluation of Processing Methods for Ballistocardiogram Peak Detection,” *40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, Hawaii Convention Center, Honolulu, HI, USA, July 17-21, 2018, KSU Graduate Student Council, July 2018, \$400.
- (Declined) Zollman, Dean, Steve Warren, and Lydia Yang. “NRT: Crosscutting Research in STEM Education (CRiSE): Training Graduate Students to Succeed in Discipline-Based Education Research,” National Science Foundation, NSF Research Traineeship (NRT) Program, 8/1/2018–7/31/2021, \$2,999,185.
- (Awarded) Betz, Amy and Emily Dringenberg. “Research Initiation: The Formation of Engineering Students' Beliefs about Intelligence,” National Science Foundation, NSF EEC 1738209, Division of Engineering Education and Centers, 9/1/2017–8/31/2019, \$199,995. Role: evaluator.
- (Withdrawn) Zollman, Dean, Michael Herman, Steve Warren, and Lydia Yang. “IGE: Providing Graduate Students with Interdisciplinary Skills to Succeed in Discipline-Based Education Research,” National Science Foundation, Innovations in Graduate Education (IGE) Program, 6/1/2018–5/31/2021, \$498,579.
- (Declined) Zacharakis, Jeffrey, Charles Rice, Andrew Wefald, and Steve Warren. “IGE Leadership Development for STEM Graduate Students,” National Science Foundation, Innovations in Graduate Education (IGE) Program, 7/1/2018–6/30/2021, \$499,637.
- (Submitted) Warren, Steve, Punit Prakash, David Thompson, Bala Natarajan, and Don Gruenbacher. “Netsmart/KSU Sensor Bed Effort,” Netsmart Corporation, 1/15/2018–1/14/2019, \$50,000.
- (Declined) Rozell, Timothy, Ashley Rhodes, and Steve Warren. “USRG: Development of Wearable Sensors to Evaluate Heat Stress in Dairy Cattle,” Kansas State University, Office of the Vice President for Research, University Small Research Grant program, 7/1/2017–6/30/2018, \$4,500.

- (Declined) Zollman, Dean, Lydia Yang, Steve Warren, and Michael Herman. “NRT-IGE: Providing Graduate Students with Interdisciplinary Skills to Succeed in Discipline Based Education Research,” National Science Foundation, National Science Foundation Research Traineeship (NRT) Program, 6/1/2017–5/31/2020, \$486,608.
- (Declined) Zacharakis, Jeffrey, Charles Rice, Andrew Wefald, and Steve Warren. “NRT-IGE Leadership Development for STEM Graduate Students,” National Science Foundation, National Science Foundation Research Traineeship (NRT) Program, 7/1/2017–6/30/2020, \$498,073.
- (Awarded) Grother, Ethan, Mei He, and Steve Warren. “Cautious Use of Mobile Device Cameras for Color Scale Assays,” *IEEE-NIH Special Topics Conference on Healthcare Innovations and Point-of-Care Technologies (HI-POCT ‘16)*, Cancun, Mexico, November 9-11, 2016. Travel award from the KSU Johnson Cancer Research Center, \$870.
- (Awarded) Carlson, Charles, Ahmad Suliman, Punit Prakash, David Thompson, Shangxian Wang, Bala Natarajan, and Steve Warren. “Bed-Based Instrumentation for Unobtrusive Sleep Quality Assessment in Severely Disabled Autistic Children,” *38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, Disney’s Contemporary Resort, Orlando, FL, USA, August 16-20, 2016, KSU Graduate Student Council, 4/28/2016, \$500.
- (Awarded) He, Mei, Steve Warren, and Matthew Campbell. “Self-Management of Ovarian Cancer Using Smartphone-Based, 3D Chip Diagnostic Strategy,” Johnson Cancer Research Center, Kansas State University, 12/1/2015–11/30/2016, \$30,000.
- (Declined) He, Mei, Marselina Tan, Steve Warren, Alisjahbana Bacht, Mukta Sharma, and Susan Brown. “Low-Cost iPOC^{3D} Diagnostic Technology for Promoting Wireless Healthcare of Anemia in Indonesia,” National Institutes of Health, NIH PAR-14-028, Mobile Health: Technology and Outcomes in Low and Middle Income Countries (R21), 10/15/2015–10/14/2017, \$375,000.
- (Declined) Zollman, Dean, et al. “Center for Research and Innovation in STEM Education: CeRISE @ KSU,” NSF Science and Technology Centers. Declined by KSU prior to submission to the NSF.
- (Awarded) Thompson, David (PI) and Steve Warren (Mentor). 2015-2016 K-State Mentoring Fellowship, Alfred E. Sloan Foundation, \$6,000.
- (Awarded) Warren, Steve, Punit Prakash, Bala Natarajan, David Thompson, and Wayne Piersel. “GARDE: Research to Quantify the Health and Development of Children with Disabilities Around the Clock,” National Science Foundation, General & Age-Related Disabilities Engineering (GARDE) Program, UNS 1512564, 8/15/2015–8/14/2018, \$399,962.
- (Declined) Warren, Steve, Jianchu Yao, Sunghan Kim, and William Hageman. “EXP: Collaborative Research: A Cyber-Physical System for Electrical Engineering Education that Generates and Grades Personalized Hands-On Assignments,” National Science Foundation, Cyberlearning and Future Learning Technologies (Cyberlearning) Program, 9/15/2014–9/14/2017, \$324,834 (KSU) + \$225,142 (ECU) = \$549,976.
- (Declined) Warren, Steve, Punit Prakash, Bala Natarajan, Minyoung Suh, and Wayne Piersel. “GARDE: Research to Quantify the Health and Development of Children with Disabilities Around the Clock,” National Science Foundation, General & Age-Related Disabilities Engineering (GARDE) Program, 6/15/2014–6/14/2017, \$299,999.
- (Declined) Warren, Steve, Jianchu Yao, and Sunghan Kim. “Collaborative Research: Cyber-Physical Tools for the MOOC Age: Personalized Hands-On Electrical Engineering Learning Experiences for Distributed Education,” National Science Foundation, EHR Core Research (ECR) Program, 1/1/2014–12/31/2016, \$149,936 (KSU) + \$150,063 (ECU) = \$299,999.
- (Awarded) Thomson, Daniel, Lindsey Hulbert, Michael Dikeman, Christopher Reinhardt, K. Olson, John Gonzales, Steve Warren, David Renter, Steve Bartle, and Daniel Frese. “Animal Health and Animal Welfare Implications of Beta Adrenergic Agonists (BAA) in Feedlot Cattle,” National Cattlemen's Beef Association (840738973), 5/1/2013–8/31/2014, \$190,455.
- (Awarded) Hatcliff, John, Patrice Chalin, Robby, and Steve Warren. “FDA SIR: Risk Assessment Techniques for Apps and Devices in Interoperable Medical Device Frameworks,”

National Science Foundation, U.S. Food and Drug Administration, Scholars In Residence Program, 10/1/2013–9/30/2015, \$104,348.

- (Declined by KSU) Rebello, N. Sanjay, Andrew G. Bennett, Eleanor Sayre, Steve Warren and Dean Zollman. “IGERT: Integrated Experiences Preparing Future Faculty in Interdisciplinary STEM Education Research,” National Science Foundation, IGERT Program. Two-page summary.
- (Awarded) Hatcliff, John M., Patrice Chalin, Robby, Eugene Y. Vasserman, Steve Warren, Insup Lee, and Oleg Sokolsky. “CPS: Synergy: Collaborative Research: Trustworthy Composition of Dynamic App-Centric Architectures for Medical Application Platforms,” National Science Foundation, Cyber-Physical Systems Program, CNS–1239543, 9/1/2012 – 8/31/2016, \$800,000.
- (Declined) Warren, Steve, Donald Gruenbacher, Balasubramaniam Natarajan, and John Hatcliff. “SHB: Type I (EXP): Hierarchical Coordination and Computation Frameworks for Reconfigurable Everyday-Carry Medical Devices and Body Area Networks,” National Science Foundation, Smart Health and Wellbeing Program, IIS–1231767, 8/15/2012 – 8/14/2015, \$600,000.
- (Accepted) Ben-Arieh, David, John Wu, Caterina Scoglio, and Steve Warren. “Operations Systems Engineering Support for the Department of Veterans Affairs MidWest Mountain Veterans Engineering Resource Center”, Veterans Administration, Department of Veterans Affairs, VA Nebraska – Western Iowa Health Care System, 8/1/2011 – 7/31/2014. Inclusion on the bidding process for upcoming short-duration healthcare system tasks. Overall project amount: \$500,000.
- (Declined) Li, Kejia and Steve Warren. “Everyday Carry Wireless Health Monitor with Customizable Surface Components,” Center for Integration of Medicine and Innovative Technology (CIMIT), Student Prize for Technology in Primary Healthcare, submitted 5/30/2011, \$150,000.
- (Awarded) Singh, Gurdip, Steve Warren, Balasubramaniam Natarajan, John Hatcliff, and Virg Wallentine. “Falls Predictor Device”, Cerner Corporation, 5/1/2011 – 8/31/2012, \$93,992, \$139,108 with KSU match.
- (Awarded) Kuhn, William, Steve Warren, Balasubramaniam Natarajan, Don Gruenbacher, Dwight Day, Thomas Barstow, and Timothy Sobering. “Biosensor Networks and Telecommunication Subsystems for Long-Duration Missions, EVA Suits, and Robotic Precursor Scout Missions: Coordinated Research Spanning Early-Stage Innovations through Game Changing Technologies,” NASA EPSCoR Program, 10/1/2011 – 6/30/2015, \$1,224,635.
- (Awarded) Li, Kejia and Steve Warren. “Everyday Carry Wireless Health Monitor with Customizable Surface Components,” Center for Integration of Medicine and Innovative Technology (CIMIT), Student Prize for Technology in Primary Healthcare, submitted 1/17/2011, 6/2011–5/2012, \$10,000.
- (Pending) Teixeira, Rodrigo, XXXX. “Enhanced Pulse Oximetry for Neonates,” National Institutes of Health, Safe and Effective Instruments and Devices for Use in the Neonatal Intensive Care Units (SBIR [R43]), RFA-HD-10-012, 6/1/2010 – 5/31/2012, \$450,000. (Steve Warren – subcontractor)
- (Awarded) Hatcliff, John, Gurdip Singh, Robby, Virgil Wallentine, and Steve Warren. NSF CNS–1065887. “An Integrated Development and Certification Environment for a Medical Device Coordination Framework,” National Science Foundation, U.S. Food and Drug Administration, Scholars In Residence Program, 1/1/2011–12/31/2012, \$80,000.
- (Awarded) Warren, Steve, John Devore, and Balasubramaniam Natarajan. “Communication and Controls Development,” Sandia National Laboratories, 10/28/2008–2/28/2011, \$122,724.
- (Awarded) Hatcliff, John, Daniel Andresen, and Steve Warren. “REU: CPS: Medium: Collaborative Research: Infrastructure and Technology Innovations for Medical Device Coordination,” National Science Foundation, CNS Cyber-Physical Systems (CPS) Program, 9/15/2009–8/31/2012, \$12,000.
- (Declined) Warren, Steve, Jianchu Yao, and Tarek Abdel-Salam. “TUES: Interactive Technology to Facilitate Collaborative Learning in Engineering Courses,” National Science

Foundation, Transforming Undergraduate Education in Science, Technology, Engineering and Mathematics (TUES) Program, 1/1/2011–12/31/2012, \$81,641 (KSU) + \$118,356 (ECU) = \$199,997.

- (Awarded) Warren, Steve, Punit Prakash, Balasubramaniam Natarajan, and Shelby Evans. “KSU Student Chapter of the IEEE EMBS as a Focal Point for Senior Design Projects to Aid Children with Disabilities,” National Science Foundation, Research to Aid Persons with Disabilities Program, CBET 1067740, 8/15/2011 – 7/31/2017, \$125,000.
- (Awarded) Teixeira, Rodrigo, Alton Reich, Jim Shaw, and Steve Malin. “Intelligent Data Extraction Algorithm for Pulse Oximetry,” National Science Foundation, SBIR Phase II, NSF08-548: Biotech and Chemical Technologies, 6/1/2010 – 5/31/2012, \$500,000. (Steve Warren – consultant)
- (Awarded) Hatcliff, John, Virg Wallentine, Robby, Daniel Andresen, Gurdip Singh, and Steve Warren. “Development of a Prototype Healthcare Intranet for Improved Health Outcomes,” Massachusetts General Hospital through the National Institutes of Health NIBIB Quantum Program, 10/1/2010–8/31/2015, \$475,000.
- (Awarded) Barstow, Thomas, Carl Ade, Steve Warren, and Chris Lewis. “Standardized “Pre-flight” Exercise Tests to Predict Performance During Extravehicular Activities in a Lunar Environment,” Step-2 proposal, Research and Technology Development to Support Crew Health and Performance in Space Exploration Missions, NASA Human Research Program, Exploration Systems Mission Directorate, Johnson Space Center, Houston, TX, 7/1/2010–5/30/2015, \$1,192,194.
- (Awarded) Warren, Steve and Jianchu Yao. NSF DUE–0942425. “CCLI: Portable Cyber-Laboratories: Virtual Instruments and Affordable Prototyping Kits to Enhance Learning and Improve Access to Electrical Engineering Education,” National Science Foundation, Course Curriculum and Laboratory Improvement (CCLI) Program, 1/1/2010–12/31/2011, \$111,899 (KSU) + \$88,100 (ECU) = \$199,999.
- Warren, Steve. Funding to support student travel to the *31st Annual Conference of the IEEE EMBS*, Minneapolis, MN, Sept. 2-6, 2009, KSU Electrical & Computer Engineering (\$1000), KSU College of Engineering (\$1550), \$2550.
- (Awarded) Hatcliff, John, Daniel Andresen, Robby, and Steve Warren. NSF CNS–0932289. “CPS: Medium: Collaborative Research: Infrastructure and Technology Innovations for Medical Device Coordination,” National Science Foundation, CNS Cyber-Physical Systems (CPS) Program, 9/1/2009–8/31/2012, \$851,548.
- Zollman, Dean, Sanjay Rebello, Andrew Bennett, and Steve Warren. NSF DRL–0816207. “Investigating Trajectories of Learning & Transfer of Problem Solving Expertise from Mathematics to Physics to Engineering,” National Science Foundation REESE Program, DRL–0816207, 9/1/2008–8/31/2011, \$999,841.
- (Declined) “Consortium for the Infusion of Technology into Rural Care Environments,” Kansas Congressional allocation for Kansas State University, 11/1/2007–10/31/2012, \$20,000,000.
- (Declined) Smith, Gregory, Virgil Wallentine, Beth Unger, and Steve Warren. “Center of Innovation for Pervasive Health Information Technology,” Kansas Bioscience Authority, 1/1/2008–6/30/2008, \$200,000.
- (Declined) Warren, Steve, Donald Gruenbacher, Caterina Scoglio, Bala Natarajan, Gayle Doll, Shing Chang, Margaret Rys, Scott DeLoach, John Hatcliff, and Daniel Andresen. “Pervasive Biomedical Technology Alliance,” Kansas State University, Targeted Excellence Program, 7/1/2008–6/30/2011, \$2,000,000.
- (Declined) Bennett, Andrew, David Allen, LaVerne Bitsie-Baldwin, Kimberly Douglas, Sanjay Rebello, and Steve Warren. “Center for Quantitative Education,” Kansas State University, Targeted Excellence Program, 7/1/2008–6/30/2011, \$2,000,000.
- (Declined) Scoglio, Caterina, and Todd Easton. “EPICENTER: The Epidemiological Modeling and Simulation Center at Kansas State University,” Kansas State University, Targeted Excellence Program, 7/1/2008–6/30/2011, \$1,000,000.

- (Declined) Polson, Cheryl, Craig Beardsley, Lisa Freeman, Charlie Griffin, Curtis Kastner, Justin Kastner, Denis Medeiros, Gurdip Singh, Marty Vanier, Virgil Wallentine, and Steve Warren. "Homeland Security Studies: Development of an Interdisciplinary Collaborative Master's Degree Program," Kansas State University, Targeted Excellence Program, 7/1/2008–6/30/2013, \$2,166,588.
- Warren, Steve. Funding to support student travel to the 30th *Annual Conference of the IEEE EMBS*, Vancouver, British Columbia, Canada, August 20-24, 2008, KSU Electrical & Computer Engineering (\$500), KSE Chemical Engineering (\$400), KSU Mechanical & Nuclear Engineering (\$500), KSU College of Engineering (\$2272), KSU Office of Student Activities (\$300), \$3972.
- Hatcliff, John, Daniel Andresen, Robby, and Steve Warren. NSF CNS-0734204. "Development of an Open Test-Bed for Application of Formal Methods to Medical Plug-and-Play Architectures," National Science Foundation, U.S. Food and Drug Administration, Scholars In Residence Program, 8/1/2007–7/31/2008, \$64,098.
- (Declined) Erickson, Howard, Dale Blasi, Joann Kouba, and Steve Warren. "Evaluation of Bio-Thermo Radio Frequency Microchips in Resting and Exercising Horses," U.S. Department of Agriculture, Equine Species Working Group, 10/1/2006–9/30/2008, \$250,000.
- (Declined) Warren, Steve (PI), Scott Deloach, Mark Haub, Balasubramaniam Natarajan, and Caterina Scoglio. "EFRI-ARESCI Preliminary Proposal: iNtelligent Occupant-Aware Home (iNOAH)," National Science Foundation, Emerging Frontiers in Research Innovation (EFRI) Program, Autonomously Reconfigurable Engineered Systems Enabled by Cyberinfrastructure (ARES-CI), 8/1/07–7/31/11, \$1,954,047.
- (Declined) Bennett, Andrew, David Allen, LaVerne Bitsie-Baldwin, Kimberly Douglas, Sanjay Rebello, and Steve Warren. "Center for Quantitative Education," Kansas State University, Targeted Excellence Program, 7/1/2007–6/30/2011, \$2,000,000.
- Bennett, Andrew, Sanjay Rebello, and Steve Warren. "Center for Quantitative Education," Kansas State University, Targeted Excellence Program, 7/1/2006–6/30/2010, \$998,498. Reduced to \$100,000 for the period of 7/1/2006–6/30/2007.
- Martinez, Angel and Steve Warren. "Ingestible Pill for Heart Rate and Core Temperature Measurement in Cattle." Travel award for the 28th *Annual Conference of the IEEE EMBS*, Marriott at Times Square, New York City, New York, Kansas State University, August 30 – September 3, 2006, Graduate Student Council, 4/28/2006, \$90.
- Warren, Steve. Funding to support student travel to the 28th *Annual Conference of the IEEE EMBS*, Marriott at Times Square, New York City, New York, August 30 – September 3, 2006, KSU Electrical & Computer Engineering (\$2824), KSE Computing & Information Sciences (\$500), KSU Mechanical & Nuclear Engineering (\$500), \$3824.
- Warren, Steve and Suzanne Dubnicka. "Light Reflectance Signals as Biometric Authenticators for the Elderly," Galichia Center on Aging, 5/15/06–9/24/06, \$2,000.
- Singh, Gurdip, Daniel Andresen, Scott Deloach, Balasubramaniam Natarajan, and Steve Warren. NSF CNS-0551626. "CRI: An Experimentation Platform for Developing Customized, Large-Scale Sensor Systems," National Science Foundation, CNS Division, Computing Research Infrastructure, 4/1/06–3/31/09, \$420,110. Reduced by NSF to \$200,000.
- (Declined) Warren, Steve (PI). "Secure Infrastructure for Wearable Monitoring Systems Applied to Cardiac Rehabilitation," National Institutes of Health, Biomedical Research Partnerships Program, \$424,982. Subcontract within the project "Body Networks of Wearable Sensors for Patient Monitoring," with Emil Jovanov, Ph.D. (PI – University of Alabama in Huntsville), Piet de Groen, M.D. (CoPI – Mayo Clinic, Rochester, MN), Felicity Enders, Ph.D. (CoPI – Mayo Clinic, Rochester, MN), Bruce Johnson, Ph.D. (CoPI – Mayo Clinic, Rochester, MN), and Aleksandar, Milenkovic, Ph.D. (CoPI – University of Alabama in Huntsville). UAH 2004-566, 8/1/2006–7/31/2010, \$2,226,631.
- Warren, Steve (PI). "Laptop Computer for NI ELVIS Platform Development," EECE, Student Equipment Fee program, \$2,000, January 2006.

- Warren, Steve (PI). “SolidWorks with COSMOS – 10 Student Licenses,” December 2005, \$1850.
- (Declined) Douglas, Kimberly, et al. “Battling the Quiet Crisis: Building the STEM Pipeline,” Kansas State University, Targeted Excellence Program, 7/1/2006–6/30/2010. (Steve Warren – Senior Personnel)
- Singh, Gurdip, Douglas McGregor, and James Edgar. First year transition funds for the “Center for Sensors and Sensor Systems,” Kansas State University, Targeted Excellence Program, 1/1/2006–12/31/2007, \$132,000. (Steve Warren – Co-PI)
- (Declined) Sawan, Edwin, et al. (WSU); Bala Natarajan, Steve Warren, et al. (KSU); Ron Hui, et al. (KU), and Vasundara Varadan (UA). “IGERT: Graduate Student Training for Research and Healthcare,” Integrative Graduate Education and Research Traineeship (IGERT) program, National Science Foundation, 7/1/2006–6/30/2009. (Steve Warren – Co-PI)
- (Declined) Singh, Gurdip, Steve Warren, Daniel Andresen, Sudharman Jayaweera, and Mark Spire. “Human and Animal Health Monitoring: Enhancing Quality of Life and Security Through Research in Distributed, Reconfigurable Sensor Networks,” National Science Foundation, Kansas NSF EPSCoR Program, 4/1/2006–3/31/2009, \$2,449,319.
- Singh, Gurdip, Douglas McGregor, and James Edgar. “Center for Sensors and Sensor Systems,” Kansas State University, Targeted Excellence Program, 7/1/2006–6/30/2010, \$1,500,000. (Steve Warren – Co-PI)
- Starrett, Shelli, Anil Pahwa, and David Soldan. “Planning Learning Environments for a More Diversified Workforce,” National Science Foundation, EEC Division – Engineering Education, 1/1/06–12/31/07, \$100,000. Department Level Reform planning grant. (Steve Warren and William Kuhn – Senior Personnel)
- Andresen, Daniel, Howard Erickson, David Poole, Mark Spire, and Steve Warren. REU Supplement to ITR–0205487. “ITR: An Infrastructure for Veterinary Telemedicine – Proactive Herd Health Management for Disease Prevention from Farm to Market,” National Science Foundation, 10/1/03–9/30/09, \$12,000.
- Lebak, Jeff and Steve Warren. “HL7-Compliant Healthcare Information System.” Travel award for the *10th Annual Meeting of the American Telemedicine Association*, Colorado Convention Center, Denver, CO, April 17–20, 2005, Kansas State University, Graduate Student Council, 2/16/2005, \$200.
- Warren, Steve (PI). “Enhanced Learning Via the National Instruments LabVIEW and NI ELVIS Platforms,” 12/1/2004–8/31/2007, \$6,004.
- Yao, Jianchu and Steve Warren. “A Novel Algorithm to Separate Motion Artifacts from Photoplethysmographic Signals Obtained with a Reflectance Pulse Oximeter.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, Kansas State University, Graduate Student Council, 10/22/04, \$184.
- Warren, Steve (PI). NSF BES–0440183. REU Supplement to NSF BES–0093916. “CAREER: Beyond Telemedicine: Realizing a Proactive Home Health Care Delivery Model Through Research and Education in Plug-and-Play Components and Novel Sensors,” Faculty Early Career Development (CAREER) Program, National Science Foundation, Division of Bioengineering and Environmental Systems, 2/1/01–1/31/06, \$6,000.
- Nagl, Luke and Steve Warren. “Experiences Using WinCE PocketPCs as Computation and Data Acquisition Platforms for Ambulatory Telemonitoring.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, Whitaker Foundation, 7/19/04, \$145.
- Krishnamurthi, Balakumar and Steve Warren. “Experiences Using WinCE PocketPCs as Computation and Data Acquisition Platforms for Ambulatory Telemonitoring.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, National Science Foundation, Kansas EPSCoR Program, 7/19/04, \$500.

- Warren, Steve (PI). “iINTERFAWARE Chameleon Licenses,” 2004, \$12,000.
- Lebak, Jeffrey and Steve Warren. “HL7-Compliant Healthcare Information System for Home Monitoring.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, National Science Foundation, Kansas EPSCoR Program, 4/27/04, \$500.
- Schoenig, Scott and Steve Warren. “Ambulatory Instrumentation Suitable for Long-Term Monitoring of Cattle Health.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, National Science Foundation, Kansas EPSCoR Program, 4/27/04, \$500.
- Yao, Jianchu and Steve Warren. “A Novel Algorithm to Separate Motion Artifacts from Photoplethysmographic Signals Obtained with a Reflectance Pulse Oximeter.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, National Science Foundation, Kansas EPSCoR Program, 4/27/04, \$500.
- Nagl, Luke and Steve Warren. “Experiences Using WinCE PocketPCs as Computation and Data Acquisition Platforms for Ambulatory Telemonitoring.” Travel award for the *26th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, San Francisco, CA, 9/1/04 - 9/5/04, National Science Foundation, Kansas EPSCoR Program, 4/27/04, \$500.
- Andresen, Daniel, Howard Erickson, David Poole, Mark Spire, and Steve Warren. REU Supplement to ITR–0205487. “ITR: An Infrastructure for Veterinary Telemedicine – Proactive Herd Health Management for Disease Prevention from Farm to Market,” National Science Foundation, 10/1/03-9/30/09, \$12,000.
- Andresen, Daniel, Howard Erickson, David Poole, Mark Spire, and Steve Warren. ITR–0205487. “ITR: An Infrastructure for Veterinary Telemedicine – Proactive Herd Health Management for Disease Prevention from Farm to Market,” National Science Foundation, 10/1/03-9/30/09, \$899,996.
- Andresen, Daniel, Steve Warren, Howard Erickson, David Poole, and Jan Sargeant. ITR–0325921. “ITR: Veterinary Telemedicine – Proactive Herd Health Management for Disease Prevention from Farm to Market,” National Science Foundation, 10/1/02-9/30/04, \$107,025.
- Bennett, Andrew and N. Sanjay Rebello. NSF DUE–0206943 “Assessing Student Transfer and Retention of Learning in Mathematics, Physics, and Engineering,” National Science Foundation, 7/15/02-7/14/07, \$500,000. (Steve Warren – Senior Personnel)
- Warren, Steve (PI). “Software Components for Research and Education,” Spatial Corporation, 7/18/01–7/10/03, \$58,500.
- Warren, Steve (PI). NSF BES–0093916. “CAREER: Beyond Telemedicine: Realizing a Proactive Home Health Care Delivery Model Through Research and Education in Plug-and-Play Components and Novel Sensors,” Faculty Early Career Development (CAREER) Program, National Science Foundation, Division of Bioengineering and Environmental Systems, 2/1/01–1/31/06. Five-year continuing award totaling \$375,000.
- Warren, Steve (PI). NSF EPS–9874732. “Beyond Telemedicine: Realizing a Proactive Home Health Care Delivery Model Through Research and Education in Plug-and-Play Components and Novel Sensors,” National Science Foundation, Kansas EPSCoR Program, University of Kansas, 2/1/01–1/31/02. Originally \$31,750; adjusted to \$5,000 upon receipt of the National Science Foundation CAREER award.
- Warren, Steve (PI). “Explore Electron Transport in a Complex Solid Model Using the Adjoint Version of ITS-CAD,” Sandia National Laboratories, Albuquerque, NM, 1/20/2000–12/31/2004, \$66,993.
- Warren, Steve. IronCAD for Teaching and Research, Visionary Design Systems, 12/22/99, \$4,995.

PROGRAM DEVELOPMENT

- 2024 Kansas HOSA Fall Leadership Conference, Future Health Professionals, Kansas State University, Manhattan, KS, October 17, 2024
- Tour of Ascension Via Christi, Manhattan, KS, TRIMEDX, BME 575 – Clinical Systems Engineering, September 20, 2024
- 2024 Kansas HOSA Spring Leadership Conference, Future Health Professionals, Kansas State University, Manhattan, KS, March 27, 2024.
- Tour of Ascension Via Christi, Manhattan, KS, TRIMEDX, BME 575 – Clinical Systems Engineering, March 6, 2024
- 2023 Kansas HOSA Fall Leadership Conference, Future Health Professionals, Kansas State University, Manhattan, KS, October 4, 2023
- Discussion with Dr. Ryan Fischer, Dr. Danial Forsha, and other research faculty at Children’s Mercy Hospital, univentricular physiology and Fontan-associated liver disease, December 18, 2023
- Discussion with Rick Haigh, Sean Hayes, Alexx Frankos, Tim Duncan, and other Garmin engineers, canine biometrics, December 6, 2023
- Senior Experience Day, Kansas State University, Carl R. Ice College of Engineering, Various Dates, Fall 2023
- Powercat Preview, Kansas State University, Carl R. Ice College of Engineering, Various Dates, Fall 2023
- Tour of the Children’s Mercy Hospital clinical engineering facility, October 26, 2023
- Matthew Dodge, Biosense Webster, Electrophysiology Immersion Program, October 26, 2023
- Reformulation of the KSU Undergraduate Biomedical Engineering Program in response to the Kansas Board of Regents (KBoR) undergraduate core mandate, Fall 2023
 - Complete curricular overhaul for the BME emphasis areas, including new BME 390 and BME 470 courses
 - Updated BME emphasis areas: Sensors & Devices, Biomaterials, Computation, and Pre-Medicine
 - Nine CurricuLog proposals related to facets of this overhaul, November 5, 2023
- ABET Accreditation, Undergraduate Biomedical Engineering Program, Summer/Fall 2023
 - Course assessment collation
 - BME self-study, June 28, 2023
 - BME student transcript analysis, July 10, 2023
 - Interactions with the BME evaluator, Dr. Kim Anderson, University of Kentucky
 - Examples of student work
 - ABET evaluator visit, October 15-17, 2023
- DaVinci surgical robot tour, Ascension Via Christi, Manhattan, KS, KSU IEEE EMBS Student Chapter, April 15, 2023
- BME 575 tour of Ascension Via Christi, Manhattan, KS, Rachelle White, TRIMEDX, February 22, 2023; March 6, 2024
- Internship and Opportunity Discussions, Biosense Webster, Kansas City, MO, October 25, 2022, January 20, 2023.
- Internship and Opportunity Discussions, TRIMEDX, Salina, KS January 20, 2023.
- Internship and Opportunity Discussion, Children’s Mercy, Kansas City, KS, July 7, 2022.
- Internship and Opportunity Discussion, Garmin, Kansas State University, Carl R. Ice College of Engineering, Manhattan, KS, April 20, 2022
- K-State Group Travel, ECE Faculty and EMBS Officers, 21st Annual Design of Medical Devices Conference, University of Minnesota, April 11-14, 2022, Minneapolis, MN.
- Interactions with University of Wisconsin - Madison faculty/staff regarding elements of their BME curriculum, Madison, WI, Oct. 26, 2018
- Interactions with Marquette faculty/staff regarding elements of their BME curriculum, Milwaukee, WI, Oct. 25, 2018

- Undergraduate degree in Biomedical Engineering, KSU College of Engineering, 2017–present.
- “Master Agreement for Research and Service Projects,” Netsmart Technologies and Kansas State University, April 2017. Supplemented with a “Business Associate Subcontractor Agreement” and a “Special Intellectual Property Assignment and Confidentiality Agreement – Student Led Learning Experience.”
- Nondisclosure agreement with Streamline Automation, January 20, 2010. Subject: analysis of reflectance pulse oximeter data to assess hemodynamic parameters noninvasively.
- Nondisclosure agreement with Orbital Outfitters, November 24, 2009. Subject: creation of a customized space suit replica for anti-gravity mobility and physiology experiments.
- Nondisclosure agreement with Raytheon Corporation, October 8, 2008. Subject: co-development of an ingestible pill for cattle health monitoring

HONORS/SCHOLARSHIPS

- Robert and Becca Reichenberger Cornerstone Teaching Scholar, Kansas State University College of Engineering, May 1, 2024 – April 30, 2027
- 2022 Team Award of Excellence, Kansas State University, Carl R. Ice College of Engineering, Development of the Biomedical Education and Innovation Laboratory
- Faculty Recognition Award, XIX Chapter of the Mortar Board Senior Honor Society, Kansas State University, November 9, 2021
- Robert and Becca Reichenberger Cornerstone Teaching Scholar, Kansas State University College of Engineering, September 1, 2019 – June 30, 2024
- 2019 Larry E. and Laurel Erickson Public Service Award, Kansas State University College of Engineering
- KSU Medical Component Design Laboratory, KSU Student Chapter of the IEEE EMBS, Best ECE Department Display, 2019 Engineering Open House, KSU College of Engineering, April 5-6, 2019
- KSU Student Chapter of the IEEE EMBS, 2019 K-State College of Engineering Outstanding Professional Student Chapter Award, KSU College of Engineering, Engineering Student Council
- KSU Student Chapter of the IEEE EMBS, 2019 E-Week Best Booth Award, KSU College of Engineering, Engineering Student Council
- KSU Student Chapter of the IEEE EMBS, 2018 K-State College of Engineering Outstanding Professional Student Chapter Award, KSU College of Engineering, Engineering Student Council
- Quality Mentoring in Undergraduate Research Award, Kansas State University Developing Scholars Program, April 2018
- Outstanding Service Award, Midwest Section of the American Society for Engineering Education, September 2017
- Promotion to Professor, Kansas State University, February 2017
- Electrocardiogram Pill for Cattle – FY2007 CISE highlight for the National Science Foundation
- ‘Making a Difference’ Award, Kansas State University Women in Engineering and Science Program, December 2005, October 2007
- Tenure, Kansas State University, February 2002
- Promotion to Associate Professor, Kansas State University, February 2002
- Nominated for the 2002 James L. Hollis Memorial Award for Excellence in Undergraduate Teaching, KSU College of Engineering
- Award for Excellence, Kansas State University, Engineering Student Council, Dec. 1999
- Graduate Faculty, Kansas State University, Membership and Certification, Dec. 1999
- Promotion to Principal Member of the Technical Staff, Sandia National Laboratories, Nov. 1998
- Sandia National Laboratories Individual Performance Award, Telemedicine effort, August 1997
- Sandia National Laboratories Award For Excellence, Personal Status Monitor effort, July 1997
- Engineering Honors Program, Kansas State University
- Tau Beta Pi, Eta Kappa Nu National Engineering Honoraries, Kansas State University
- Phi Kappa Phi, Golden Key National Honoraries, Kansas State University

- 1st Place - Kansas State University Open House Poster Contest, 1992
- Special Citation for Outstanding Musicianship, International Association of Jazz Educators, Wichita Jazz Festival “20” College Big Band Competition, April 26, 1991
- Certificate of Recognition, University of Missouri, Kansas City, Conservatory of Music, Kansas City Jazz Commission, 1990 Kansas City Jazz Band Festival, March 9, 1990
- Graduate Research Assistantship, The University of Texas at Austin, 6 semesters, 1992-1994
- Graduate Teaching Assistantship, The University of Texas at Austin, 2 semesters, 1991-1992
- Graduate Research/Teaching Assistantship, Kansas State University, 4 semesters, 1989-1991
- Bandsman of the Year, Kansas State University, 1986
- Dean of Engineering Scholarship, Kansas State University, 1984-1989
- Dean's Honor Roll and National Dean's List, Kansas State University, 10 semesters, 1984-1989

COURSES TAUGHT

- **BME 001 – New Student Assembly**, Kansas State University, F18, F19, F20, F21, F22, F23
- **BME 200 – Introduction to Biomedical Engineering**, Kansas State University, F18, F19, F20, F21, F22, F23
- **BME 575 – Clinical Systems Engineering**, Kansas State University, S22, S23, S24, F24
- **DEN 398 – EMBS Learning Community**, Kansas State University, F05
- **ECE 499 – Honors Research**, Kansas State University, F99–present
- **ECE 510 – Circuit Theory I**, Kansas State University, S07, S08
- **ECE 511 – Circuit Theory II**, Kansas State University, F99, F00, F10, F16
- **ECE 512 – Linear Systems**, Kansas State University, F01, S02, F02, S03, S04, S05, F05, S06, F06, S07, F07, S08, F08, S09, F09, S10, S11, F11, S12, F12, S13, F13, F14, F15, F16, F17, F18, F19, F20, S21, F21, S22, F22, S23, F23, S24, F24
- **ECE 571 – Introduction to Biomedical Engineering**, Kansas State University, S00, S01, S02, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14, S15, S16, S17, S18, S19, S20, S21
- **ECE 636 – Introduction to Computer Graphics**, Kansas State University, F03, S05, S06, S09
- **ECE 690/890 – Problems in Electrical & Computer Engineering**, Kansas State University, F05, S07, S09, S12, F12, S13, F13, S14, F14
- **ECE 772 – Bioinstrumentation**, Kansas State University, F00, F01, F02, F03, F04, F05, F06, F08, F09, F11, F12, F17, F18
- **ECE 840 – Computer Engineering Methods for Analysis, Simulation, and Design**, Kansas State University, S00, S01, S03, S04

PROFESSIONAL DEVELOPMENT

- **James R. Coffman Leadership Institute**, August 9–11, 2006, Rock Springs 4-H Center
- **Feeding and Caring For Graduate Students**, The Graduate School, Kansas State University, December 8, 2001
- **What Matters Most**, Kansas State University, April 21, 2001. Time management and life leadership workshop sponsored by Raytheon Corporation
- **Kansas State University New Faculty Workshop**, Emporia, KS, February 17, 2001
- **Engineering LEA/RN Program**, Kansas State University, 1999-2004
- **Academic Advising**, Kansas State University, Campus Collaboration to Foster Retention, satellite teleconference, November 4, 1999
- **Management of Sandia Technical Projects**, Erica Jones Associates, Inc., Coronado Club, Albuquerque, NM, February 9-11, 1999
- **Marketing and Market Research Workshop**, sponsored by the Sandia New Ventures Program, Technology Ventures Corporation, and the Albuquerque Chamber of Commerce, Lockheed Martin Building, Albuquerque, NM, January 14, 1999

- **Structuring Group Relationships to Achieve Technical Success.** Workshop on high-performance teams and workgroups, sponsored by Sandia National Laboratories, January 12, 1999.
- **On the Fast Track: Recruit, Retain, Become a High-Potential Employee,** Advanced Technology and Management Programs, National Technological University, sponsored by the U.S. Chamber of Commerce, November 12, 1997
- **Seven Habits of Highly Effective People,** Covey Leadership Center, Coronado Club ZIA Room, Albuquerque, NM, January 24-26, 1995
- **Lessons in Leadership,** Covey Leadership Center, based on the book **First Things First,** Albuquerque Convention Center, Albuquerque, NM, November 16, 1994

TECHNICAL DEVELOPMENT

- **Hands-On Education: The Future of Engineering Education Made Possible by Portable Electronics,** U223 Workshop, *2012 Conference of the American Society for Engineering Education*, San Antonio Convention Center, San Antonio, TX, June 10, 2012.
- **Point-of-Care Medical Device Communications,** IEEE 1073 – Medical Information Bus, Westin Galleria, Houston, TX, October 22, 2002.
- **Enhancement and Revision of the IEEE 1451.2 and 1451.5 (Smart Sensors) Standards,** May 20-21, 2002, Sensors Expo 2002, San Jose McEnery Convention Center, San Jose, CA
- **Spatial User's Conference 2002.** Tutorial and application workshop in support of the ACIS solid modeling toolkit, May 13-15, 2002, Westminster, CO.
- **Spatial User's Conference 2000.** Three-day tutorial and application workshop in support of the ACIS solid modeling toolkit and customer applications (i.e., the Sandia/KSU work on radiation transport through complex solid models), June 21-23, 2000, Boulder, CO.
- **Medical Applications of Sensors, Transmitters, Devices, and Instrumentation.** 3-day short course on biomedical instrumentation offered through the UCLA Extension, Department of Engineering, Information Systems, and Technical Management, Los Angeles, CA, January 31 – February 2, 2000.
- **Introduction to CORBA Programming.** 4-day short course in CORBA/C++ implementation taught by Black & White Corporation, sponsored by Sandia National Laboratories, Research Park, September 8-11, 1998.
- **Development of ACIS-Based Applications.** Short course on the use of the ACIS geometric modeler, offered by Spatial Technology, Boulder, CO, November 25-26, 1996.
- **The Wavelet Transform and Applications.** Short course in applied optics jointly sponsored by the Department of Electrical and Computer Engineering and the Department of Physics at New Mexico State University, May 17, 1996.
- **Electrical Modeling, Simulation, and Design of Electronic Packages.** Class sponsored by the University of Illinois at Urbana-Champaign and North Carolina State University, Radisson Hotel, San Jose, CA, June 26-29, 1995.
- **Electrical Modeling with the Saber Circuit Simulator.** Class Sponsored by Analogy Corporation, Beaverton, OR, January 14-17, 1995.

SOFTWARE EXPERIENCE

- **Office/Productivity:** Microsoft Office, Microsoft Project, Adobe Acrobat, Lotus 123, Quattro-Pro, Graphics Workshop, Paint Shop Pro, Wordstar, TeX, LaTeX, PiCTeX, Graftool, and DesignCAD2D.
- **Computational Methods:** MATLAB/Simulink, Derive, Maple V, Numerical Recipes in C, ACIS 3D Modeler, IronCAD, Ansoft Maxwell Eminance, LINPACK, EISPACK, ODEPACK, IMSL, 3D/Eye Trispectives Technical, Microsoft Visual C++, Borland Turbo C/C++, Microsoft Fortran, DEC Fortran, SAS, SPICE/PSPICE, Analogy Saber Circuit Simulator, and NeXT Application Builder

- **Programming Languages:** C/C++, CORBA IDL, Pascal, Fortran, Forth, Objective C, and Basic
- **Operating Systems:** Microsoft Windows 3.1/95/98/NT, MS-DOS, Sun Solaris, NeXT, Hewlett-Packard, and Macintosh
- Software Reviewer for *Circuits and Devices Magazine*, 1994-1995

CONFERENCE SERVICE

- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Charlotte, North Carolina, June 21–24, 2026.
- **Technical Program Committee Member**, *47th International Conference of the IEEE EMBS*, Copenhagen, Denmark, July 14–17, 2025.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Montreal, Canada, June 22–25, 2025.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Portland, OR, June 23–26, 2024.
- **Associate Editor: Theme 7 – Biomedical Sensors and Wearable Systems**, *45th International Conference of the IEEE EMBS*, Sydney, Australia, July 24–28, 2023.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Baltimore, MD, June 24–27, 2023.
- **Associate Editor: Theme 7 – Biomedical Sensors and Wearable Systems**, *44th International Conference of the IEEE EMBS*, Glasgow, Scotland, UK, July 11–15, 2022.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Minneapolis, Minnesota, June 26–29, 2022.
- **Associate Editor: Theme 7 – Biomedical Sensors and Wearable Systems**, *43rd International Conference of the IEEE EMBS*, Guadalajara, Mexico, October 31 – November 4, 2021.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Long Beach, California, June 27–30, 2021.
- **Associate Editor: Theme 7 – Biomedical Sensors and Wearable Systems**, *42nd International Conference of the IEEE EMBS*, in conjunction with the 43rd Annual Conference of the Canadian Medical and Biological Engineering Society, Montréal, Québec, Canada, July 20–24, 2020.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Montréal, Québec, Canada, June 21–24, 2020.
- **Associate Editor: Theme 7 – Biomedical Sensors and Wearable Systems**, *41st International Conference of the IEEE EMBS*, CityCube Berlin, Messedamm 26, 14055 Berlin, Germany, July 23–27, 2019.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Tampa, FL, June 16–19, 2019.
- **Associate Editor: Theme 7 – Biomedical Sensors and Wearable Systems**, *40th International Conference of the IEEE EMBS*, Hilton Hawaii Village Waikiki Beach Resort, Hawaii Convention Center, Honolulu, HI, July 17–21, 2018.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, Salt Lake City, UT, June 24–27, 2018.
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies**, *39th International Conference of the IEEE EMBS*, International Convention Center, Jeju Island, Korea, July 11–15, 2017.
- **Technical Program Committee Member**, *2017 ASEE Midwest Section Conference*, Stillwater, OK, September 24–26, 2017.
- **Chair**, *2016 ASEE Midwest Section Conference*, Manhattan, KS, September 25–27, 2016.
- **Technical Program Committee Member**, *Annual Conference of the ASEE*, New Orleans, LA, June 26–29, 2016.
- **Technical Program Committee Member**, *2016 ASEE Midwest Section Conference*, Manhattan, KS, September 25–27, 2016.

- Manuscript Review, *Annual Conference of the ASEE*, 2016
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies**, 38th *International Conference of the IEEE EMBS*, Disney’s Contemporary Resort, Orlando, FL, August 16–20, 2016.
- **Technical Program Committee Member**, 2015 *ASEE Zone III Conference*, Springfield, MO, September 23–25, 2015.
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies**, 37th *International Conference of the IEEE EMBS*, Milano Conference Center, Milan, Italy, August 25–29, 2015.
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies**, 36th *International Conference of the IEEE EMBS*, Sheraton Chicago Hotel and Towers, Chicago, IL, August 26–30, 2014.
- **Invited Session Organizer (Co-Organizer: Punit Prakash)**. “Technologies for Around-the-Clock Health Assessment of Special Needs Children” ; Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies); 36th *International Conference of the IEEE EMBS*, Sheraton Chicago Hotel and Towers, Chicago, IL, August 26–30, 2014.
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies; Chair: Track 3.6 – Wearable Sensor Technology and Instrumentation**, 35th *International Conference of the IEEE EMBS*, Osaka International Convention Center, Osaka, Japan, July 3–7, 2013.
- **Technical Program Committee Member**, *Frontiers in Education 2012 (FIE 2012)*, Seattle, WA, October 3–6, 2012.
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies**, 34th *International Conference of the IEEE EMBS*, IEEE EMBS Conference Editorial Board, San Diego, CA, August 28 – September 1, 2012.
- **Technical Program Committee Member**. 2nd *International Conference on Computer Science and Automation Engineering (CSAE 2012)*, Zhangjiajie, China, May 25–27, 2012.
- **Co-Chair: Session FrB19 – Wireless Sensors and Systems**, September 2, 2011, 33rd *Annual Conference of the IEEE EMBS*, Boston Marriott Copley Place, Boston, MA, Aug. 30 – Sept. 3, 2011.
- **Associate Editor: Theme 3 – Bioinstrumentation, Biosensors, and Bio-Micro/Nano Technologies**, 33rd *International Conference of the IEEE EMBS*, IEEE EMBS Conference Editorial Board, Boston, MA, August 30 – September 3, 2011.
- **Technical Program Committee Member**, 2009 *International Conference on Computer Communications and Networks (ICCCN)*, San Francisco, CA, August 2–6, 2009.
- **Technical Program Committee Member**, 2009 *International Conference on Communications (IEEE ICC)*, Dresden, Germany, June 14–18, 2009.
- **Technical Program Committee Member**, 2008 *International Conference on Computer Communications and Networks (ICCCN)*, St. Thomas, U.S. Virgin Islands, August 4–7, 2008.
- **Panel Member: Session SaD10 – Students: The Movers and Shakers of EMBS**, September 2, 2006, 28th *Annual Conference of the IEEE EMBS*, Marriott Times Square, New York City, NY, Aug. 30 – Sept. 3, 2006.
- **Chair: Session 7.8.1 – Medical Information Systems I**, September 4, 2004, 26th *Annual Conference of the IEEE EMBS*, Westin St. Francis Hotel, San Francisco, CA, Sept. 1–5, 2004.
- **Workshop on Home Care Technologies for the 21st Century**. Sponsored by the National Science Foundation, the Catholic University of America, and the U.S. Food and Drug Administration. Department of Health and Human Services (DHHS), Rockville, MD, April 7–9, 1999. Session chair: Topic F – Smart Health Care Systems and the Home of the Future, pp. 58–63, ©1999, Catholic University of America.

PROFESSIONAL ACTIVITIES

Short Courses and Workshops

- “Wearable Sensors and Devices,” Winter Camp for Research Experience, sponsored by the Korea Institute for Advancement of Technology, Kansas State University, Carl R. Ice College of Engineering, Jan. 5–10, 2020.
- Warren, Steve. “The Health Care Facelift: How Telecommunications Technology, Distributed Computing, and Intelligent Medical Devices will Redefine the Patient-Physician Relationship,” **Four-hour short course on smart health care technology** prepared for the Department of Biomedical Engineering, University of Minnesota, Twin Cities Campus, December 1, 1999.

Textbook, Proposal, and Manuscript Review

- Guest Editor, *Sensors*, Special Issue: “Recent Applications of Non-Invasive Biomedical Sensors,” ISSN 1424-8220, Manuscripts due March 10, 2022.
- Oxford University Press, Linear Systems and Signals Textbook (3rd edition) review, 2018
- National Science Foundation, Disability and Rehabilitation Engineering (DARE) Program, Proposal Review Panel B – *Biomechanics, Biomaterials, Computational Modeling, & Design*, January 22-23, 2018, Alexandria, VA
- CRC, Biomedical Engineering Textbook Feedback, 2017
- zyBooks Interactive Textbook Review, 2017
- Elsevier, Introduction to Biomedical Engineering Textbook Review, 2015
- Addison-Wesley Computer Graphics Textbook Review, 2004
- Science Foundation Ireland (SFI), SFI Investigators Programme, 2013.
- National Science Foundation, General and Age-Related Disabilities Engineering (GARDE) Proposal Panel Review, December 3–4, 2012, Arlington, VA
- National Science Foundation, Smart Health and Wellbeing (SHB) Proposal Panel Review, May 6-8, 2011, Arlington, VA
- National Institutes of Health, National Institute of Biomedical Imaging and Bioengineering (NIBIB) Panel Review, Point-of-Care (POC) Technology Research Network (U54) Proposals, July 13, 2007, Bethesda, MD
- National Science Foundation, Biomedical Engineering CAREER Proposal Panel Review, November 6–7, 2001, November 13–15, 2002, November 11–13, 2003, October 5–6, 2005, Arlington, VA
- Proposal Review, U.S. Department of Veterans Affairs, 2006
- Proposal Review, KSU Targeted Excellence Program, 2005
- Proposal Review, Louisiana Health Excellence Fund, 2000
- Proposal Review, Department of Energy Small Business Innovation Research (SBIR) Program, FY 1998/1999 Awards, Biological and Environmental Research – Medical Applications
- Proposal review, National Science Foundation, 1993
- Manuscript Review, *IEEE Transactions on Learning Technologies*, 2017
- Manuscript Review, *Computers and Electrical Engineering*, 2017
- Manuscript Review, *Annual Conference of the ASEE*, 2016–2020
- Manuscript Review, *ASEE Midwest Section Conference*, 2015–2018
- Manuscript Review, *ASEE Zone III Meeting, American Society for Engineering Education*, Springfield, MO, 2015
- Manuscript Review, *Healthcare Innovations and Point-of-care Technologies Conference of the IEEE Engineering in Medicine and Biology Society*, Seattle, Washington, 2014
- Manuscript Review, *IEEE-EMBS International Conferences on Biomedical and Health Informatics*, Valencia, Spain, 2014
- Manuscript Review, *2nd Middle East Conference on Biomedical Engineering, Qatar*, 2014
- Manuscript Review, *Frontiers in Education Conference*, 2012
- Manuscript Review, *IEEE ICC*, 2009
- Manuscript Review, *IEEE ICCCN*, 2008, 2009
- Manuscript Review, *Annual Conference of the IEEE EMBS*, 2007–2020
- Manuscript Review, *Biomedical Signal Processing and Control*, 2016
- Manuscript Review, *Expert Review of Medical Devices*, 2015

- Manuscript Review, *International Journal of Distributed Sensor Networks*, 2014
- Manuscript Review, *IEEE Microwave and Wireless Components Letters*, 2014
- Manuscript Review, *Sensors*, 2014
- Manuscript Review, *IEEE Transactions on Biomedical Circuits and Systems*, 2012–present
- Manuscript Review, *IEEE Communications Magazine*, 2011
- Manuscript Review, *EURASIP Journal on Wireless Communications and Networking*, 2011
- Manuscript Review, *Journal of Healthcare Engineering*, 2010
- Manuscript Review, *IEEE Transactions on Biomedical Engineering*, 2005–present
- Manuscript Review, *IEEE Journal of Biomedical and Health Informatics*, formerly *IEEE Transactions on Information Technology in Biomedicine*, 2000–2003, 2005–present
- Manuscript Review, *Journal of Neurorehabilitation Research*, 2007
- Manuscript Review, *Journal of Biomedical Optics*, 2005
- Manuscript Review, *Optical Spectroscopy*, 1994
- Manuscript Review, *Applied Optics* (Optical Society of America), 1997

Professional Societies and Industry Workgroups

- Institute of Electrical and Electronics Engineers (IEEE), 2001–present
- IEEE Engineering in Medicine & Biology Society, 2001–present
- American Society for Engineering Education (ASEE), 2001–present
- Founding chair, EMBS Chapter for the IEEE Kansas City section, 2006–2007
- IEEE 11073 Personal Health Devices Working Group, 2009–present
- IEEE 11073 Appeals Committee, 2010
- American Telemedicine Association, 1997–1998, 2005–2006
- National Instruments myDAQ Beta Program, 2010–2012
- Microsoft Health User’s Group, 1997–1998
- Health Level 7 message standards organization, 1997–1998
- Order of the Engineer, 2000–present
- Southwest Biomedical Industry Coalition, 1995
- Sandia National Laboratories Biomedical Engineering Focus Group, 1994–1996

Expert Witness

- Expert Witness, Physiological Monitoring Hardware and Software, 2013–present
- Expert witness engagement listing: provided upon request depending on need

Leadership Service

- Program Coordinator, Undergraduate Biomedical Engineering Degree Program, KSU College of Engineering, 2017–present
- Chair, Midwest Section of the American Society for Engineering Education, 2015–2016
- Chair Elect, Midwest Section of the American Society for Engineering Education, 2014–2015
- American Society for Engineering Education Representative for the Kansas State University College of Engineering, 2014–present

Committees

- Search committee, Dean of the College of Health and Human Sciences, Kansas State University, 2021–2022
- Curriculum subcommittee, B.S. in Public Health, Kansas State University, Department of Kinesiology, 2019–present
- Advisory Panel, Midwest Institute for Comparative Stem Cell Biology, Kansas State University, 2019–present
- Faculty Senator, Kansas State University, College of Engineering, 2019–present
- Academic Affairs Committee, Faculty Senate, Kansas State University, 2020–2022
- Faculty Affairs Committee, Faculty Senate, Kansas State University, 2019–2020
- College Assessment Review Committee, Kansas State University College of Engineering, 2018–present
- Honors & Awards Committee, Kansas State University College of Engineering, 2018–2020

- Chair, Biomedical Engineering Professor-of-Practice Search Committee, 2018–2019
- Chair, Biomedical Engineering Curriculum Task Force, Kansas State University, Electrical & Computer Engineering, 2016–2017
- KSU College of Engineering Phase IV Biomedical Suite and Resource Contributions, 2013–present
- Kansas State University Internal Review Board (Human Subjects Research), 2000–present
- Heartspring Board of Trustees, 2009–2015
- Kansas State University Goldwater Nominating Committee, 2009–present
- Women in Engineering and Science Program, Search Committee, 2003–2004
- Electrical and Computer Engineering, Education Research Committee, 2013–2015
- Electrical and Computer Engineering, Recruitment Committee, 2013–2015
- Electrical and Computer Engineering, Department Committee on Planning, 2011–2014, 2020–2022
- Electrical and Computer Engineering, College Committee on Planning, 2020–2023
- Electrical and Computer Engineering, Internship Program, 2000–2001
- Chair, Electrical and Computer Engineering, Quality of Life Committee, 2004–2007
- Electrical and Computer Engineering, Academic Progress Committee, 2001–2004
- Electrical and Computer Engineering, Advising Committee, 2009–2010
- Electrical and Computer Engineering, Department Head Search Committee, 2005–2006, 2006–2007
- Electrical and Computer Engineering, Faculty Search Committee, 2008, 2010–2011, 2011–2012
- College of Engineering, Senior Biomedical Researcher Search Committee, 2011–2012
- Electrical and Computer Engineering, Awards Committee, 2005–2007
- Electrical and Computer Engineering, Course and Curriculum Committee, 2002–2005, 2011, 2015–2016, 2020–2022
- Electrical and Computer Engineering, Department Appearance Committee, 1999–2001
- Electrical and Computer Engineering, Equipment Committee, 1999, 2003–2004, 2007–2010
- Gerontology Academic Affairs Committee, 2010
- College of Engineering Academic Reinstatement Committee, 2003
- KSU Research Excellence Award Committee, 2001–2003
- K-State Olathe Innovation Campus [KOIC] Faculty Advisory Council, 2009–2011
- K-State Olathe Innovation Campus [KOIC] Academic Planning Committee Member, 2009–2011
- Kansas State University Health IT Working Group, 2008–2011

Community Service

- Infrastructure Improvement and Vision Committee, Flint Hills Christian School, 2018–2019
- Manhattan High School Robotics Team Exchange, Medical Component Design Laboratory, KSU Electrical & Computer Engineering, November 30, 2017
- Laboratory demonstration, *DEN 161 – Engineering Problem Solving*, March 8, 2016
- Judge, Fall 2015 Engineering Undergraduate Research Poster Forum, December 3, 2015
- Bioengineering Session, E3 Workshop for Science Teachers Hosted by the KSU College of Engineering, Bette Grauer, Organizer, May 31, 2012; May 30, 2013
- Lecturer, Kansas State University Presidential Lecture Series, 2011–2013
- Girls Researching Our World (GROW) Workshops, 2001, 2002, 2007, 2008, 2016, 2018, 2019, 2020, 2022, 2023, 2024
- “Engineering the Body” Workshop, Medical Component Design Laboratory, Women in Engineering and Science Program, November 8, 2008
- Helping Hands Workshop Support, “Vital Signs Shirt,” Medical Component Design Laboratory, Women in Engineering and Science Program, November 10, 2007
- Judge, Flint Hills Christian School Creative Arts Fair, March 4–6, 2003, Manhattan, KS.
- Community Health Council, 2002–2004, Information Systems Management Committee

- Contributor, *Information for Tomorrow* home telehealth web site, <http://www.informationfortomorrow.com/>, 1999–2000
- Contributor, Expert Panel on Telemedicine Technology, Last Acts Web Site, <http://www.lastacts.org/>, 1999
- Judge, New Mexico Junior Academy of Sciences paper/oral competition, University of New Mexico, 1997–98
- Judge, 36th National Junior Science and Humanities Symposium, sponsored by the United States Army, Navy, and Air Force, Albuquerque, NM, May 3, 1998

Academic Advising/Mentoring

- Powercat Preview, Kansas State University, Carl R. Ice College of Engineering, Fall 2023
- Prospective Student Visits/Tours, KSU Biomedical Engineering Program, 2017–present
- Faculty Advisor, KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society (EMBS), 2005–present
- Faculty Advisor, KSU Student Ambassadors for Biomedical Research Enhancement (SABRE) Initiative, 2006–2008
- Faculty Advisor, KSU Future Engineers in Medicine and Law, 2005–2006
- Organizer, Learning Community for the KSU Student Chapter of the IEEE Engineering in Medicine and Biology Society (EMBS), College of Engineering Learning Community Initiative, 2005–2006
- Faculty Liaison – KSU University Recognition Program, 2006 (Mark Freyenberger), 2007 (Heather Arnold)
- Faculty Advisor, KSU Tau Beta Pi student chapter, 2001–2004
- Faculty Advisor, Triangle Fraternity, 2000–2001
- KSU Engineering LEA/RN Program, 1999–2004
- College of Engineering – Advising for entering EE and CMPEN freshmen, Summer 2006/2007
- Teaching mentor for new faculty (Scoglio – 2005; Prakash – 2012)
- Kansas State University, Graduate Faculty, 1999–present

Other

- Dossier review, Texas Tech University, 2021
- Dossier review, University of Missouri Kansas City, 2018
- Dossier review, University of North Florida, 2017
- Dossier review, University of Alabama, Huntsville, 2015
- Academic Program Consultant, East Carolina University, General Engineering Department, 2009–2010
- USB pulse oximeter module shipment to Howard Apple, GlucoLight Corporation, Feb. 3, 2007
- No-fee consultant, Sandia National Laboratories, Telemedicine and Radiation Transport project teams, 1999–2001
- No-fee consultant, U.S. Food and Drug Administration, Center for Devices and Radiological Health, 2001–2004
- The University of New Mexico, Albuquerque, NM. Interim Instructor, *Engr. F-122, Introduction to Engineering*, 1994

PERSONAL ACTIVITIES

Physical Service/Missions

- Mission Trip, Santiago do Los Caballeros, Dominican Republic, University Christian Church, July 28 – August 4, 2017
- Mission Trip, Marouge, Haiti, Mole, Haiti, University Christian Church, July 19–26, 2013
- Disaster Response Team, Co-Organizer, University Christian Church, Manhattan, KS, 2009–present. Manhattan, KS tornado cleanup – June 2008. Reading, KS tornado cleanup – June May 2011; Oklahoma City tornado cleanup – June 2013
- Manhattan Emergency Shelter Meals (2006–present)

- Habitat for Humanity home construction – Kansas State University (2000,2002,2005), Lockheed Martin, Sandia National Laboratories (1998)
- Development Team, King Solomon Christian Service Camp, 2017–present
- Tree and field work, King Solomon Christian Service Camp, 2010–2013, 2015
- Inner-City Mission and Construction Work, University Christian Church, Manhattan, KS: Dallas, TX (2011), Cincinnati, OH (2012)
- Grow Our School Committee, Flint Hills Christian School, 2004–2005
- Grounds Maintenance, Flint Hills Christian School, Manhattan, KS, 2007–present
- Residential tree cleanup and firewood, 2008–present

Music

- University Christian Church Orchestra, Manhattan, KS. 1999–present
- Manhattan Municipal Band, 2000–2001, 2003
- Hoffmantown Orchestra, Albuquerque, NM. Trumpet player and soloist, 1994–1999. President, 1995–1997, 1999
- Trumpet instructor, Hoffmantown Academy of Music, 1995–1996
- Austin Community Jazz Ensemble. Lead trumpet, 1993–1994
- University of Texas Jazz Ensembles. Lead trumpet, 1991–1993
- Kansas State University Marching Band. Trumpet section leader (1986–1989), first chair trumpet/soloist (1985–1991)
- Kansas State University Men's Basketball Pep Band. First-chair trumpet/section leader, 1984–1991
- Kansas State University Concert Jazz Ensemble. Lead trumpet/soloist, 1984–1991
- Vice President, National Association of Jazz Educators Student Chapter (Kansas State University), 1988.

Church

- University Christian Church, Manhattan, KS. Elder, 2002–2004, 2009–2012
- Chinese Bible Study Meals (2009–present)
- University Christian Church, Manhattan, KS. Periodic classroom instructor, 1999–2000
- Promise Keepers Organization, 1995–1999
- Hoffmantown Church, Albuquerque, NM. Connection class care group leader, 1997–1999
- Hoffmantown Church, Albuquerque, NM. Ordained deacon, 1999

STUDENTS SUPPORTED AS MAJOR PROFESSOR

- Nazmun Khan, Ph.D. EE, December 2025 (GTA). Co-major professor: David Thompson. Dissertation title: *Advancing EEG-Based Brain-Computer Interfaces Through Performance Modeling, Decoding Validation, and Methodological Optimization: a Multi-Study Approach*
- Chase Harvey, Ph.D. EE anticipated December 2026 (GTA). Co-major professor: David Thompson
- Bitar Movazzafgharehbagh, M.S. EE, August 2025 (GTA) Thesis Title: *Integration of Wearable Sensors into Sensory Room Therapies for Children with Developmental Challenges*
- Dong Ren, M.S. EE August 2019 (GTA) Thesis Title: *Simultaneous Pulse Rate Estimation for Two Individuals that Share a Sensor-Laden Bed*
- Charles Carlson, Ph.D. EE May 2019 (NSF GARDE grant; GTA) Dissertation Title: *Development of a Bed-Based Nighttime Monitoring Toolset*
- Kim Fowler, Ph.D. EE December 2018 (GTA)
- Taehee Lee, M.S. EE August 2020 (GTA) Thesis title: *Pen-Type Electrodermal Activity Sensing System for Stress Detection Based on Likelihood Ratios*
- Ethan Grother, M.S. EE May 2017 (KSU JCRC grant; GTA) Thesis Title: *Mobile Device Reference Apps to Monitor and Display Biomedical Information*
- Tianyu Lin, M.S. EE August 2017 (NSF GARDE grant) Thesis Title: *Wireless Biomedical Sensor Network Reference Design Based on the Intel® Edison Platform*

- Phillip Kuehl, M.S. EE 5/2015 (NASA Crew Performance grant) Thesis Title: *Real-Time Processing of Electromyograms in an Automated Hand-Forearm Data Collection and Analysis System*
- Shiwei Luan, M.S. EE 8/2015 (NSF CPS Synergy grant) Thesis Title: *Modularized PCA Pump Design for an ICE-Informed Medical Device Coordination Framework*
- Xiongjie Dong, M.S. EE 8/2014 (NASA EPSCoR grant) Thesis Title: *A ZigBee-Based Wireless Biomedical Sensor Network as a Precursor to an In-Suit System for Monitoring Astronaut State of Health*
- Steve Rubenthaler, M.S. EE, 8/2014 (NSF RAPD grant; GTA) Report Title: *Unobtrusive Ballistocardiography Using an Electromechanical Film to Obtain Physiological Signals from Children with Autism Spectrum Disorder*
- Wen Song, M.S. EE 5/2013 (NASA Crew Performance grant) Thesis title: *Planetary Navigation Activity Recognition Using Wearable Accelerometer Data*
- Dana Gude, M.S. EE 8/2013 (NASA Crew Performance grant) Thesis title: *Automated Hand-Forearm Ergometer Data Acquisition and Analysis System*
- Devon Krenzel, M.S. EE anticipated 5/2013 (Cerner grant)
- Kejia Li, Ph.D. EE 12/2012 (NSF CPS grant) Dissertation title: *Custom Biomedical Sensors for Application in Wireless Body Area Networks and Medical Device Integration Frameworks*. ECE Outstanding GRA 2012
- Kejia Li, M.S. EE 5/2010 (KSU TE grant; GTA) Thesis title: *Wireless Reflectance Pulse Oximeter Design and Photoplethysmographic Signal Processing*
- Kentaro Takamatsu, M.S. EE 8/2009 (KSU Plant Pathology) Thesis title: *System for Greenhouse Climate Monitoring in Three Dimensions*
- William Hitchcock, M.S. EE 12/2008 (KSU TE grant) Report title: *Low-Power Wireless Sensor Network Data Collection*
- Chen Jia, Ph.D. EE 5/2015 (Grain Science; NSF REESE; GTA). ECE Outstanding GTA 2013. Dissertation title: *Higher-Level Learning in an Electrical Engineering Linear Systems Course*.
- Daniel Wilson, M.S. EE anticipated 12/2008 (NSF ITR grant)
- Tim Sobering, Ph.D. EE anticipated 12/2009 (KSU TE grant)
- Lara Pickel, M.S. EE 5/2008 (NIH grant – Vet Med) Thesis title: *Study of the Role of the Angiotensin II (Ang II) Type 2 Receptor (AT2) in Lung Tumorigenesis*
- Seth Hoskins, M.S. EE 5/2011 (NSF ITR grant) Thesis title: *A Physiological Sensor Network Supported by an Inductive Communication Link*
- Angel Martinez, M.S. EE 5/2007 (NSF ITR grant) Thesis title: *Acquisition of Heart Rate and Core Body Temperature in Cattle Using Ingestible Sensors*
- Kevin Smith, M.S. EE 8/2006 (NSF ITR grant) Thesis title: *A Wearable Cattle Health Monitoring System with an Emphasis on Motion-Based Behavior Assessment*
- Scott Schoenig, M.S. EE 12/2007 (NSF ITR grant) Final Project: *A Signal Processing System for Long-Term Acquisition of Heart-Rate in Cattle*
- Keil Regehr, M.S. EE 8/2006 (Anatomy & Physiology NIH grant (external)); Recipient – 2005 Benjamin Zweifach Student Award from the Microcirculatory Society. Thesis title: *Analysis of Ca^{2+} Sparks in the Gerbil Spiral Modiolar Artery*
- Jeffrey Lebak, M.S. EE 12/2005 (NSF CAREER grant; ECE matching funds) Thesis title: *The Design and Implementation of a Standards-Based, Home-to-Remote Health Care Information System*
- Yong Yang, M.S. EE 5/2005 (NSF math assessment grant) Thesis title: *Computer-Based Homework Generation and Assessment for Linear Systems*
- Luke Nagl, M.S. EE 12/2004 (NSF ITR grant) Thesis title: *The Design and Implementation of a Cattle Health Monitoring System*
- Jianchu Yao, Ph.D. EE 8/2005 (NSF CAREER grant) Dissertation title: *Design of Standards-Based Medical Components and a Plug-and-Play Home Health Monitoring System*

- Matthew Martin, M.S. EE 5/2004 (Sandia Monte Carlo grant) Thesis title: *Improved Geometry Representations for Monte Carlo Radiation Transport*
- Ryan Schmitz, M.S. EE 12/2003 (NSF CAREER grant) Thesis title: *Bluetooth Sensor Networking for Distributed Medical Systems*
- G. Edward Barnes, M.S. EE 2/2002 (NSF CAREER grant; startup funds) Thesis title: *Development of a Wearable, Bluetooth-Enabled System for Home Health Care*

STUDENTS SUPPORTED AS A COMMITTEE MEMBER OR CO-ADVISOR

- Taylor Gregory-Sweet, Ph.D. EE anticipated 5/2025. Major professor: David E. Thompson.
- Nooshin Zeinali, Ph.D. EE 8/2024. Major professor: Punit Prakash.
- Allison Adams, M.S. ME 8/2021. Major professor: Amy Betz.
- Nazmun Khan, Ph.D. EE anticipated 5/2025. Major professor: David E. Thompson.
- Md Rakibul Mowla, Ph.D. EE, 5/2020. Major professor: David E. Thompson.
- Farrukh Ali, Ph.D. EE 5/2024. Major professor: William Hsu.
- Ghina Zia, Ph.D. EE 12/2021. Major professor: Punit Prakash.
- Shuvo Islam, Ph.D. CE 8/2019. Major professor: Mustaque Hossain. (External Chair)
Dissertation title: *Implementation of AASHTOWare Pavement ME Design Software for Pavement Rehabilitation*
- Hojjat Fallahi, Ph.D. EE 12/2019. Major professor: Punit Prakash.
- Alaleh Alivar, Ph.D. EE 8/2019. Major professor: Bala Natarajan. Dissertation Title: *Inferencing with Motion Artifact-Bed Based BCG and MRI Case Studies*
- Reza Barazideh, Ph.D. EE 5/2019. Major professor: Bala Natarajan. Dissertation title: *Impulsive Noise Detection and Mitigation in Communication Systems*
- Tra Thi Thanh Huynh, Ph.D. Physics 12/2020. Major professor: Eleanor Sayre.
- Pegah Faridi, Ph.D. EE 8/2020. Major professor: Punit Prakash.
- Dinaz Alaei, Ph.D. Physics 5/2018. Major professor: Eleanor Sayre.
- Nandana Liyanage, Ph.D. Physics 8/2019. Major professor: Eleanor Sayre.
- Mengjiao Wu, Ph.D. Statistics, 5/2019. Major professor: Suzanne Dubnicka. Dissertation title: *Equivalence testing for identity authentication using pulse waves from photoplethysmograph*
- Ahmad Suliman, Ph.D. EE 8/2019. Major professor: David Thompson. Dissertation title: *Quality of Heart Rate Variability Features Obtained from Ballistocardiogram*
- Yubaraj Malakar, Ph.D. Physics 5/2018. Major professor: Artem Rudenko. (External Chair)
- Hao Qian, Ph.D. CIS 12/2015. Major professor: Daniel Andresen. Dissertation Title: *Extending the Battery Life of Mobile Devices by Computation Offloading*
- Mohammed Taj Eldin, Ph.D. EE 5/2015. Major professor: William Kuhn. Dissertation Title: *Wireless Body Area Networks for Intra-Space Suit Communications: Modeling, Measurements and Wearable Antennas*
- Kim Fowler, Ph.D. EE August 2017. Major professor: Stephen Dyer.
- Sam Procter, Ph.D. CIS 8/2016. Major professor: John Hatcliff. Dissertation Title: *A Development Environment for Medical Application Platform Apps*
- Phillip Schumm, Ph.D. EE 12/2013. Major professor: Caterina Scoglio. Dissertation Title: *Characterizing Epidemics in Metapopulation Cattle Systems Through Analytic Models and Estimation Methods for Data-Driven Model Inputs*
- Jordan Bever, M.S. IE 5/2013. Major professor: John Wu.
- Amelia Hodges, M.S. EE 5/2013. Major professor: William Kuhn. Thesis Title: *Investigation of Antennas and Energy Harvesting Methods for Use with a Microtransceiver in a Biosensor Network*
- Timothy Mourlam, M.S. MNE 12/2012. Major professor: Dale Schinstock. Thesis Title: *Developing a Mobile Reduced Gravity Simulator*
- James Strieby, M.S. IE 12/2012. Major professor: Todd Easton. Thesis Title: *Heuristically Optimizing Crane Schedules on Multiple Bays in Conjunction with Continuous Time Simulation*

- Joshua Weese, M.S. CIS 12/2013. Major professor: William Hsu. Thesis Title: *A Convolutional Model for Polyphonic Instrument Identification and Pitch Detection Using Combined Classification*
- Aaron Chavez, Ph.D. CIS 5/2012. Major professor: David Gustafson. (External Chair) Dissertation Title: *Image Classification with Dense Sift Sampling: An Exploration of Optimal Parameters*
- Scott Bell, Ph.D. CIS 8/2014. Major professor: Eugene Vasserman. Dissertation Title: *Low Overhead Methods for Improving Education Capacity and Outcomes in Computer Science*
- Robert Richards, M.S. IE 12/2011. Major professor: John Wu.
- Dehui Hu, Ph.D. Physics 12/2013. Major professor: Sanjay Rebello. Dissertation Title: *Understanding Introductory Students' Application of Integrals in Physics from Multiple Perspectives*
- Yu Jin Kim, Ph.D. CIS 5/2014. Major professor: John Hatcliff.
- Rui Zhang, Ph.D. CIS 5/2015. Major professor: Scott Deloach. Dissertation Title: *A Theory for Understanding and Quantifying Moving Target Defense*
- Nidhi Tare, M.S. EE, 12/2009 (KSU TE grant; GTA) Major professor: Caterina Scoglio. Thesis Title: *A Comparative Performance Analysis of GENI Control Framework Aggregates*
- Dong-Hai Nguyen, Ph.D. Physics 12/2011. Major professor: Sanjay Rebello. Dissertation Title: *Facilitating Students' Application of the Integral and the Area Under the Curve Concepts in Physics Problems*
- Daniel Allegre, M.S. EE 5/2009. Major professor: William Kuhn. Thesis Title: *Reducing Phase and Spurious Tones in Fractional-N Synthesizers*
- Matthew Miller, Ph.D. CIS 12/2010. Major professor: Scott Deloach.
- Rajet Krishnan, M.S. EE 12/2009. Major professor: Balasubramaniam Natarajan.
- Justin Dugger, M.S. CIS 5/2007. Major professor: Daniel Andresen.
- Joseph Lutz, M.S. CIS 5/2007. Major professor: Daniel Andresen.
- Santosh Bejjamshety, M.S. CIS 5/2008 (KSU TE grant) Major professor: Daniel Andresen.
- Scott Harmon, Ph.D. CIS 5/2012. Major professor: Scott Deloach. Dissertation Title: *MASSPEC: Multiagent System Specification Through Policy Exploration and Checking*
- Herve Oyenon, Ph.D. CIS 5/2010. Major professor: Scott Deloach. Dissertation Title: *An Algebraic Framework for Compositional Design of Autonomous and Adaptive Multiagent Systems*
- Oksana Tkachuk, Ph.D. CIS 12/2006. Major professor: John Hatcliff. Dissertation Title: *Domain-Specific Environment Generation for Modular Software Model Checking*
- Pavan Nayini, M.S. CIS, 5/2006. Major professor: Daniel Andresen. Report title: *Applying HL7 Messages for Bovine Data*
- Balakumar Krishnamurthi, M.S. CIS 8/2005 (NSF ITR grant). Major professor: Daniel Andresen.
- Dominic Gelinas, M.S. CIS, 5/2005 (NSF ITR grant). Major professor: Daniel Andresen. Report title: *Bovine Mobile Observation Operation Device Software Engineering Report*
- Shilpa Vaze, Ph.D. EE 12/2007. Major professors: Jim DeVault & Prakash Krishnaswami. Dissertation title: *Integrated Formulation-Solution-Design Scheme for Nonlinear Multidisciplinary Systems in the MIXEDMODELS Platform*
- Daniel Stevens, M.S. EE 12/2004. Major professor: Sanjoy Das (Partial Kinesiology support). Thesis title: *Biologically Inspired Algorithms for DS-CDMA Code Design*
- Colin Christy, Ph.D. EE, 12/2004. Major professor: Stephen Dyer. Dissertation title: *Estimation of Soil Properties Using a Combination of Spectral and Scalar Sensor Data*
- Rajaravi Kollarapu, M.S. CIS 12/2003 (NSF ITR grant). Major professor: Daniel Andresen. Report Title: *Windows CE Based Application for Telemonitoring Cattle*
- Chadd May, M.S. EE, 5/2003. Major professor: Kenneth Carpenter. Thesis title: *A Comparison of Equations of State for Use with Exploding Conductors*

- Jason Wertz, M.S. EE, 5/2003. Major professor: Don Gruenbacher. Thesis title: *Modeling Specialized High Speed Wide Area Networks*
- William Hageman, M.S. EE, 12/2002. Major professor: Andrew Rys. Thesis title: *Electrical Characterization of Aluminum Nitride on Silicon Carbide Metal Insulator Semiconductor Structures*
- Xin He, M.S. EE, 5/2002. Major professor: Stephen Dyer. Thesis title: *A Comparison of Various Methods of Approximation to Real and Complex Data by Rational Functions*
- Nesreen Alsoub, M.S. EE, 12/2001. Major professor: William Kuhn. Report title: *Verilog Design of Digital FM Demodulator*
- Brayden Wilbeck, M.S. EE, 5/2001. Major professor: Kenneth Carpenter. Thesis title: *Design and Implementation of a Graphical User Interface to Construct Input Files for a Hydrodynamics Code*

UNDERGRADUATE STUDENTS SUPPORTED

- Halle Eastling, B.S. BME anticipated May 2028 (Honors Program)
- Bailey Bliss, B.S. BME anticipated May 2028 (Honors Program)
- Damian Garcia, B.S. EE anticipated December 2024 (Developing Scholars Program)
- Dallas Rice, B.S. EE anticipated December 2021 (Developing Scholars Program)
- Sophie Braynock, B.S. BME anticipated May 2022 (KSU ECE U/G Research)
- Kelsey Warren, B.S. BME anticipated May 2022 (KSU ECE U/G Research)
- Kala Keusler, B.S. EE December 2020 (Women in Engineering Program)
- Derrius Washington, B.S. EE anticipated December 2021 (Developing Scholars Program)
- Kendra Schuette, B.S. BAE May 2017 (NSF GARDE Summer Research)
- Martin Ortega, B.S. EE May 2021 (KS-LSAMP RiPS U/G Research)
- Andrew McKittrick, B.S. EE May 2018 (NSF GARDE U/G Research)
- Li Wang, B.S. EE May 2017 (NSF GARDE U/G Research)
- Taishan Li, B.S. EE December 2016 (NSF GARDE U/G Research)
- Shangxian Wang, B.S. EE May 2018 (NSF GARDE U/G Research)
- Kaleb Anderson, B.S. EE 5/2014 (NSF RAPD Independent Study)
- Sarah Carr, B.S. EE 5/2014 (NSF RAPD Independent Study)
- Ethan LaForge, B.S. EE 12/2013 (NSF RAPD Independent Study)
- Ethan Grother, B.S. EE 12/2014 (NSF RAPD Independent Study)
- Yuwei Wang, B.S. EE 12/2014 (NSF RAPD Independent Study)
- Emily Schnell, B.S. EE 5/2014 (NSF RAPD, ECE 773)
- Larissa Hall, B.S. BSE 5/2013 (NSF RAPD, BAE 636)
- Yu He, B.S. BSE 5/2013 (NSF RAPD, BAE 636)
- Dakota Bixler, B.S. EE 5/2014 (NSF RAPD, ECE 773)
- Devin Rood, B.S. EE 5/2014 (NSF RAPD, ECE 773)
- Amanpreet Singh, B.S. EE 5/2014 (NSF RAPD, ECE 773)
- Sergio Ortiz, B.S. EE 5/2016 (Developing Scholars Program)
- Shuo Ouyang, B.S. EE 5/2013 (NCBA; Beef Cattle Institute)
- Tianyu Lin, B.S. EE 5/2014 (Entomology; NCBA; Beef Cattle Institute)
- James Thoennes, B.S. BIOL 12/2014 (Independent Study)
- Neelou Hadavandifard, B.S. EE 12/2014 (Honors Research, NSF RAPD Independent Study)
- Austin White, B.S. EE 5/2016 (Developing Scholars Program; NSF GARDE; ECE 690)
- Hannah Gray, B.S. BAE anticipated 5/2015 (Developing Scholars Program)
- Kevin Garman, B.S. BAE 5/2014 (K-INBRE Scholarship)
- Tommy Nelson, B.S. EE 12/2011 (NASA Crew Performance)
- Jacob Nagely, B.S. BAE 5/2012 (NASA Crew Performance)
- Chenyu Zheng, B.S. EE 12/2012 (NSF CCLI; NASA Crew Performance)
- Xiongjie Dong, B.S. EE 12/2011 (NSF CCLI)

- Ashvin Nagaraja, B.S. EE 5/2010 (NSF CCLI)
- Adam Casada, B.S. EE anticipated 5/2012 (NSF CCLI)
- Kekoa Mendenhall, B.S. EE antic. 5/2013 (NIH Bridges)
- Jim Groening, B.S. EE 5/2010 (Honors research)
- Derek Woodman, B.S. CMPEN 5/2009 (Honors research; NSF ITR REU)
- Neal Dettmer, B.S. EE 12/2007 (Honors research)
- Adriann Sullivan, B.S. EE 5/2010 (Honors research)
- Nicholas Rome, B.S. EE 12/2009 (CSI)
- Paul Edgington, B.S. EE 12/2009 (TE)
- Jeffrey Newlin, B.S. EE, 5/2007 (Honors research)
- Paul Creekmore, B.S. EE, 12/2006 (Honors research)
- David Thompson, B.S. EE 5/2006 (NSF CAREER grant; Galichia Mini-Grant; Honors research)
- Suzanne Newman, B.S. EE 5/2006 (NSF CAREER grant)
- Patrick York, B.S. EE 12/2006 (NSF ITR grant)
- Matthew Spexarth, B.S. EE 5/2006 (National Instruments; Honors research)
- Albert Zhang, B.S. EE 12/2005 (Honors research)
- Roland Craddolph, B.S. CIS 5/2006 (NSF ITR REU grant)
- Federico Lopez-Bertoni, B.S. EE 12/2005 (Honors research)
- Austin Wareing, B.S. EE 5/2005 (NSF CAREER REU grant; Honors research)
- Jonathan Hicks, B.S. EE, B.S. ME, 12/2004 (Honors research)
- Jonathon Noble, B.S. CIS 5/2005 (NSF ITR grant)
- Craig Nelson, B.S. EE 12/2003 (Honors research)
- Scott Schoenig, B.S. EE 12/2003 (NSF CAREER & ITR grants; work study)
- Matthew R. Martin, B.S. EE 5/2003 (Sandia Monte Carlo grant)
- Jeffrey W. Lebak, B.S. EE 5/2003 (NSF CAREER grant)
- John McConeghey, B.S. EE 12/2002 (NSF CAREER grant)
- John Turner, B.S. EE 5/2004 (NSF CAREER grant; work study)
- Benjamin D. Young, B.S. EE 5/2002 (Honors research)
- Shelly A. Allison, B.S. EE 12/2002 (Honors research)
- Matthew D. Ownby, B.S. EE, B.S. ME, 12/2002 (Honors research)

EXPERT WITNESS ENGAGEMENT SUMMARY

Total Engagements: 22

Total Cases: 23

Active Engagements: 13

Depositions: 9

Hearing/Trial Testimonies: 4

Expert Reports and Declarations: 35

Engagements on Behalf of Plaintiffs: 10

Engagements on Behalf of Defendants: 12

EXPERT WITNESS ENGAGEMENTS

<A given engagement was/is on behalf of the company or companies **identified in bold.**>

- Expert Witness, Confidential Matter, November 2025 – present.
- Expert Witness, Confidential Matter, October 2025 – present.
- Expert Witness, *Infusion System Functionality*, Wilmer Cutler Pickering Hale and Dorr LLP (WilmerHale), **Becton, Dickinson and Company, CareFusion 303, Inc., CareFusion Manufacturing, LLC, CareFusion Solutions, LLC, and BD Switzerland S.A.R.L.** v. Baxter International, Inc. and Baxter Healthcare Corporation, Case No. 1:25-cv-00591, Filed May 12, 2025, United States District Court for the District of Delaware, April 2025 – present.
- Expert Witness, Confidential Matter, March 2025 – present.
- Expert Witness, Confidential Matter, March 2025 – present.
- Expert Witness, Confidential Matter, October 2024 – present.
- Expert Witness, *Animal Management System*, Dechert LLP through IMS Expert Services, ST Reproductive Technologies, LLC d/b/a/ STinsight v. **smaXtec Inc. and smaXtec Animal Care GmbH**, Case No. 3:24-cv-00037-JDP, Filed January 18, 2024, United States District Court for the Western District of Wisconsin, October 2024 – present.
- Expert Witness, *Inter Partes Review of U.S. Patent No. 7,969,307*, Perkins Coie LLP through IMS Expert Services, Abbott Laboratories, Inc. and Abbott Diabetes Care, LLC v. **Gentag, Inc. and Altivera, LLC**, Case No. IPR2024-00425, Filed January 16, 2024, Instituted July 22, 2024, United States Patent and Trademark Office, Patent Trial and Appeal Board, August 2024 – present. Deposition – Manhattan, KS, December 19, 2024.
- Expert Witness, Confidential Matter, April 2024 – present.
- Expert Witness, *Smartwatch Wearer Physiological Functions*, Finnegan, Henderson, Farabow, Garrett & Dunner, LLP (Finnegan) through IMS Expert Services, The Research Foundation for the State University of New York, University of Connecticut, and Worcester Polytechnic Institute v. **Samsung Electronics Co., Ltd., Samsung Electronics America, Inc., Samsung Austin Semiconductor, LLC, and Samsung Semiconductor, Inc.**, Case No. 2:23-cv-00141, Filed April 3, 2023, United States District Court for the Eastern District of Texas, Marshall Division, February 2024 – May 2025. Deposition – Washington, D.C., March 5-6, 2025.
- Expert Witness, *Wearable Devices*, Desmarais LLP and Wilmer Cutler Pickering Hale and Dorr LLP (WilmerHale), **Apple, Inc.** v. Masimo Corporation and Sound United, LLC, Case Nos. 1:22-cv-01377-UNA (Filed October 20, 2022; Design Patents) and 1:22-cv-01378-UNA (Filed October 20, 2022; Utility Patents), United States District Court for the District of Delaware, May 2023 – present. Various reports. Deposition – New York City, NY, January 23, 2024. Various reports and declarations. Jury Trial Testimony – Wilmington, DE, October 24, 2024.
- Expert Witness, *Disposable Wireless Sensors*, Perkins Coie LLP through IMS Expert Services, **Gentag, Inc. and Altivera, LLC** v. Abbott Laboratories, Inc. and Abbott Diabetes Care, LLC, Case

No. 1:23-cv-00310, Filed January 19, 2023, United States District Court, Northern District of Illinois, Eastern Division, May 2023 – present.

- Expert Witness, *Percutaneous Hair-Like Sensors*, Through Cahn Litigation, LLC, **Jane Doe v. University of Southern CA**, Case No. 20CV373489, Filed July 12, 2021, Superior Court of the State of California, County of Santa Clara, April 2022 – October 2022.
- Expert Witness, *Oscillometric Blood Pressure Monitor*, Foley & Lardner LLP, The Trustees of Purdue University v. **Omron Corporation, Omron Healthcare Company, Limited, and Omron Healthcare, Inc.**, Case No. 1:20-cv-05443, Filed December 4, 2020, United States District Court for the Northern District of Illinois, Eastern Division, September 2021 – May 2022.
- Expert Witness, *In the Matter of Certain Light-Based Physiological Measurement Devices and Components Thereof*, Wilmer Cutler Pickering Hale and Dorr LLP (WilmerHale), Masimo Corporation and Cercacor Laboratories, Inc. v. **Apple, Inc.**, ITC Inv. No. 337-TA-1276, Filed June 30, 2021, United States International Trade Commission, Washington, D.C., August 2021 – present. Various reports. Deposition – Boston, MA, April 1, 2022. Hearing testimony – Washington, D.C., June 10, 2022.
- Expert Witness, *Peripheral Arterial Tonometry*, Foley & Lardner LLP through IMS Expert Services, **Itamar Medical Ltd.** v. Ectosense nv and Virtuox, Inc., Case No. 0:20-cv-60719-WPD, Filed April 8, 2020, United States District Court, Southern District of Florida, Fort Lauderdale Division, April 2021 – April 2022.
- Expert Witness, *Diabetes Management Technology*, Wilmer Cutler Pickering Hale and Dorr LLP (WilmerHale), **Roche Diabetes Care, Inc.** v. Insulet Corporation, Case No. 1:20-cv-00825-MN, Filed June 17, 2020, United States District Court for the District of Delaware, November 2020 – August 2022.
- Expert Witness, *Wearable Pulse Oximetry*, Wilmer Cutler Pickering Hale and Dorr LLP (WilmerHale), Gibson, Dunn & Crutcher LLP, Masimo Corporation and Cercacor Laboratories, Inc. v. **Apple, Inc.**, Case No. 8:20-cv-00048-DOC-DFM, Filed January 9, 2020, United States District Court for the Central District of California, Southern Division, March 2020 – present. Various reports. Deposition – Boston, MA, December 9-10, 2022. Jury Trial Testimony – Costa Mesa, CA, April 20, 2023. Direct Witness Testimony – October 23, 2024. Bench Trial Testimony – Costa Mesa, CA, November 13, 2024. Deposition – Kansas City, MO, July 19, 2025.
- Expert Witness, *Certain Wearable Monitoring Devices, Systems, and Components Thereof*, Foley & Lardner LLP through Rubin Anders Scientific, **Philips North America LLC f/k/a Philips Electronics North America Corporation and Koninklijke Philips N.V.** v. Fitbit, Inc., Garmin International, Inc., Garmin Ltd. d/b/a Garmin Switzerland, Ingram Micro, Inc., Maintek Computer (Suzhou) Co., Ltd., and Inventec Appliances (Pudong), ITC Inv. No. 337-TA-1190, Filed December 10, 2019, United States International Trade Commission, Washington, D.C., March 2020 – October 2020. Various reports and witness statements. Deposition – August 11, 2020 via Zoom.
- Expert Witness, *Certain Electrochemical Glucose Monitoring Systems and Components Thereof*, Goodwin Procter LLP through Scitemex LLC, Dexcom, Inc. v. **AgaMatrix, Inc.**, ITC Inv. No. 337-TA-1075, Civil Action No. 1:17-cv-01310-UNA, Filed September 18, 2017, United States International Trade Commission, Washington, D.C., January 2018 – June 2018. Various reports. Deposition – Boston, MA, May 10, 2018.
- Expert Witness, *Home Telemedicine Technology*, Merchant & Gould, P.C. through Expert Search Group, LLC, Robert Bosch Healthcare Systems, Inc. v. **CardioCom, LLC and Abbott Diabetes Care**, Civil Action No. 2:13-cv-00349-JRG, M&G No. 12771.0106USZA, Filed April 26, 2013, United States District Court for the Eastern District of Texas, Marshall Division, January 2014 – January 2015.
- Expert Witness, *Pulse Oximetry Signal Processing*, Mayor Brown LLP through IMS Expert Services, Masimo Corporation v. **Philips Electronics North America Corporation and Philips Medizin Systeme Boblingen GMBH**, Civil Action No. 1:09-cv-00080-LPS-MPT, Filed Feb. 3, 2009, United States District Court for the District of Delaware, August 2013 – November 2016. Various reports. Deposition – New York, New York, May 27, 2016.

EXPERT WITNESS REPORTS, DECLARATIONS, AND DIRECT TESTIMONY

- Warren, Steve. “Reply Expert Report of Steven Warren, Ph.D.,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEx), June 20, 2025, 233 pages.
- Warren, Steve. “Rebuttal Expert Report of Steven Warren, Ph.D.,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEx), May 30, 2025, 45 pages plus appendices.
- Warren, Steve. “Opening Expert Report of Steven Warren, Ph.D. Regarding U.S. Patent Nos. 8,457,703 and 10,433,776,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEx), April 11, 2025, 536 pages plus appendices.
- Warren, Steve. “Rebuttal Expert Report of Dr. Steven Warren Regarding Non-Infringement of U.S. Patent Nos. 8,417,326 and 9,986,921,” The Research Foundation for the State University of New York, University of Connecticut, and Worcester Polytechnic Institute v. Samsung Electronics Co., Ltd., Samsung Electronics America, Inc., and Samsung Semiconductor, Inc., U.S District Court for the Eastern District of Texas, Marshall Division, Case No. 2:23-CV-00141-RWS-RSP, February 21, 2025, 113 pages plus exhibits.
- Warren, Steve. “Expert Report of Dr. Steven Warren Regarding Invalidity of U.S. Patent Nos. 8,417,326 and 9,986,921,” The Research Foundation for the State University of New York, University of Connecticut, and Worcester Polytechnic Institute v. Samsung Electronics Co., Ltd., Samsung Electronics America, Inc., and Samsung Semiconductor, Inc., U.S District Court for the Eastern District of Texas, Marshall Division, Case No. 2:23-CV-00141-RWS-RSP, January 31, 2025, 189 pages plus exhibits.
- Warren, Steve. “Direct Testimony of Steven Warren, Ph.D.,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEx), October 23, 2024, 71 pages.
- Warren, Steve. “Declaration of Steven Warren, Ph.D.,” Abbott Laboratories and Abbott Diabetes Care Inc., Petitioner v. GenTag, Inc. and Altivera LLC, Patent Owner., United States Patent and Trademark Office, Patent Trial and Appeal Board, Case No. IPR2024-00425, Patent No. 7,969,307 B2, October 15, 2024, 99 pages.
- Warren, Steve. “Declaration of Steven Warren, Ph.D.,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, February 19, 2024, 512 pages.
- Warren, Steve. “Reply Expert Report of Steven Warren, Ph.D.,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, January 5, 2024, 564 pages.
- Warren, Steve. “Rebuttal Expert Report of Steven Warren, Ph.D.,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, December 15, 2023, 552 pages.
- Warren, Steve. “Expert Report of Steven Warren, Ph.D. Regarding Invalidity of U.S. Patent Nos. 8,190,223, 10,736,507, & 10,984,911,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, November 22, 2023, 1,011 pages.
- Warren, Steve. “Sur-Reply Declaration of Steven Warren, Ph.D. Regarding Claim Construction Brief,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, August 24, 2023, 24 pages.
- Warren, Steve. “Declaration of Steven Warren, Ph.D. Regarding Plaintiff Apple Inc.’s Reply Claim Construction Brief,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, August 17, 2023, 347 pages.
- Warren, Steve. “Declaration of Steven Warren, Ph.D. Regarding Claim Construction,” Apple Inc. v. Masimo Corporation and Sound United, LLC, U.S District Court, District of Delaware, Case Nos. 22-1377-MN-JLH and 22-1378-MN-JLH, August 10, 2023, 171 pages.

- Warren, Steve. “Supplemental Rebuttal Expert Report of Steve Warren, Ph.D. Regarding Certain of Plaintiffs’ Alleged Technical Trade Secrets,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEX), December 2, 2022, 11 pages.
- Warren, Steve. “Rebuttal Expert Report of Steve Warren, Ph.D. Regarding Certain of Plaintiffs’ Alleged Technical Trade Secrets,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEX), November 4, 2022, 395 pages. Appendices: 111 pages.
- Warren, Steve. “Expert Report of Steve Warren, Ph.D., Regarding Certain of Plaintiffs’ Alleged Technical Trade Secrets,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., U.S District Court, Central District of California, Southern Division, Case No. 8:20-cv-00048-JVS (JDEX), September 23, 2022, 442 pages.
- Warren, Steve. “Supplement to Expert Report of Dr. Steven Warren Regarding Invalidity for U.S. Patent Nos. 10,912,501, 10,912,502 and 10,945,648,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, March 25, 2022, 6 pages.
- Warren, Steve. “Rebuttal Expert Report of Dr. Steven Warren Regarding Non-Infringement of U.S. Patent Nos. 10,912,501, 10,912,502 and 10,945,648,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, March 23, 2022, 177 pages.
- Warren, Steve. “Rebuttal Expert Report of Steven Warren, Ph.D. and Majid Sarrafzadeh, Ph.D. Regarding the Domestic Industry Technical Prong for U.S. Patent Nos. 10,687,745, 10,912,501, 10,912,502 and 10,945,648,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, March 23, 2022, 401 pages.
- Warren, Steve. “Amendment to Expert Report of Dr. Steven Warren Regarding Invalidity for U.S. Patent Nos. 10,912,501, 10,912,502 and 10,945,648,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, March 21, 2022, 6 pages.
- Warren, Steve. “Expert Report of Dr. Steven Warren Regarding Invalidity for U.S. Patent Nos. 10,912,501, 10,912,502 and 10,945,648,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, March 4, 2022, 858 pages.
- Warren, Steve. “Rebuttal Expert Claim Construction Report of Steven Warren, Ph.D.,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, January 25, 2022, 14 pages.
- Warren, Steve. “Initial Expert Claim Construction Report of Steven Warren, Ph.D.,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., ITC Inv. No. 337-TA-1276, January 18, 2022, 100 pages.
- Warren, Steve. “Declaration of Dr. Steve Warren in Support of Apple’s Portion of the Joint Stipulation Regarding Plaintiffs’ Motion for Protective Order,” Masimo Corporation and Cercacor Laboratories, Inc. v. Apple, Inc., Case No. 8:20-cv-00048-D0C-DFM, November 9, 2020, 79 pages.
- Warren, Steve. “Rebuttal Witness Statement of Steve Warren,” Philips North America LLC f/k/a Philips Electronics North America Corporation and Koninklijke Philips N.V. v. Fitbit, Inc., Garmin International, Inc., Garmin Ltd. d/b/a Garmin Switzerland, Ingram Micro, Inc., Maintek Computer (Suzhou) Co., Ltd., and Inventec Appliances (Pudong), ITC Inv. No. 337-TA-1190, September 18, 2020, 52 pages.
- Warren, Steve. “Direct Witness Statement of Steve Warren,” Philips North America LLC f/k/a Philips Electronics North America Corporation and Koninklijke Philips N.V. v. Fitbit, Inc., Garmin International, Inc., Garmin Ltd. d/b/a Garmin Switzerland, Ingram Micro, Inc., Maintek Computer (Suzhou) Co., Ltd., and Inventec Appliances (Pudong), ITC Inv. No. 337-TA-1190, September 8, 2020, 74 pages.
- Warren, Steve. “Rebuttal Expert Report of Steve Warren Regarding the Validity of U.S. Patent No. 9,820,698,” Philips North America LLC f/k/a Philips Electronics North America Corporation and Koninklijke Philips N.V. v. Fitbit, Inc., Garmin International, Inc., Garmin Ltd. d/b/a Garmin Switzerland, Ingram Micro, Inc., Maintek Computer (Suzhou) Co., Ltd., and Inventec Appliances (Pudong), ITC Inv. No. 337-TA-1190, August 7, 2020, 86 pages.

- Warren, Steve. “Supplemental Expert Report of Steve Warren Concerning Infringement by Respondents and the Technical Prong of the Domestic Industry Requirement,” Philips North America LLC f/k/a Philips Electronics North America Corporation and Koninklijke Philips N.V. v. Fitbit, Inc., Garmin International, Inc., Garmin Ltd. d/b/a Garmin Switzerland, Ingram Micro, Inc., Maintek Computer (Suzhou) Co., Ltd., and Inventec Appliances (Pudong), ITC Inv. No. 337-TA-1190, July 31, 2020, 42 pages.
- Warren, Steve. “Expert Report of Steve Warren Concerning Infringement by Respondents and the Technical Prong of the Domestic Industry Requirement,” Philips North America LLC f/k/a Philips Electronics North America Corporation and Koninklijke Philips N.V. v. Fitbit, Inc., Garmin International, Inc., Garmin Ltd. d/b/a Garmin Switzerland, Ingram Micro, Inc., Maintek Computer (Suzhou) Co., Ltd., and Inventec Appliances (Pudong), ITC Inv. No. 337-TA-1190, July 22, 2020, 350 pages.
- Warren, Steve. “Expert Report of Steven Warren, Ph.D. Concerning the Technical Prong of Domestic Industry of U.S. Patent Nos. 9,724,045 and 9,750,460,” Dexcom, Inc. v. AgaMatrix, Inc., ITC Inv. No. 337-TA-1075, April 26, 2018, 198 pages.
- Warren, Steve. “Reply Expert Report of Dr. Steve Warren Regarding Masimo’s Expert Report of Joe Kiani Dated April 29, 2016,” Masimo Corporation v. Philips Electronics North America Corporation and Philips Medizin Systeme Böblingen GmbH, May 23, 2016, 11 pages.
- Warren, Steve. “Reply Expert Report of Dr. Steve Warren Regarding the Invalidity of U.S. Patent No. 7,509,154,” Masimo Corporation v. Philips Electronics North America Corporation and Philips Medizin Systeme Böblingen GmbH, April 29, 2016, 48 pages.
- Warren, Steve. “Expert Report of Dr. Steve Warren Regarding the Noninfringement of U.S. Patent No. 7,509,154,” Masimo Corporation v. Philips Electronics North America Corporation and Philips Medizin Systeme Böblingen GmbH, March 28, 2016, 72 pages.
- Warren, Steve. “Expert Report of Dr. Steve Warren Regarding the Invalidity of U.S. Patent No. 7,509,154,” Masimo Corporation v. Philips Electronics North America Corporation and Philips Medizin Systeme Böblingen GmbH, February 12, 2016, 101 pages.

EXHIBIT 6

GREGORY FRANCIS WELCH

Curriculum Vitae — February 6, 2026

CONTACT INFORMATION

University of Central Florida

College of Nursing

12201 Research Parkway

Orlando, FL 32826-3298 (USA)

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4000 Central Florida Boulevard

Orlando, FL 32816-2362 (USA)

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Orlando, FL 32826-3281 (USA)

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PROFESSIONAL EXPERIENCE (SUMMARY)

- 2011–present **University of Central Florida**
 Pegasus Professor (2020)
 AdventHealth Endowed Chair in Healthcare Simulation (2013)
 Professor, College of Nursing, Academic Health Sciences Center (2013)
 Professor, Computer Science, College of Engineering and Computer Science (2011)
 Professor, Institute for Simulation & Training (2011)
 Co-Director, IST Synthetic Reality Lab (2011)
 Faculty, Graduate Certificate in Mixed Reality Engineering (2020)
 Faculty, Modeling & Simulation Graduate Program (2011)
 Faculty, Interactive Computing Experiences Research Cluster (2014)
- 1996–2020 **University of North Carolina at Chapel Hill**
 Adjunct Professor, Computer Science (2012–2020)
 Research Professor (Assistant, Associate), Computer Science (1996–2012)
 UNC Site Coordinator, NSF Graphics and Visualization Center (1996–1998)
- 1990–1992 **Northrop Defense Systems Division**
 Senior Engineer, Airborne Electronic Countermeasures, Digital Systems Group
- 1987–1990 **NASA Jet Propulsion Laboratory (California Institute of Technology)**
 Voyager Spacecraft Project, Flight Command and Data Management Section

EDUCATION

- May 1997 **Ph.D., Computer Science**
 University of North Carolina at Chapel Hill, Chapel Hill, NC
 Under the direction of Gary Bishop
- May 1995 **M.S., Computer Science**
 University of North Carolina at Chapel Hill, Chapel Hill, NC
- May 1986 **B.S. with *Highest Distinction*, Electrical Engineering Technology**
 Purdue University, West Lafayette, IN

PERSONAL AWARDS

- **IEEE Virtual Reality Academy**, IEEE Visualization and Graphics Technical Committee, March 13, 2022 during the 28th IEEE Virtual Reality Conference in Christchurch, New Zealand.
 - **Scroll & Quill Society**, the University of Central Florida, October 1, 2021. The award recognizes “sustained and outstanding achievements in research and/or creative activities” at UCF.
 - **Pegasus Professor**, the University of Central Florida, April 1, 2020. The award recognizes faculty excellence in the three primary areas of academic endeavor: teaching, research and service. There were approximately 50 Pegasus Professors at UCF in 2020.
 - **IEEE Virtual Reality Technical Achievement Award**, IEEE Visualization and Graphics Technical Committee, “in recognition for his contributions to human motion tracking and to mixed reality applications in medicine and training,” March 21, 2018 during the 25th IEEE Virtual Reality Conference in Reutlingen, Germany.
 - **Luminary Award**, University of Central Florida, for “exceptional faculty whose work is advancing their discipline and making a difference,” October 18, 2017.
 - **Research Incentive Award (RIA)**, University of Central Florida, for “outstanding research, scholarly, or creative activity that advances the body of knowledge in a particular field, including interdisciplinary research and collaborations,” March 7, 2017.
 - **Outstanding Performance as Co-Chairman of the IEEE VR 2014 Conference**, IEEE Computer Society, April 3, 2014.
 - **Outstanding Performance as Co-Chairman of the IEEE VR 2013 Conference**, IEEE Computer Society, March 18, 2013.
 - **Rock StAR Award**, 15th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2016), for teaching role in an introductory course on Augmented Reality (AR) for approximately 40 Mexican students interested in AR, September 20, 2016. Mérida, México.
 - **Excellence in Teaching Award**, UNC-Chapel Hill Computer Science Student Association, for 3D Computer Modeling and Animation, Spring 2007.
 - **Outstanding Senior Design Project Award**, “The Easy Chair: A Microprocessor-Controlled Wheelchair for Children With Muscular Disorders,” E.T., Purdue University, 1986
-

JOINT AWARDS

- **Virtual Simulation Innovation Award** to the UCF CON STIM Center, January 13, 2025.
- **Best Paper Award**, “Intuitive user interfaces for real-time magnification in augmented reality,” with Ryan Schubert and Gerd Bruder, 29th ACM Symposium on Virtual Reality Software and Technology (VRST), October 9, 2023.
- **TechConnect Innovation Award**, “Intelligent Object Magnification for Augmented Reality Displays,” with Dr. Gerd Bruder, Zubin Choudhary, and Dr. Kangsoo Kim. 2021 TechConnect Innovation Showcase, October 18-21, 2021, Washington, DC. USA
- **Best Demo Award**, “Real Time Magnification in Augmented Reality,” with Zubin Choudhary, Jesus Ugarte, and Gerd Bruder. ACM Spatial User Interaction 2021, November 9–10, 2021 (Virtual).
- **Best Demo Audience Choice Award**, “Towards Interactive Virtual Dogs as a Pervasive Social Companion in Augmented Reality,” with Nahal Norouzi, Kangsoo Kim, and Gerd Bruder. International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE 2020).

- **Best Long Paper**, “Effects of shared gaze parameters on visual target identification task performance in augmented reality,” with Nahal Norouzi, Austin Erickson, Kangsoo Kim, Ryan Schubert, Joeseoph J. LaViola Jr., Gerd Bruder, in *Proceedings of the ACM Symposium on Spatial User Interaction (SUI)*, page 11, New Orleans, LA USA, October 20 2019. Association of Computing Machinery.
- **Best Paper**, “The physical-virtual table: Exploring the effects of a virtual human’s physical influence on social interaction,” with Myungho Lee, Nahal Norouzi, Gerd Bruder, and Pamela J. Wisniewski. 24th ACM Symposium on Virtual Reality Software and Technology; Tokyo, Japan; December 1, 2018.
- **Best Paper Honorable Mention**, “Blowing in the Wind: Increasing Copresence With a Virtual Human via Airflow Influence in Augmented Reality,” with Kangsoo KIM and Gerd BRUDER. ICAT-EGVE 2018 - International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments; Limassol, Cyprus; November 9 2018.
- **Best Student Paper**, 2017 ACM Symposium on Virtual Reality Software (VRST 2017), “Exploring the effects of observed physicality conflicts on real-virtual human interaction in augmented reality,” by Kangsoo Kim (student advisee), Gerd Bruder, and Greg Welch; Gothenberg, Sweden; November 10, 2017.
- **Long Lasting Impact Paper**, 15th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2016), for a pair of 1998 papers on “Spatial Augmented Reality” (SAR), with co-authors Prof. Ramesh Raskar, Prof. Henry Fuchs, and Deepak Bandyopadhyay; Mérida, México; September 20, 2016.
- **Second Best Paper** at 5th ACM/IEEE International Conference on Distributed Smart Cameras; Ghent, Belgium; August 25, 2011.
- **Best Paper**, ACM Symposium on Virtual Reality Software and Technology 1999 (VRST 99); London, England, United Kingdom; December 22, 1999.

ADVISEE AWARDS (WHEN ADVISEES)

- Jonathan Jules (undergraduate advisee), **National Action Council for Minorities in Engineering (NACME) Scholarship** for 2019–2020, September 30, 2019.
- Salam Daher (Ph.D. advisee), **Collegiate Award** from the National Center for Women in Technology (NCWIT) — four recipients total, May 15, 2018, Grapevine, TX USA.
- Salam Daher (Ph.D. advisee), **RADM Fred Lewis Scholarship Award**, Interservice/Industry Training, Simulation and Education Conference (I/ITSEC), August 5, 2017.
- Salam Daher (Ph.D. advisee), **Link Foundation Fellowship in Modeling, Simulation, and Training** for the 2016–2017 academic year, March 16, 2016.
- Joshua Morton (undergraduate advisee), **Rodney F. Hood Undergraduate Research Award** in support of “Tremor Sensing and Quantification” (November 15, 2010–June 30, 2011).

OTHER HONORS (SELECTED)

- **Keynote Talks** and **Other Invited Talks** as indicated below.
- **Fellow**, Institute of Electrical and Electronics Engineers (IEEE), January 1, 2022.
- **Fellow**, National Academy of Inventors (NAI), December 7, 2021.
- Internationally-recognized **Kalman filter** web site (<http://www.cs.unc.edu/~welch/kalman/>) and associated article “An Introduction to the Kalman Filter,” UNC Technical Report TR95-041, **cited over 12,250 times** according to Google as of 09 MAY 2025.

- **Prof. Rudolph Kalman personally invited Prof. Welch** to attend the National Academy of Engineering ceremony where he was awarded the 2008 Charles Stark Draper Prize for the development of what became known as the **Kalman Filter**.
- **IEEE Technical Expert** for Virtual, Augmented and Mixed Reality, IEEE Public Visibility Initiative, 2018–present.
- **Member**, World Economic Forum (WEF), Metaverse Governance Working Group (2022–present).
- **Member**, World Economic Forum (WEF), Global Future Councils, Council on Virtual Reality and Augmented Reality (2018–2022).
- **Member**, National Council of Architectural Registration Boards (NCARB), Futures Task Force (2017–2019).
- **Certificate of Appreciation** for two-day course on an “Introduction to the Kalman Filter” for Autonomy Incubator Short Course Series (AISCs), NASA Langley Research Center (LaRC), Hampton, Virginia, November 5-6, 2014,
- **Featured exhibit**, Collaborations: Humanities, Arts & Technology (CHAT) Festival, “The Bathysphere: Motion Capture as Art,” with Prof. Francesca Talenti, February 16–20, 2010, UNC-Chapel Hill.

PROFESSIONAL SERVICE

- Professional Associations
 - Fellow, Institute of Electrical and Electronics Engineers (IEEE)
 - Fellow, National Academy of Inventors (NAI)
 - Vice Chair for Conferences, IEEE Visualization and Graphics Technical Community (VGTC) Executive Committee
 - Member, Association for Computing Machinery (ACM)
 - Pioneer, ACM Special Interest Group on Computer Graphics and Interactive Techniques (SIGGRAPH)
 - Member, European Association for Extended Reality (EuroXR)
 - Member, European Association for Computer Graphics (Eurographics)
 - Member, International Virtual Reality Healthcare Association (Founding Member)
 - Member, Society for Simulation in Healthcare (SSIH)
 - Member, International Nursing Association of Clinical Simulation & Learning (INACSL)
 - Member, Southern Nursing Research Society (SNRS)
- Event Organization
 - Dagstuhl Seminar on “Beyond XR” (Seminar 27151), co-organized with Misha Sra (University of California - Santa Barbara, US), Jeanine Stefanucci (University of Utah - Salt Lake City, US), and Frank Steinicke (Universität Hamburg, DE), April 11–16, 2027, The International Conference and Research Center for Computer Science, Dagstuhl, Germany.
 - AIxVR 2027, General Co-Chair with Wolfgang Hürst and Phil Sheu, January–February 2027, Vancouver, Canada.
 - SIBGRAPI 2025, with Gustavo Carneiro, João Paulo Papa, and Manuel M. Oliveira. September 30 – October 3, 2025, Salvador, Bahia, Brazil.
 - IEEE Virtual Reality 2025 (VR 2025), 1st Workshop on XR Accessibility (XRACCESS), with Bobby Bodenheimer, Gerd Bruder, Dylan Fox, Anne-Hélène Olivier, Jeanine Stefanucci, and Daniel Zielasko. March 9, 2025, Saint-Malo, France.

- IEEE Virtual Reality 2025 (VR 2025), 1st International Workshop on Real And Virtual Spaces Influences (REVISI), with Laura Battistel, Kangsoo Kim, Riccardo Parin, Massimiliano Zampini, and Gerd Bruder. March 8, 2025, Saint-Malo, France.
- IEEE Virtual Reality 2024 (VR 2024), Co-General Chair with Carolina Cruz-Neira (University of Central Florida), March 16–21, 2024, Orlando, FL, USA.
- EuroXR 2023 General Co-Chair with Patrick Bourdot (University Paris-Saclay, France) and Benjamin de Witt (Utrecht, The Netherlands), November 28–30, 2023, Rotterdam, The Netherlands.
- Workshop on Inclusive and Generalizable Human Subject Research in XR, ISMAR 2022, with Stevie Carnell, Gerd Bruder, Jeanine Stefanucci, Sun Joo (Grace) Ahn, Valerie Taylor, Tabitha Peck, and Jeremy Bailenson. October 17, 2022, Singapore, Singapore (and virtual).
- Workshop on Considerations for Research Studies in XR, with Nahal Norouzi (University of Central Florida), Advertising in the Metaverse, 2022 American Academy of Advertising Annual Conference, March 24, 2022, St. Petersburg, FL USA.
- IEEE Virtual Reality 2020, Best Paper awards committee, November 22–26, 2020. Atlanta, Georgia, USA. (Held on-line due to Covid-19.)
- Extended Reality for Good (XR4Good) workshop, 18th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2019), co-organized with Arindam Dey (University of Queensland), Mark Billinghurst (University of South Australia), Gun Lee (University of South Australia), and Stephan Lukosch (TU Delft). October 2019, Beijing, China.
- VR 2019 Program Committee Co-Chair, IEEE Virtual Reality 2019 (Osaka, Japan), with Bruce Thomas (University of South Australia in Adelaide, Australia), Torsten Kuhlen (RWTH Aachen University, Aachen, Germany), and Rob Teather (Carleton University, Ottawa, Canada).
- Augmented Reality Visioning Workshop (ARVW 2018), with Gerd Bruder (University of Central Florida) and Peter Squire (Office of Naval Research), July 30–August 3, 2018, Warrenton, VA, USA.
- VR 2018 Program Committee Co-Chair, IEEE Virtual Reality 2018 (Germany), with Kiyoshi Kiyokawa (Osaka University, Japan), Frank Steinicke (University of Hamburg, Germany), and Bruce Thomas (University of South Australia in Adelaide, Australia).
- Virtual and Augmented Reality for Good (VAR4Good) workshop, 25th IEEE Conference on Virtual Reality and 3D User Interfaces (VR 2018), co-organized with Bruce Thomas, Arindam Dey, and Mark Billinghurst (all University of South Australia in Adelaide, Australia), March 18, 2018, Reutlingen, Germany.
- Augmented Reality for Good (AR4Good) workshop, 16th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2017), co-organized with Arindam Dey and Mark Billinghurst (both University of South Australia in Adelaide, Australia), October 9, 2017, Nantes, France.
- Combined 26th International Conference on Artificial Reality and Telexistence (ICAT 2016) and 21st Eurographics Symposium on Virtual Environments (ICAT-EGVE 2016) *Best Paper Award* committee, December 7–9, 2016. Little Rock, Arkansas, USA.
- ACM Symposium on Virtual Reality Software and Technology (VRST 2016) *Best Paper Award* committee, November 2–4, 2016. Munich, Germany.
- 15th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2016), co-organized “AR101,” an introductory course on Augmented Reality (AR) for a group of approx. 40 Mexican students interested in AR. September 20, 2016. Mérida, México.

- 15th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2016), **Best Paper Award** committee, September 19–23, 2016. Mérida, México.
- ISMAR 2016 Outreach Co-Chair, IEEE International Symposium on Mixed and Augmented Reality (ISMAR2016), Mérida, México, September 19–23, 2016.
- Workshop “Human Perception and Psychology in Augmented Reality (HPPAR),” with Bruce H. Thomas (The University of South Australia Mawson Lakes) and James Baumeister (University of South Australia), IEEE International Symposium on Mixed and Augmented Reality (ISMAR2015), September 29, 2015, Fukuoka, Japan.
- Dagstuhl Seminar on “Virtual Realities” (Seminar 13241), co-organized with Guido Brunnett (TU Chemnitz, Germany), Sabine Coquillart (INRIA Rhône-Alpes, France), and Robert van Liere (Center for Mathematics and Computer Science, Netherlands), June 10–14, 2013, The International Conference and Research Center for Computer Science, Dagstuhl, Germany.
- IEEE Virtual Reality 2013 (VR 2013), Co-General Chair with Ben Lok (University of Florida), March 16–23, 2013, Orlando, FL, USA.
- 11th International Symposium on Mixed and Augmented Reality (ISMAR 2012), Co-General Chair with Blair MacIntyre (Georgia Tech), November 5–8, 2012, Atlanta, Georgia, USA.
- Joint Virtual Reality Conference 2011 (JVRC 2011), International Program Committee Co-chair with Sabine Coquillart (INRIA Rhône-Alpes, France) and Anthony Steed (University College of London, UK).
- 10th International Symposium on Mixed and Augmented Reality (ISMAR 2011), Program Committee Co-Chair with Gerhard Reitmayr (Graz University of Technology) and Jun Park (Hongik University).
- VR 2011 Research Demos Co-Chair, IEEE Virtual Reality 2011 (Singapore), with Dirk Reiners (University of Louisiana at Lafayette, USA), Henry Duh (National University of Singapore), and Jiung-Yao Huang (National Taipei University, Taiwan).
- VR 2010 Research Demos Co-Chair, IEEE Virtual Reality 2010 (Waltham, MA, USA), with Yoshifumi Kitamura (Osaka University, Japan), Laura Monroe (Los Alamos National Laboratories, USA), Dirk Reiners (University of Louisiana at Lafayette, USA), and Simon Richir (Laval Virtual, France).
- Dagstuhl Seminar on “Virtual Realities” (Seminar 08231), co-organized with Guido Brunnett (TU Chemnitz, Germany) and Sabine Coquillart (INRIA Rhône-Alpes, France), June 1–6, 2008, The International Conference and Research Center for Computer Science, Dagstuhl, Germany.
- VR 2007 Local Arrangements Co-Chair, IEEE Virtual Reality 2007 (Charlotte, NC, USA), with Zachary Wartell (UNC-Charlotte, USA), Sabarish Babu (UNC-Charlotte, USA), and Regis Kopper (Virginia Tech, USA).
- Workshops/Tutorials Chair, 5th IEEE and ACM International Symposium on Mixed and Augmented Reality (Santa Barbara, CA, USA).
- Co-Chair with Chris Jaynes (University of Kentucky) of IEEE CVPR 2006 International Workshop on Projector-Camera Systems, (ProCams 2006), New York, NY, USA.
- 3DPVT 2006 Organizing Committee, Third International Symposium on 3D Data Processing, Visualization and Transmission (Chapel Hill, NC, USA).
- EDT 2006 Co-chair, IEEE Virtual Reality 2006 International Workshop on Emerging Display Technologies (Alexandria, VA, USA), with Mark Bolas (USC and Fakespace Labs) and Andreas Simon (FH-Aargau).

- ICAT 2005 Program Co-Chair, 15th International Conference on Artificial Reality and Telexistence (Christchurch, New Zealand), with Sang Chul Ahn (KIST, Korea), and Haruo Noma (ATR, Japan).
- ProCams 2005 Posters Chair, IEEE CVPR 2005 International Workshop on Projector-Camera Systems (San Diego, CA, USA).
- EDT 2005 Co-chair, IEEE Virtual Reality 2005 International Workshop on Emerging Display Technologies (Bonn, Germany), with Mark Bolas (USC and Fakespace Labs) and Andreas Simon (Fraunhofer IMK).
- I3D 2001 Registration Chair, ACM Symposium on Interactive 3D Graphics (Chapel Hill, NC, USA).
- Boards and Committees
 - Member, World Economic Forum (WEF), Metaverse Governance Working Group (2022–present).
 - Member, World Economic Forum (WEF), Global Future Councils, Council on Virtual Reality and Augmented Reality (2018–2022)
 - Founding Member, International Virtual Reality Healthcare Association, Advisory Board (2019–present)
 - Emeritus Member, International Symposium for Mixed & Augmented Reality, Steering Committee (2012–2014, Emeritus 2014–present)
 - Member, UCF IST Modeling & Simulation Graduate Program, Advisory Committee (2012–present)
 - Member, UCF Information Technology Resource, Advisory Committee (2014–present)
 - Member, UCF Office of Research, COVID-19 Lab Reopening Task Force (2020–2021)
 - Member, National Council of Architectural Registration Boards (NCARB), Futures Task Force (2017–2019)
 - Senior Advisor, TrakMark Working Group — SIG MR and VR Society of Japan (past)
 - Member, Board of Directors, HiBall Tracker, Inc. (past)
 - Member, UNC Committee on Student Conduct (past)
 - Member, UNC Computer Science Committee on Curriculum and Planning (past)
 - Member, UNC Computer Science Committee on Publications (past)
- Reviewer, Committee, and Editorial Activity
 - Reviewer/panelist, National Science Foundation (NSF), U.S. Department of Energy (DOE), European Research Council, and other funding organizations/units
 - Reviewer, NSF Computing Innovation Fellows
 - Associate Editor, PRESENCE: Virtual and Augmented Reality (2002–present)
 - Associate Editor, Frontiers in Virtual Reality (2014–present)
 - Editorial Board, Virtual Reality & Intelligent Hardware (2019–present)
 - Editorial Board, Journal of Virtual Reality (past)
 - Reviewer, ACM: International Conference on Graphics and Interactive Techniques (SIGGRAPH); Symposium on Virtual Reality Software and Technology (VRST); Symposium on Interactive 3D Graphics; Multimedia; User Interface Software and Technology (UIST); Transactions on Mathematical Software;
 - IEEE: Computer Graphics and Applications (CG&A); International Symposium on Mixed and Augmented Reality (ISMAR) — Area Chair; Virtual Reality (VR); Transactions on Visualization and Computer Graphics; Transactions on Aerospace and Electronic Systems; Conference on Computer Vision and Pattern Recognition (CVPR)

- Guest Editor, Special Issue, Special Section on Pattern Recognition Letters with the Best Papers from SIBGRAPI 2025, Elsevier, 2025.
- International Symposium on 3D Data Processing, Visualization, and Transmission (3DPVT)
- International Symposium on Ubiquitous Virtual Reality (ISUVR)
- IEEE/ACM International Symposium on Wearable Computers
- The combined International Conference on Artificial Reality & Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE)
- EURASIP Journal of Applied Signal Processing; Transactions on Automatic Control
- Journal of Optimal Control Applications and Methods
- Eurographics Workshop on Virtual Environments
- New Immersive Displays for the Near Future (Laval Virtual 2011 Symposium)
- Developer of popular free Mac OS X Mail plugins MailFollowUp and MailRecent
- Testifying intellectual property expert witness / consultant in areas including human-computer interaction, virtual and augmented reality, handheld devices, position and motion estimation, sensor fusion, interactive electronic systems, and related technology.

KEYNOTE/PLENARY TALKS

- SIBGRAPI 2025, Opening Keynote, Salvador, Brazil, October 1, 2025.
- Purdue University, Purdue Polytechnic, Dean's Lecture Series, West Lafayette, IN, April 17, 2025.
- ISTE C Distinguished Lecture, Colorado State University, Fort Collins, CO, November 13, 2023.
- IEEE Life Member Affinity Group Inaugural Meeting, Orlando, FL, December 21, 2022.
- 2nd Workshop on Replication in Extended Reality (WoR XR), ISMAR 2022, Singapore, Singapore, October 17, 2022.
- AI Colloquium, Pusan National University, December 2, 2021.
- Hoag Advances in Clinical Virtual Reality Symposium, October 1, 2021.
- Training XR: Workshop on 3D Content Creation for Simulated Training in eXtended Reality, IEEE Virtual Reality 2020, on line (originally Atlanta, GA USA), March 22, 2020.
- ChinaVR 2019, Shenzhen, China, November 24, 2019.
- 6th ACM Symposium on Spatial User Interaction, Berlin, Germany, October 13–14, 2018.
- New Knowledge and Innovations, Sarasota Memorial Health Care System, Sarasota, Florida, May 4, 2018.
- International Conference on Computer-Aided Design and Computer Graphics (CAD/Graphics), Zhangjiajie, China, August 26, 2017.
- Society for Design and Process Science conference (SDPS), Orlando, Florida, December 5, 2016.
- Annual Meeting of the Association for Information Science and Technology (ASIS&T), Copenhagen, Denmark, October 16, 2016.
- 22nd Annual Research Day, Sigma Theta Tau International Honor Society of Nursing, Theta Epsilon chapter, Winter Park, Florida, USA, April 8, 2014.
- Society for Computer Science, Workshop on Virtual Reality-Augmented Reality (GI VR-AR 2013), Würzburg, Germany, September 19, 2013. .
- International Symposium on Ubiquitous Virtual Reality (ISUVR), Daejeon, South Korea, August 23, 2012.

- Joint Virtual Reality Conference (JVRC), Fellbach, Germany, September 29, 2010.
- Winter Augmented Reality Meeting (WARM), Graz, Austria, February 24–25, 2010.
- Real Action, Virtual Environments (RAVE), Barcelona, Spain, March 4, 2009.
- Digital City Monitoring and Emergency Management (D2D), Shenzhen, China, October 25, 2008.
- Eurographics Symposium on Virtual Environments (EGVE), Eindhoven, The Netherlands, May 29, 2008.

OTHER INVITED TALKS (SELECTED)

- “Participant Data Considerations for the Virtual Experience Research Accelerator (VERA),” University of Maryland, Department of Computer Science, Guest Lecture, May 9, 2024.
- “The Virtual Experience Research Accelerator (VERA),” ISTeC Distinguished Lecture, Colorado State University, Fort Collins, CO, November 13, 2023.
- “Universal User Data (UUD) for Virtual Experience Research,” Computer Science Department, Colorado State University, Fort Collins, CO, November 13, 2023.
- “The Virtual Experience Research Accelerator (VERA) and Universal User Data (UUD),” the 22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 3rd Workshop on Replication in Extended Reality (WoR XR), Sydney, Australia, October 16, 2023.
- “The Virtual Experience Research Accelerator (VERA) and Universal User Data (UUD),” the 22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2nd Workshop on Inclusion, Diversity, Equity, Accessibility, Transparency, and Ethics in XR (IDEA-TE XR), Sydney, Australia, October 20, 2023.
- “The Virtual Experience Research Accelerator (VERA),” The 5th XR Access Symposium, New York City, USA, June 15, 2023.
- “Embodied Agents for Simulating, Comforting, and Assisting Humans,” 7th Annual IVRHA Virtual Reality and Healthcare Global Symposium, University of Pennsylvania Medicine, Philadelphia, PA. March 4, 2023.
- “How to XR: Hardware & Technology,” World Economic Forum (WEF), Knowledge Lab with the Global Future Council on AR/VR 2021, April 29, 2021.
- “Personal Agents for Workplace (Nursing) Skills and Emotional Support,” HCI and the Future of Work and Wellbeing: A Series of Conversations. May 7, 2020, online (Longwood, FL USA).
- “Physical-Virtual Training for Healthcare,” International Women’s Club of Gothenburg, May 26, 2020, Gothenburg, Sweden (via video during pandemic).
- “Aware and Influential Virtual Humans and Animals,” Naval Postgraduate School, MOVES lecture, April 13, 2020, Monterey, CA (via video during pandemic).
- “Vera Real: Stroke assessment using a Physical Virtual Patient (PVP)” with Laura Gonzalez and Salam Daher (both UCF), 2019 Conference of the International Nursing Association for Clinical Simulation and Learning (INACSL), Phoenix, AZ, June 21, 2019.
- “Enhanced Perception & Cognition in Augmented Reality (EPiC AR),” Naval Postgraduate School, MOVES Open House, May 23, 2019, Monterey, CA.
- “Identifying User Physical States,” XR Advance Webinar Series, Iowa State University, April 22, 2019. Lecture series was a follow-up to a June 2017 VR/AR Workshop sponsored by NSF.
- “Physical-Virtual Patients: Human Shape, Appearance, Awareness, and Influence,” Virtual Reality & Health, Tucson, AZ, March 7, 2019.

- “Human Surrogates: Real, Virtual, and Augmented Shape, Appearance, and Intelligence,” Rubin Center for Healthy Aging, Sarasota, Florida, March 3, 2019.
- “Awareness and Influence of Human Surrogates in Augmented Reality,” 2017 International Conference on Computer-Aided Design and Computer Graphics (CAD/Graphics), Zhangjiajie, China, August 26, 2017.
- “Bridging the Telepresence Valley,” 2016 Annual Meeting of the Association for Information Science and Technology (ASIS&T), Copenhagen, Denmark, October 16, 2016.
- “Human Surrogates and their Effects on Humans,” Naval Postgraduate School, MOVES Institute, Simulation and Training Course, Monterey, CA, September 8, 2016.
- “Human Surrogates and their Effects on Humans,” Technische Universität München (TU Munich), Garching b. Munich, Germany, June 28, 2016.
- “Interactive Rear-Projection Physical-Virtual Patient Simulators,” Mini-Lecture at NextMed/MMVR22 (Medicine Meets Virtual Reality), Los Angeles, CA, April 7, 2016.
- “Making the Virtual Real: Virtual Reality, Augmented Reality, Physical-Virtual Reality,” Examination Development & Research committee of the National Council of Architectural Registration Boards (NCARB), Phoenix, AZ, January 30, 2016.
- “My patient is a dummy — so why should I care?” classroom lecture, Rollins College (Winter Park, FL), hosted by Darin Hughes, November 11, 2015.
- “Making the Virtual Real: Physical Manifestations of Dynamic Virtual People, Objects, and Places,” Qualcomm/Vienna University of Technology (Vienna, Austria), September 15, 2014.
- “Making the Virtual Real: Physical Manifestations of Dynamic Virtual People, Objects, and Places,” Graz University of Technology (Graz, Austria), September 17, 2014.
- “Past and Current Research in Modeling and Reproducing Humans for Warfighter Training,” Orlando VA Medical Center (Orlando, FL), August 26, 2014.
- “Making the Virtual Real: Physical Manifestations of Dynamic Virtual People, Objects, and Places,” Harris Corporation, Melbourne, FL, February 18, 2014.
- “Face-to-Face: Interactive Simulation Using a Mobile Physical-Virtual Patient,” work with Steven Talbert, Mary Lou Sole, and Karen J. Aroian, SNRS 28th Annual Conference, San Antonio, TX, February 15, 2014.
- “Physical-Virtual Humans for Healthcare,” the University of Texas at San Antonio, San Antonio, TX, February 13, 2014.
- “Physical-Virtual Humans for Training and Teleportation,” Max Planck Institute, Tübingen, Germany, September 17, 2013, and Bonn-Rhein-Sieg University of Applied Sciences, Bonn, Germany, September 20, 2013.
- “Interactive Shader Lamps: Projector-Based Graphics in Motion,” GI VR-AR (Würzburg, Germany), September 19, 2013.
- “Physical-Virtual Humans for Training and Teleportation,” TU Chemnitz (Chemnitz, Germany), June 6, 2013.
- “Physical-Virtual Humans: Challenges and Opportunities,” KIST (Seoul, South Korea) and KAIST (ISUVR 2012, Daejeon, South Korea), August 22–23, 2012.
- “Physical Manifestations of Virtual Humans,” Technische Universität München (TU Munich); Garching b. Munich, Germany; April 11, 2011.
- “Crossing Realities: The Convergence of Physical and Virtual Worlds,” Microsoft Research; Redmond, WA; May 25, 2010.
- “Crossing Realities: An Overview of Related Work at UNC-Chapel Hill,” Joint Virtual Reality Conference (JVRC); Lyon, France; December 7, 2009.
- “Virtual Artifacts and Experiences in Real Environments,” second RAVE 2009 workshop (Real Action, Virtual Environments); Barcelona, Spain; March 4, 2009.

- “A Living, Breathing, Dynamic Digital City,” D2D 2008 workshop (Digital City Monitoring and Emergency Management); Shenzhen, China; October 25, 2008.
- “Immersive Display Research at UNC-CH: A Sampling,” with Henry Fuchs; Max Planck Institute, Tübingen, Germany; June 9, 2008..
- “Motion Tracking as an Epic Battle Between Information and Uncertainty,” 14th Eurographics Symposium on Virtual Environments (EGVE 2008); Eindhoven, The Netherlands; May 29, 2008..
- “Improving, Expanding and Extending 3D Telepresence,” International Workshop on Advanced Information Processing for Ubiquitous Networks, ICAT 2005; Christchurch, New Zealand; December 8, 2005..
- “3D Telepresence for Off-Line Surgical Training and On-Line Remote Consultation,” 2nd CREST Symposium on Telecommunication, Teleimmersion and Telexistence; University of Tokyo; December 9–10, 2004..
- “Immersive Telepresence for Surgical Teaching and Remote Consultation: Extending Medical Expertise Over Time and Space,” University of Florida, Department of Computer and Information Science and Engineering; “Geometry, Graphics, Vision, Visualization/ Simulation (G2V2)” seminar series; October 17, 2003. .
- “Sensor Information Efficiency for Pose Estimation,” University of Rochester, Department of Computer Science; April 13, 2001..
- “Inside the HiBall Tracking System,” Siemens Corporate Research, Imaging and Visualization Department; Princeton, NJ; October 25, 2000..
- “Tracking for Interactive Computer Graphics: To the Hallway and Beyond,” the Boeing Corporation, Virtual Reality Group, Seattle, WA, April 21, 1999. (Also presented at Microsoft Research; Seattle, WA; April 21, 1999.).
- “Spatially-Augmented Visualization,” Walt Disney Imagineering, Virtual Reality Studio; Burbank, CA; September 1998..

INVITED WORKSHOP PARTICIPATION

- The 22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 3rd Workshop on Replication in Extended Reality (WoRXR), Sydney, Australia, October 16, 2023.
- The 22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2nd Workshop on Inclusion, Diversity, Equity, Accessibility, Transparency, and Ethics in XR (IDEATEXR), Sydney, Australia, October 20, 2023.
- ACM SIGGRAPH Frontiers Workshop: Digital Avatars: Risks, Harms, Barriers, Opportunities, with Eakta Jain (Co-Organizer), Apu Kapadia (Co-Organizer), Theodore Kim, and Taslima Akter. August 6, 2023, Los Angeles, CA.
- Anticipating the Metaverse: The Future Of Mixed Reality, invitation-only Workshop at the University of Southern California Institute for Creative Technologies, Los Angeles, CA, March 24, 2023.
- ACM SIGGRAPH Frontiers Workshop: Content Generation for Workforce Training, with Holly Rushmeier (Organizer), Carolina Cruz-Neira, Eakta Jain, Rajesh Jha, Jerome Solomon, and Louise Yarnall. July 28, 2019, Los Angeles, CA.
- National Science Foundation (NSF) Principal Investigator workshop on the Future of Work at the Human-Technology Frontier (FW-HTF), “Panel: Augmented Reality in the Future of Work,” with Joe Gabbard (moderator), Virginia Tech; Doug Bowman, Virginia Tech; Steve Feiner, Columbia University; and Henry Fuchs, University of North Carolina at Chapel Hill. April 5, 2019, NSF Headquarters (Alexandria, VA).

- Computing Community Consortium (CCC) Visioning Workshop: Content Generation for Workforce Training, March 14–15, 2019, Atlanta, GA.
- Computing Community Consortium (CCC) Computing Research Symposium: Addressing National Priorities and Societal Needs 2017, October 23–24, 2017, Washington, DC.
- National Science Foundation (NSF) Virtual Reality (VR)/Augmented Reality (AR) Workshop, July 17–18, 2017, Washington, DC.
- Computing Community Consortium (CCC) Visioning Workshop on Discovery & Innovation in Smart & Pervasive Health, December 5–6, 2016, Washington, DC.
- National Research Council (NRC) Mapping Science Committee meeting, October 3, 2012, National Academies Keck Center, Washington, DC, USA.
- Computing Community Consortium (CCC) Visioning Workshop on Spatial Computing, September 10–11, 2012, National Academies Keck Center, Washington, DC, USA.
- IMMERSCOM 2009 (2nd International Conference on Immersive Telecommunications), Panel with Jaron Lanier (Microsoft), Randy Harrell (CISCO), Frantz Lohier (Logitech), William C. Wickes (Hewlett Packard), and Zhengyou Zhang (Microsoft), May 29, 2009.

PANEL PARTICIPATION

- **Greg Welch**, Mindi Anderson, Desiree Díaz, Frank Guido-Sanz, and Elizabeth Wells-Beede, “Patents for Healthcare Simulation Technology: A Multidisciplinary Perspective,” the 2024 International Meeting for Simulation in Healthcare, San Diego, CA, January 23, 2024.
- **Greg Welch**, Jeni Paay, Tim Chen, Bruce Thomas, Kim Vincs, Mark Billingham, and Aaron Quigley. “Why is XR still so hard to use?” The 22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2023), October 16, 2023, Sydney, Australia.
- **Greg Welch**, Tabitha Peck, Jonathan Beever, and Christine Hemphill. “The Virtual Experience Research Accelerator (VERA),” The 5th XR Access Symposium, New York City, USA, June 15, 2023.
- **Greg Welch**, Nahal Norouzii, Rosario Casas. “Approaching the Metaverse — An Overview & the State of the Art,” Advertising in the Metaverse, 2022 American Academy of Advertising Annual Conference, March 24, 2022, St. Petersburg, FL USA.
- **Greg Welch**, Salam Daher, Laura Gonzalez, Jacob Stuart. “Workshop on Collaborations with Domain Experts for Virtual Agent Research (CoVAR),” Intelligent Virtual Agents (IVA 2020), October 18, 2020, online workshop (Longwood, FL USA).
- **Greg Welch**, Laura Gonzalez, Juan Cendan, Mindi Anderson. “Simulating Patients for Healthcare Training: Appearance, Shape, Influence, and Awareness,” Serious Play Conference, Healthcare Track, July 25, 2019, Orlando, FL USA.
- Joe Gabbard (Virginia Tech), Doug Bowman (Virginia Tech), Steve Feiner (Columbia University), Henry Fuchs (University of North Carolina at Chapel Hill), and **Greg Welch** (UCF). “Augmented Reality in the Future of Work,” National Science Foundation (NSF) Principal Investigator workshop on the Future of Work at the Human-Technology Frontier (FW-HTF), April 5, 2019, NSF Headquarters (Alexandria, VA).
- **Greg Welch** (UCF), Albert “Skip” Rizzo (University of Southern California), Maria Sanchez-Vives (Institute of Biomedical Research August Pi i Sunyer), and Petr Legkov (University of Osnabrück), “The Future Impact of Neuroscience and Cognitive Psychology on Virtual Environments,” IEEE Conference on Virtual Reality and 3D User Interfaces (VR 2018), Reutlingen, Germany.
- Anthony Steed (University College London), Doron Friedman (IDC Herzliya), **Greg Welch** (UCF), Susumu Tachi (The University of Tokyo), and moderator Mel Slater (University of

- Barcelona). “Instantaneous Beaming to Distance Places — A Possible and Desirable Future?” IEEE Virtual Reality 2017, March 21, Los Angeles, CA USA.
- **Greg Welch** (UCF), Charles Hughes (UCF), Linda Gibson-Young (UCF), Chait Renduchintala (UCF), Peter Kincaid (UCF), Sarah Matthews (FL DOH), Tom Herring (FL DOH), “Technology & Public Health Use Now and in the Future,” 2014 Florida Public Health Educational Meeting, Orlando, FL, August 1, 2014.
 - Betty Mohler (Max Planck Institute for Biological Cybernetics), **Greg Welch** (UNC-CH), Daniel Thalmann (Nanyang Technological University and EPFL), and Victoria Interrante (University of Minnesota), “Avatars in Virtual Environments,” IEEE Virtual Reality 2011 (VR 2011), Singapore, March 21, 2011.
 - Randy Harrell (CISCO), Jaron Lanier (Microsoft), Frantz Lohier (Logitech), **Greg Welch** (UNC-CH), William C. Wickes (Hewlett Packard), and Zhengyou Zhang (Microsoft Research), IMMERSCOM 2009 (2nd International Conference on Immersive Telecommunications), May 29, 2009.
 - Henry Fuchs, Bernd Fröhlich (Bauhaus-Universität Weimar), and **Greg Welch**, “Display Technology,” 14th Eurographics Symposium on Virtual Environments (EGVE 2008), Eindhoven, The Netherlands, May 30, 2008.
 - Barbara Hayes-Roth, Austin Henderson, Ramesh Jain, Lev Manovich, **Greg Welch**, and Gopal Pingali, “Experiential Telepresence: How Can Telepresence Research be Guided Towards Better End User Experience?” ACM SIGMM 2003 Workshop on Experiential Telepresence 2003 (ETP 2003), November 7, 2003, Berkeley, CA USA.

GRADUATED PH.D. ADVISEES

- Austin Erickson, 2023: Understanding, Modeling, and Simulating the Discrepancy Between Intended and Perceived Image Appearance On Optical See-Through Augmented Reality Displays
- Nahal Norouzi, 2021: The Social and Behavioral Influences of Interactions with Virtual Dogs as Embodied Agents in Augmented and Virtual Reality
- Jason Hochreiter, 2019: Multi-Touch Detection and Semantic Response on Non-Parametric Rear-Projection Surfaces
- Myungho Lee, 2019: Mediated Physicality: Inducing Illusory Physicality of Virtual Humans Via Their Interactions With Physical Objects
- Salam Daher, 2018: Physical-Virtual Patient Simulators: Bringing Tangible Humanity to Simulated Patients
- Kangsoo Kim, 2018: Environmental Physical-Virtual Interactivity to Improve Social Presence With a Virtual Human in Mixed Reality
- Peter Lincoln, 2017 (UNC Chapel Hill, with Henry Fuchs): Low Latency Displays for Augmented Reality
- Christoph Resch, 2016 (TU München, with Gudrun Klinker): Enhancing Projective Spatial Augmented Reality in Industry: A Model Based Approach for Registration and Calibration
- Feng Zheng, 2015: Spatio-Temporal Registration in Augmented Reality
- Jinghe Zhang, 2013: Uncertainty-Driven Adaptive Estimation with Applications in Electrical Power Systems
- Peter Keitler, 2011 (TU München, with Gudrun Klinker): Management of Tracking and Tracking Accuracy in Industrial Augmented Reality Environments
- Adrian Ilie, 2010: On-Line Control of Active Camera Networks

- Hua Yang, 2008: Differential Tracking Through Sampling and Linearizing the Local Appearance Manifold
- B. Danette Allen: 2007, Hardware Design Optimization for Human Motion Tracking Systems
- Aditi Majumder, 2003: A Practical Framework To Achieve Perceptually Seamless Multi-Projector Displays
- Ruigang Yang, 2003: View-Dependent Pixel Coloring — A Physically-Based Approach for 2D View Synthesis
- Ramesh Raskar, 2001: Projector-Based Three Dimensional Graphics

SUPERVISED POSTDOCTORAL SCHOLARS

- Dr. Nahal Norouzi, University of Central Florida, January 2022–May 2022.
- Dr. Kangsoo Kim, University of Central Florida, January 2019–December 2020.
- Dr. Salam Daher, University of Central Florida, January 2019–July 2019.
- Dr. Arjun Nagendran, University of Central Florida, August 2011–January 2013, with Dr. Charles Hughes (primary advisor/mentor).

GRADUATED M.S. AND B.S. HONORS THESIS ADVISEES

- Camila Acevedo Carillo, 2023 (UCF B.S. Computer Science, Advisor)
- Angelica Garcia, 2017 (UCF M.S. Modeling & Simulation, Advisor)
- Justin Grace, 2016: Recognizing Pain Using Novel Simulation Technology (UCF B.S.N. Honors in the Major, Committee Member)
- Krishna Kollu, 2012: Modeling the Neural Pathology of Parkinson’s Disease (UNC B.S. C.S. Honors, Advisor)
- Caitlyn Losee, 2011: The Bathysphere: Motion Capture and Immersive Projection (UNC B.S. C.S. Honors, Advisor)

STUDENT ADVISEES (ALL)

Ahinya Alwin (current); Armin Delmo (current); Jasmine Joyce DeGuzman (current); Corey Clements (current); Lam Nguyen (current); Stephanie Liebel (current); Hiroshi Furuya (current); Zubin Choudhary (current); Matthew Gottsacker (current); Tiffany Losekamp (current, College of Nursing, with Mindi Anderson); Akhil Pavan Sai Machavaram (2024, MS CS); Nicholas Piazza (2024); Juanita Benjamin (2023); Austin Erickson (2023, Ph.D.); Camila Acevedo Carillo (2023); Nahal Norouzi (2021, Ph.D.); Faith Sauber (2021); Indira Avendano (2021); Nafisa Mostofa (2021); Alyssa Feagans (2020); Jonathan Jules (2020); Alexis Lambert (2020); Yazdan Jamshidi (2019, M.S.); Jason Hochreiter (2019, Ph.D.); Brandon Belna (2018); Ryan Schubert (2018); Salam Daher (2018, Ph.D.); Kangsoo Kim (2018, Ph.D.); Stephanie Mutter (JUN–AUG 2018, NSF REU); Brandon Belna (JUN–AUG 2017, NSF REU); Peter Lincoln (2017, Ph.D., with Henry Fuchs); Christoph Resch (2016, Ph.D., TU München, with Gudrun Klinker); Kiran Pandit (JUN–AUG 2016, NSF REU); Feng Zheng (2015, Ph.D.); Ryan Patrick (2013, M.S.); Jinghe Zhang (2013, Ph.D.); Krishna Kollu (2012, B.S. Undergraduate Honors); Caitlyn Losee (2011, B.S. Undergraduate Honors); Peter Keitler (2011, Ph.D., TU München, with Gudrun Klinker); Adrian Ilie (2010, Ph.D.); Tianren Wang (2009–2011); Hua Yang (2008,

Ph.D.); Danette Allen (2007, Ph.D.); Vincent Noel (2006, M.S.); Michael Noland (2006, M.S.); Aditi Majumder (2003, Ph.D.); Ruigang Yang (2003, Ph.D.); and Ramesh Raskar (2001, Ph.D.).

STUDENT COMMITTEES

Samuel Cosgrove (current, M.S., Advisor Joe LaViola); Xinyu Hu (current, Ph.D., Advisor Ryan McMahan); Kyle Martin (current, Ph.D., Advisor Joe LaViola); Alec Moore (2022, Ph.D., Advisor Ryan McMahan); Kevin Pfeil (2022, Ph.D., Advisor Joe LaViola); Andres Vargas (2022, Ph.D., Advisor Joe LaViola); Sarah Braeger (2021, Ph.D., Advisor Hassan Foroosh); Corey Pittman (2021, Ph.D., Advisor Joe LaViola); Min Wang (2020, Ph.D., Advisor Hassan Foroosh); Alex Blate (2019, Ph.D., UNC, Advisor Henry Fuchs); Baoyuan Liu (2016, Ph.D., Amazon, Advisor Hassan Foroosh); Rick Skarbez (2016, Ph.D., UNC, Advisor Frederick P. Brooks and Mary Whitton); Alexandra Carson (2016, BSN Honors in the Major); Justin Grace (2016, BSN Honors in the Major, Advisor Kelly Allred); Monika Schaeffer (2011, M.S., Duke University, Advisor Ron Parr); Brian Clipp (2010, Ph.D., UNC, Advisor Marc Pollefeys); David Gallup (2010, Ph.D., UNC, Advisor Marc Pollefeys); Andrew Nashel (2010, Ph.D., UNC, Advisor Henry Fuchs); Tyler Johnson (2009, Ph.D., UNC, Advisor Henry Fuchs); Seon Joo Kim (2008, Ph.D., UNC, Advisor Marc Pollefeys); Philip McLeod (2008, Ph.D., University of Otago, Advisor Geoff Wyvill); Patrick Quirk (2006, M.S., UNC, Advisor Henry Fuchs); Kok-Lim Low (2005, Ph.D., UNC, Advisor Anselmo Lastra); Lindsey McCarthy (2011, M.S. UNC Dentistry, Advisor Lucia Cevdanes); Michael Rosenthal (2005, Ph.D., UNC, Advisor Henry Fuchs); Benjamin Lok (2002, Ph.D., UNC, Advisor Frederick Brooks); Nicholas Vallidis (2002, Ph.D., UNC, Advisor Gary Bishop); Gopi Meenakshisundaram (2001, Ph.D., UNC, Advisor Dinesh Manocha); Mark Livingston (1998, Ph.D., UNC, Advisor Henry Fuchs); and Hans Weber (UNC, Advisor Gary Bishop).

PH.D. THESIS EXAMINER (INTERNATIONAL)

- Ylva Hansdotter, “The Affordances of Immersive Virtual Reality for Stimulating Prosocial Behaviour: A Mixed-Methods Pro-Environmental Intervention Study,” University College Dublin; Advisor: Prof. Lizbeth Goodman; 2022.
 - Susanne Schmidt, “Blended Spaces: Perception and Interaction in Projection-Based Spatial Augmented Reality Environments,” Universität Hamburg; Advisor: Dr. Simone Frintrop; 2020.
 - Iana Podkosova, “Walkable Multi-User VR: The Effects of Physical and Virtual Colocation,” Technische Universität Wien; Advisor: Dr. techn. Hannes Kaufmann, Priv.-Doz.; Second advisor: Dr. techn. Peter Ferschin, Ass. Prof; 2019.
 - Michael Robert Marner, “Physical-Virtual Tools for Interactive Spatial Augmented Reality,” The University of South Australia; Principal Supervisor: Prof. Bruce Hunter Thomas; Associate Supervisor Prof. Christian Sandor; 2013.
 - Philip Grant McLeod, “Fast, Accurate Pitch Detection Tools for Music Analysis,” the University of Otago; 2008.
-

CLASSROOM TEACHING

- *Augmented, Virtual, and Mixed Realities in Healthcare*, UCF, 2025 (Spring), with Mindi Anderson.
- *3D Computer Modeling and Animation*, First Year Seminar, UNC-CH, 2003–2007. Created the course, which was chosen by UNC from among competing First Year Seminar proposals. Awarded Computer Science Student Association *Excellence in Teaching* award, Spring 2007
- *Introduction to Programming (Java)*, UNC-CH, 2002–2003
- *Team Software Engineering*, UNC-CH, 2001 (Spring)
- *Exploring Virtual Worlds*, UNC-CH, 1997–1998, with Henry Fuchs.
- NSF STC Center-Wide Lecture Series, coordinated within the Center, 1997
- *Computers: Power Tools for the Mind*, UNC-CH, 1995

TUTORIALS/SHORT COURSES

- AR101 (An Introduction to Augmented Reality), 15th IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2016), September 18, 2016. Mérida, México. **Greg Welch** with Mark Billinghurst, Tobias Höllerer, Dieter Schmalstieg, Gerhard Reitmayr, Kiyoshi Kiyokawa, and Steve Feiner.
- **Greg Welch**. “An Introduction to the Kalman Filter,” NASA Langley Research Center, National Institute of Aerospace, Hampton, VA USA, November 5–6, 2014.
- **Greg Welch**. “Tracking: Beyond 15 Minutes of Thought” and “An Introduction to the Kalman Filter,” University of Würzburg, Interactive Intelligent Spaces Summer School, Würzburg, Germany, September 13–14, 2013.
- **Greg Welch** and Gary Bishop. “An Introduction to the Kalman Filter,” ACM SIGGRAPH 2001 tutorial, August 12, 2001, Los Angeles, CA USA.
- **Greg Welch**, Gerhard Reitmayr (Graz Univ. of Tech.), Vincent Lepetit (Swiss Federal Institute of Tech.), and Brian Clipp (UNC-Chapel Hill), “Tracking for AR Tracking Researchers,” International Symposium on Mixed and Augmented Reality (ISMAR 2009), October 19, 2009.
- **Greg Welch** and Gary Bishop. “An Introduction to the Kalman Filter,” ACM SIGGRAPH 2001 tutorial, August 12, 2001, Los Angeles, CA USA.
- B. Danette Allen, Gary Bishop and **Greg Welch**. “Tracking: Beyond 15 Minutes of Thought,” ACM SIGGRAPH 2001 short course, August 12, 2001, Los Angeles, CA USA.
- **Greg Welch**. “An Introduction to the Kalman Filter,” multi-day seminars offered periodically at UNC Chapel Hill, 1996–2009.

PATENTS

- US 12,464,309, “Spatially Explicit Auditory Cues for Enhanced Situational Awareness,” with Gerd Bruder and Zubin Choudhary.
- US 12,008,917, “Physical-Virtual Patient System,” with Gerd Bruder, Salam Daher, Jason Hochreiter, Mindi Anderson, Laura Gonzalez, and Desiree A. Diaz.
- US 11,954,900, “Audiovisual Detection of Expectation Violations in Disparate Home Automation Systems,” with Gerd Bruder, Ryan Schubert, and Austin Erickson.
- US 11,798,127, “Spatial Positioning of Targeted Object Magnification,” with Gerd Bruder, Kangsoo Kim, and Zubin Choudhary.

- US 11,729,448, “Intelligent Digital Interruption Management,” with Matthew Gottsacker, Nahal Norouzi, and Gerd Bruder.
- US 11,557,216, “Adaptive Visual Overlay for Anatomical Simulation,” with Joseph LaViola II, Francisco Guido-Sanz, Gerd Bruder, Mindi Anderson, and Ryan Schubert.
- US 11,550,470, “Grammar Dependent Tactile Pattern Invocation,” with Gerd Bruder and Ryan McMahan.
- US 11,535,261, “Medical Monitoring Virtual Human with Situational Awareness,” with Gerd Bruder.
- US 11,467,399, “Augmentation of Relative Pose In Co-located Devices,” with Gerd Bruder.
- US 11,410,270, “Intelligent Object Magnification for Augmented Reality Displays,” with Gerd Bruder, Kangsoo Kim, and Zubin Choudhary.
- US 11,287,971, “Visual-Tactile Virtual Telepresence,” with Gerd Bruder and Ryan McMahan.
- US 11,148,671, “Autonomous Systems Human Controller Simulation,” with Gerd Bruder.
- US 11,106,357, “Low Latency Tactile Telepresence,” with Ryan McMahan and Gerd Bruder.
- US 11,042,028, “Relative Pose Data Augmentation of Tracked Devices in Virtual Environments,” with Gerd Bruder.
- US 10,854,098, “Adaptive Visual Overlay Wound Simulation,” with Joseph LaViola II, Francisco Guido-Sanz, Gerd Bruder, Mindi Anderson, and Ryan Schubert.
- US 10,803,761, “Multisensory Wound Simulation,” with Joseph LaViola II, Francisco Guido-Sanz, Gerd Bruder, Mindi Anderson, and Ryan Schubert.
- US 10,410,541, “Physical-Virtual Patient Bed,” with Karen Aroian, Steven Talbert, Kelly Allred, Patricia Weinstein, Arjun Nagendran, and Remo Pillat.
- US 10,380,921, “Physical-Virtual Patient Bed System,” with Arjun Nagendran, Mary Lou Sole, and Laura Gonzalez.
- US 10,365,711, “Methods, Systems, and Computer Readable Media for Unified Scene Acquisition and Pose Tracking in a Wearable Display,” with Henry Fuchs, Mingsong Dou, and Jan-Michael Frahm.
- US 10,321,107, “Methods, Systems, and Computer Readable Media for Improved Illumination of Spatial Augmented Reality Objects,” with Henry Fuchs.
- US 9,808,549, “System for Detecting Sterile Field Events and Related Methods,” with Arjun Nagendran, Jason Hochreiter, Laura Gonzalez, and Hassan Foroosh.
- US 9,792,715, “Methods, Systems, and Computer Readable Media for Utilizing Synthetic Animatronics,” with Kurtis Keller, Andrei State, Henry Fuchs, and Ryan Schubert.
- US 9,679,500, “Physical-Virtual Patient Bed,” with Karen Aroian, Steven Talbert, Kelly Allred, Patricia Weinstein, Arjun Nagendran, and Remo Pillat.
- US 9,538,167, “Methods, Systems, and Computer Readable Media for Shader-Lamps Based Physical Avatars of Real and Virtual People,” with Henry Fuchs, Peter Lincoln, Andrew Nashel, and Andrei State.
- US 8,849,408, “Methods for Electronic Directionality of Deep-Brain Stimulation,” with Richard Gilson and Nizam Razack.
- US 7,068,274, “System and Method for Animating Real Objects With Projected Images,” with Kok-Lim Low and Ramesh Raskar.
- US 6,930,681, “System and Method for Registering Multiple Images with Three-Dimensional Objects,” with Ramesh Raskar and Kok-Lim Low.
- US 6,677,956, “Method for Cross-Fading Intensities of Multiple Images of a Scene for Seamless Reconstruction,” with Ramesh Raskar and Kok-Lim Low.

- US 5,870,136, “Dynamic Generation of Imperceptible Structured Light for Tracking and Acquisition of Three Dimensional Scene Geometry and Surface Characteristics in Interactive Three Dimensional Computer Graphics Applications,” with Gary Bishop, Henry Fuchs, and Mark Livingston.

BOOKS, CHAPTERS, AND SECTIONS

- Gabriel Zachmann, Krzysztof Walczak, Omar A. Niamut, Kyle Johnsen, Wolfgang Stuerzlinger, Mariano Alcañiz-Raya, **Greg Welch**, and Patrick Bourdot. *Virtual Reality and Mixed Reality: 20th EuroXR International Conference, EuroXR 2023, Rotterdam, The Netherlands, November 29 - December 1, 2023, Proceedings*. Lecture Notes in Computer Science. Springer Cham, 2023.
- Kangsoo Kim, Nahal Norouzi, Dongsik Jo, Gerd Bruder, and **Greg Welch**. *The Augmented Reality Internet of Things: Opportunities of Embodied Interactions in Transreality*, pages 797–829. Springer International Publishing, Cham, 2023.
- **Greg Welch**, Kalman Filter. In: Ikeuchi K. (eds) *Computer Vision*. Springer, Cham. https://doi.org/10.1007/978-3-030-03243-2_716-1
- Nahal Norouzi, Gerd Bruder, Brandon Belna, Stephanie Mutter, Damla Turgut, and **Greg Welch**. A Systematic Review of the Convergence of Augmented Reality, Intelligent Virtual Agents, and the Internet of Things, *Artificial Intelligence in IoT*, page 37. Springer, 2019.
- Bruce H. Thomas, **Greg Welch**, Pierre Dragicevic, Niklas Elmqvist, Pourang Irani, Yvonne Jansen, Dieter Schmalstieg, Aurélien Tabard, Neven A. M. ElSayed, Ross T. Smith, Wesley Willett. Situated analytics. In K. Marriott, F. Schreiber, T. Dwyer, K. Klein, N. H. Riche, T. Itoh, W. Stuerzlinger, and B. H. Thomas, editors, *Immersive Analytics*, Lecture Notes in Computer Science book series (LNCS, volume 11190), chapter 7, pages 185–220. Springer, Cham, 2018.
- Guido Brunnett, Sabine Coquillart, Robert van Liere, **Greg Welch**, and Libor Váša, editors. *Virtual Realities: International Dagstuhl Seminar, Dagstuhl Castle Germany; June 9–14, 2013; Revised Selected Papers*. Number 8844 in Lecture Notes in Computer Science. ISBN: 978-3-319-17042-8, Springer, 2015.
- Charles Hughes, Arjun Nagendran, Lisa Dieker, Michael Hynes, and **Greg Welch**. Applications of avatar mediated interaction to teaching, training, job skills and wellness. In Brunnett, G., Coquillart, S., van Liere, R., Welch, G., and Váša, L., editors, *Virtual Realities*, volume 8844 of *Lecture Notes in Computer Science*, pages 133–146. Springer International, 2015.
- Arjun Nagendran, **Greg Welch**, Charles Hughes, and Remo Pillat. Technical report: Exploring human surrogate characteristics. In Brunnett, G., Coquillart, S., van Liere, R., Welch, G., and Váša, L., editors, *Virtual Realities*, volume 8844 of *Lecture Notes in Computer Science*, pages 215–228. Springer International, 2015.
- **Greg Welch**, Kalman Filter. In Katsushi Ikeuchi, editor, *Computer Vision: A Reference Guide*, pp. 435–437. Boston, MA: Springer US, 2014.
- Sabine Coquillart, Guido Brunnett, and **Greg Welch**, eds., *Virtual Realities: Dagstuhl Seminar 2008*. 1st Edition., 2011, XIII, 251 p. 78 illus. ISBN: 978-3-211-99177-0, Springer, 2011.
- **Greg Welch** and Larry Davis. Tracking for Training in Virtual Environments: Estimating the Pose of People and Devices for Simulation and Assessment. In J. Cohn, D. Nicholson, and D. Schmorow, editors, *PSI Handbook of Virtual Environments for Training and Education: Developments for the Military and Beyond*, chapter 30. Praeger Security International, 2008.
- **Greg Welch**, Ruigang Yang, Bruce Cairns, Herman Towles, Andrei State, Adrian Ilie, Sasch Becker, Dan Russo, Jesse Funaro, Diane Sonnenwald, Ketan Mayer-Patel, Bonnie Danette

Allen, Hua Yang, Eugene Freid, Andries van Dam, and Henry Fuchs. 3D Telepresence for Off-Line Surgical Training and On-Line Remote Consultation. In S. Tachi, editor, *Telecommunication, Teleimmersion and Telexistence II*, pages 113–152. IOS Press (English) and Ohmsha (Japanese), 2005.

REFEREED JOURNAL, CONFERENCE, AND SYMPOSIUM PUBLICATIONS

- [183] M. Gottsacker, Y. Hmaiti, M. Maslych, H. Furuya, J. J. DeGuzman, G. Bruder, G. F. Welch, and J. J. L. Jr., “From one world to another: Interfaces for efficiently transitioning between virtual environments,” in *ACM CHI Conference on Human Factors in Computing Systems (CHI ’26)*, pp. 1–17, 2026.
- [182] A. Alwin, M. Gottsacker, H. Furuya, A. Delmo, L. Lisle, G. Bruder, and G. F. Welch, “[poster] Can you see it? Evaluating color visibility in simulated outdoor environments,” in *Proceedings of IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR) 2026.*, 2026.
- [181] H. Furuya, J. J. DeGuzman, Z. D. Choudhary, M. Gottsacker, A. Alwin, L. Lisle, G. Bruder, and G. F. Welch, “Effects of multisensory feedback and real-world priming on presence and copresence in mixed reality drone simulation,” in *Proceedings of the 2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (IEEE VRW): 2nd Annual Workshop on Real and Virtual Spaces Influences (ReVISI)*, pp. 1–7, 2026.
- [180] A. Alwin, M. Gottsacker, H. Furuya, L. Lisle, G. Bruder, and G. F. Welch, “Moving towards visibility evaluation in augmented reality using simulated environments and continuous psychophysics,” in *Proceedings of the 2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (IEEE VRW): Third International Workshop on Perception-driven Graphics and Displays for VR and AR (PerGraVAR)*, 2026.
- [179] Z. Choudhary, J. Ugarte, G. Bruder, and G. Welch, “Real-time magnification in augmented reality,” in *Proceedings of the 2021 ACM Spatial User Interaction, SUI 2021*, pp. 1–2, ACM, 2021.
- [178] M. Gottsacker, Y. Hmaiti, M. Maslych, H. Furuya, G. Bruder, G. F. Welch, and J. J. L. Jr., “XR-First Design For Productivity: A Conceptual Framework for Enabling Efficient Task Switching in XR,” in *IEEE International Symposium on Mixed and Augmented Reality (ISMAR) Adjunct Proceedings*, 2025.
- [177] M. Gottsacker, Y. Hmaiti, M. Maslych, H. Furuya, G. Bruder, G. F. Welch, and J. J. L. Jr., “From Alt-Tab To World-Snap: Exploring Different Metaphors for Swift and Seamless VR World Switching,” in *IEEE International Symposium on Mixed and Augmented Reality*, pp. 1–2, 2025.
- [176] Gerd Bruder, Ryan Schubert, Mike Browne, Austin Erickson, Zubin Choudhary, Matthew Gottsacker, Hiroshi Furuya, and **Greg Welch**, “Exploring how augmented reality display features affect training system performance,” *HCI International (HCII), Lecture Notes in Computer Science (LNCS)*, pp. 1–17, 2025.
- [175] Matthew Gottsacker, Nels Numan, Anthony Steed, Gerd Bruder, **Greg Welch**, and Steve Feiner, “Decoupled Hands: An Approach for Aligning Perspectives in Collaborative Mixed Reality,” in *ACM CHI Conference on Human Factors in Computing Systems (CHI ’25)*, 2025.

- [174] Hiroshi Furuya, Jasmine Joyce DeGuzman, Zubin Choudhary, Matthew Gottsacker, Gerd Bruder, and **Greg Welch**, “How can real-world feedback and priming affect trust in simulated autonomous agents?,” in *Proceedings of the 2025 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (IEEE VRW): 1st Annual Workshop on Real and Virtual Spaces Influences (ReVISI)*, pp. 1–2, 2025.
- [173] Hiroshi Furuya, Zubin Choudhary, Jasmine J. DeGuzman, Matthew Gottsacker, Gerd Bruder, and **Greg Welch**, “Using simulated real-world terrain in VR to study outdoor AR topographic map interfaces,” in *Proceedings of the International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE 2024)*, Tsukuba, Japan, December 1-3, 2024, pp. 1–10, 2024.
- [172] Zubin Choudhary, Laura Battistel, Raiffa Syamil, Hiroshi Furuya, Ferran Argelaguet, Gerd Bruder, and **Greg Welch**, “Examining the effects of teleportation on semantic memory of a virtual museum compared to natural walking,” in *Proceedings of the International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments (ICAT-EGVE 2024)*, Tsukuba, Japan, December 1-3, 2024, pp. 1–12, 2024.
- [171] Hiroshi Furuya, Laura Battistel, Zubin Choudhary, Matthew Gottsacker, Gerd Bruder, and **Greg Welch**. (2024). Difficulties in perceiving and understanding robot reliability changes in a sequential binary task. In *Proceedings of the 2024 ACM Symposium on Spatial User Interaction*, pages 1–11.
- [170] Matthew Gottsacker, Hiroshi Furuya, Zubin Choudhary, Austin Erickson, Ryan Schubert, Gerd Bruder, Michael P. Browne, and **Greg Welch**, “Investigating the relationships between user behaviors and tracking factors on task performance and trust in augmented reality,” *Elsevier Computers & Graphics*, vol. 123, pp. 1–14, 2024.
- [169] Matthew Gottsacker, Hiroshi Furuya, Laura Battistel, Carlos P. Jimenez, Nicholas LaMontagna, Gerd Bruder, and **Greg Welch**. Exploring spatial cognitive residue and methods to clear users’ minds when transitioning between virtual environments. In *IEEE International Symposium of Mixed and Augmented Reality (ISMAR)*, pages 1–11, 2024.
- [168] Gerd Bruder, Ryan Schubert, Michael Browne, Austin Erickson, Zubin Choudhary, Matthew Gottsacker, Hiroshi Furuya, and **Greg Welch**. Exploring how augmented reality display features affect training system performance. In *Proceedings of 2024 Interservice/Industry Training, Simulation, and Education Conference (I/ITSEC)*, 2024.
- [167] Michael P. Browne, **Greg Welch**, Gerd Bruder, and Ryan Schubert. Understanding the impact of trust on performance in a training system using augmented reality. In *Proceedings of SPIE Conference 13051: Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications VI*, 2024.
- [166] Gerd Bruder, Michael Browne, Zubin Choudhary, Austin Erickson, Hiroshi Furuya, Matthew Gottsacker, Ryan Schubert, and **Greg Welch**. Visual factors Influencing Trust and Reliance with Augmented Reality Systems. *Journal of Vision Abstracts — Vision Sciences Society (VSS) Annual Meeting*, 2024.
- [165] Juanita Benjamin, Austin Erickson, Matthew Gottsacker, Gerd Bruder, and **Greg Welch**. Evaluating Transitive Perceptual Effects Between Virtual Entities in Outdoor Augmented Reality. In *Proceedings of IEEE Virtual Reality (VR)*, pages 1–11, 2024.
- [164] Laura Battistel, Matthew Gottsacker, **Greg Welch**, Gerd Bruder, Massimiliano Zampini, and Riccardo Parin. Chill or Warmth: Exploring Temperature’s Impact on Interpersonal Boundaries in VR. In *Adjunct Proceedings of the IEEE Conference on Virtual Reality and 3D User*

- Interfaces (IEEE VR): 2nd Annual Workshop on Multi-modal Affective and Social Behavior Analysis and Synthesis in Extended Reality (MASSXR)*, pages 1–3, 2024.
- [163] Juanita Benjamin, Gerd Bruder, Carsten Neumann, Dirk Reiners, Carolina Cruz-Neira, and **Greg Welch**. Perception and proxemics with virtual humans on transparent display installations in augmented reality. In *Proceedings of the IEEE Conference on International Symposium on Mixed and Augmented Reality (ISMAR) 2023.*, pages 1–10, 2023.
 - [162] Zubin Choudhary, Gerd Bruder, and **Greg Welch**. Visual facial enhancements can significantly improve speech perception in the presence of noise. *IEEE Transactions on Visualization and Computer Graphics*, Special Issue on the 22nd IEEE International Symposium on Mixed and Augmented Reality (ISMAR), 2023.
 - [161] Ryan Schubert, Gerd Bruder, and **Greg Welch**. Testbed for intuitive magnification in augmented reality. In *Proceedings IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pages 1–2, 2023.
 - [160] Ryan Schubert, Gerd Bruder, and **Greg Welch**. Intuitive user interfaces for real-time magnification in augmented reality. In *Proceedings of the ACM Symposium on Virtual Reality Software and Technology (VRST)*, pages 1–10, 2023. **Best Paper Award**.
 - [159] Zubin Choudhary, Gerd Bruder, and **Greg Welch**. Visual hearing aids: Artificial visual speech stimuli for audiovisual speech perception in noise. In *Proceedings of the 29th ACM Symposium on Virtual Reality Software and Technology, 2023*, October 9 2023.
 - [158] Zubin Choudhary, Nahal Norouzi, Austin Erickson, Ryan Schubert, Gerd Bruder, and **Greg Welch**. Exploring the social influence of virtual humans unintentionally conveying conflicting emotions. In *Proceedings of the 30th IEEE Conference on Virtual Reality and 3D User Interfaces*, IEEE Virtual Reality 2023, 2023.
 - [157] Zubin Choudhary, Austin Erickson, Nahal Norouzi, Kangsoo Kim, Gerd Bruder, and **Greg Welch**. Virtual big heads in extended reality: Estimation of ideal head scales and perceptual thresholds for comfort and facial cues. *ACM Transactions on Applied Perception*, November 10, 2022.
 - [156] Matthew Gottsacker, Nahal Norouzi, Ryan Schubert, Frank Guido-Sanz, Gerd Bruder, and **Greg Welch**. Effects of environmental noise levels on patient handoff communication in a mixed reality simulation. In *28th ACM Symposium on Virtual Reality Software and Technology (VRST '22)*, pages 1–10, 2022.
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- [33] Austin Erickson, Gerd Bruder, **Greg Welch**, Isaac Bynum, Tabitha Peck, and Jessica Good. Perceived Humanness Bias in Additive Light Model Displays. In *Vision Sciences Society (VSS) Abstracts* (Poster Presentation), 2022.
- [32] Jesus Ugarte, Nahal Norouzi, Austin Erickson, Gerd Bruder, and **Greg Welch**. Distant Hand Interaction Framework in Augmented Reality. In *Proceedings of IEEE Virtual Reality (VR)*, pages 1–2, 2022.
- [31] Zubin Choudhary, Jesus Ugarte, Gerd Bruder, and **Greg Welch**. Real-Time Magnification in Augmented Reality. In *Proceedings of ACM Symposium on Spatial User Interaction (SUI)*, pages 1–2, 2021. **Best Demo Award**.
- [30] Zubin Choudhary, Gerd Bruder, and **Greg Welch**. Scaled User Embodied Representations in Virtual and Augmented Reality. In *Proceedings of the Workshop on User-Embodied Interaction in Virtual Reality (UIVR)*, pages 1–3, 2021.
- [29] Austin Erickson, Dirk Reiners, Gerd Bruder, and **Greg Welch**. Augmenting Human Perception: Mediation of Extrasensory Signals in Head-Worn Augmented Reality. In Poster Proceedings of the *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pages 1–4, 2021.
- [28] Francisco Guido-Sanz, Mindi Anderson, Desiree Díaz, **Greg Welch**, and Laura Gonzalez. (2021). POSTER: Using XR Technology to Innovate Healthcare Education. In *2021 International Nursing Association for Clinical Simulation and Learning Conference (INACSL 2021)*.
- [27] Nafisa Mostofa, Indira Avendano, Ryan P. McMahan, and **Greg Welch**. (2020). POSTER: Tactile Telepresence for Isolated Patients. In Kulik, A., Sra, M., Kim, K., and Seo, B.-K., editors, *ICAT-EGVE 2020 - International Conference on Artificial Reality and Telexistence and Eurographics Symposium on Virtual Environments - Posters and Demos*, pages 7–8. The Eurographics Association.
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- [24] Kangsoo Kim, Austin Erickson, Alexis Lambert, Gerd Bruder, and **Greg Welch**. Effects of dark mode on visual fatigue and acuity in optical see-through head-mounted displays. In *32nd ACM User Interface Software and Technology Symposium (UIST 2019)*, New Orleans, LA USA, October 2019.
- [23] Nahal Norouzi, Austin Erickson, Kangsoo Kim, Ryan Schubert, Joeseeph LaViola, Gerd Bruder, and **Greg Welch**. Effects of shared gaze parameters on visual target identification task performance in augmented reality. In *32nd ACM User Interface Software and Technology Symposium (UIST 2019)*, New Orleans, LA USA, October 2019.
- [22] Kendra Richards, Nikhil Mahalanobis, Kangsoo Kim, Ryan Schubert, Myungho Lee, Salam Daher, Nahal Norouzi, Jason Hochreiter, Gerd Bruder, and **Greg Welch**. Analysis of peripheral vision and vibrotactile feedback during proximal search tasks with dynamic virtual entities in augmented reality. In *32nd ACM User Interface Software and Technology Symposium (UIST 2019)*, New Orleans, LA USA, October 2019.
- [21] Salam Daher, Jason Hochreiter, Nahal Norouzi, Ryan Schubert, Gerd Bruder, Laura Gonzalez, Mindi Anderson, Desiree Diaz, Juan Cendan, and **Greg Welch**. Matching vs. non-matching visuals and shape for embodied virtual healthcare agents. *IEEE VR 2019 (Osaka, Japan)*, March 27, 2019.
- [20] Ladda Thiamwong, Nahal Norouzi, and **Greg Welch**, “Fear of falling and eye movement behavior in young adults and older adults during walking: A case study,” in *39th Annual Southern Gerontological Society Meeting*, (Buford, GA USA), April 11–14 2018.
- [19] Dongsik Jo, Kangsoo Kim, **Greg Welch**, Woojin Jeon, Yongwan Kim, Ki-Hong Kim, Gerard Kim, “The Impact of Avatar-Owner Visual Similarity on Body Ownership in Immersive Virtual Reality,” *ACM Symposium on Virtual Reality Software and Technology (VRST)*, November 8–10, Gothenburg, Sweden.
- [18] Myungho Lee, Gerd Bruder, and **Greg Welch**, “Exploring the effect of vibrotactile feedback through the floor on social presence in an immersive virtual environment,” *Vision Sciences Society* 2017.
- [17] Ryan Schubert, Gerd Bruder, and **Greg Welch**, “Mitigating perceptual error in synthetic animatronics using visual feature flow,” *Vision Sciences Society* 2017.
- [16] Richard Skarbez, **Greg Welch**, Frederick P. Brooks Jr., and Mary C. Whitton, “Coherence changes gaze behavior in virtual human interactions,” in *Proceedings of IEEE Virtual Reality 2017*, pp. 287–288, March 2017.
- [15] Salam Daher, Kangsoo Kim, Myungho Lee, Gerd Bruder, Ryan Schubert, Jeremy Bailenson, and **Greg Welch**, “Can social presence be contagious? Effects of social presence priming on interaction with virtual humans,” in *Proceedings of IEEE Symposium on 3D User Interfaces 2017*, (Los Angeles, CA), March 18–19 2017.
- [15] Kangsoo Kim, Ryan Schubert, and **Greg Welch**. Poster: Exploring the impact of environmental effects on social presence with a virtual human. In *Proceedings of the 16th International Conference on Intelligent Virtual Agents (IVA 2016)*, volume 10011, pages 470–474, Los Angeles, CA, September 20–23, 2016.

- [14] **Greg Welch**, Salam Daher, Jason Hochreiter and Laura Gonzalez. Interactive Rear-Projection Physical-Virtual Patient Simulators. Poster presented at the 22nd Medicine Meets Virtual Reality (NextMed / MMVR) conference, Los Angeles, CA, April 7–9, 2016.
- [13] Salam Daher and **Greg Welch**. Humanikins: Humanity Transfer to Physical Manikins. Poster presented at the 22nd Medicine Meets Virtual Reality (NextMed / MMVR) conference, Los Angeles, CA, April 7–9, 2016.
- [12] Salam Daher, Kangsoo Kim, Myungho Lee, Andrew Raij, Ryan Schubert, Jeremy Bailenson, and **Greg Welch**. Poster: Exploring social presence transfer in real-virtual human interaction. In *Proceedings of IEEE Virtual Reality 2016*, Greenville, SC, USA, March 19–23, 2016.
- [11] Salam Daher, Laura Gonzalez, **Greg Welch**. Preliminary Assessment of Neurologic Symptomatology Using an Interactive Physical-Virtual Head with Touch. Poster presented at the 17th International Meeting on Simulation in Healthcare (IMSH 2016), January 16–20, 2016.
- [10] Feng Zheng, Ryan Schubert, and **Greg Welch**, “A General Approach for Closed-Loop Registration in AR,” in *ISMAR '12: Proceedings of the Eleventh IEEE International Symposium on Mixed and Augmented Reality (ISMAR'12)*, (Atlanta, GA, USA), November 2012.
- [9] **Greg Welch**, Diego Rivera-Gutierrez, Peter Lincoln, Mary Whitton, Juan Cendan, David A. Chesnutt, Henry Fuchs, Ben Lok, and Rick Skarbez. Physical manifestations of virtual patients. Poster presented at the 12th International Meeting on Simulation in Healthcare (IMSH 2011), 2011.
- [8] Jinghe Zhang, **Greg Welch**, Gary Bishop, and Zhenyu Huang. Adaptive kalman filtering for robust power system state tracking. Poster presented at the DOE Applied Mathematics Program meeting, October 17–18 2011.
- [7] Brian Clipp, **Greg Welch**, Jan-Michael Frahm, and Marc Pollefeys. Structure from motion via a two-stage pipeline of extended kalman filters. Proceedings of the British Machine Vision Conference (BMVC 2007), September 10–13 2007.
- [6] Hua Yang and **Greg Welch**. Illumination Insensitive Model-Based 3D Object Tracking and Texture Refinement. In Proceedings of the Third International Symposium on 3D Data Processing, Visualization and Transmission (3DPVT 2006), The University of North Carolina at Chapel Hill, Chapel Hill, NC USA, June 14-16, 2006.
- [5] Ruigang Yang, Liang Wang, **Greg Welch**, and Marc Pollefeys. Stereovision on GPU. Poster presentation at the 2006 Workshop on Edge Computing Using New Commodity Architectures (EDGE 2006), May 23–24 (Chapel Hill, NC, USA).
- [4] **Greg Welch**, Hua Yang, Andrei State, Vincent Noel, Adrian Ilie, Ruigang Yang, Marc Pollefeys, and Henry Fuchs. GPU-Based View Synthesis Using an Orbital Reconstruction Frustum. Poster presentation at the 2006 Workshop on Edge Computing Using New Commodity Architectures (EDGE 2006), May 23–24 (Chapel Hill, NC, USA).
- [3] Adrian Ilie and **Greg Welch**. Ensuring Color Consistency Across Multiple Cameras. International Conference on Computer Vision (ICCV), October, 2005 (Beijing, China).
- [2] Bruce A. Cairns, **Greg Welch**, Adrian Ilie, Ruigang Yang, Kok-Lim Low, Anselmo Lastra, Henry Fuchs, and Anthony Meyer. Three Dimensional (3D) Acquisition and Display of Reality: The Potential for a ‘Holodeck’ in Trauma Surgery. Presented at *The American Association for the Surgery of Trauma 2003 Annual Meeting*.

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TECHNICAL REPORTS

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- [12] Ramesh Raskar, Kok-Lim Low and **Greg Welch**. Shader Lamps: Animating Real Objects with Image-Based Illumination. Technical Report TR00-027, University of North Carolina at Chapel Hill, Department of Computer Science, 2000.
- [11] Ruigang Yang and **Greg Welch**. Automatic Display Surface Estimation using Everyday Imagery. Technical Report TR00-015, University of North Carolina at Chapel Hill, Department of Computer Science, 2000.
- [10] Ramesh Raskar, Henry Fuchs, **Greg Welch**, Adam Lake, and Matt Cutts. 3D Talking Heads: Image Based Modeling at Interactive Rates Using Structured Light Projection. Technical Report TR98-017, University of North Carolina at Chapel Hill, Department of Computer Science, 1998.
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- [8] **Greg Welch** and Gary Bishop. One-Step-at-a-Time Tracking. Technical Report TR96-021, University of North Carolina at Chapel Hill, Department of Computer Science, 1996.
- [7] **Greg Welch**. Hybrid Self-tracker: An Inertial/Optical Hybrid Three-Dimensional Tracking System. Technical Report TR95-048, University of North Carolina at Chapel Hill, Department of Computer Science, 1995.
- [6] **Greg Welch** and Gary Bishop. An Introduction to the Kalman Filter. Technical Report TR95-041, University of North Carolina at Chapel Hill, Department of Computer Science, 1995. **Cited over 12,250 times** according to Google as of 09 MAY 2025. Translated into Chinese (modern and traditional) by others.
- [5] **Greg Welch** VERSICOM: Versatile communications software. NASA Jet Propulsion Laboratory, July 13, 1989.
- [4] **Greg Welch** and James P. Williams. The easy chair: A microprocessor-controlled wheelchair for children with muscular disorders. Purdue University, E.E.T. 490/491 Senior Design Project, Final Report, May 1986.
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- [1] James Williams and **Greg Welch**. The pressure sensitive touch-pad. Purdue University, E.E.T. 454 Project Report, April 30, 1985.

PH.D. DISSERTATION

Title: SCAAT: Incremental Tracking with Incomplete Information
 Date: October 1996
 Advisor: Gary Bishop
 Committee: Gary Bishop, Henry Fuchs, Leandra Vicci (née Vernon Chi), Anselmo Lastra, Russell Taylor, and John Poulton
 PDF: http://www.cs.unc.edu/~welch/media/pdf/scaat_dissertation.pdf

ACTIVE FUNDING

- ONR Awards# N00014-25-1-2245 for “Visualization Enhancements for Augmented Reality (VEAR),” **Greg Welch** (PI) and Gerd Bruder (Co-PI). Dr. Peter Squire, Program Manager. \$700,000 total for April 2025–March 2027.
 - ONR DURIP Award# N00014-25-1-2159 for “Hybrid Visualization Testbed for Warfighter Augmented Reality Head-Worn Display Research with High-Ambient Photometric Environmental Properties,” **Greg Welch** (PI) and Gerd Bruder (Co-PI). Dr. Peter Squire, Program Manager. \$233,247 for February 2025–January 2027.
 - NSF Award# 2235066 for “Collaborative Research: CCRI: Grand: Virtual Experience Research Accelerator (VERA),” **Greg Welch** (Lead PI) and Gerd Bruder (PI) at the University of Central Florida; Shiri Azenkot (PI, NSF Award# 2235067) at Cornell Tech; Jeremy Bailenson at Stanford University (PI, NSF Award# 2235070); Tabitha Peck at Davidson College (PI, NSF Award# 2235068); and Valerie Jones Taylor at Rutgers University (PI, NSF Award# 2350377). NSF PO: Ephraim P. Glinert (IIS CHS), \$4,900,000 total (\$4,192,490 to UCF and \$707,510 to collaborative partners) for April 2023–March 2027. <https://veraccel.org>
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PAST FUNDING

- ONR Awards# N00014-21-1-2578 for “Enhanced Information Comprehension via Augmented Reality (EICAR),” **Greg Welch** (PI) and Gerd Bruder (Co-PI). Dr. Peter Squire, Program Manager. \$1,290,291 total for June 2021–May 2025.
- NSF Award# 1800961 for “CHS: Medium: Collaborative Research: Augmented Reality Agents with Pervasive Awareness, Appearance, and Abilities,” **Greg Welch** (PI), Gerd Bruder (Co-PI), and Damla Turgut (Co-PI) at UCF; Jeremy Bailenson (Co-PI) at Stanford University; and Benjamin Lok (Co-PI) at the University of Florida. NSF PO: Ephraim P. Glinert (IIS CHS), \$674,264 for August 15, 2018–July 31, 2024
- Vision Products, Inc., Trust and Situational Awareness in Augmented Reality Soldier Training, SBIR-Phase II. PIs Greg Welch and Gerd Bruder. PM: Ted Harmer. \$299,318. 2022–2023.
- ONR DURIP Award# N00014-21-1-2882 for “Testbed for Enhanced Perception and Cognition via Augmented Reality (EICAR),” **Greg Welch** (PI). Dr. Peter Squire, Program Manager. \$177,484 for August 2021–May 2024.
- NSF Award# 1852002 for “REU Site: Research Experiences for Undergraduates Site on Internet of Things (IoT),” Damla Turgut (PI), Student Mentors: Damla Turgut, Ladislau Boloni, **Greg Welch**, Gerd Bruder, Samiul Hasan, Qun Zhou, Hyoung Jin Cho, and Andrew Dickerson at the University of Central Florida. NSF PO: Jonathan Sprinkle, \$323,945 for May 1, 2019–April 30, 2023

- NSF Award# 1564065 for “CHS: Medium: Physical-Virtual Patient Bed for Healthcare Training and Assessment,” **Greg Welch** (PI), Juan Cendan (Co-PI), and Laura Gonzalez (Co-PI), NSF PO: Ephraim P. Glinert (IIS CHS), \$894,431 for August 1, 2016–July 31, 2022
- Soar Technology, Inc., “Handoff Training for Combat Casualty Care (HTC3) CCHAT System PHASE II” **Greg Welch** (PI), Gerd Bruder (Co-PI), and Francisco Guido-Sanz (Co-PI). SoarTech PI: Alyssa Tanaka. \$463,949 for June 15, 2019–April 30, 2022
- Florida High Tech Corridor Council (FHTCC), Handoff Training for Combat Casualty Care (HTC3), PIs: **Greg Welch** (PI), Gerd Bruder (Co-PI), and Francisco Guido-Sanz (Co-PI). SoarTech PI: Alyssa Tanaka. \$248,928 for December 16, 2019–April 30, 2022.
- University of Central Florida, Exploratory Effort Aimed at a Center for Virtual and Augmented Reality for Healthcare, PIs **Greg Welch** and Carolina Cruz-Neira (CS), Co-PIs Mindi Anderson (CON), Gerd Bruder (IST), Juan Cendan (COM), Lisa Dieker (CCIE), Charlie Hughes (CS), and Ben Noel (FIEA). \$40K for 2020–2021.
- ONR Award# N00014-18-1-2927 for “Enhanced Perception and Cognition in Augmented Reality (EPiC AR),” **Greg Welch** (PI) and Gerd Bruder (Co-PI). Dr. Peter Squire, Program Manager. \$1,175,000 total for January 2018–December 2021.
- SA Photonics, Inc. “Blended Reality Solution for Live, Virtual, and Constructive Field Training PHASE II.” **Greg Welch** (PI) and Gerd Bruder (Co-PI). SA Photonics PI: Michael Browne. \$156,138.50 for December 21, 2018–May 22, 2021.
- Soar Technology, Inc., “Intelligent Diagnostic Trauma Algorithms for a Ruggedized Autonomous Combat Casualty Care Capability.” Francisco Guido-Sanz (PI) and **Greg Welch** (Co-PI). SoarTech PI: Alyssa Tanaka. \$7,500 for September 4, 2018–April 3, 2019
- Soar Technology, Inc., “Combat Casualty Care Augmented Reality Intelligent Training System - C3ARESIS Phase II,” **Greg Welch** (PI), Gerd Bruder (Co-PI), and Francisco Guido-Sanz (Co-PI). SoarTech PI: Glenn Taylor. \$145,900 for April 6, 2018–April 5, 2019.
- Florida High Tech Corridor Council, “FHTCC: Combat Casualty Care Augmented Reality Intelligent Training System (C3ARESIS),” **Greg Welch** (PI), Gerd Bruder (Co-PI). \$49,760 for April 6, 2018–April 5, 2019.
- REU Grant (Award# 1560302) for “REU Site: Research Experiences in the Internet of Things,” led by Damla Turgut (PI) and Yier Jin (Co-PI), **Greg Welch** (Investigator) et al. \$359,912.00 for June 2016–May 2019
- ONR DURIP grant for “Transportable Human-Surrogate Interaction System (THuSIS),” **Greg Welch** (PI). Dr. Peter Squire, Program Manager. \$148,216 for September 15, 2016–March 14, 2017.
- Lockheed Martin Corporation grant for “Improving Augmented Reality Technology,” Joseph LaViola (PI), **Greg Welch** (Co-PI), and Malcolm Butler (Co-PI), \$100,000, September 1, 2016–August 29, 2017.
- UCF Office of Research and Commercialization, Mentoring Program Award, “Fear of Falling in Older Adults,” Ladda Thiamwong (PI) and **Greg Welch** (Mentor), \$3,000, July 1, 2017–June 30, 2018.
- Soar Technology, Inc., “Handoffs for Joint Service Casualty Care (HJSCC),” Francisco Guido-Sanz (PI) and **Greg Welch** (Co-PI). SoarTech PI: Alyssa Tanaka. \$59,763.00 for November 15, 2017–June 14, 2018.
- Soar Technology, Inc., “Handoff Training for Combat Casualty Care (HTC3),” **Greg Welch** (PI), Francisco Guido-Sanz (Co-PI), and Gerd Bruder (Co-PI). SoarTech PI: Alyssa Tanaka. \$60,000.00 for December 17, 2017–July 17, 2018.
- SA Photonics, Inc., “Blended Reality Solution for Live, Virtual, and Constructive Field Training,” **Greg Welch** (PI) and Gerd Bruder (Co-PI). SoarTech PI: Michael Browne. \$31,000.00 for November 16, 2017–June 1, 2018.

- Soar Technology, Inc., “A16-076 Augmented/Mixed Reality for Force-on-Force Combat Casualty Care Training- Phase 1 Option,” **Greg Welch** (PI). SoarTech PI: Glenn Taylor. \$6,000.00 for November 27, 2017–March 26, 2018.
- ONR grant for “Human-Surrogate Interaction,” **Greg Welch** (PI) at UCF; and Jeremy Bailenson at Stanford University. Dr. Peter Squire, Program Manager. \$2,312,188 total for March 2014–December 2017.
- SoarTech/Department of the Army SBIR grant for “Combat Casualty Care Augmented Reality Intelligent Training System (C3ARESIS),” **Greg Welch** (UCF PI) and Frank Guido-Sanz (UCF), \$18,000 + \$6,000 option, July 1, 2016–December 31, 2016
- ONR DURIP grant for “A Testbed for Evaluating Human Surrogates for Live-Virtual Training,” **Greg Welch** (PI) and Arjun Nagendran (Co-PI). Dr. Peter Squire, Program Manager. \$178,437 for November 2014–April 2016.
- ONR DURIP grant for “A Physical-Virtual Human-Robot Interaction System for Training, Education, and Rehabilitation,” **Greg Welch** (PI), Charlie Hughes (Co-PI), and Arjun Nagendran (Investigator). Dr. Peter Squire, Program Manager. \$268,598. July 2012–July 2014.
- ONR grant for “3D Display and Capture of Humans for Live-Virtual Training,” led by **Greg Welch** (PI) and Charlie Hughes (Co-PI) at UCF, Jeremy Bailenson at Stanford University, and Mel Slater at the University of Barcelona. Dr. Peter Squire, Program Manager. \$1,316,334 total for October 2011–September 2014.
- University of Central Florida, Major Research Equipment grant for “Comparative Evaluations of Display and Control Paradigms for Surrogate Humans,” **Greg Welch** and Charlie Hughes, \$57,500, February 2013–June 2013.
- University of Central Florida, Institute for Simulation & Training, Simulation for Healthcare Education and Learning Laboratory (SHELL) grant for “Simulating Subtle Clinical Changes and Realistic Conversation: Going Beyond Mannequins,” Kelly Allred (PI), Patricia Weinstein, **Greg Welch**, Steve Talbert, Erica Hoyt, Karen J. Aroian, Anne E. Norris, Mary Lou Sole, Susan K. Chase. \$125,000 in October 2012.
- DOE grant for “Advanced Kalman Filter for Real-Time Responsiveness in Complex Systems,” led by **Greg Welch** (PI) at UNC and Zhenyu Huang (PI) at the Pacific Northwest National Laboratory. \$320,502 total for September 2009–August 2013.
- ARA/DARPA contract for “ULTRA-Vis Phase 3: Sensor Fusion Methods/Software to Achieve Robust Pose Estimation,” **Greg Welch** (PI). Alberico Menozzi, TPOC (ARA). \$73,438 total for June 2012–June 2013.
- ONR grant for “3D Display and Capture of Humans for Live-Virtual Training,” led by **Greg Welch** (lead PI) and Henry Fuchs (PI) at UNC, and Amela Sadagic (PI) at the Naval Postgraduate School. Roy Stripling and Clarke Lethin, Program Managers. \$2.3M total for May 2009–September 2012. Welch led on the proposal and leads the subsequent research at UNC.
- NSF CRI:IAD grant for “Integrated Projector-Camera Modules for the Capture and Creation of Wide-Area Immersive Experiences,” led by Henry Fuchs (PI), **Greg Welch** (Co-PI), Leonard McMillan (Co-PI), Mary Whitton (Co-PI), and Svetlana Lazebnik (Co-PI). \$310K total for April 2008–March 2012. Welch led on the proposal at UNC.
- ONR contract for “Behavior Analysis and Synthesis for Intelligent Training (BASE-IT),” led by **Greg Welch** (PI) at UNC, Amela Sadagic (PI) at the Naval Postgraduate School, and Rakesh Kumar (PI) and Hui Cheng (Co-PI) at Sarnoff. Roy Stripling, Ph.D., Program Manager. \$2.2M total for February 2008–September 2011. Welch led on the proposal and leads the subsequent research at UNC.
- ONR SBIR Phase II contract for “Deployable Intelligent Projection Systems for Training,” led by Henry Fuchs (PI) and **Greg Welch** (PI), sub-contract from Renaissance Sciences Corporation

(RSC), Jeff Clark, CEO, \$374,920 for March 2009–February 2011. Welch led on the proposal and leads the subsequent research at UNC.

- UNC-Chapel Hill “Scholarship, Creative Activity or Research in the Humanities and Fine Arts” led by Francesca Talenti (PI) and **Greg Welch** (Co-PI), \$10K for June 2009–May 2010.
- IARPA A-SpaceX contract for “Mockup Future Analyst Workspace (A-Desk),” led by **Greg Welch** (lead PI) and Henry Fuchs (PI). Jeff Morrison, Ph.D., Program Manager. \$260K for April 2008–December 2008. Welch led the proposal and the subsequent research at UNC.
- ONR SBIR contract for “Deployable Intelligent Projection Systems for Training,” led by **Greg Welch** (PI) at UNC and Karl Matias (PI) of Renaissance Sciences Corporation, sub-contract to UNC-Chapel Hill (Henry Fuchs, PI), \$80K for Phase 1, June 2007–September 2008. Welch led on the proposal and led the subsequent research at UNC.
- IARPA VACE contract for “3D Content Extraction from Video Streams,” led by Marc Pollefeys (PI) and **Greg Welch** (Co-PI), with Jan-Michael Frahm. Dan Aldridge, Program Manager. \$660K October 2006–September 2008.
- Cisco Systems grant for “Telepresence Wall: Research Exhibit,” with Henry Fuchs. \$439K for August 2007–July 2008. Welch co-led the proposal and the subsequent research at UNC.
- ONR STTR contract for “Deployable Intelligent Projection Systems for Training: Enhanced Integrated Pose Estimation Technologies,” led by **Greg Welch** (PI) at UNC and Karl Matias (PI) of Renaissance Sciences Corporation, sub-contract to UNC-Chapel Hill (Henry Fuchs, PI), \$70K for Phase 1, August 2007–February 2008. Welch led on the proposal and led the subsequent research at UNC.
- Cisco Systems grant for “Prototype for Two-station, Four-Person, Proper Eye-Gaze Telepresence System,” with Henry Fuchs. \$376K total for August 2006–July 2007. Welch co-led on the proposal and co-led the subsequent research at UNC.
- National Library of Medicine contract for “3D Telepresence for Medical Consultation” led by **Greg Welch** (lead PI) and Prof. Henry Fuchs (PI) at UNC, Prof. Bruce Cairns, M.D. (Co-PI) at UNC, Prof. Ketan Mayer-Patel (Co-PI) at UNC, and Prof. Diane Sonnenwald (Co-PI) at Göteborg University and the University College of Borås. \$2.5M total for October 2003–December 2007. Welch led on the proposal and led the subsequent research at UNC.
- ONR VIRTE contract for “Front-Projective Display For Virtual Environments: Phase 2,” with Henry Fuchs and Herman Towles. Dylan Schmorrow, Ph.D. CDR MSC USN, Program Manager. \$560K total for October 2004–December 2007.
- Office of Naval Research DURIP 2006 grant for “Computing for Real World Acquisition, Display and Immersive Training,” with Henry Fuchs and Marc Pollefeys. \$136K total.
- DARPA DSO contract for “Wide Area Visuals for a Simulator in a Box,” with Henry Fuchs and Herman Towles. Ralph Chatham, DARWARS Program Manager. \$1.2M total for 2003–2006.
- NSF ITR grant “Electronic Books for the Tele-Immersion Age: A New Paradigm for Teaching Surgical Procedures,” led by **Greg Welch** (PI) at UNC and Andy van Dam (PI) at Brown University, \$609K (UNC) for Sep 2001–Aug 2005. Welch led on the proposal and led the subsequent research at UNC.
- DOE contract “Front-Projection Display Wall, Group Tele-Immersion, and Tracking,” with Henry Fuchs and Herman Towles, \$1.8M for September 2001–August 2004.
- “3D Tele-Immersion Over Next Generation Internet,” UNC/UPenn DARPA contract with Henry Fuchs, Herman Towles, Kostas Danillidis, and Ruzena Bajcsy, \$550K (UNC) for June 2001–December 2002.
- Argonne National Laboratories contract “Compensating for Color Variations Across Multi-Projector Displays,” \$61K for September 2001–August 2002. Led by **Greg Welch** (PI).

- NSF grant “High-Fidelity Tele-Immersion for Advanced Surgical Training,” UNC/UPenn/Brown, with Henry Fuchs, Kostas Danillidis, and Andy van Dam, \$750K (UNC) for January 2001–December 2001.
- Naval Research Lab contract “Technology for Full-Body Tracking,” with Gary Bishop. Larry Rosenblum, NRL program manager. \$100,271 over October 2000–September 2001.
- NSF Research Infrastructure grant for the NSF Graphics and Visualization Science and Technology Center. The entire Center participated in the proposal process, **Greg Welch** played a significant role in the proposal content/realization. This grant supported equipment for Center-wide televideo and several related research projects. Support under this grant totaled \$1.3M for the Center over October 1998–September 2001. The UNC portion was \$304K.
- “The National Tele-Immersion Initiative,” with Henry Fuchs. Gift from Advanced Network and Services, Inc., Al Weiss, Terry Rogers, and Jaron Lanier. \$750K per year over January 1998–December 2000.
- NSF Graphics and Visualization Science and Technology Center, 5-year Renewal. The entire Center participated in the renewal proposal process, including a major site visit at the University of Utah, summer 1997. In addition to contributing to the written proposal, **Greg Welch** played a significant role in the Utah site visit. Total Center support under this renewal totals approximately \$13.7M for the entire Center for the final four years (February 1998–January 2002). The UNC portion totaled approximately \$2.7M over those four years.
- DARPA contract “Geospatially Registered Information for Dismounted Infantry” (GRIDS), with Gary Bishop and Vernon Chi. Joint effort with Raytheon Defense Systems, Hughes Research Labs, UNC, and the University of Southern California. Total support over two years (May 1997–April 1999) was approximately \$3.1M, UNC portion was approximately \$1.2M.

ACTIVITIES AS A STUDENT

- President, UNC-Chapel Hill Computer Science Student Association, 1994–1995
- Graduated with *Highest Distinction* from Purdue University (among three-tenths of the baccalaureate graduates having the highest graduation indexes)
- *Outstanding Senior Design Project*, “The Easy Chair: A Microprocessor-Controlled Wheelchair for Children With Muscular Disorders,” E.T., Purdue University, 1986
- *Distinguished Student*, Purdue University, 1982–1986 (all semesters)
- Residence Hall Counselor, Purdue University, 1985–1986
- Tau Alpha Pi National Honor Society, Pi Alpha Chapter, 1986
- Golden Key National Honor Society, 1984
- Phi Eta Sigma National Honor Society, 1983

PROFESSIONAL EXPERIENCE (MORE DETAIL)

University of Central Florida, Orlando, Florida

AdventHealth Endowed Chair in Healthcare Simulation, October 2013–present

Professor, College of Nursing, October 2013–present

Professor, Computer Science Division of CECS, November 2013–present

Professor, Institute for Simulation & Training, November 2013–present

Co-Director, IST Synthetic Reality Lab (SREAL), August 2011–present

Co-Director, Interactive Computing Experiences Research Cluster of Excellence (2014–2017)

Research Professor, Institute for Simulation & Training (IST), August 2011–October 2013

Research Professor, Department of Computer Science, CECS, October 2011–October 2013

University of North Carolina at Chapel Hill, Chapel Hill, North Carolina

Adjunct Research Professor, Computer Science, September 2012–September 2020

Research Professor, Computer Science, July 2010–August 2012

Research Associate Professor, Computer Science, September 2001–June 2010

Adjunct Associate Professor, Applied & Materials Sciences, 2005–2008

Adjunct Assistant Professor, Applied & Materials Sciences, 2000–2003

Research Assistant Professor, Computer Science, September 1996–August 2001

UNC Site Coordinator, NSF Graphics and Visualization STC, September 1996–August 1998

- Coordinate UNC research for the National Science Foundation’s Science and Technology Center for Graphics and Visualization.
- STC members included Brown University (PI Andy van Dam, Co-PI John Hughes), The California Institute of Technology (PI Al Barr), Cornell University (PI Don Greenberg), The University of Utah (PI Rich Riesenfeld, Co-PI Elaine Cohen), and UNC (PI Henry Fuchs).
- Primary goals of the STC, which existed from 1991–2002, included build a stronger scientific foundation for computer graphics and scientific visualization, and to helping create a basic framework for future interactive graphical environments.

University College Dublin (2011–2016)

Visiting Professor, School of Information and Library Studies

Keio University (2010–2012)

Visiting Professor, Graduate School of Media Design

Renaissance Sciences Corporation, Chandler, Arizona

Principal Investigator II, June 2007–2008

- Part-time research and development efforts related to “intelligent projector units” for deployable training systems.
- Work with Jeff Clark and Karl Mathias at RSC to coordinate UNC-RSC joint efforts.

Northrop Defense Systems Division, Rolling Meadows, Illinois

Senior Software Engineer, Digital Systems, June 1990–June 1992 (Secret Clearance)

- Developed, integrated, and maintained embedded software for the AN/ALQ-135, the electronic (radar) countermeasures system currently deployed on the USAF F-15 Eagle.
- Responsible for the AN/ALQ-135 Receiver-Transmitter Compatibility Feature which ensures that radar countermeasures do not interfere with radar or threat acquisitions.

NASA Jet Propulsion Laboratory (Caltech), Pasadena, California

CCS/COMSIM Programmer and Analyst, Voyager Sequence Team, January 1987–May 1990

- Enhanced and maintained the software Command Simulator for the Voyager spacecraft. This simulator is (still) used to validate all instructions sent to the spacecraft.
- Programmed and maintained the Voyager spacecraft Computer Command Subsystem (CCS), the master processor on both Voyager spacecraft.
- Simulated and validated Voyager spacecraft command sequence activity. Responsible for the simulation and validation of all sequenced commanding during Voyager II closest approach to the planet Neptune.

Significant Projects (developed under own initiative)

- Developed MEMMAN, a ground-based software system that optimizes the allocation of the limited Voyager spacecraft memory. Savings during the Voyager II Neptune encounter (August, 1989) enabled additional planetary imaging.
- Developed specialized IBM PC/AT telecommunications software for the Voyager Flight Team. Software submitted to NASA for distribution under COSMIC. Still used today.

Purdue University, West Lafayette, Indiana

Undergraduate, Electrical Engineering Technology, August 1982–May 1986

- Co-developed “The Easy Chair: A Microprocessor-Controlled Wheelchair for Children With Muscular Disorders.” The wheelchair included an ultrasonic “bumper” system, a custom infrared touch-pad, motor controllers, and a user-programmable 8085-based control unit.
- The Easy Chair project was awarded *Outstanding Senior Design Project* by the Purdue University Department of Electrical Engineering Technology, May 1986.
- Residence Hall Counselor (Resident Assistant) at Cary Quadrangle, August 1985–May 1986.

EXHIBIT 7

Curriculum Vitae

Darrin J. Young

Email: darrin_young@yahoo.com

Phone: 510-282-3638

Education:

University of California at Berkeley

Ph.D. Electrical Engineering and Computer Sciences, May 1999.

University of California at Berkeley

M.S. Electrical Engineering and Computer Sciences, May 1993.

University of California at Berkeley

B.S. Electrical Engineering and Computer Sciences, May 1991.

Research Interests:

Integrated microsystem technologies (based on MEMS device design, fabrication, and integrated circuit design) for industrial sensing, wireless sensing, biomedical implant, wearable health monitoring, environmental sensing, navigation, power and energy scavenging, and RF communication applications.

Professional Experiences:

2009-Present

U-STAR Associate Professor, Electrical and Computer Engineering Department

The University of Utah, Salt Lake City, Utah.

2013-2020

Associate Chair, Electrical and Computer Engineering Department

The University of Utah, Salt Lake City, Utah.

2005-2009

Associate Professor, Electrical Engineering and Computer Science Department

Case Western Reserve University, Cleveland, Ohio.

1999-2005

Assistant Professor, Electrical Engineering and Computer Science Department

Case Western Reserve University, Cleveland, Ohio.

1991-1999

Research Assistant, Electrical Engineering and Computer Sciences Department
University of California, Berkeley, California.

1997-1999

Participating Researcher, Lawrence Livermore National Laboratory, Livermore,
California.

1997 -1998

Analog RF IC Design Intern, Rockwell Semiconductor Systems, Newport Beach,
California.

1991-1993

Intern, Hewlett-Packard Laboratories, Palo Alto, California.

Honors:

- Recognition and Appreciation of Valued Services and Contributions as IEEE EDS Microelectromechanical Systems Chair, December 2011
- Nomination for 2003 Carl F. Wittke Award at Case Western Reserve University for Excellence in Undergraduate Teaching
- Bachelor of Science Degree with Honors in EECS at U.C. Berkeley, 1991
- Scholastic All American Collegiate Award, 1988

Professional Service:

- Administrative Committee of IEEE Sensors Council, 2002-2016
- Technical Program Session Chair, The 12th International Conference on Solid-State Sensors, Actuators, and Microsystems, Boston, MA, 2003
- MEMS Plenary Session Chair, Emerging Information Technology Conference, Princeton University, Princeton, NJ, 2003
- MEMS Session Chair, Emerging Information Technology Conference, Princeton University, Princeton, NJ, 2004
- Advisor Board, Emerging Information Technology Conference, Princeton University, Princeton, NJ, 2004

- Technical Program Committee for IEEE Sensors Conference, Vienna, Austria, 2004
- Technical Program Committee for International Electron Device Meeting, 2004
- Technical Program Session Chair, International Electron Device Meeting, 2004
- Technical Program Committee for the 1st International Workshop on Networked Sensing Systems, University of Tokyo, Tokyo, Japan, 2004
- Organizing Committee for the 7th International Conference on Solid-State and Integrated-Circuit Technology, Beijing, China, 2004
- Technical Program Session Chair, The 13th International Conference on Solid-State Sensors, Actuators, and Microsystems, Seoul, Korea, 2005
- Conference MEMS Session Chair, Emerging Information Technology Conference, National Taiwan University, Taiwan, R.O.C., 2005
- Advisor Board, Emerging Information Technology Conference, National Taiwan University, Taiwan, R.O.C., 2005
- Technical Program Committee for IEEE Sensors Conference, Irvine, California, USA, 2005
- Technical Program Session Chair, IEEE Sensors Conference, Irvine, California, USA, 2005
- Technical Program Committee for International Electron Device Meeting, Washington, D.C. 2005
- Technical Program Session Chair, International Electron Device Meeting, Washington, D.C. 2005
- Technical Program Committee for the 2nd International Workshop on Networked Sensing Systems, San Diego, California, USA, 2005
- Technical Program Session Chair, International Workshop on Networked Sensing Systems, San Diego, California, USA, 2005
- Technical Program Committee for IEEE International Symposium on Circuits and Systems, Island of Kos, Greece, 2006
- Technical Program Committee for IEEE International Conference on Nano/Micro Engineered and Molecular Systems, Zhuhai, China, 2006

- Program Vice-Chair for the 3rd International Conference on Networked Sensing Systems, Chicago, Illinois, USA, 2006
- Technical Program Committee for IEEE Sensors Conference, Korea, 2006
- Conference MEMS Session Chair, Emerging Information Technology Conference, University of Dallas, Dallas, Texas, 2006
- Advisor Board, Emerging Information Technology Conference, University of Dallas, Dallas, Texas, 2006
- Technical MEMS Program Committee of Electronic Device Society, 2006
- Technical Program Committee for MEMS/MOEMS Technologies and Applications Conference as part of the Photonics Asia Symposium, Beijing, China, 2007
- **Associated Editor for the IEEE Journal of Solid-State Circuits, 2006 to 2011**
- IEEE Electron Devices Society MEMS Committee Member, January 2006 to December 2009
- **Chair of the IEEE Electron Devices Society MEMS Committee, January 2010 to 2014**
- Technical Program Committee Member and Conference Publicity Co-Chair for the 4th International Conference on Networked Sensing Systems, Braunschweig, Germany, 2007
- Steering Committee and Technical Program Committee Member the 5th International Conference on Networked Sensing Systems, Kanazawa, Japan, 2008
- Organizing Committee Member and Session Chair for the 9th International Conference on Solid-State and Integrated-Circuit Technology, Beijing, China, 2008
- Steering Committee Member for 2008 Emerging Information Technology Conference
- Technical Program Committee Member and MEMS-Track Co-Chair for the 21st Annual IEEE International SOC Conference, 2008
- Technical Program Committee Member and Industry-Track Co-Chair for the 6th International Conference on Networked Sensing Systems, Pittsburgh, Pennsylvania, 2009
- Technical Program Committee Member for the 22nd Annual IEEE International SOC Conference, 2009

- Medical IC Session Chair, Emerging Information Technology Conference, M.I.T., Cambridge, MA, August 2009
- Technical Program Session Chair, The 5th Annual IEEE International Conference On Nano/Micro Engineered and Molecular Systems, Xiamen, China, 2010
- Steering Committee and Technical Program Committee Member the 7th International Conference on Networked Sensing Systems, Kassel, Germany 2010
- Technical Program Committee Member for the 10th International Conference on Solid-State and Integrated-Circuit Technology, Shanghai, China, 2010
- Technical Program Committee Member of the 22nd Annual IEEE International SOC Conference, 2010
- Technical Program Committee Member and Session Chair of the 24th IEEE International Conference on Micro Electro Mechanical Systems, Cancun, Mexico, 2011
- Technical Program Committee Member and Session Chair of the 25th IEEE International Conference on Micro Electro Mechanical Systems, Paris, France, 2012
- Technical Program Committee Member of the 8th International Conference on Networked Sensing Systems, Penghu, Taiwan, 2011
- Technical Program Session Chair for the 8th International Conference on Networked Sensing Systems, Penghu, Taiwan, 2011
- Technical Program Committee Member of the 9th International Conference on Networked Sensing Systems, Germany, 2012
- Guest Editor for the Special Issue of ACM Journal of Emerging Technologies in Computing Systems (JETC) – Special Issue on Implantable Electronics, Volume 8 Issue 2, June 2012
- Member of Editorial Board of the Journal of Sensor and Actuator Networks, 2012
- Technical Program Committee Member of the 24th Annual IEEE International SOC Conference, 2012
- Technical Session Chair for the 6th Asia-Pacific Conference on Transducers and Micro/Nano Technologies, Nanjing, China, July 2012
- Technical Track Chair and Session Chair of the 34th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, 2012

- Technical Program Committee Member and Session Chair of the International Conference on BioElectronics, BioSensors, BioMedical Devices, BioMEMS/NEMS and Applications, Singapore, November 2012
- IEEE Sensors Council Meritorious Award Committee Member, 2012-2013
- Technical Program Committee Member of the 8th IEEE International Conference on Nano/Micro Engineered and Molecular Systems in 2013
- Chair of IEEE Sensors Council Meritorious Award Committee, 2014
- Technical Program Committee Member of the 9th IEEE International Conference on Nano/Micro Engineering and Molecular Systems, Hawaii, USA, 2014
- Technical Program Committee Member of the 9th IEEE International Conference on Intelligent Sensors, Sensor Networks and information Processing, Singapore, 2014
- Technical Program Committee Member and Session Chair of the International Conference on BioElectronics, BioSensors, BioMedical Devices, BioMEMS/NEMS and Applications, Shanghai China, November 2014
- Technical Program Committee and Track Chair for Circuits and Systems for Biomedical Applications of the IEEE 58th International Midwest Symposium on Circuits and Systems, 2015
- Special Sessions Co-Chair, IEEE Sensors Conference, 2015
- Chair of IEEE Sensors Council Meritorious Award Committee, 2015
- Technical Session Chair of the 29th IEEE International Conference on Micro Electro Mechanical Systems, Shanghai, China, January 2016
- IEEE Sensors Council Technical Achievement Award Committee Member, 2017-2019
- Member of IEEE Sensors Conference Venue Selection Committee, 2018
- IEEE Sensors Council Meritorious Award Committee Member, 2020 and 2021
- Technical Session Chair of the IEEE Sensors Conference, Dallas, Texas, 2022
- Technical Session Chair of the IEEE PowerMEMS Conference, Salt Lake City, Utah, 2022

- Member of Awards Committee of the IEEE PowerMEMS Conference, Salt Lake City, Utah, 2022
- Technical Track Co-Chair of the IEEE Sensors Conference, Kobe, Japan, 2024
- **Editor for the IEEE Transactions on Electron Devices, 2015 to 2024**
- NSF Review Panel for Major Research Instrumentation (MRI) Program, 2004
- NSF Review Panel for Sensor Networks, RF Communications, and VLSI Program, 2004
- NSF Review Panel for SBIR Program, 2003 and 2004
- NSF Review Panel for Sensor Networks, 2005
- NSF CAREER Panel, 2005 and 2008
- NSF Review Panel for Biomedical Engineering / Research to Aid Persons with Disabilities (BME/RAPD) Program, 2006
- NSF Review Panel, 2006
- NSF Review Panel, 2007
- NSF MNS Horizon 2040 Workshop-Health Care Panelist, June 2009
- NSF Biomedical Engineering CAREER Panel, October 2009
- NSF Biomedical Engineering Review Panel, May 2010
- NSF Biomedical Engineering Review Panel, January 2011
- NSF Biomedical Engineering Review Panel, May 2011
- NSF SitS Review Panel, June 2020
- Review for San Francisco State University Faculty Small Grant for Research and Creative Activities, May 2013
- Journal Review for IEEE Journal of MicroElectroMechanical Systems
- Journal Review for IEEE Transactions on Electron Devices
- Journal Review for IEEE Electron Devices Letter

- Journal Review for IEEE Sensors Journal
- Journal Review for Journal of Micromechanics and Microengineering
- Journal Review for IEEE Transaction on Biomedical Engineering
- Journal Review for IEEE Transaction on Biomedical Circuits and Systems
- Journal Review for Sensors & Actuators: A. Physical
- Journal Review for IEEE Transactions on Microwave Theory and Techniques
- Journal Review for Journal of Micromechanics
- Journal Review for Journal of Microelectronics Engineering
- Journal Review for Biosensors
- Journal Review for Journal of Sensors and Sensor Systems
- Journal Review for Microsystems & Nanoengineering

Consulting Service:

MEMSGEN, Burbank, California (2001)

Impact Technologies, LLC, Rochester, New York (2008)

L-3 Communications, Salt Lake City, Utah (2010)

Alfred E. Mann Foundation for Scientific Research (2013)

Expert witness for Alfred E. Mann Foundation for Scientific

Alfred E. Mann Foundation for Scientific Research v. Cochlear Corporation, Cochlear Limited

Patent Infringement Litigation

Nike, Inc. (2014-2015)

Expert witness for Nike, Inc.

Nike, Inc. v. Mayfonk, Inc.

Inter Partes Review

AliphCom d/b/a Jawbone and Bodymedia, Inc. (2016-2017)

Expert witness for AliphCom d/b/a Jawbone and Bodymedia, Inc.

Fitbit, Inc. v. AliphCom d/b/a Jawbone and Bodymedia, Inc.

Inter Partes Review

Boston Scientific (2018-2021)
Expert witness for Boston Scientific
Boston Scientific v. Nevro
Patents Infringement Litigation & Inter Partes Review

Murata Manufacturing Co., Ltd. (2018-2019)
Expert witness for Murata Manufacturing Co., Ltd.
Murata Manufacturing Co., Ltd. v. Inergetic AB
Inter Partes Review

Philips North America, LLC and Koninklijke Philips N.V. (2020-2021)
Expert witness for Philips North America, LLC and Koninklijke Philips N.V.
Philips North America, LLC and Koninklijke Philips N.V. v. Fitbit and Garmin
International Trade Commission (ITC) Case & Inter Partes Review

Sonova AG (2020-2021)
Expert witness for Sonova AG
MED-EL Elektromedizinische Geräte Ges.m.b.H. v. Sonova AG
Inter Partes Review

Momentum Dynamics Corp. (2021-2022)
Expert witness for Momentum Dynamics Corp.
WiTricity v. Momentum Dynamics Corp.
Inter Partes Review

DexCom, Inc. (2022)
Expert witness for DexCom, Inc.
DexCom v. Abbott Diabetes Care, Inc.
Patent Infringement Litigation

Echologics, Inc (2022-2023)
Expert witness for Echologics, Inc,
Echologics v. Orbis Intelligent Systems, Inc.
Patent Infringement Litigation

Masimo Corporation (2023)
Expert witness for Masimo Corporation
Apple, Inc. v. Masimo Corporation
Patent Infringement Litigation

Nintendo Corporation (2024)
Expert witness for Nintendo Corporation
Nintendo Co. v. American GNC Corporation
Inter Partes Review

Bragi GmbH (2024)
Expert witness for Bragi GmbH
K/S HIMPP v. Bragi GmbH
Inter Partes Review

SMAXTEC INC (2024)
Expert witness for SMAXTEC INC. AND SMAXTEC ANIMAL CARE GMBH
SMAXTEC INC v. ST REPRODUCTIVE TECHNOLOGIES, LLC
Inter Partes Review

Mueller Systems, LLC (2024)
Expert witness for Mueller Systems, LLC
SIPCO, LLC v. Mueller Systems, LLC
Patent Infringement Litigation Within License Dispute

Publications:

Journals

1. D. J. Young, B. E. Boser, V. Malba, A. F. Bernhardt, "A Micromachined RF Low Phase Noise Voltage-Controlled Oscillator for Wireless Communication," *International Journal of RF and Microwave Computer-Aided Engineering*, pp. 285-300, 2001.
2. J. Du, W. H. Ko, D. J. Young, "Single Crystal Silicon MEMS Fabrication Based On Smart-Cut Technique," *Sensors and Actuators A*, vol. 112, issue 1, pp. 116-121, 2004.
3. M. Suster, W. H. Ko, D. J. Young, "An Optically-Powered Wireless Telemetry Module for High-Temperature MEMS Sensing and Communication," *IEEE Journal of Microelectromechanical Systems*, June, pp. 536-541, 2004.
4. D. J. Young, J. Du, C. A. Zorman, W. H. Ko, "High-Temperature Single Crystal 3C-SiC Capacitive Pressure Sensor," *IEEE Sensors Journal, Special Issue on Microsensors and Microactuators*, August, pp. 464-470, 2004.
5. R. Wang, W. H. Ko, and D. J. Young, "SiC-Carbide-MESFET-Based High-Temperature MEMS Sensing and Data Telemetry Module," *IEEE Sensors Journal*, December, pp. 1389-1394, 2005.
6. Peng Cong and Darrin J. Young, "Single Crystal 6H-SiC MEMS Fabrication Based On Smart-Cut Technique," *Journal of Micromechanics and Microengineering*, 15, pp. 2243-2248, 2005.
7. M. Suster, J. Guo, N. Chaimanonart, W. H. Ko, and D. J. Young, "A High-Performance MEMS Capacitive Strain Sensing Microsystem," *IEEE Journal of Microelectromechanical Systems*, Vol. 15, issue 5, pp. 1069-1077, 2006.

8. N. Chaimanonart and D. J. Young, "Remote RF Powering for Industrial Strain Sensing Microsystem," *IEEE Sensors Journal*, Volume 6, Issue 2, April, pp. 484 – 489, 2006.
9. W. H. Ko, D. J. Young, J. Guo, M. Suster, H-I Kuo, and N. Chaimanonart, "A High-Performance MEMS Capacitive Strain Sensing System," *Sensors and Actuators A*, vol. 133, pp. 272-277, 2007.
10. D. J. Young, M. A. Zurcher, T. Trang, C. A. Megerian, W. H. Ko, "Characterization of Ossicular Chain Vibration at the Umbo: Implications for a Middle Ear Microelectromechanical System Design," *Ear Nose Throat Journal*; 89(1), pp. 21-26, January 2010.
11. P. Cong, W. H. Ko, and D. J. Young, "Wireless Implantable Blood Pressure Sensing Microsystem Design for Small Laboratory Animals Monitoring," *Sensors and Materials*, Vol. 20, Issue 7, pp. 327-340, 2008.
12. P. Cong, N. Chaimanonart, W. H. Ko, and D. J. Young, "A Wireless and Batteryless 10-bit Implantable Blood Pressure Sensing Microsystem with Adaptive RF Powering for Real-time Genetically Engineered Mice Monitoring," *IEEE Journal of Solid-State Circuits*, Vol. 44, No. 12, pp. 3631-3644 (Special Issue), December 2009.
13. W. H. Ko, R. Zhang, P. Huang, J. Guo, X. Ye, D. J. Young, C. A. Megerian, "Studies of MEMS Acoustic Sensors as Implantable Microphones for Totally Implantable Hearing Aid Systems," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 3, no. 5, pp. 277-285, October 2009.
14. D. J. Young, I. E. Pehlivanoglu, and Christian A. Zorman "Silicon Carbide MEMS Resonator Based Oscillator," *Journal of Micromechanics and Microengineering*, 19, 2009, 115027.
15. P. Cong, W. H. Ko, and D. J. Young, "Wireless Batteryless Implantable Blood Pressure Monitoring Microsystem for Small Laboratory Animals," *IEEE Sensors Journal*, Vol. 10, No. 2, pp. 243-254, 2010.
16. M. Z. Zurcher, M. Semann, C. A. Megerian, W. H. Ko, D. J. Young, "A MEMS Capacitive Accelerometer Design as Middle Ear Microphone Based on Ossicular Chain Micromechanics Characterization at Umbo for Fully Implantable Cochlear Prosthesis," *Sensors and Materials*, Volume 22, Number 6, pp. 297-312, 2010.
17. N. Chaimanonart, M. A. Suster, and D. J. Young, "Two-Channel Passive Data Telemetry with Remote RF Powering for High-Performance Wireless and Batteryless Strain Sensing Microsystem Applications," *IEEE Sensors Journal*, Volume 10, Issue 8, pp. 1375-1382, 2010.
18. O. Bebek, M. A. Suster, S. Rajgopal, M. J. Fu, X. Huang, M. C. Cavusoglu, D. J. Young, M. Mehregany, A. J. van den Bogert, and C. H. Mastrangelo, "Personal

Navigation via High-Resolution-Gait-Corrected Inertial Measurement Units,” *IEEE Transactions on Instrumentation and Measurement*, Volume 59, Number 11, pp. 3018-3027, 2010.

19. R. Wang, P. Cheng, F.Xie, D. Young, and Z. Hao, “A Multiple-Beam Tuning-Fork Gyroscope with High Quality Factors,” *Sensors and Actuators A* 166, pp. 22-33, 2011.

20. D. J. Young, M. Z. Zurcher, M. Semann, C. A. Megerian, W. H. Ko, “MEMS Capacitive Accelerometer-Based Middle Ear Microphone,” *IEEE Transactions On Biomedical Engineering*, Vol. 59, No. 12, pp. 3283-3292, December 2012.

21. Xing Liu, Hui Chen, Qing-An Huang, Darrin J. Young, “A MEMS-Based Intraoperative Monitoring System for Improved Safety in Lumbar Surgery”, *IEEE Sensors Journal*, Volume 13, Issue 5, pp. 1541-1548, 2013.

22. R. Surapaneni, Q. Guo, Y. Xie, D. J. Young and C. H, Mastrangelo, "A three-axis high-resolution capacitive tactile imager system based on floating comb electrodes," *Journal of Micromechanics and Microengineering*, Vol. 23, No. 7, 075004, 2013.

23. Q. Guo, M. A. Suster, R. Surapaneni, C. H, Mastrangelo, D. J. Young, “High-Performance Interface Electronic System for a 13 x 13 Flexible Biomechanical Ground Reaction Sensor Array Achieving a Gait Ground Velocity Resolution of 100um/sec,” *IEEE Sensors Journal*, Vol. 13, No. 11, pp. 4496-4505, 2013.

24. R. Yang, Z. Wang, J. Lee, K. Ladhane, D. J. Young and P. X.-L. Feng, “6H-SiC microdisk torsional resonators in a “smart-cut” technology,” *Applied Physics Letters* 104, 091906, 2014.

25. Y.-P. Hsu and D. J. Young, “Skin-Coupled Personal Wearable Ambulatory Pulse Wave Velocity Monitoring System Using Microelectromechanical Sensors,” *IEEE Sensors Journal, Special Issue*, Vol. 14, No. 10, pp. 3490-3497, 2014.

26. D. J. Young, P. Cong, M. A. Suster, M. Damaser, “Implantable Wireless Battery Recharging System for Bladder Pressure Chronic Monitoring,” *Lab on a Chip*, DOI: 10.1039/C5LC00821B, 2015, 15, 4338-4347.

27. Q. Guo, W. Deng, O. Bebek, C. Cavusoglu, C. Mastrangelo, and D. Young, “Personal Inertial Navigation System Assisted by MEMS Ground Reaction Sensor Array and Interface ASIC for GPS-Denied Environment,” *IEEE Journal of Solid-State Circuits*, Vol. 53, No. 11, pp. 3039-3049 (Special Issue), November 2018.

28. H. Basaeri, Y. Yu, D. Young, and S. Roundy, “A MEMS-Scale Ultrasonic Power Receiver for Biomedical Implants,” *IEEE Sensors Letters*, 2019.

29. D. Young and P. Cong, "Wireless Implantable Sensors: From Lab to Technology Breakthrough Ambitions," *Sensors and Actuators A*, 2019.
30. H. Bsaeri, Y. Yu, D. Young, and S. Roundy "Acoustic Power Transfer for Biomedical Implants Using Piezoelectric Receivers: Effects of Misalignment and Misorientation," *Journal of Micromechanics and Microengineering* (Special Issue), 2019.
31. Y. Yu, T. Nguyen, P. Tathireddy, S. Roundy, and D. J. Young, "Wireless Inductive Sensing and Robust Packaging for Future Implantable Hydrogel-based Glucose Monitoring Applications," *IEEE Sensors Journal*, 2020.
32. W. Luo, V. Sharma, and D. J. Young, "A Paper-Based Flexible Tactile Sensor Array for Low-Cost Wearable Human Health Monitoring," *IEEE Journal of Microelectromechanical Systems*, Vol. 29, No. 5, pp. 825-831, 2020.
33. Y. Yu, T. Nguyen, P. Tathireddy, S. Roundy, and D. J. Young, "A Wireless Battery-less and Moisture-Resistant Packaged Glucose Sensing System Employing Hydrogel-based Inductive Sensing Technique and Low-power ASIC for Long-term Glucose Monitoring," *Sensors and Actuators A*, July, 2022.
34. J. M. Arteaga, J. Sanchez, F. Elsakloul, M. Marin, C. Zesiger, N. Pucci, G. J. Norton, D. J. Young, D. E. Boyle, E. M. Yeatman, P. D. Hallett, S. Roundy, and P. D. Mitcheson, "High Frequency Inductive Power Transfer Through the Soil for Agricultural Applications", *IEEE Transactions on Power Electronics*, 2023. DOI: 10.1109/TPEL.2023.3305642
35. W. Luo, V. Sharma, and D. J. Young, "A sock-embedded wireless sensing system employing paper-substrate-based MEMS tactile sensors array and low power ASIC for accurate and comfortable AFib monitoring," *IEEE Sensors Journal*, Volume; 23, Issue: 24, December, 2023. DOI: 10.1109/JSEN.2023.3329100
36. J. Sanchez, J. M. Arteaga, C. Zesiger, P. Mitcheson, D. Young, S. Roundy, "Inductive Power Transfer Coil Misalignment Perception and Correction for Wirelessly Recharging Underground Sensors," *Sensors* 2025, 25(2), 309; <https://doi.org/10.3390/s25020309>
37. S. Ding, S. Roundy, R. Goel, C. Zesiger, and D. J. Young, "Wireless Battery-Free Active and Passive Underground Soil Moisture Sensing System Design and Field Characterization for Future Agricultural," Submitted to *IEEE Sensors Journal*, 2025.

Peer Reviewed Conference Papers

1. D. J. Young, B. E. Boser, "A Micromachined Variable Capacitor for Monolithic Low-Noise VCOs," *Technical Digest, IEEE Solid-State Sensor and Actuator Workshop*, Hilton Head Island, SC. June 1996, pp. 86-89.

2. D. J. Young, B. E. Boser, "A Micromachine-Based RF Low-Noise Voltage-Controlled Oscillator," *Technical Digest, IEEE Custom Integrated Circuits Conference*, Santa Clara, CA. May 1997, pp. 431-434.
3. D. J. Young, V. Malba, J. J. Ou, A. F. Bernhardt, B. E. Boser, "Monolithic High-Performance Three-Dimensional Coil Inductors for Wireless Communication Applications," *Technical Digest, International Electron Devices Meeting*, Washington, D.C. December 1997, pp. 67-70.
4. V. Malba, D. J. Young, J. J. Ou, A. F. Bernhardt, B. E. Boser, "High-performance RF coil inductors on silicon," *48th IEEE Electronic Components & Technology Conference*, 1998, pp. 252-255.
5. D. J. Young, V. Malba, J. J. Ou, A. F. Bernhardt, B. E. Boser, "A Low-Noise RF Voltage-Controlled Oscillator Using On-chip Three-Dimensional Coil Inductor and Micromachined Variable Capacitor," *Technical Digest, IEEE Solid-State Sensor and Actuator Workshop*, Hilton Head Island, SC. June 1998. pp. 128-131.
6. D. J. Young, J. L. Tham, B. E. Boser, "A Micromachine-Based Low Phase-Noise GHz Voltage-Controlled Oscillator for Wireless Communication," *Technical Digest, The 10th International Conference on Solid-State Sensors and Actuators (Transducers '99)*, Sendai, Japan, June 1999. pp. 1386-1389.
7. D. J. Young, "Micromachining for RF Communications," *MRS Bulletin on Microelectromechanical Systems: Technology and Applications*, Volume 26, No. 4, pp.331-332, April 2001.
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Invited Book Chapters

1. D. J. Young, “Analog Front-End and Telemetry Systems”, *Chapter 3 of Bioimplantable Systems: Design Principles and Applications*. Elsevier Inc. 2015.
2. D. J. Young and H. Kim, “Microelectromechanical Systems (MEMS),” *Chapter 18 of Guide to State-of-the-Art Electronic Devices*, Jonh Wiley & Sons Ltd, 2013.
3. C-K Lu and D. J. Young, "Wireless MEMS Accelerometer for Small Laboratory Animal Activity Monitoring," *VDM Verlag Dr. Muller Aktiengesellschaft & Co. KG*, 2000 (ISBN: 978-3-8364-7595-2).
4. D. J. Young, C. A. Zorman, and M. Mehregany, “MEMS/NEMS Devices and Applications,” in *The Springer Handbook of Nanotechnology – 2nd Edition*, B. Bhushan, ed., Springer-Verlag, Berlin, 2006.
5. D. J. Young, “MEMS for communication systems,” in *Standard Handbook of Electronic Engineering*, Donald Christiansen and Charles K. Alexander, McGraw-Hill, NY, NY, 5th Edition, 2002, Chapter 17.8, pp. 17.144-17.147.
6. S. Rajgopal, C.A. Zorman, D.J. Young, M. Mehregany, “Reliability and MEMS,” *The CRC Handbook of Mechanical Engineering*, F. Kreith and D. Y. Goswami, Second Edition, Boca Ration [Fla.]: CRC Press, 2005, pp. 15.68-15.83.
7. P. Cong, W. H. Ko, and D. J. Young, “Integrated Electronic System Design for Implantable Wireless Batteryless Blood Pressure Sensing Microsystem,” *IEEE Communications Magazine*, April 2010.
8. D. Young and H. Kim, “A Review of MEMS Innovation.” In: Joachim Burghartz (Editors in Chief), *the IEEE EDS Yearbook: 50th Anniversary Review of Electrical Revolutions*, Chapter 18: IEEE, Amsterdam: Elsevier, 2013.

Patents

US 8,634,924 “MEMS Implanted Acoustic Sensor”

US 8,602,999 “Implantable Flat Blood Pressure Sensing Cuff Structure And Implantable Blood Pressure Monitoring Device Using The Cuff Structure”

Presentations

“Wireless Sensing Technology with Example Applications” presented at ECE Department Graduate Seminar, University of Utah, February 4, 2026.

“Microsystem Research Overview and Graduate Programs in ECE Department at University of Utah” presented at School of Electronics and Information Engineering, Soochow University, March 27, 2018.

“About Graduate School” presented at the ECE Department Junior Seminar, University of Utah, Salt Lake City, Utah, March 3, 2017

“Smart and Robust Sensing Systems Enabled by MEMS Technology and Low Power Electronics”, presented at the graduate seminar, Electrical and Computer Engineering Department, University of Utah, Salt Lake City, Utah, August 29, 2016.

“MEMS-Based Personal Navigation System Under GPS-Denied Environment”, presented at the Mechanical Engineering Department Seminar, National Taiwan University, Taiwan, R.O.C., June, 6, 2016.

“Smart and Robust Sensing Systems Enabled by MEMS Technology and Low Power Electronics”, presented at Shanghai Institute of Microsystem and Information Technology, Chinese Academy of Sciences, May 20, 2016.

“Bionic Human”, panel discussion of Leo After Hours at the Leonardo Museum, Salt Lake City, Utah, March 10, 2016.

“Smart and Robust Sensing Systems Enabled by MEMS Technology and Low Power Electronics” presented at GE HealthCare, Salt Lake City, Utah, February 10, 2016.

“Smart and Robust Sensing Systems Enabled by MEMS Technology and Low Power Electronics” presented at School of Electronics and Information Engineering, Soochow University, January 22, 2016.

“Smart and Robust Sensing Systems Enabled by MEMS Technology and Low Power Electronics” presented at Vimicro Corporation, Beijing, China, December 17, 2015.

“Smart and Robust Sensing Systems Enabled by MEMS Technology and Low Power Electronics” presented at the Institute of Microelectronics, Tsing Hua University, December 16, 2015.

“Smart and Robust Sensing Systems Enabled by MEMS Technology and Low Power Electronics” presented at the institute of Electronics, Chinese Academy of Sciences (IECAS), Beijing, China, December 13, 2015.

“Smart and Robust Sensing Systems Enabled by MEMS Technology and Low Power Electronics” presented at School of Electronics Engineering and Computer Science, Peking University, December 8, 2015.

“Smart Sensing Applications Enabled by Micro-Electro-Mechanical Sensors and Low-Power Integrated Circuits,” presented at the Computer Engineering Students Seminar,

Univeristy of Utah, November 25, 2014.

“Skin-Coupled Wearable Ambulatory Health Monitoring System Using MEMS Sensors,” presented as an invited talk at the International conference on BioSensors, BioElectronics, BioMedical Devcies, BioMEMS/NEMS and Applications 2014 & The 5th Symposium on Green Microsystem Technology, Shanghai, China, November 17, 2014,

“Skin-Coupled Wearable Ambulatory Pulse Wave Velocity Monitoring System Using MEMS Sensors” RF & Wireless Day, University of Utah, September 16, 2014.

“High Performance Microsystems Enabled by MEMS and Integrated Circuits,” presented at the Electrical and Computer Engineering Department Graduate Seminar, Univeristy of Utah, September 16, 2013.

“Biomedical Implant System Design Enabled by MEMS and CMOS IC,” presented at the International Conference on BioElectronics, BioSensors, BioMedical Devices, BioMEMS/NEMS and Applications, Singapore, November 20, 2012.

“Wireless Microsystem Technologies and Future Direction”, presented at National Taiwan University, October 31, 2012.

“MEMS-Based Biomedical Implant Research and Commercialization Opportunities”, presented at the 6th Asia-Pacific Conference on Transducers and Micro/Nano Technologies, Nanjing, China, July 10, 2012.

“Wireless Microsystem Technologies for Biomedical Implant Applications”, presented at the Electrical and Computer Engineering Department Graduate Seminar, Univeristy of Utah, September 16, 2011.

“MEMS-Based Biomedical Implant Development and Commercialization Opportunity”, presented at the 4th International Workshop on Innovation and Commercialization of Micro & Nano Technology (ICMAN 2010), Shanghai, China, 11-23-2010.

“Low-Interference Sensing Electronics for High-Resolution Error-Correcting Biomechancial Ground Reaction Sensor”, presented at the IEEE Sensors Conference, Big Island, Hawaii, 11-2-2010.

“Wireless Biomedical Microsystems”, presented at the Bioengineering Seminar, Unvieirty of Utah, 10-6-2010.

“Low-Interference Sensing Electronics for High-Resolution Error-Correcting Biomechancial Ground Reaction Sensor”, presented at the Nanotechnology session of the Emerging Information Technology Conference, Stanford, CA, 8-14-2010.

“A Wireless and Batteryless Blood Pressure Sensor for Laboratory Mice *In Vivo* Real-Time Monitoring”, presented at the CMOS Emerging Technology Workshop, Whistler, British Columbia, Canada, May 20, 2010.

“Interface Electronics for MEMS-Based Wireless Sensing Applications”, presented at the *IEEE International Symposium on VLSI Design, Automation and Test*, Hsinchu, Taiwan, R.O.C., April, 28, 2010.

“Wireless Implantable Microsystem Applications In Animal And Human” presented at Zhe Jiang University, March 25, 2010.

“Wireless Biomedical Microsystems”, presented at the Bioengineering Seminar, Unvierity of Utah, February 17, 2010.

“Wireless Batteryless Less-Invasive Blood Presusre Sensing Microsystem for Advanced Biomedical Application,” presented at the 23rd IEEE International Conference on Micro Electro Mechanical Systems, Hong Kong, China, January, 2010

“Wireless Power Recharging for Implanatble Bladder Pressure Chronic Monitoring,” presented at the 5th IEEE International Conference on Nano/Micro Engineered and Molecular Systems, Xiamen, China, January 2010.

“Wireless Microsystem Technology For Biomedical Applications,” presented at the Bioengineering Department Seminar, Univeristy of Utah, November 20, 2009.

“Development of Wireless BatterylessImplantanble Blood Pressure-EKG-Core Boby Temperature Sesning Microsystem For Genetically Engineerd Mice Real Time Monitoring,” presented at the 3rd IEEE International Conference on Nano/Molecular Medicine and Engineering, Taiwan, R.O.C. October 21, 2009.

“Wireless Microsystem Technology With Focus on Biomedical Applications”, presented at the graduate seminar of Mechanical Engineering Department, National Taiwan University, October 19, 2009.

“Wireless MEMS-Based Strain Sensing Microsystem” presented at the Department of Information and Elctronic Engineering, Zhejiang University, Hangzhou, China, 9-22-2009.

“Wireless and Batteryless Biomedical Implant Microsystem” presented at the Shanghai Institute of Microsystem and Information Technology, Shanghai, China, 9-21-2009.

“Wireless MEMS Strain Sensing Technology for Military Applications” presented at the Worshop on the Research and Evaluation of NEMS/MEMS sponsored by US Army AMRDEC, DARPA, and SMDC, Redstone Arsenal, Alabama, 9-9-2009.

“Wireless Powering and Data Telemetry for Biomedical Implants”, presented in *the 31st*

Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC-2009), Minneapolis, Minnesota, 9-4-2009.

“Wireless Batteryless 10-bit Implantable Blood Pressure Sensing Microsystem for Real-Time Genetically Engineered Mice Monitoring”, presented at the Medical IC session of the Emerging Information Technology Conference, Cambridge, MA, 8-6-2009.

“Wireless Microsystem Research”, presented at EECS Department Seminar at Case Western Reserve University, Cleveland, Ohio, 1-20-2009.

“Wireless Microsystem Technology for Biomedical Applications”, presented at the ECE Department Seminar at University of Utah, Salt Lake City, Utah, 12-1-2008.

“Wireless Implantable EMG Sensor for Powered Prosthesis Control”, presented in *the 9th International Conference on Solid-State and Integrated-Circuit Technology*, Beijing, China, 10-22-2008.

“CMOS Cross-Coupled VCO Architecture Comparison at 2GHz and 16GHz”, presented in *the 9th International Conference on Solid-State and Integrated-Circuit Technology*, Beijing, China, 10-22-2008.

“Wireless Microsystem Technology for Biomedical Applications”, presented at State Key Laboratory of Precision Measuring Technology & Instruments, Tianjin University, China, 10-20-2008

“Wireless Microsystem Technology for Biomedical Interface”, presented at the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, 10-16-2008.

“Wireless Microsystem Technology for Biomedical Applications”, presented at the ECE Department graduate seminar, University of Utah, September 23, 2008.

“MEMS Middle Ear Microphone for Fully Implantable Cochlear Prosthesis”, presented at National Chiao Tung University, Taiwan, R.O.C., 6-26-2008.

“Wireless Implantable Less-Invasive Blood Pressure Sensing Microsystem for Small Laboratory Animal Real-Time Monitoring”, presented at the Asia-Pacific Conference on Transducers and Micro-Nano Technology, Taiwan, R.O.C., 6-25-2008.

“Wireless Less-Invasive Blood Pressure Sensing Microsystem for Small Laboratory Animal *In Vivo* Real-Time Monitoring”, presented at the 5th International Conference on Networked Sensing System (INSS 2008), Kanazawa, Japan, 6-18-2008.

“MEMS Middle Ear Microphone for Fully Implantable Cochlear Prosthesis”, presented at Tsinghua University, Beijing, China, 2-26-2008.

“Wireless Less-Invasive Blood Pressure Sensing Microsystem for Small Laboratory Animal *In Vivo* Real-Time Monitoring”, presented at the Institute of Microelectronics, Beijing University, Beijing, China, 2-25-2008.

“Wireless Less-Invasive Blood Pressure Sensing Microsystem for Small Laboratory Animal *In Vivo* Real-Time Monitoring”, presented at Southeast University, Nanjing, China, 2-18-2008.

“Wireless Less-Invasive Long-Term Implantable Blood Pressure Sensing System for Small Animal Real-Time Monitoring”, presented at the 2007 International Electron Devices and Materials Symposia (IEDMS), Taiwan, R.O.C., 11-29-2007.

“Wireless Implantable EMG Sensing Microsystem”, presented at the 2007 Biomedical Engineering Society Annual Fall Meeting, Los Angeles, California, 9-29-2007.

“MEMS Accelerometer with Low Noise Interface Circuits as Middle Ear Microphone for Fully Implantable Cochlear Prosthesis”, presented at the College of Mechanical Engineering, Dalian Technical University, Dalian, China, 10-27-2006.

“Microsystem Technology for Fully Implantable Cochlear Prosthesis”, presented at the 6th Emerging Information Technology Conference, University of Texas at Dallas, Dallas, Texas, 8-11-2006.

“Wireless Strain Sensors”, presented at The Timken Company, Canton, Ohio, 2-28-2006.

“Wireless Strain Sensing Microsystem Technology”, presented at Southeast University, Nanjing, China, 9-6-2005.

“Microsystem Technology for Totally Implantable Cochlear Prosthesis”, presented at National Cheng Kung University, Tainan, Taiwan, R.O.C., 8-18-2005.

“Wireless Strain Sensing Microsystem”, presented as a plenary session talk at the 5th Emerging Information Technology Conference, National Taiwan University, Taipei, Taiwan, R.O.C., 8-18-2005.

“Microsystem Technology for Totally Implantable Cochlear Prosthesis”, presented at Neural Prosthesis Seminar at Case Western Reserve University, Cleveland, Ohio, 7-15-2005

“Strain Sensing Microsystem with Remote RF Powering and Data Telemetry”, presented at ECE Department Seminar, Carnegie Mellon University, Pittsburgh, Pennsylvania, 3-18-2005.

“High-Temperature Wireless Sensing Technology and Its Future Directions”, presented at Beijing University, Beijing, China, 10-21-2004.

“A High-Performance Strain Sensing Microsystem with Remote RF Power Capability,” presented at the 7th International Conference on Solid-State and Integrated-Circuit Technology (ICSICT-2004), Beijing, China, 10-19-2004.

“High-Performance MEMS Capacitive Strain Sensing Microsystem”, presented at Army Research Office MEMS Meeting, Arlington VA. 6-24-2004.

“Single Crystal SiC Capacitive Pressure Sensor at 400 °C”, presented at International Electron Devices Meeting, Washington D.C., 12-10-2003.

“3D-MEMS Approaches to Varactors”, presented at IEEE International Microwave Symposium, Philadelphia, Pennsylvania, 6-9-2003.

“MEMS for RF Applications”, presented at MEMGen Corporation, Burbank, CA, 11-20-2002.

“RF MEMS and Its Applications”, presented at Institute of Microelectronics, TsingHua University, Beijing, China, 7-30-2002.

“Micromachined RF Low Phase Noise Voltage-Controlled Oscillator”, presented at Shanghai Institute of Microsystem and Information Technology, Shanghai, China, 7-15-2002.

“MEMS Technologies for Radio Frequency (RF) and Optical Applications” Short Course Presented at Adriatic Research Institute, Berkeley, California, 4-26-2002.

“Micromachined Low Phase Noise Voltage-Controlled Oscillator”, EE Department Seminar, University of Southern California, Los Angeles, California, 3-19-2002.

“MEMS for RF Communications”, Tutorial Session Presented at Material Research Society (MRS) Fall Meeting, Boston, Massachusetts, 11-26-2001.

“Micromachined Low Phase Noise RF Voltage-Controlled Oscillator”, EECS Department Seminar, University of Michigan, Ann Arbor, Michigan, 11-5-2001.

“MEMS for Wireless Communication”, presented at National Taiwan University, Taipei, Taiwan. R. O. C. on 10-29-2001.

“Key Technology for MEMS”, Short Course Presented at Precision Instrument Development Center, National Science Council, The Executive Yuan, Taiwan, R. O. C., 10-22-2001 to 10-25-2001.

“MEMS & Communications - A Complimentary System” presented at Glennan Microsystem Initiative Meeting, Independence, Ohio on 6- 9-2000.

“RF Voltage-Controlled Oscillator (VCO) with MEMS Capacitors and Inductors” presented at Naval Research Lab workshop on Military Application of RF MEMS, Washington, D.C. on 2-8-2000.

“MEMS for Wireless Communications” presented at Symposium of Micromachine and Microtechnology, Bio-Medical, and Industrial Prospects at Case Western Reserve University, Cleveland, Ohio on 11-12-1999.

“A Low-Noise RF VCO Using On-Chip High-Q 3-D Coil Inductor & Micromachined Variable Capacitor” presented at Solid-State Circuit Technology Committee Workshop, Arlington, Virginia on 11-15-1999.

Research Grants and Contracts:

- “Robotic Platform For Precision Irrigation Management Using Passive Zero Maintenance, Long-Life, Buried Sensors” (Co-PI)
USDA, September 1, 2024 to August 30, 2027
- “SitS NSF-UKRI: Wireless In-Situ Soil Sensing Network for Future Sustainable Agriculture” (PI)
NSF, January 1, 2020 to December 31, 2024
- “Acoustic Power and Communication Platform for a Deeply Implanted Intraluminal Glucose Sensor” (Co-PI)
NSF, September 2014 to August 2018
- “Implantable Particle-Based Wireless Neurorecording Probes for Minimally-Intrusive 3D Mapping of Brain Signals” (Co-PI)
NSF, September 2015 to August 2017
- “Soldier Navigation via High-Resolution-Gait-Corrected IMUs” (Co-PI)
DARPA, July 2008 to December 2013
- “Robust Design of High Performance MEMS Resonators” (Co-PI)
NSF, September 2008 to August 2011
- “Wireless Implantable Pressure Monitor for Clinical Applications” (Co-PI)
U.S. Department of Veteran Affairs, September 2007 to August 2010
- “Intelligent Sensor Network for In-Vivo Real Time Biological Dynamic Systems” (PI)
NSF, September 2003 to August 2007
- “RF Data Transmitter and Receiver Modules for MEMS Strain Sensors” (PI)
Army Research Office, June 2002 to May 2007
- “Wireless EMG Sensing Microsystem” (PI)

Veteran Affairs Medical Center, Cleveland, Ohio, September 2005 to August 2006

- “Harsh Environment MEMS Sensing and Communications” (PI)
NASA, October 1999 to October 2002
- “MEMS Tunable Antennas for Power-Efficient Wireless Communications” (PI)
NASA, October 2001 to March 2003
- “Nerve Reshaping” (Co-PI)
NIH, July 2005 to June 2010
- “Micro Sensor for Cochlear and Middle Ear Implant Systems” (Co-PI)
NIH, December 2003 to November 2005
- “Microfabricated Strain Sensor Module with Interface Circuits” (Co-PI)
Army Research Office, June 2002 to May 2007
- “Fast Bonding Techniques for Attaching Strain Sensors to Steel Substrates” (Co-PI)
Army Research Office, June 2002 to May 2007
- “Multi-Range Pressure Sensor Arrays For Harsh Environment” (Co-PI)
NASA, October 1999 to October 2002

Graduate Research Students Advising:

- Weijie Luo
Ph.D. Dissertation Title: “A Wireless Wearable Atrial Fibrillation Monitoring System Employing A Paper-Based MEMS Tactile Sensors Array And A Low-Power ASIC”
Completion Date: December 2021
- Yuechuan Yu
Ph.D. Dissertation Title: “A Wireless and Battery-less Hermetically Packaged Glucose Sensor Employing Hydrogel-Based Inductive Sensing Scheme and Low-Power ASIC for Future Long-Term Implantable Glucose Monitoring”
Completion Date: March 2020
- Qingbo Guo
Ph.D. Dissertation Title: “Personal Inertial Navigation System Assisted By MEMS Wearable Ground Reaction Sensor Array and Sensing Interface ASIC”
Completion Date: December 2018
- Xing Chen (Post-Doc, 2015-2016)
Packaging Technology for Biomedical Implant Applications
- Michael Suster

Ph.D. Dissertation Title: “Wireless Strain Sensing Microsystem with External RF Powering and Two-Channel Data Telemetry Capability”
Completion Date: July 2011

- Nattapon Chaimanonart
Ph.D. Dissertation Title: “Wireless Implantable Multi-channel Bio-sensing Network with Adaptive RF Power Control for Untethered Laboratory Animal Real-time Monitoring”
Completion Date: July 2009
- Peng Cong
Ph.D. Dissertation Title: “Wireless Less-invasive Implantable Blood Pressure Sensing System for Small Animal Real-time Monitoring”
Completion Date: December 2008
- Kalyan Ladhane
M.S. Project Title: “SiC MEMS”
Completion Date: December 2013
- Bradley Farnsworth
M.S. Thesis Title: “Wireless Implantable EMG Sensor”
Completion Date: May 2010
- Mark Zimmerman
M.S. Thesis Title: “*In Vivo* RF Powering for Advanced Biological Research”
Completion Date: June 2008
- Engin Pehlivanoglu
M.S. Thesis Title: “Silicon Carbide MEMS Oscillator”
Completion Date: January 2008
- Mark Zurcher
M.S. Thesis Title: “Development of a MEMS Middle Ear Acoustic Sensor for a Fully Implantable Cochlear Prosthesis”
Completion Date: November 2007
- Matthew Cross
M.S. Thesis Title: “Architectural Comparison Study for RF and Microwave CMOS Voltage-Controlled Oscillators”
Completion Date: August 2007
- Cheng-Kuan Lu
M.S. Thesis Title: “Wireless MEMS Accelerometer for Real-time Small Laboratory Animal Activity Monitoring”
Completion Date: July 2007

- Steven Mallin
M.S. Thesis Title: “Phase Noise Processes and Suppression Techniques in RF LC Voltage-Controlled Oscillators”
Completion Date: May 2006
- Nattapon Chaimanonart
M.S. Thesis Title: “Remote Radio-Frequency Powering System for Wireless MEMS Strain Sensor”
Completion Date: January 2006
- Jiangang Du
M.S. Thesis Title: “Single Crystal Silicon Carbide Capacitive Pressure Transducer”
Completion Date: May 2005
- Run Wang
M.S. Thesis Title: “Silicon-Carbide-MESFET-Based 400 °C MEMS Sensing and Data Telemetry”
Completion Date: August 2004
- Michael Suster
M.S. Thesis Title: “High-Temperature MEMS Wireless Sensing and Telemetry”
Completion Date: August 2004
- Brian Quach
M.S. Thesis Title: “Micromachined Touch Mode Tunable Capacitor Design and Fabrication for RF Communication Applications”
Completion Date: August 2003

Teaching and Courses Development:

- ECE 4900 Senior Clinic Project (Undergraduate Course)
- ECE 6720 Analog Integrated Circuit Design (Graduate Course)
- ECE 6722 Analog Integrated Circuit Testing (Graduate Course)
- ECE 6960 Electrical Interface for MEMS (Graduate Course)
- ECE 6730 Radio Frequency Integrated Circuits Design (Graduate Course)
- ECE 2280 Fundamentals of Engineering Electronics (Undergraduate Course)
- ECE 3110 Electronic Circuits-II (Undergraduate Course)
- EECS 426 MEMS for Sensing and Communication (Graduate Course)

- EECS 600 Radio-Frequency Integrated Circuits Design (Graduate Course)
- EBME 418 Applied Electronics for Biomedical Engineering (Selected Lectures, Graduate Course in BME)
- EECS 344 Microelectronic Design and Analysis (Undergraduate Course)
- EECS 245 Electronic Circuits (Undergraduate Course)
- ENRG 210 Introduction to Circuits and Instruments (Undergraduate Course)

EXHIBIT 8

Matthew J. Schurman, Ph.D

5 Purdue Drive • Richboro, PA • 18954

732-266-4155

matthewjschurman@gmail.com

I am a seasoned technologist that has worked across a wide variety of technologies and industries and have remained hands on in research and product development despite holding many VP and CTO positions. I have utilized my strong communication talents to raising money for technology companies, writing successfully awarded grants, and providing expert witness support in complex IP litigation.

Education

- Ph.D. Materials Science and Engineering, Rutgers University , New Brunswick, NJ, 1996
- BA Physics, Franklin and Marshall College, Lancaster PA, 1993

Areas of Expertise

Medical Devices: glucose monitoring (spot, continuous, non-invasive), statistical methods for evaluation of glucose monitors, insulin delivery, optical diagnostics (i.e. pulse oximetry, peripheral perfusion) with particular expertise in optical skin coupling and light / tissue interaction, Continuous glucose monitor sensor design (materials and structure), clinical design for validation of devices (working with clinicians and IRBs on study design and approval), Hydrogen Peroxide sterilization.

Optics and Optoelectronics: Optical system design, laser and detector technologies, Light Emitting Diodes, Interferometry, spectrometry, low coherence interferometry, optical coherence tomography, fiber optic sensors and communication, machine vision, image processing and analysis

Materials Science and Engineering: semiconductors (silicon, compound semiconductors), metals and metallurgy (particularly for medical devices), silicones and adhesives (particularly for space applications)

Semiconductors: semiconductor fabrication (lithography, etching, metal deposition) for optical semiconductors, optical chips, semiconductor material growth (metal organic vapor deposition MOCVD), optical semiconductor device design

Data Analysis and Algorithm Development: Data and image analysis, analog / digital filtering, signal extraction, machine learning (Python, MATLAB)

Computer Languages: Python, MATLAB, C / C++ (embedded microcontrollers), Labview. All utilized in development and final deployment in products. Experienced with all major operating systems (Windows, Mac, Linux)

System Design and Automation: optical, mechanical, electrical, and control design and implementations for capital equipment through small battery powered embedded systems, test equipment automation, PLCs, motor and motion control.

Work Experience

- **CEO and Co-Founder**, Optical Roasting Technologies, 2022 to Present. Co-Inventor of innovative new roasting technology. Launched a new coffee brand, Outer Worlds Coffee, based on the technology and began shipping coffee to customers roasted with our custom designed and built equipment.
- **Principal / Consultant**, 2009 to Present.
 - **Managing Partner, newForge Technologies.** Co-founder and managing partner of an interdisciplinary engineering team serving the engineering and research needs of the semiconductor, alternative energy, medical device, and specialty lighting industries
 - **Medical Device Subject Matter Expert:** Reviewed 1000s of lines of code in embedded consumer medical device. Provided analysis of algorithm used and demonstrated the part of the code that executed it enabling client's successful defense during a patent litigation.
 - **Medical Device Subject Matter Expert:** Successfully defended 80% of patent claims from 4 patents during Patent Office **Inter Partes Review** as part of a broader patent litigation. This enabled the continuation and eventually settlement of the litigation. Provided expert witness testimony during depositions.
 - **Grove Instruments.** Managed team of 12 scientists and engineers in the development of novel non-invasive technology. Introduced Agile development to the team, turned around failing program and helped raise over \$3 million in funding
 - **Helmsley Charitable Trust.** Consulted on trust funded emerging technologies for the management of type one diabetes. Developed road map, solicitation, web content and intellectual property terms and conditions for grant.
- **Chief Technology Officer**, GlucoLight Corporation, 2002 to 2009. \$12M biomedical device company. Took on raw technology for non-invasive glucose monitor, developing into viable product. Executed due diligence, writing, evaluating and securing patent portfolio. Managed operation that included 13 employees. Conducted pre-clinical and clinical trials.
- **VP – Engineering / Director of Operations**, Qusion Technologies, 2001 to 2002. Turned around \$11M start-up telecom firm with depleted financial resources. Refined failing technology. Slashed overhead 40%. Started from scratch, developing successful state-of-the-art clean room. Secured \$500K from secondary financial backer. Managed 10 employees.
- **Director, Application & Advanced Research Group**, Emcore Corporation, 1994 to 2001. \$200M technology corporation. Hired as staff scientist while completing Ph.D. Directed team of 30 scientists, engineers and technicians. Researched capital equipment, solid state lighting and optoelectronics. Oversaw field engineering and training.

Publications In Last 10 Years

Application: 63/373,283

Patent US10791971B2

Patent US20190216370A1

UV Degradation of Space Solar Cell Assemblies under High Temperature and Irradiance, PVSC Talk 2018

Path Dependent Power Degradation via UV Exposure for the Parker Solar Probe, Space Power Workshop 2018

Characterizing UV Degradation of Silicone in Solar Cells: Lessons Learned from the Parker Solar Probe and the Game Changing Development for Extreme Environments Solar Power Programs, Space Power Workshop 2021

EXHIBIT 9

Gregory Kesden

5605 Pocusset St, Pittsburgh, PA 15217, 412-818-7813 gkesden@gmail.com,
<http://www.andrew.cmu.edu/~gkesden>

ACADEMIC EXPERIENCE

July 2020 ***Teaching Professor and MS.ECE Advisor***

Present **Electrical and Computer Engineering Dept, Carnegie Mellon University (Pittsburgh, PA)**

- Develop and deliver ugrad and grad courses, especially intro, software systems, and embedded system including:
 - 18-613/18-213/15-213/18-213/14-513: Computer Systems
 - 18-349: Embedded Systems
 - 18-344: Hardware-Software Interface
 - 18-100: Introduction to Electrical and Computer Engineering
- Advise masters students in the MS in ECE programs and certain high-risk/complex undergraduates
- Various University- and department- level committee service
- Recent promotion to full *Teaching Professor* from *Assoc. Teaching Professor* effective 7/1/2026

July 2017 ***Associate Teaching Professor and Academic Advisor***

July 2020 **Information Networking Institute, Carnegie Mellon University (Pittsburgh, PA)**

- Developed and delivered graduate courses, especially those in cloud computing infrastructure, distributed systems, networking, and other areas of software systems, especially for audiences in ECE and INI Masters programs. Recent examples include:
 - 14-513: Computer Systems
 - 14-712: Operating Systems
 - 14-736: Distributed Systems
 - 14-740: Networks
 - 14-760: Adv. Real-World Data Networks
 - 14-848: Cloud Infrastructure
 - 14-642/18-349: Embedded Systems
- Courses included those for local, bicoastal, remote, and hybrid constituencies.
- Advise masters students in the MSIN program (Distributed Systems, Networking, Cloud)
- Departmental faculty senate representative
- MS admissions committee member

June 2015 ***Senior Lecturer and Director of Masters Programs***

July 2017 **Computer Science and Engineering, University of California at San Diego**

- Developed and delivered undergraduate and graduate courses, especially in systems, software engineering, and at the intro-level. Examples include:
 - CSE 110: Software Engineering (Upper-division undergrad)
 - CSE 120: Operating Systems (Spring 2016, Upper-division undergrad)
 - CSE 291: Distributed Systems (Spring 2016, Graduate, Mostly Masters)
 - CSE 291: Cloud Computing (Fall 2016, Graduate, Mostly Masters)
 - CSE 8A: Intro to Computer Science in Java (Non-major and no-experience undergrads)
- Provide leadership for the department's Masters programs, notably the MS in CS/CE program. Primary responsibility included collaborating to implement year-over-year growth from a class size of ~80 students to ~250 students (program size from ~200 students to ~500 students over 2 years), with the goal of simultaneously improving program quality, as well as serving as chair/co-chair of the MS Committee, and serving on the MS Admissions Committee.

July 1999 **Associate Teaching Professor** (2005 – 2015)
 May 2015 **Carnegie Mellon University - School of Computer Science**
 Director of Educational Computing/Undergraduate Laboratories (2004 – 2015)
 Associate/Assistant Director, MS in CS Program (2012 – 2015)
 Teaching Assistant Program Coordinator (2012 – 2015)
 Academic Advisor (2009 – 2015)
 Lecturer (1999 – 2005)

- Developed and delivered courses at all levels of the curriculum, especially systems courses, first year and lower-division courses, and eBusiness courses. Mentored large student projects, including summer-long industry-sponsored practicums. Taught and supported pipeline programs. Examples included:
 - 08-705: Networks (Masters)
 - 08-706; Web Services Integration (Masters)
 - 15-640/440: Distributed Systems (Masters + Advanced undergrads)
 - 15-615/415: Databases (Masters + Advanced undergrads)
 - 15-641/441: Networks (Masters + Advanced undergrads)
 - 15-513/213: Computer Systems (Masters + Undergrads)
- Developed and led collaborations with industry and alumni to increase investment in academic programs, improve student access to opportunities in industry, and increase cross-pollination. Accomplishments included the development of the Citadel Commons, industry sponsored collaborative computer clusters, and sponsorship for the programming team and contests.
- Led educational initiatives, including the growth of 15-213 into the summer and to include 15-513 (to specially reach out to MS students), then to include distance offerings, then to include online (but proctored) exams for all students (local and distance).
- Provided leadership for, and management of, college-level educational computing, including servers and services, virtualization, public laboratories, public printing, robots, laptops, desktops, etc, as well as certain cross-campus and inter-campus services as per collaborative agreements. Fostered and worked as part of cross-campus and inter-campus collaborations
- Managed student-side administration of the MS in CS program, including advising, registration, waitlist management, academic auditing and academic actions; serve on admissions committee; designed and maintain handbook; defined thesis option; maintained policies with evolving goals and landscape.
- Coordinated teaching assistant program for computer science, including assignment, development program, talk series, pay rates, policies; chair advisory committee, etc.
- Advised students. Undergraduates (2009-2014). MS in CS students (2012 –2015).
- Coached Carnegie Mellon's world-renowned ACM-ICPC programming team and direct our own regionally acclaimed spring contest (2000-2010)

August 1999 **Lecturer**

July 1999 **Computer Science Department, Clemson University**

- Delivered courses and coordinated multi-section labs at the 100, 200, and 300 levels.

INDUSTRY EXPERIENCE

March 1998 **Consulting Forensic Computer Scientist**

Present **Pittsburgh, PA**

- I apply my education and experience assisting in criminal, civil, and administrative cases in federal, state, and local jurisdictions, including patent, copyright, trade secret cases, and class action cases.
- Matters have spanned technical domains including, but not limited to, source code and software architecture, data centers and infrastructure, network and file system, mobile devices, reconstruction of runtime environments, complex systems and large scale environments, security and integrity, robustness/availability, software engineering and development processes, IT systems integration, software development and IT project management, requirements analysis and specification, operations, and routine tool-based forensic analysis and evidence collection.
- I have been retained by plaintiffs, defendants, law firms, courts, and those considering or fearful of future legal action.
- I have testified in federal and state jurisdictions.
- Please see my consulting list for information about cases and roles.

ACADEMIC-INDUSTRY COLLABORATIONS

- Intellectual collaboration with, as well as monetary gifts from, *Dropbox* in support of the renovation of a technology-enhanced collaborative space as well as other programs/projects. Dropbox invested \$150,000 and significant intellectual collaboration. (Carnegie Mellon, 2015) Collaboration with *Apple*, including their donation of \$160,000 equipment and service contracts, in support and improvement of an existing student collaboration and laboratory facility (Carnegie Mellon, 2015)
- Intellectual collaboration with, as well as equipment gifts and monetary gifts from, *Citadel* in support of the design and construction of a new collaborative commons, new server infrastructure for big data and machine learning, two new computer laboratories, and one upgraded computer laboratory. Citadel has invested \$750,000.00+ in cash (\$400,000), equipment (\$100,000+), furniture (\$250,000), and significant intellectual collaboration. (Carnegie Mellon, 2014)
- Intellectual collaboration with, and financial support from, *IMC Financial Markets* to support in support of competition programming. *IMC* invested \$90,000 in monetary gifts for the development of Carnegie Mellon's programming teams, as well as in the regional and global travel of the teams for competition. Additionally, *IMC* collaborated in the planning, staffing, and operating of regional programming contest and workshop, for which they additionally provided student prizes. (Carnegie Mellon, 2010 – 2012)
- Investment from *Red Hat* in support of undergraduate computer laboratories, including a \$100,000 monetary gift, as well as technical collaboration and providing speakers to campus to speak about Open Source software. (Carnegie Mellon, 2010)

WORKSHOPS, PANELS, AND PUBLICATIONS

- CSCNE '09 Panel: Integrating DISC Into the Computer Science Curriculum (April 2009)
- Justin Carlson, Gregory Kesden, "Virtual Scrimmaging", ACM-ICPC CLIS (April 2009)
- G. Kesden, E. Fink, and D. Sleator, "Training...for the ACM Programming Competition", ACM-ICPC CLIS (April 2009)
- Non-traditional publication include extensive library of open, Web-accessible coursenotes.
- Operating and Distributed Systems notes frequently referenced by those outside of the University.
- DeCSS tutorial, even years after its authoring, continues to be an especially frequently cited and heavily referenced work.
- Reviewer for Wiley's The Handbook of Computer Networks articles on *Distributed Systems* and also *Personal Communication Services*

EDUCATION

- **Clemson University**, Clemson, South Carolina
M.S., Computer Science (1999)
- **Clemson University**, Clemson, South Carolina
B.S., *Cum Laude*, Computer Science (1996)

RECENT HONORS

- The Benjamin Richard Teare Teaching Award (with Jimmy Zhu, CIT, CMU, 2023-2024)
 - The CIT college-level annual teaching award, "made to a faculty member within the College of Engineering in recognition of excellence in engineering education."
- Joel and Rut Spira Excellence in Teaching Award (with Jimmy Zhu, ECE, CMU, 2020-2021)
 - The once-per-year department-level award recognizing major accomplishments and outstanding qualities and strengths in teaching.
- The A. Nico Habermann Education Service (SCS, CMU, 2014-2015)
 - The one-per-year college-level award for educational service other than direct teaching.
- ACM-ICP Coach Award (ACM-ICPC, 2010)
 - Award recognizing a coach for consistently bringing teams to the ACM-ICPC world finals.
- *Andrew Carnegie Society*, "Remember Your Professor" Award of Merit (CMU, 2007)
 - An annual alumni recognition
- The *Simon Award for Teaching Excellence* (SCS, CMU, 2004-2005)
 - The one-per-year college-level award for outstanding teaching

GREGORY KESDEN
Consulting List 2/27/2026

- Shirokov et al v. Ardatovskiy et al (3:25-cv-02701-EMC, CAND) [Filed : 3/21/2025]
 - I was retained by the plaintiff
 - My role involved the analysis of source code.
- Truesight v LGL (2:24-CV-00031-JRG, TXED) [Filed: 1/22/2024]
 - I was retained by the plaintiff
 - My role involved the analysis of source code.
- LiTL v HP / Microsoft v LiTL (1:23-cv-00120-RGA, DED) [Filed: 10/16/2023]
 - I was retained by Microsoft, an Intervenor-Plaintiff, in a patent infringement case
 - My role involved analysis of source code.
- Thentia v. Julian Cardarelli, GovCore Canada (CV-24-00726669-0000, Ontario Super. Ct.)
 - I am retained by the plaintiff in a trade secret case
 - My role involves the analysis of source code and other technical materials.
- FaceTec v iProov/iProov v FaceTech (2:21-cv-02252-ART-BNW, NVD). [Filed: December 28, 2021]
 - I am retained by the defendant/counterclaimant in a patent infringement case
 - My role involves the analysis of source code and other technical materials.
- Autoscribe Corporation v. Repay Holdings Corporation et a, (2-23-cv-00349, TXED) [Filed July 27, 2023]
 - I was retained by the defendant in a patent infringement case
 - My role was to review source code and other technical material
- UUSI, LLC v. Apple (3:18-cv-04637, CAND). [Filed: August 2, 2018]
 - I am retained by the plaintiff in a patent infringement and trade secret case
 - My role involves the analysis of source code and other technical materials.
- UUSI, LLC v. Samsung (15-cv-00146, MIWD). [Filed: February 13, 2015]
 - I am retained by the plaintiff in a patent infringement and trade secret case
 - My role involves the analysis of source code and other technical materials.
- SitePro v Waterbridge (6:2023cv00115, TXWD) [Filed: February 13, 2023]
 - I am retained by the plaintiff in a patent infringement and trade secret case
 - My role involves the analysis of source code, including ladder diagrams, among other technical materials.
- CosmoKey GmbH v Duo LLC (1:18-cv-01477-CFC, DED) [Filed: July 24, 2020]
 - I am retained by the plaintiff in a patent infringement case
 - My role involves the analysis of source code
- SurgeTech LLC v Uber Technologies, Inc. (1:2022cv00882, DED) [Filed: June 29, 2022]
 - I am retained by the plaintiff in a patent infringement case
 - My role involves the analysis of source code
- Playvuu, Inc. v. Snap, Inc. (2:2022cv06019, CACD) [Filed: August 24, 2022]
 - I am retained by the plaintiff in a patent infringement case
 - My role involves the analysis of source code

- Better Browsing LLC v Opera Norway AS (2:23-cv-00020, TXED) [Filed: January 17, 2023]
 - I am retained by the defendant in a patent infringement case.
 - My role involves providing expert analysis and consulting w.r.t. Internet architecture, infrastructure, and services.
- IOEngine LLC v Roku Inc (6:21-cv-01296, TXWD) [Filed: December 14, 2021]
 - I am retained by the plaintiff in a patent infringement case
 - My role involves the analysis of source code
- OANDA v Gain Capital (3:20-cv-05784-BRM-DEA, DNJ) [Filed: May 11, 2020]
 - I am retained by the plaintiff in a patent infringement case
 - My role involves the analysis of source code
- Meishe v. TikTok Inc. et al (6:21-cv-00504, TXWD, now CAND) [Filed: May 18, 2021]
 - I am retained by the plaintiff in a patent infringement case
 - My role involves the analysis of source code
- Versata Software, Inc. f/k/a Trilogy Software, Inc. and Versata Development Group, Inc. v. Configit A/S, (2:20-cv-09019-JAK-MRW, C.D. Cal.) [Filed: October 1, 2020]
 - I am retained by the plaintiff in a patent infringement case
 - My role involves the analysis of source code
- Aqua Connect, Inc. v. TeamViewer US, LLC (No. 1:2018cv01572, D.Del) [Filed: May 15, 2020]
 - I am retained by the plaintiff in a patent infringement case
 - My role involves the analysis of source code
- Tennessee v. Patrick Albano (No. C2004171, Shelby County, TN) [Filed: January 23, 2020]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics and review of discovery related to computer, mobile, network systems, and cloud storage systems
- Alabama v. Benny Rex Smith [No. 67200324001, Alabama 25th Circuit) [Filed: April 9, 2020]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics and review of discovery related to computer, mobile, network systems, and cloud storage systems
- Alabama v. Veneziano (No. CC-2020-000218.00, Coffee County Circuit, AL) [Filed: 11/20/2020]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics and review of discovery related to computer, mobile, network systems, and cloud storage systems
 - I testified at a motions hearing.
- Fortinet, Inc. v. Forescout Technologies, Inc (No. 3:2020cv03343, D.Del) [Filed: October 11, 2018]
 - I am retained by the plaintiff in a patent infringement case
 - My role involves the analysis of source code
- Fintiv v. Apple (No. 6:18-cv-372-ADA, W.D. Tex) [Filed: December 21, 2018]
 - I am retained by the plaintiff in a patent infringement case
 - My role involves the analysis of source code

- Tennessee v. Patrcik Albano (No. 1:2018cv01572, D.Del) [Filed: October 11, 2018]
 - I am retained by the plaintiff in a patent infringement case
 - My role involves the analysis of source code
- Election Systems & Software, LLC v. Smartmatic USA Corp. (1:18-cv-01259-UNA,DED) [Filed: August 17, 2018]
 - I am retained by the plaintiff in a patent infringement case
 - My role involves the analysis of source code
- Beal v. Outfield Brewhouse LLC (2:18-cv-4028, W.D. Missouri) [Filed: February 13, 2018]
 - I am retained by the plaintiff in a TCPA class action case
 - My role involves the analysis of source code, system infrastructure and configuration, deployable logs and other records and system artifacts.
- State of Missouri vs. Bryan Austin Prather (18PU-CR00375-01, Pulaski Cty) [March 23, 2018]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics and review of discovery related to computer and network systems
- State of Missouri vs Maumau V. Benner (18PU-CR00374-01, Pulaski Cty) [March 23, 2018]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics and review of discovery related to computer and network systems
- State of Alabama v Kirt Brown (CC-20-18-001998.00, Houston Cty) [January 23, 2018]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics and review of discovery related to computer and network systems
 - I testified as an expert in a motions hearing
- State of Alabama v Justin Romine (44-CC-2014-000747.00, Lauderdale Cty) [August 29, 2014]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics
 - Testified in pre-trial hearing.
- State of Alabama v Jeffery Dale Hunt (41-CC-2017-000215.00, Lauderdale Cty) [March 23, 2017]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics and review of discovery related to computer and network systems
- US v Hesham Jamaledin (4:2018-cr-00503-162780, MOED) [June 14, 2018]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics and review of discovery related to computer and network systems
 - I testified as an expert at trial.
- State of Oklahoma vs Thomas Keith Meyers (CF-2017-2523, Tulsa Cty) [Filed: April 27, 2017]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics and review of discovery related to computer and network systems
- Commonwealth of Kentucky v. Swart (16-CR-146, Oldham Cty) [Filed: August 12, 2016]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics
- Commonwealth of Kentucky v. Ronaghi (ECB-16-0258 / KSP 35-16-0051, Jefferson Cty) [October 26, 2016]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics

- United States v. Guy Rosenschein (16-CR-4571, NMD) [Filed: November 9, 2016]
 - I am retained by the defendant in a criminal matter
 - My role involves digital forensics
- State of Florida vs Ronald Scott Sherman (90-1801-007263, Broward Cty) [Filed: January 31, 2018]
 - I am retained by the defendant in a criminal matter
 - My role involves review of discovery related to computer and network systems
- State of Alabama vs Wayde Slocumb (CC18-5218 RAH, Madison Cty) [Filed: December 15, 2017]
 - I am retained by the defendant in a criminal matter
 - My role involves review of discovery related to computer and network systems
 - Testified as an expert at trial.
- Commonwealth of Kentucky vs Patrick Voegelé (18-Cr-00023, Owen Cty) [Filed: May 14, 2018]
 - I am retained by the defendant in a criminal matter
 - My role involves review of discovery related to computer and network systems
- Semcon v. Huawei (Case No. 2:16-CV-00437-JRG-RSP, EDT) [Filed: April 25, 2016]
 - I am retained by the plaintiff
 - My role involves software-related patents, and especially the analysis of software and technical documents
- Intellectual Ventures I LLC v. Nextel Operations, Inc. and Sprint Spectrum L.P. (DED)
Intellectual Ventures I LLC v. T-Mobile USA, Inc. et al., (DED)
 - Intellectual Ventures I LLC v. United States Cellular Corporation (DED) [Filed: October 7, 2013]
 - I am retained by the defendant in a software patent matter
 - My role involves software-related patents, and especially the analysis of source code
- The You Business [Engaged: 9/10/2015 – 10/30/2015]
 - Provided advice about software development
 - This work was unrelated to litigation.
- United States v. Matthew Graves (KYWD) [Filed: 3/21/2016]
 - I was retained by the defense in a criminal matter
 - My role involves system and network analysis, as well as data carving, data recovery, and digital forensics, as well as supervising others I the same.
- United States v. Jason Dion Johnson (TXSD) [Filed: 11/10/2016]
 - I was retained by the defense in a criminal matter
 - My role involves system and network analysis, as well as data carving, data recovery, and digital forensics, as well as supervising others I the same.
- Illinois v. Norman Eisenberg (Cook County, Illinois) [Filed: 12/16/2014]
 - I was retained by the defense in a criminal matter
 - My role involves system and network analysis, as well as data carving, data recovery, and digital forensics, as well as supervising others I the same.

- United States v. Christopher Welshans (PAWD)
 - I was retained by the defense in a criminal matter
 - I work pro bono for the local county and federal public defenders offices when requested and able.
 - My role in this matter involved digital forensics, including end user hosts and peer-to-peer file sharing systems.
- Twin Peaks Software, Inc. vs IBM Corporation (CAND) [Filed: August 29 2014]
 - I was retained by the defendant in a software patent matter.
 - My role involved software-related patents, and especially the analysis of software systems and distributed file system design and evolution.
- Audatex v. Mitchell (CASD) [Filed: July 1, 2013]
 - I was retained by the plaintiff in a software patent matter.
 - My role involved software-related patents, and especially the analysis of source code
- TLI Communications MDL 2534 [Filed: February 26, 2014]
 - I was retained by the plaintiff in a matter involving one or more software patents
 - My role involved assisting the plaintiff in various technical capacities.
- GlobeRanger v. Software AG (TXND) [Filed: March 1, 2011]
 - I was retained by the plaintiff in a matter involving intellectual property, trade secrets, software, and RFID devices.
 - My role involves analyzing source code.
- Commonwealth of PA v. Proctor (MJ-37301-CR44-2014 Warren Co., PA) [Filed: March 22, 2014]
 - I was retained by the defense in a criminal matter involving activity on a mobile device
 - My role involved the analysis of a mobile phone w.r.t. the installed and installable software, use cases representing activities of interest, and the particulars of the mobile device's UI and likely user behaviors with respect to those interfaces.
- In Re Provider Meds, L.P. (TXNB) [Filed: February 6, 2013]
 - I was retained by as a court-appointed expert in a bankruptcy matter
 - My role involved the analysis of source code to determine the similarity between two bodies of work.
- eDirectPublishing, Inc v. LiveCareer, Ltd, et al. (PAWD) [Filed: May 8, 2012]
 - I was retained by the plaintiff in a software patent matter.
 - My role involved software-related patents, and especially the analysis of source code and computer systems
- Leslie Arrington, et al v. ColorTyme, Inc, et al (PAWD) [Filed: October 25, 2012]
 - I was retained by the defense
 - The nature of the matter involves wiretapping
 - My role involved advising counsel on issues involving the analysis of network and service infrastructure as well as data carving, data recovery, and end- system digital forensics.
- US v. Hawkins (PAWD) [Filed: September 13, 2013]
 - I was retained by the defense in a criminal child pornography case
 - My role involves system and network analysis, as well as data carving, data recovery, and digital forensics.

- US v. Cunningham (PAWD) [Filed: October 25, 2012]
 - I was retained by the defense in a criminal child pornography case
 - I work pro bono for the local county and federal public defenders offices when requested and able.
 - My role involved system and network analysis, as well as data carving, data recovery, and digital forensics.
- PeopleKeys, Inc., et al, v. e-Psyence, et al (OHND) [Filed: July 1,2013]
 - I was retained as a court-appointed independent expert
 - The nature of the matter involves intellectual property expressed software, copyright protection and/or trade secrets.
 - My role involved the analysis of source code and other evidence related to the development of software, as well as the analysis of elements of the user interface and user interaction.
- EC Data Systems, Inc. v j2 Global, Inc. and Advanced Messaging Technologies, Inc. (CACD) [Filed: February 21, 2012]
 - I was retained by the plaintiff
 - The nature of the matter involved intellectual property expressed in software and related patents.
 - My role involved the analysis of source code.
- Illinois v. Lawrence Verdick (Whiteside County, IL) [Filed: January 3, 2012]
 - I was retained by the defendant
 - My role involved system and network analysis, as well as data carving, data recovery, and digital forensics.
- Telebuyer, LLC v. Amazon, et al (WAWD) [Filed: July 22, 2013]
 - I was retained by the plaintiff in a patent matter
 - My role involved the analysis of computer systems, including user interfaces and source code.
- Dynamic Advances v. Apple (NYND) [Filed: October 19, 2012]
 - I was retained by the plaintiff in a software patent matter.
 - My role involved software-related patents, and especially the analysis of source code
- Athena Health Inc. v. AdvancedMD Software (MAD) [Filed: July 18,2011]
 - I was retained by the plaintiff in a software patent matter.
 - My role involved software-related patents, and especially the analysis of source code
- Quad/Tech v. QIPC, et al (PAED) [Filed 06/05/2009]
 - I was retained by the plaintiff in a false advertising case
 - My role involved analyzing the capabilities of a computerized printing system, and the especially the capabilities and the correctness of integrated software.
- DVSI v. University of Phoenix (VAED) [Filed 03/03/2008]
 - I was retained by the plaintiff in a software patent matter.
 - My role involved analyzing the software and system architecture, and associated source code w.r.t related patents.
- Ariba v. Coupa (CAND) [Filed 03/23/2012]
 - I was retained by the plaintiff in a software patent matter.
 - My role involved software-related patents, and especially the analysis of source code
 -

- Rogers v. Brindle (Cobb County, GA) [Filed 09/19/2012]
 - I am retained by the plaintiff in a civil matter related to sexual harassment in the workplace.
 - My role involves the management of electronic evidence and digital forensics.
- Rovi Corporation (DED, CAND, ITC) [Approximate file date: 12/2011]
 - I was retained by the plaintiff in several software patent matters opposing Amazon.com, LG Electronics, Mitsubishi, Netflix, Roku, Vizio, Apple, and related to Hulu.
 - My role involved software-related patents, and especially the analysis of source code
- Realtime Data, LLC v Morgan Stanley, et al; Realtime Data, LLC v CME Group, et al (NYSD) [Approximate file date: 9/26/2011]
 - I was retained by the plaintiff in a software patent matter.
 - My work on the infringement team involved software-related patents, and especially the analysis of source code
 - My work on the damages team was as the testifying expert for the technical basis of the damages model, and involved analysis of low-latency, long haul networking, ata centers, and co- location.
- Yardi v. RealPage (CACD) [Filed: 1/24/2011]
 - I was retained by the defendant, who also asserted counterclaims, in a trade secret matter involving the delivery of cloud-based services.
 - My role involved analyzing cloud and data center management techniques.
- Venango County Board of Elections – Forensic Investigation [Investigationordered:5/26/2011]
 - I was retained by the election board to analyze the security profile and position of the host environment for the systems managing the county election.
 - My roles involved incident response, forensic investigation, system analysis, and security consulting.
- Janel v. World Logistics (SDNY) [Filed: 2/11/2008]
 - I was retained by the defendant, who also asserted counterclaims, in a securities fraud matter
 - My role involves analyzing software intellectual property, as well as source code and system and software architecture.
- CVS v. Mildon Bus Lines (PAWD) [Filed: 11/30/2009]
 - I am retained by the defendant in a TCPA matter
 - My role involves system and network analysis, the analysis of a production environment, and digital forensics.
 - Testified at deposition.
- Illinois v. Shank (Whiteside County, IL) [Filed: 10/18/2006]
 - I was retained by the defense in a criminal child pornography case
 - My role involved system and network analysis, as well as data carving, data recovery, and digital forensics.

EXHIBIT 10

Daniel Lanovaz

Precedia Associates LLC

104 Calabasas Rd
Freedom, CA 95076
(408) 499-9908

daniel@precedia.com
<http://www.precedia.com>

SUMMARY

Technical consultant, Computer Scientist, Entrepreneur, Animator, Graphic Artist, and Spherographer

- ▶ Software Patent Litigation Support experience
- ▶ High-tech startup experience
- ▶ Object-oriented technology and language runtimes
- ▶ Computer and network security, Digital Rights Management
- ▶ Technical Writing
- ▶ 3D Computer Animations, Graphic Artwork

POSITIONS

Precedia Associates LLC

Freedom, CA

Principal and Co-Founder

1999–present

Performed all aspects of a legal consulting and litigation support business: technical analyses and litigation support for the client, client management, contract review, business development, and accounting.

Legal Consulting and Litigation Support Expertise

- ▶ Support infringement and non-infringement mappings
- ▶ Claim construction development/analysis; mapping construed claims to software systems
- ▶ Technical analyses of products including hardware disassembly and network sniffing
- ▶ Production review, analyses, and summary
- ▶ Creating technology tutorials at Markman hearings and Trial
- ▶ Copyright analyses relating to software
- ▶ Patent portfolio analyses
- ▶ Pre-litigation infringement analyses, invalidity, claim mappings
- ▶ Prior art analyses
- ▶ Trade-Secret analyses
- ▶ Source Code analyses

Legal Consulting Patent Technology Areas

- ▶ Cloud computing
- ▶ Browsers, Web application servers, HTTP, TCP/IP
- ▶ Storage devices
- ▶ Cellular devices (including LTE)
- ▶ Content streaming devices (set-top boxes)
- ▶ Packet-scanning and content-scanning appliances
- ▶ Google Android and Apple iOS internals including OS kernel
- ▶ H.264 image compression algorithms
- ▶ Digital Rights Management (DRM)
- ▶ Cryptographic toolkits, SSL/TLS

- ▶ Sputtering Machines
- ▶ Language runtimes (Java, .NET, JavaScript)
- ▶ Software platforms: Linux, OS X, Windows, .NET, Android, iOS
- ▶ Software languages: C, C++, Java, JavaScript, Objective-C, Assembly Language (x86, MIPS, ARM), Lisp, Smalltalk, C#

Legal Clients

Google, Inc.	Salesforce.com
Carr & Ferrell LLP	White & Case LLP
Bunsow De Mory Smith & Allison LLP	Oblon Spivak McClelland Maier and Neustadt LLP
Keker & Van Nest LLP	Winston Strawn LLP
Dickstein Shapiro LLP	Durie Tangri
Quinn Emanuel Urquhart & Sullivan, LLP	Akin Gump Strauss Hauer & Feld LLP
Finnegan Henderson Farabow Garrett & Dunner LLP	Wilmer Cutler Pickering Hale and Dorr
Marshall, Gerstein & Borun LLP	The Lanier Law Firm
Morgan Lewis & Bockius	Day Casebeer Madrid & Batchelder LLP
Cooley Godward LLP	Vinson & Elkins LLP
InterTrust Technologies	Arnold & Porter Kaye Scholer LLP
Hytera Communications	Waymo
Qualcomm	RPT Legal Strategies, PC
Calfee, Halter & Griswold	Mei & Mark LLP
Step toe & Johnson LLP	Dovel & Luner LLP
Friedman Kaplan Seiler & Adelman LLP	Spotify, Inc.
IBM Corporation	Fortis LLP
Perkins Coie LLP	Commvault Systems, Inc.
Paul Hastings	Kilpatrick

Software Clients

- ▶ **Virtual Health Networks**; supplied security architecture insight for a physician-patient messaging system; developed secure messaging client; scrubbed business plan and summary
- ▶ **Bigvine.com**; rescued the product release process for a Kleiner-Perkins funded 30 person WebLogic engineering-production team
- ▶ **Xythos Software**; supported under-resourced development team by off-loading creation of WebDAV installation software
- ▶ **Tendara**; software development and business development for a handheld XML database start-up
- ▶ **Si2Si, Italy**; solved SSL/TLS integration issues for a network security appliance start-up

SPYRUS**Santa Clara, CA**

Technical Director, PKI Products

1998 - 1999

- ▶ Principal security consultant for Visa Open Platform Version 2.0 Java Card™ standard
- ▶ Drove implementation of the public-key smart card authentication products: secure e-mail, SSL authentication, Windows 95/NT login authentication

Technologies: Java, SSL/TLS, SET, PKI, smart cards, PKCS, C/C++

Day Casebeer Madrid & Batchelder LLP**Cupertino, CA**

Legal Technical Expert

1999

- ▶ Provided copyright infringement analysis of Java source code

Technologies: Java, C/C++, Tcl

Terisa Systems**Los Altos, CA**

Engineering Manager and Senior Software Engineer

1995–1998

Recruited as employee number 4 to this Internet security start-up:

- ▶ Architect/engineer of a Secure Electronic Transaction (SET) Java cardholder application
- ▶ Implemented a C++ Secure HTTP protocol extension to Netscape Navigator; reverse engineered Navigator's user-interface and key store for tight plug-in integration

Technologies: SSL/TLS, SET, Java servlets, HTTP, TCP/IP, smart cards, C/C++, Perl, Solaris, Windows

ParcPlace Systems**Sunnyvale, CA**

Engineering Manager, Computer Scientist

1991–1995

Joined Xerox PARC spin-off:

- ▶ Promoted to Smalltalk Language Kernel group manager
- ▶ Project leader, architect, and developer of Smalltalk's Object Linking and Embedding (OLE) frameworks; wrote 200 page manual
- ▶ Designed and implemented Smalltalk's native language interface; wrote 200 page manual
- ▶ Smalltalk for NeXTSTEP Port Engineer; designed an Objective-C multi-threaded VM

Technologies: Smalltalk, C, Assembly, X11, Unix, graphics, and internationalization

Artificial Intelligence Lab, U. of A.**Edmonton, Canada**

Research Scientist

1988-1991

- ▶ Implemented a Prolog-based non-monotonic reasoning system

- Implemented efficient intelligent backtracking for Prolog

EDUCATION**University of Alberta****Edmonton, Canada**

Master of Science, Computing Science

Thesis: *GÖDEL, A Graphically Oriented Development Environment for Logic Programming*, <https://bit.ly/3uPR6aU>

Bachelor of Science, Computing Science

REFEREED PUBLICATIONS

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CERTIFICATE OF SERVICE

Investigation No. 337-TA-1478

The undersigned certifies that on April 8, 2026, true and correct copies of the foregoing document were served upon the following parties as indicated below:

The Honorable Lisa R. Barton Secretary to the Commission U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, SW, Room 112 Washington, DC 20436	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☒ Via Electronic Filing (EDIS) ☐ Via Electronic Mail
The Honorable Doris Johnson Hines Administrative Law Judge U.S. INTERNATIONAL TRADE COMMISSION 500 E Street, S.W. Washington, D.C. 20436 Email: JohnsonHines1478@usitc.gov	☐ Via Hand Delivery ☐ Via Overnight Delivery ☐ Via First Class Mail ☒ Via Electronic Filing (EDIS) ☒ Via Electronic Mail
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